Main Category	Category	Terms
Disease	OMIM_DISEASE	Breast cancer, somatic
Disease	OMIM_DISEASE	Migraine, susceptibility to
Disease	OMIM_DISEASE	Myocardial infarction, susceptibility to
Disease	OMIM_DISEASE	Estrogen resistance
Disease	UP_KW_DISEASE	Disease variant
Disease	UP_KW_DISEASE	Osteoporosis
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription regulation
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Golgi apparatus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Membrane
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cell membrane
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Activator
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	DNA-binding
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Receptor
Functional Annotations	UP_KW_PTM	Glycoprotein
Functional Annotations	UP_KW_PTM	Lipoprotein
Functional Annotations	UP_KW_PTM	Methylation
Functional Annotations	UP_KW_PTM	Palmitate
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation
Functional Annotations	UP_SEQ_FEATURE	CARBOHYD:O-linked (GlcNAc) serine

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic and acidic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Low complexity
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Polar residues
Functional Annotations	UP_SEQ_FEATURE	DNA_BIND:Nuclear receptor
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Estrogen receptor N-terminal
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:NR LBD
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Nuclear receptor
Functional Annotations	UP_SEQ_FEATURE	LIPID:S-palmitoyl cysteine
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:C->A: Loss of hormone binding capacity and temperatu sensitive loss in DNA-binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:I->C: Loss of transmembrane localization, no effect on peripheral membrane localization. Impairs activation of estrogen-i activation of NOS3 and production of nitric oxide. No effect on bin ERES.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:L->A: Abolishes interaction with NCOA1, NCOA2 and NC
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:L->P: Impairs AF-1 transactivation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:R->A,K: Loss of methylation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: Decreases phosphorylation and transactivation a Abolishes AF-1 transactivation. Insensitive to PPP5C inhibition of transactivation activity.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: Loss of cyclin A-dependent induction of transcrip activation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->E: Enhances transactivation activity. Insensitive to P inhibition of transactivation activity.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:V->E: Has higher transcriptional activity in the absence type ER. Inhibits transcriptional activity when coexpressed with the type receptor.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->A: Reduces PELP1-mediated activation of transcript activity.
Functional	UP_SEQ_FEATURE	MUTAGEN:Y->P: Impairs AF-1 transactivation.

Main Category	Category	Terms
Annotations		
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Hinge
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with AKAP13
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with DDX5; self-association
Functional Annotations	UP_SEQ_FEATURE	REGION:Mediates interaction with DNTTIP2
Functional Annotations	UP_SEQ_FEATURE	REGION:Modulating (transactivation AF-1); mediates interaction w MACROD1
Functional Annotations	UP_SEQ_FEATURE	REGION:Required for interaction with NCOA1
Functional Annotations	UP_SEQ_FEATURE	REGION:Self-association
Functional Annotations	UP_SEQ_FEATURE	REGION:Transactivation AF-2
Functional Annotations	UP_SEQ_FEATURE	ZN_FING:NR C4-type
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	antral ovarian follicle growth
Gene Ontology	GOTERM_BP_DIRECT	epithelial cell development
Gene Ontology	GOTERM_BP_DIRECT	chromatin remodeling
Gene Ontology	GOTERM_BP_DIRECT	regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	signal transduction
Gene Ontology	GOTERM_BP_DIRECT	phospholipase C-activating G protein-coupled receptor signaling p
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cytosolic calcium ion concentration
Gene	GOTERM_BP_DIRECT	androgen metabolic process

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_BP_DIRECT	male gonad development
Gene Ontology	GOTERM_BP_DIRECT	regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	nuclear receptor-mediated steroid hormone signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	estrogen receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	response to estradiol
Gene Ontology	GOTERM_BP_DIRECT	regulation of toll-like receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of smooth muscle cell apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of canonical NF-kappaB signal transduction
Gene Ontology	GOTERM_BP_DIRECT	steroid hormone receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of DNA-binding transcription factor activity
Gene Ontology	GOTERM_BP_DIRECT	response to estrogen
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of nitric oxide biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	fibroblast proliferation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of fibroblast proliferation
Gene Ontology	GOTERM_BP_DIRECT	stem cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	regulation of inflammatory response
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of nitric-oxide synthase activity
Gene	GOTERM_BP_DIRECT	positive regulation of DNA-binding transcription factor activity

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_BP_DIRECT	RNA polymerase II preinitiation complex assembly
Gene Ontology	GOTERM_BP_DIRECT	uterus development
Gene Ontology	GOTERM_BP_DIRECT	vagina development
Gene Ontology	GOTERM_BP_DIRECT	prostate epithelial cord elongation
Gene Ontology	GOTERM_BP_DIRECT	prostate epithelial cord arborization involved in prostate glandular morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	regulation of branching involved in prostate gland morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	mammary gland branching involved in pregnancy
Gene Ontology	GOTERM_BP_DIRECT	mammary gland alveolus development
Gene Ontology	GOTERM_BP_DIRECT	epithelial cell proliferation involved in mammary gland duct elonga
Gene Ontology	GOTERM_BP_DIRECT	protein localization to chromatin
Gene Ontology	GOTERM_BP_DIRECT	cellular response to estrogen stimulus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to estradiol stimulus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to oxygen-containing compound
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of miRNA transcription
Gene Ontology	GOTERM_BP_DIRECT	regulation of epithelial cell apoptotic process
Gene Ontology	GOTERM_CC_DIRECT	chromatin
Gene Ontology	GOTERM_CC_DIRECT	euchromatin
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	transcription regulator complex
Gene	GOTERM_CC_DIRECT	cytoplasm

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_CC_DIRECT	Golgi apparatus
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene Ontology	GOTERM_CC_DIRECT	plasma membrane
Gene Ontology	GOTERM_CC_DIRECT	membrane
Gene Ontology	GOTERM_CC_DIRECT	protein-containing complex
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase II cis-regulatory region sequence-specific DNA bi
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription factor activity, RNA polymerase II-specif
Gene Ontology	GOTERM_MF_DIRECT	TFIIB-class transcription factor binding
Gene Ontology	GOTERM_MF_DIRECT	transcription coregulator binding
Gene Ontology	GOTERM_MF_DIRECT	transcription corepressor binding
Gene Ontology	GOTERM_MF_DIRECT	transcription coactivator binding
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription activator activity, RNA polymerase II-spe
Gene Ontology	GOTERM_MF_DIRECT	DNA binding
Gene Ontology	GOTERM_MF_DIRECT	chromatin binding
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription factor activity
Gene Ontology	GOTERM_MF_DIRECT	nuclear steroid receptor activity
Gene Ontology	GOTERM_MF_DIRECT	nuclear receptor activity
Gene Ontology	GOTERM_MF_DIRECT	steroid binding
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	calmodulin binding
Gene	GOTERM_MF_DIRECT	beta-catenin binding

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_MF_DIRECT	zinc ion binding
Gene Ontology	GOTERM_MF_DIRECT	lipid binding
Gene Ontology	GOTERM_MF_DIRECT	TBP-class protein binding
Gene Ontology	GOTERM_MF_DIRECT	enzyme binding
Gene Ontology	GOTERM_MF_DIRECT	protein kinase binding
Gene Ontology	GOTERM_MF_DIRECT	nitric-oxide synthase regulator activity
Gene Ontology	GOTERM_MF_DIRECT	nuclear estrogen receptor activity
Gene Ontology	GOTERM_MF_DIRECT	nuclear estrogen receptor binding
Gene Ontology	GOTERM_MF_DIRECT	estrogen response element binding
Gene Ontology	GOTERM_MF_DIRECT	identical protein binding
Gene Ontology	GOTERM_MF_DIRECT	sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	metal ion binding
Gene Ontology	GOTERM_MF_DIRECT	ATPase binding
Gene Ontology	GOTERM_MF_DIRECT	14-3-3 protein binding
Gene Ontology	GOTERM_MF_DIRECT	sequence-specific double-stranded DNA binding
General Annotations	CHROMOSOME	6
General Annotations	CYTOGENETIC_LOCATION	6q25.1-q25.2
General Annotations	ENTREZ_GENE_SUMMARY	This gene encodes an estrogen receptor and ligand-activated transcription factor. The canonical protein contains an N-terminal l independent transactivation domain, a central DNA binding domai hinge domain, and a C-terminal ligand-dependent transactivation d The protein localizes to the nucleus where it may form either a homodimer or a heterodimer with estrogen receptor 2. The protein encoded by this gene regulates the transcription of many estrogen inducible genes that play a role in growth, metabolism, sexual development, gestation, and other reproductive functions and is

Main Category	Category	Terms
		expressed in many non-reproductive tissues. The receptor encode this gene plays a key role in breast cancer, endometrial cancer, and osteoporosis. This gene is reported to have dozens of transcript va due to the use of alternate promoters and alternative splicing, how the full-length nature of many of these variants remain uncertain. [provided by RefSeq, Jul 2020]
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative promoter usage, Alternative splicing; Named isoforms=4; Name=1; Synonyms=Long, hER-alpha66, ER66; IsoId=P03372-1; Sequence=Displayed; Name=2; Synonyms=Short; IsoId=P03372-2; Sequence=VSP_003680; Name=3; Synonyms=hER alpha46, ER46; IsoId=P03372-3; Sequence=VSP_042460; Name=4 Synonyms=hER-alpha36, ER36; IsoId=P03372-4; Sequence=VSP_ VSP_042461;
General Annotations	UP_COMMENT_DISEASE	Estrogen resistance (ESTRR) [MIM:615363]: A disorder characteriz partial or complete resistance to estrogens, in the presence of elev estrogen serum levels. Clinical features include absence of the pul growth spurt, delayed bone maturation, unfused epiphyses, reduce mineral density, osteoporosis, continued growth into adulthood an tall adult stature. Glucose intolerance, hyperinsulinemia and lipid abnormalities may also be present. {ECO:0000269|PubMed:23841 ECO:0000269|PubMed:27754803}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DOMAIN	Composed of three domains: a modulating N-terminal domain, a D binding domain and a C-terminal ligand-binding domain. The modu domain, also known as A/B or AF-1 domain has a ligand- independ transactivation function. The C-terminus contains a ligand- depend transactivation domain, also known as E/F or AF-2 domain which c with the ligand binding domain. AF-1 and AF-2 activate transcriptio independently and synergistically and act in a promoter- and cell-s manner. AF-1 seems to provide the major transactivation function differentiated cells.
General Annotations	UP_COMMENT_FUNCTION	Nuclear hormone receptor. The steroid hormones and their recepto involved in the regulation of eukaryotic gene expression and affect cellular proliferation and differentiation in target tissues. Ligand-dependent nuclear transactivation involves either direct homodime binding to a palindromic estrogen response element (ERE) sequen association with other DNA-binding transcription factors, such as A Jun, c-Fos, ATF-2, Sp1 and Sp3, to mediate ERE- independent signa Ligand binding induces a conformational change allowing subsequ combinatorial association with multiprotein coactivator complexes through LXXLL motifs of their respective components. Mutual transrepression occurs between the estrogen receptor (ER) and NF kappa-B in a cell-type specific manner. Decreases NF-kappa- B DNA binding activity and inhibits NF-kappa-B-mediated transcription fro IL6 promoter and displace RELA/p65 and associated coregulators the promoter. Recruited to the NF-kappa-B response element of the and IL8 promoters and can displace CREBBP. Present with NF-kap components RELA/p65 and NFKB1/p50 on ERE sequences. Can al synergistically with NF-kappa-B to activate transcription involving respective recruitment adjacent response elements; the function in CREBBP. Can activate the transcriptional activity of TFF1. Also me membrane-initiated estrogen signaling involving various kinase ca Essential for MTA1-mediated transcriptional regulation of BRCA1 a BCAS3 (PubMed:17922032). Maintains neuronal survival in respon ischemic reperfusion injury when in the presence of circulating est (17-beta-estradiol/E2) (By similarity). {ECO:0000250|UniProtKB:P0 ECO:0000269|PubMed:10681512, ECO:0000269|PubMed:1081657

Main Category	Category	Terms
		ECO:0000269|PubMed:11477071, ECO:0000269|PubMed:1168262 ECO:0000269|PubMed:14764652, ECO:0000269|PubMed:1507887 ECO:0000269|PubMed:15891768, ECO:0000269|PubMed:1604335 ECO:0000269|PubMed:16617102, ECO:0000269|PubMed:1668477 ECO:0000269|PubMed:17922032, ECO:0000269|PubMed:1793210 ECO:0000269|PubMed:18247370, ECO:0000269|PubMed:1935053 ECO:0000269|PubMed:20074560, ECO:0000269|PubMed:2070561 ECO:0000269|PubMed:21330404, ECO:0000269|PubMed:2208395 ECO:0000269|PubMed:37478846, ECO:0000269|PubMed:7651415 ECO:0000269|PubMed:9328340}.
General Annotations	UP_COMMENT_FUNCTION	The steroid hormones and their receptors are involved in the regula eukaryotic gene expression and affect cellular proliferation and differentiation in target tissues. {ECO:0000256|ARBA:ARBA000038
General Annotations	UP_COMMENT_FUNCTION	The steroid hormones and their receptors are involved in the regula eukaryotic gene expression and affect cellular proliferation and differentiation in target tissues. {ECO:0000256|PIRNR:PIRNR00252
General Annotations	UP_COMMENT_FUNCTION	[Isoform 3]: Involved in activation of NOS3 and endothelial nitric ox production (PubMed:21937726). Isoforms lacking one or several functional domains are thought to modulate transcriptional activity competitive ligand or DNA binding and/or heterodimerization with length receptor (PubMed:10970861). Binds to ERE and inhibits iso (PubMed:10970861). {ECO:0000269|PubMed:10970861, ECO:0000 PubMed:21937726}.
General Annotations	UP_COMMENT_INTERACTION	P03372; Q12802: AKAP13; NbExp=3; IntAct=EBI-78473, EBI-13738 P03372; Q03989: ARID5A; NbExp=9; IntAct=EBI-78473, EBI-948603 P03372; Q6XD76: ASCL4; NbExp=3; IntAct=EBI-78473, EBI-102547 P03372; Q9Y294: ASF1A; NbExp=4; IntAct=EBI-78473, EBI-749553, P03372; Q8IXJ9: ASXL1; NbExp=2; IntAct=EBI-78473, EBI-1646500 P03372; Q6PL18: ATAD2; NbExp=5; IntAct=EBI-78473, EBI-659845 P03372; P18846: ATF1; NbExp=3; IntAct=EBI-78473, EBI-852794; P P62952: BLCAP; NbExp=2; IntAct=EBI-78473, EBI-3895726; P0337 P38398: BRCA1; NbExp=14; IntAct=EBI-78473, EBI-349905; P0337 P20290-2: BTF3; NbExp=5; IntAct=EBI-78473, EBI-1054703; P0337 Q86Y37: CACUL1; NbExp=5; IntAct=EBI-78473, EBI-8168227; P033 P29279: CCN2; NbExp=7; IntAct=EBI-78473, EBI-2835375; P03372 P17676: CEBPB; NbExp=2; IntAct=EBI-78473, EBI-969696; P03372; Q99966: CITED1; NbExp=3; IntAct=EBI-78473, EBI-2624951; P0337 Q9H467: CUEDC2; NbExp=2; IntAct=EBI-78473, EBI-1248228; P033 Q9NV06: DCAF13; NbExp=2; IntAct=EBI-78473, EBI-7402939; P033 O00429: DNM1L; NbExp=2; IntAct=EBI-78473, EBI-724571; P03372 O60869: EDF1; NbExp=3; IntAct=EBI-78473, EBI-781301; P03372; P EGFR; NbExp=2; IntAct=EBI-78473, EBI-297353; P03372; P03372: E NbExp=11; IntAct=EBI-78473, EBI-78473; P03372; Q12778: FOXO1 NbExp=2; IntAct=EBI-78473, EBI-1108782; P03372; O43524: FOXO NbExp=7; IntAct=EBI-78473, EBI-1644164; P03372; Q9Y3R0: GRIP1 NbExp=2; IntAct=EBI-78473, EBI-5349621; P03372; O00165: HAX1 NbExp=2; IntAct=EBI-78473, EBI-357001; P03372; Q6NYC1: JMJD6 NbExp=8; IntAct=EBI-78473, EBI-8464037; P03372; O15054: KDM6 NbExp=2; IntAct=EBI-78473, EBI-2831260; P03372; O14686: KMT2 NbExp=3; IntAct=EBI-78473, EBI-996065; P03372; O95835: LATS1; NbExp=2; IntAct=EBI-78473, EBI-444209; P03372; Q6ZQX7-4: LIAT NbExp=3; IntAct=EBI-78473, EBI-25830459; P03372; Q96Q04: LMT NbExp=3; IntAct=EBI-78473, EBI-720814; P03372; Q9BQ69: MACR NbExp=6; IntAct=EBI-78473, EBI-5324932; P03372; Q00987: MDM2 NbExp=2; IntAct=EBI-78473, EBI-389668; P03372; Q15648: MED1; NbExp=3; IntAct=EBI-78473, EBI-394459; P03372; P60660: MYL6; NbExp=3; IntAct=EBI-78473, EBI-300817; P03372; Q15788: NCOA1

Main Category	Category	Terms
		NbExp=8; IntAct=EBI-78473, EBI-455189; P03372; Q15596: NCOA2 NbExp=5; IntAct=EBI-78473, EBI-81236; P03372; Q9Y6Q9: NCOA3; NbExp=4; IntAct=EBI-78473, EBI-81196; P03372; Q9UN36: NDRG2; NbExp=2; IntAct=EBI-78473, EBI-3895741; P03372; PRO_0000030 [P19838]: NFKB1; NbExp=3; IntAct=EBI-78473, EBI-697771; P0337 Q96RI1-3: NR1H4; NbExp=2; IntAct=EBI-78473, EBI-10921781; P03 Q9BZ95: NSD3; NbExp=3; IntAct=EBI-78473, EBI-3390132; P03372 Q9BTK6: PAGR1; NbExp=5; IntAct=EBI-78473, EBI-2372223; P0337 Q96AQ6-1: PBXIP1; NbExp=3; IntAct=EBI-78473, EBI-15606280; P0 P06401: PGR; NbExp=20; IntAct=EBI-78473, EBI-78539; P03372; P06401-1: PGR; NbExp=4; IntAct=EBI-78473, EBI-12590474; P0337 Q99623: PHB2; NbExp=4; IntAct=EBI-78473, EBI-358348; P03372; P36873: PPP1CC; NbExp=3; IntAct=EBI-78473, EBI-356283; P0337 P53041: PPP5C; NbExp=4; IntAct=EBI-78473, EBI-716663; P03372 P55345: PRMT2; NbExp=9; IntAct=EBI-78473, EBI-78458; P03372; P60763: RAC3; NbExp=5; IntAct=EBI-78473, EBI-767084; P03372; RBFOX2; NbExp=4; IntAct=EBI-78473, EBI-746056; P03372; Q0420 RELA; NbExp=9; IntAct=EBI-78473, EBI-73886; P03372; Q14151: SA NbExp=2; IntAct=EBI-78473, EBI-352869; P03372; Q96HI0: SENP5; NbExp=2; IntAct=EBI-78473, EBI-3895753; P03372; P08047: SP1; NbExp=2; IntAct=EBI-78473, EBI-298336; P03372; Q02447: SP3; N IntAct=EBI-78473, EBI-348158; P03372; P12931: SRC; NbExp=12; IntAct=EBI-78473, EBI-621482; P03372; O15164: TRIM24; NbExp=3 IntAct=EBI-78473, EBI-2130378; P03372; Q16881-4: TXNRD1; NbE IntAct=EBI-78473, EBI-9080335; P03372; Q9UBK9: UXT; NbExp=2; IntAct=EBI-78473, EBI-357355; P03372; O76024: WFS1; NbExp=3; IntAct=EBI-78473, EBI-720609; P03372; Q8N895: ZNF366; NbExp= IntAct=EBI-78473, EBI-2813661; P03372; P59598: Asxl1; Xeno; Nb IntAct=EBI-78473, EBI-5743705; P03372; P05627: Jun; Xeno; NbEx IntAct=EBI-78473, EBI-764369; P03372; Q9JLI4: Ncoa6; Xeno; NbE IntAct=EBI-78473, EBI-286271; P03372; Q60974: Ncor1; Xeno; NbE IntAct=EBI-78473, EBI-349004; P03372; P62962: Pfn1; Xeno; NbEx IntAct=EBI-78473, EBI-647096; P03372; P00523: SRC; Xeno; NbExp IntAct=EBI-78473, EBI-848039; P03372-1; P03372-1: ESR1; NbExp= IntAct=EBI-15606245, EBI-15606245; P03372-1; Q92731: ESR2; Nb IntAct=EBI-15606245, EBI-78505; P03372-1; Q92993: KAT5; NbExp IntAct=EBI-15606245, EBI-399080; P03372-1; Q96AQ6-1: PBXIP1; NbExp=5; IntAct=EBI-15606245, EBI-15606280; P03372-4; P00533 NbExp=4; IntAct=EBI-4309277, EBI-297353; P03372-4; P29353: SH NbExp=2; IntAct=EBI-4309277, EBI-78835; P03372-4; P12931: SRC NbExp=2; IntAct=EBI-4309277, EBI-621482;
General Annotations	UP_COMMENT_MISCELLANEOUS	Selective estrogen receptor modulators (SERMs), such as tamoxife raloxifene, toremifene, lasofoxifene, clomifene, femarelle and ormeloxifene, have tissue selective agonistic and antagonistic effe the estrogen receptor (ER). They interfere with the ER association coactivators or corepressors, mainly involving the AF- 2 domain.
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform 3]: Produced by alternative promoter usage. {ECO:000030
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform 4]: Produced by alternative splicing of isoform 3. {ECO:00
General Annotations	UP_COMMENT_POLYMORPHISM	Genetic variations in ESR1 are correlated with bone mineral density (BMD). Low BMD is a risk factor for osteoporotic fracture. Osteopo characterized by reduced bone mineral density, disruption of bone microarchitecture, and the alteration of the amount and variety of collagenous proteins in bone. Osteoporotic bones are more at risk fracture. {ECO:0000269|PubMed:10942433}.

Main Category	Category	Terms
General Annotations	UP_COMMENT_PTM	Dimethylated by PRMT1 at Arg-260. The methylation may favor cytoplasmic localization (PubMed:18657504, PubMed:24498420). Demethylated by JMJD6 at Arg-260 (PubMed:24498420). {ECO:00 PubMed:18657504, ECO:0000269|PubMed:24498420}.
General Annotations	UP_COMMENT_PTM	Glycosylated; contains N-acetylglucosamine, probably O-linked. {ECO:0000269|PubMed:8999954}.
General Annotations	UP_COMMENT_PTM	Palmitoylated (isoform 3). Not biotinylated (isoform 3). {ECO:0000 PubMed:22031296}.
General Annotations	UP_COMMENT_PTM	Palmitoylated by ZDHHC7 and ZDHHC21. Palmitoylation is require plasma membrane targeting and for rapid intracellular signaling vi and AKT kinases and cAMP generation, but not for signaling media the nuclear hormone receptor. {ECO:0000269|PubMed:22031296}.
General Annotations	UP_COMMENT_PTM	Phosphorylated by cyclin A/CDK2 and CK1. Phosphorylation proba enhances transcriptional activity. Self-association induces phosphorylation. Dephosphorylation at Ser-118 by PPP5C inhibits transactivation activity. Phosphorylated by LMTK3 in vitro. {ECO:00 PubMed:10428798, ECO:0000269|PubMed:14764652, ECO:00002 PubMed:19339517, ECO:0000269|PubMed:21602804, ECO:00002 PubMed:7476978, ECO:0000269|PubMed:7539106, ECO:0000269 PubMed:7838153}.
General Annotations	UP_COMMENT_PTM	Ubiquitinated; regulated by LATS1 via DCAF1 it leads to ESR1 proteasomal degradation (PubMed:21602804, PubMed:28068668) Deubiquitinated by OTUB1 (PubMed:19383985). Ubiquitinated by S CHIP; in the CA1 hippocampal region following loss of endogenou circulating estradiol (17-beta-estradiol/E2) (By similarity). Ubiquitin UBR5, leading to its degradation: UBR5 specifically recognizes and ligand-bound ESR1 when it is not associated with coactivators (NC (PubMed:37478846). In presence of NCOAs, the UBR5- degron is n accessible, preventing its ubiquitination and degradation (PubMed:37478846). {ECO:0000250|UniProtKB:P06211, ECO:0000 PubMed:19383985, ECO:0000269|PubMed:21602804, ECO:00002 PubMed:28068668, ECO:0000269|PubMed:37478846}.
General Annotations	UP_COMMENT_SEQUENCE_CAUTION	Sequence=AAB00115.1; Type=Miscellaneous discrepancy; Note=c an in-frame duplication of exons 6 and 7.; Evidence={ECO:0000305
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the nuclear hormone receptor family. NR3 subfamily. {ECO:0000256|ARBA:ARBA00005413, ECO:0000256| PIRNR:PIRNR002527}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the nuclear hormone receptor family. NR3 subfamily. {ECO:0000256|ARBA:ARBA00005413}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the nuclear hormone receptor family. NR3 subfamily. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000256|ARBA:ARBA00004123, ECO:0000256| PIRNR:PIRNR002527}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000256|ARBA:ARBA00004123, ECO:0000256| RuleBase:RU004334}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000256|ARBA:ARBA00004123}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus. Golgi apparatus. Cell membrane. Note=Colocalizes with ZDHHC7 and ZDHHC21 in the Golgi apparatus where most probab

Main Category	Category	Terms
		palmitoylation occurs. Associated with the plasma membrane whe palmitoylated.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Isoform 1]: Nucleus {ECO:0000255|PROSITE- ProRule:PRU00407, ECO:0000269|PubMed:12682286, ECO:0000269|PubMed:2007456 Cytoplasm {ECO:0000269|PubMed:12682286, ECO:0000269| PubMed:24498420}. Cell membrane {ECO:0000269|PubMed:1268 Peripheral membrane protein {ECO:0000269|PubMed:12682286}; Cytoplasmic side {ECO:0000269|PubMed:12682286}. Note=A min fraction is associated with the inner membrane.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Isoform 3]: Nucleus. Cytoplasm. Cell membrane; Peripheral memb protein; Cytoplasmic side. Cell membrane; Single- pass type I mem protein. Note=Associated with the inner membrane via palmitoylat (Probable). At least a subset exists as a transmembrane protein w terminal extracellular domain. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBUNIT	Binds DNA as a homodimer. Can form a heterodimer with ESR2. In with FOXC2, MAP1S, SLC30A9, UBE1C and NCOA3 coactivator (By similarity). Interacts with PELP1, the interaction is enhanced by 17 estradiol, the interaction increases ESR1 transcriptional activity (PubMed:11481323, PubMed:14963108). Interacts with EP300; the interaction is estrogen-dependent and enhanced by CITED1. Intera CITED1; the interaction is estrogen-dependent. Interacts with NCO NCOA6 coactivators. Interacts with NCOA7; the interaction is a liga inducible. Interacts with PHB2, and UBE1C. Interacts with AKAP13 Interacts with CUEDC2. Interacts with KDM5A. Interacts with SMA Interacts with HEXIM1. Interacts with PBXIP1. Interaction with MU stimulated by 7 beta-estradiol (E2) and enhances ESR1-mediated transcription. Interacts with DNTTIP2, FAM120B and UIMC1. Intera with isoform 4 of TXNRD1. Interacts with KMT2D/MLL2. Interacts ATAD2 and this interaction is enhanced by estradiol. Interacts with KIF18A and LDB1. Interacts with RLIM (via C-terminus). Interacts w MACROD1. Interacts with SH2D4A and PLCG. Interaction with SH2 blocks binding to PLCG and inhibits estrogen-induced cell prolifera Interacts with DYNLL1. Interacts with CCDC62 in the presence of estradiol/E2; this interaction seems to enhance the transcription o genes. Interacts with NR2C1; the interaction prevents homodimeri of ESR1 and suppresses its transcriptional activity and cell growth Interacts with DNAAF4. Interacts with PRMT2. Interacts with RBFO Interacts with STK3/MST2 only in the presence of SAV1 and vice-v Binds to CSNK1D. Interacts with NCOA2; NCOA2 can interact with AF-1 and AF-2 domains simultaneously and mediate their transcrip synergy. Interacts with DDX5 (PubMed:11682626). Interacts with N the interaction seems to require a self- association of N-terminal a terminal regions. Interacts with ZNF366, DDX17, NFKB1, RELA, SP SP3. Interacts with NRIP1 (By similarity). Interacts with GPER1; the interaction occurs in an estrogen-dependent manner. Interacts wit CLOCK and the interaction is stimulated by estrogen. Interacts wit BCAS3. Interacts with TRIP4 (ufmylated); estrogen dependent. Inte with LMTK3; the interaction phosphorylates ESR1 (in vitro) and pro against proteasomal degradation. Interacts with CCAR2 (via N-terr in a ligand- independent manner. Interacts with ZFHX3. Interacts w SFR1 in a ligand-dependent and -independent manner (PubMed:23874500). Interacts with DCAF13, LATS1 and DCAF1; re ESR1 ubiquitination and ubiquitin-mediated proteasomal degradat (PubMed:28068668). Interacts (via DNA-binding domain) with POU terminus); this interaction increases the estrogen receptor ESR1 transcriptional activity in a DNA- and ligand 17-beta-estradiol-indep manner (By similarity). Interacts with ESRRB isoform 1 (PubMed:19755138). Interacts with UBE3A and WBP2

Main Category	Category	Terms
		(PubMed:16772533). Interacts with GTF2B (PubMed:1517211). Int with RBM39 (By similarity). In the absence of hormonal ligand, inte with TACC1 (PubMed:20078863). Interacts with PI3KR1 or PI3KR2 PTK2/FAK1 (PubMed:18657504). Interacts with SRC (PubMed:149 PubMed:18657504). Interacts with BAG1; the interaction is promot the absence of estradiol (17-beta- estradiol/E2) (By similarity). Inte with and ubiquitinated by STUB1; the interaction is promoted in the absence of estradiol (17- beta-estradiol/E2) (By similarity). Interac NEDD8 (By similarity). {ECO:0000250, ECO:0000250|UniProtKB:P0 ECO:0000250|UniProtKB:P19785, ECO:0000269|PubMed:1035981 ECO:0000269|PubMed:10567404, ECO:0000269|PubMed:1068151 ECO:0000269|PubMed:10816575, ECO:0000269|PubMed:1097086 ECO:0000269|PubMed:11113208, ECO:0000269|PubMed:1126575 ECO:0000269|PubMed:11358960, ECO:0000269|PubMed:1147707 ECO:0000269|PubMed:11481323, ECO:0000269|PubMed:1158116 ECO:0000269|PubMed:11682626, ECO:0000269|PubMed:1187510 ECO:0000269|PubMed:11971969, ECO:0000269|PubMed:1203995 ECO:0000269|PubMed:12093804, ECO:0000269|PubMed:1255477 ECO:0000269|PubMed:12917342, ECO:0000269|PubMed:1496310 ECO:0000269|PubMed:15047147, ECO:0000269|PubMed:1517211 ECO:0000269|PubMed:15199063, ECO:0000269|PubMed:1589176 ECO:0000269|PubMed:15940264, ECO:0000269|PubMed:1642701 ECO:0000269|PubMed:16603732, ECO:0000269|PubMed:1677253 ECO:0000269|PubMed:17006958, ECO:0000269|PubMed:1704323 ECO:0000269|PubMed:17085477, ECO:0000269|PubMed:1731181 ECO:0000269|PubMed:17347654, ECO:0000269|PubMed:1750505 ECO:0000269|PubMed:17587566, ECO:0000269|PubMed:1765848 ECO:0000269|PubMed:17914104, ECO:0000269|PubMed:1793210 ECO:0000269|PubMed:17998543, ECO:0000269|PubMed:1856371 ECO:0000269|PubMed:18657504, ECO:0000269|PubMed:1911799 ECO:0000269|PubMed:19339517, ECO:0000269|PubMed:1935053 ECO:0000269|PubMed:19383985, ECO:0000269|PubMed:1942355 ECO:0000269|PubMed:19636373, ECO:0000269|PubMed:1971258 ECO:0000269|PubMed:19718048, ECO:0000269|PubMed:1974915 ECO:0000269|PubMed:19755138, ECO:0000269|PubMed:2007456 ECO:0000269|PubMed:20078863, ECO:0000269|PubMed:2072001 ECO:0000269|PubMed:21104395, ECO:0000269|PubMed:2160280 ECO:0000269|PubMed:23160374, ECO:0000269|PubMed:2387450 ECO:0000269|PubMed:25219498, ECO:0000269|PubMed:2806866 ECO:0000269|PubMed:9328340, ECO:0000269|PubMed:9627117}
General Annotations	UP_COMMENT_SUBUNIT	Binds DNA as a homodimer. {ECO:0000256|PIRNR:PIRNR002527}.
General Annotations	UP_COMMENT_SUBUNIT	[Isoform 3]: Probably homodimerizes or heterodimerizes with isofo and ESR2. {ECO:0000269|PubMed:10970861}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Widely expressed (PubMed:10970861). Not expressed in the pituit gland (PubMed:10970861). {ECO:0000269|PubMed:10970861}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	[Isoform 3]: Widely expressed, however not expressed in the pituita gland. {ECO:0000269|PubMed:10970861}.
Identifiers	ENTREZ_GENE_ID	2099
Identifiers	ENSEMBL_GENE_ID	ENSG00000091831
Identifiers	UNIPROT_ID	Q9UE35_HUMAN
Identifiers	UNIPROT_ID	Q9UBT1_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	Q9H2M1_HUMAN
Identifiers	UNIPROT_ID	Q9H1Z6_HUMAN
Identifiers	UNIPROT_ID	Q9H2M2_HUMAN
Identifiers	UNIPROT_ID	Q5T5H8_HUMAN
Identifiers	UNIPROT_ID	Q0H8T6_HUMAN
Identifiers	UNIPROT_ID	L8E8A8_HUMAN
Identifiers	UNIPROT_ID	K7R989_HUMAN
Identifiers	UNIPROT_ID	K4PXX0_HUMAN
Identifiers	UNIPROT_ID	K4Q331_HUMAN
Identifiers	UNIPROT_ID	K4PX46_HUMAN
Identifiers	UNIPROT_ID	H0Y4W6_HUMAN
Identifiers	UNIPROT_ID	G4XH65_HUMAN
Identifiers	UNIPROT_ID	ESR1_HUMAN
Identifiers	UNIPROT_ID	E3WH19_HUMAN
Identifiers	UNIPROT_ID	E3WH18_HUMAN
Identifiers	UNIPROT_ID	C9E0F3_HUMAN
Identifiers	UNIPROT_ID	C9E0F1_HUMAN
Identifiers	UNIPROT_ID	C9E0F0_HUMAN
Identifiers	UNIPROT_ID	C9E0F2_HUMAN
Identifiers	UNIPROT_ID	C9E0E9_HUMAN
Identifiers	UNIPROT_ID	C8CD01_HUMAN
Identifiers	UNIPROT_ID	C8CJL5_HUMAN
Identifiers	UNIPROT_ID	C8CJK8_HUMAN
Identifiers	UNIPROT_ID	C8CJL1_HUMAN
Identifiers	UNIPROT_ID	C8CJL2_HUMAN
Identifiers	UNIPROT_ID	C8CJL4_HUMAN
Identifiers	UNIPROT_ID	C8CJK9_HUMAN
Identifiers	UNIPROT_ID	C8CJL6_HUMAN
Identifiers	UNIPROT_ID	C8CJL0_HUMAN
Identifiers	UNIPROT_ID	C8CJL3_HUMAN
Identifiers	UNIPROT_ID	C7FFR5_HUMAN
Identifiers	UNIPROT_ID	C0LLI7_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	C0LLI8_HUMAN
Identifiers	UNIPROT_ID	C0LJA9_HUMAN
Identifiers	UNIPROT_ID	C0LJA8_HUMAN
Identifiers	UNIPROT_ID	C0LLI5_HUMAN
Identifiers	UNIPROT_ID	C0LJA7_HUMAN
Identifiers	UNIPROT_ID	C0L927_HUMAN
Identifiers	UNIPROT_ID	C0LLI9_HUMAN
Identifiers	UNIPROT_ID	C0LLI6_HUMAN
Identifiers	UNIPROT_ID	B6ZGU3_HUMAN
Identifiers	UNIPROT_ID	B6DU67_HUMAN
Identifiers	UNIPROT_ID	B6DU66_HUMAN
Identifiers	UNIPROT_ID	B6DU68_HUMAN
Identifiers	UNIPROT_ID	B6DU70_HUMAN
Identifiers	UNIPROT_ID	B6DU69_HUMAN
Identifiers	UNIPROT_ID	B6DP75_HUMAN
Identifiers	UNIPROT_ID	B6DP72_HUMAN
Identifiers	UNIPROT_ID	B6DP74_HUMAN
Identifiers	UNIPROT_ID	B6DP71_HUMAN
Identifiers	UNIPROT_ID	B6DP73_HUMAN
Identifiers	UNIPROT_ID	B5TKC1_HUMAN
Identifiers	UNIPROT_ID	B5TZ82_HUMAN
Identifiers	UNIPROT_ID	B5TKC0_HUMAN
Identifiers	UNIPROT_ID	B5TZ83_HUMAN
Identifiers	UNIPROT_ID	B5TKC2_HUMAN
Identifiers	UNIPROT_ID	B5TZ80_HUMAN
Identifiers	UNIPROT_ID	B5LY05_HUMAN
Identifiers	UNIPROT_ID	B5TZ79_HUMAN
Identifiers	UNIPROT_ID	B5TZ78_HUMAN
Identifiers	UNIPROT_ID	B5TZ77_HUMAN
Identifiers	UNIPROT_ID	B5TZ81_HUMAN
Identifiers	UNIPROT_ID	B5M0Y2_HUMAN
Identifiers	UNIPROT_ID	B0QYW7_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	A8KAF4_HUMAN
Identifiers	UNIPROT_ID	A0A679EM68_HUMAN
Identifiers	UNIPROT_ID	A0A4P8D7D1_HUMAN
Identifiers	UNIPROT_ID	A0A3G1VCY8_HUMAN
Identifiers	UNIPROT_ID	A0A125SXW3_HUMAN
Identifiers	UNIPROT_ID	A0A125SXW2_HUMAN
Identifiers	UNIPROT_ID	A0A125SXV7_HUMAN
Identifiers	UNIPROT_ID	A0A125SXV6_HUMAN
Identifiers	UNIPROT_ID	A0A125SXW1_HUMAN
Identifiers	UNIPROT_ID	A0A125SXW0_HUMAN
Identifiers	UNIPROT_ID	A0A125SXV8_HUMAN
Identifiers	UNIPROT_ID	A0A125SXV9_HUMAN
Interactions	DRUGBANK	Diethylstilbestrol
Interactions	DRUGBANK	Chlorotrianisene
Interactions	DRUGBANK	Conjugated estrogens
Interactions	DRUGBANK	Desogestrel
Interactions	DRUGBANK	Levonorgestrel
Interactions	DRUGBANK	Progesterone
Interactions	DRUGBANK	Lindane
Interactions	DRUGBANK	Raloxifene
Interactions	DRUGBANK	Toremifene
Interactions	DRUGBANK	Medroxyprogesterone acetate
Interactions	DRUGBANK	Testosterone
Interactions	DRUGBANK	Mitotane
Interactions	DRUGBANK	Estrone
Interactions	DRUGBANK	Tamoxifen
Interactions	DRUGBANK	Hexachlorophene
Interactions	DRUGBANK	Estradiol
Interactions	DRUGBANK	Ethynodiol diacetate
Interactions	DRUGBANK	Dobutamine
Interactions	DRUGBANK	Clomifene
Interactions	DRUGBANK	Dienestrol

Main Category	Category	Terms
Interactions	DRUGBANK	Fulvestrant
Interactions	DRUGBANK	Norgestimate
Interactions	DRUGBANK	Ethinylestradiol
Interactions	DRUGBANK	Melatonin
Interactions	DRUGBANK	Trilostane
Interactions	DRUGBANK	Naloxone
Interactions	DRUGBANK	Fluoxymesterone
Interactions	DRUGBANK	Estramustine
Interactions	DRUGBANK	Mestranol
Interactions	DRUGBANK	Danazol
Interactions	DRUGBANK	Oxybenzone
Interactions	DRUGBANK	Allylestrenol
Interactions	DRUGBANK	Zinc
Interactions	DRUGBANK	Genistein
Interactions	DRUGBANK	Prasterone
Interactions	DRUGBANK	Benzophenone
Interactions	DRUGBANK	Equilin
Interactions	DRUGBANK	Compound 19
Interactions	DRUGBANK	Resveratrol
Interactions	DRUGBANK	Compound 18
Interactions	DRUGBANK	Phthalic Acid
Interactions	DRUGBANK	Pyrazole
Interactions	DRUGBANK	Stanolone
Interactions	DRUGBANK	Naringenin
Interactions	DRUGBANK	Compound 4-D
Interactions	DRUGBANK	1-[4-(Octahydro-Pyrido[1,2-a]Pyrazin-2-Yl)-Phenyl]-2-Phenyl-1,2,3,4-Tetrahydro-Isoquinolin-6-Ol
Interactions	DRUGBANK	Quercetin
Interactions	DRUGBANK	Afimoxifene
Interactions	DRUGBANK	2-Phenyl-1-[4-(2-Piperidin-1-Yl-Ethoxy)-Phenyl]-1,2,3,4-Tetrahydro-Isoquinolin-6-Ol
Interactions	DRUGBANK	Estriol
Interactions	DRUGBANK	Estrone sulfate

Main Category	Category	Terms
Interactions	DRUGBANK	Quinestrol
Interactions	DRUGBANK	Phenolphthalein
Interactions	DRUGBANK	Permethrin
Interactions	DRUGBANK	Ospemifene
Interactions	DRUGBANK	AP1081
Interactions	DRUGBANK	Custirsen
Interactions	DRUGBANK	NP-50301
Interactions	DRUGBANK	CHF 4227
Interactions	DRUGBANK	TAS-108
Interactions	DRUGBANK	Lasofoxifene
Interactions	DRUGBANK	Arzoxifene
Interactions	DRUGBANK	Elacestrant
Interactions	DRUGBANK	Bazedoxifene
Interactions	DRUGBANK	Methyltestosterone
Interactions	DRUGBANK	beta-Naphthoflavone
Interactions	DRUGBANK	17-METHYL-17-ALPHA-DIHYDROEQUILENIN
Interactions	DRUGBANK	4-(2-amino-1-methyl-1H-imidazo[4,5-b]pyridin-6-yl)phenol
Interactions	DRUGBANK	[5-HYDROXY-2-(4-HYDROXYPHENYL)-1-BENZOFURAN-7-YL]ACETONITRILE
Interactions	DRUGBANK	4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-8-METHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL
Interactions	DRUGBANK	4-[(1S,2S,5S,9R)-5-(HYDROXYMETHYL)-8,9-DIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL
Interactions	DRUGBANK	4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL
Interactions	DRUGBANK	(2R,3R,4S)-3-(4-HYDROXYPHENYL)-4-METHYL-2-[4-(2-PYRROLIDIN YLETHOXY)PHENYL]CHROMAN-6-OL
Interactions	DRUGBANK	(3AS,4R,9BR)-2,2-DIFLUORO-4-(4-HYDROXYPHENYL)-1,2,3,3A,4,9B-HEXAHYDROCYCLOPENTA[C]CHROMEN-8-OL
Interactions	DRUGBANK	(9ALPHA,13BETA,17BETA)-2-[(1Z)-BUT-1-EN-1-YL]ESTRA-1,3,5(10) TRIENE-3,17-DIOL
Interactions	DRUGBANK	(9BETA,11ALPHA,13ALPHA,14BETA,17ALPHA)-11-(METHOXYMETHYL)ESTRA-1(10),2,4-TRIENE-3,17-DIOL
Interactions	DRUGBANK	3-CHLORO-2-(4-HYDROXYPHENYL)-2H-INDAZOL-5-OL
Interactions	DRUGBANK	3-ETHYL-2-(4-HYDROXYPHENYL)-2H-INDAZOL-5-OL
Interactions	DRUGBANK	Hexestrol

Main Category	Category	Terms
Interactions	DRUGBANK	dimethyl (1R,4S)-5,6-bis(4-hydroxyphenyl)-7-oxabicyclo[2.2.1]hepta diene-2,3-dicarboxylate
Interactions	DRUGBANK	Erteberel
Interactions	DRUGBANK	N-[(1R)-3-(4-HYDROXYPHENYL)-1-METHYLPROPYL]-2-(2-PHENYL-INDOL-3-YL)ACETAMIDE
Interactions	DRUGBANK	(3AS,4R,9BR)-4-(4-HYDROXYPHENYL)-6-(METHOXYMETHYL)-1,2,3,3A,4,9B-HEXAHYDROCYCLOPENTA[C]CHROMEN-8-OL
Interactions	DRUGBANK	4-[1-allyl-7-(trifluoromethyl)-1H-indazol-3-yl]benzene-1,3-diol
Interactions	DRUGBANK	4-(6-HYDROXY-1H-INDAZOL-3-YL)BENZENE-1,3-DIOL
Interactions	DRUGBANK	DIETHYL (1R,2S,3R,4S)-5,6-BIS(4-HYDROXYPHENYL)-7-OXABICYCLO[2.2.1]HEPT-5-ENE-2,3-DICARBOXYLATE
Interactions	DRUGBANK	2-Amino-1-methyl-6-phenylimidazo(4,5-b)pyridine
Interactions	DRUGBANK	4-[(1S,2R,5S)-4,4,8-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL
Interactions	DRUGBANK	(3AS,4R,9BR)-4-(4-HYDROXYPHENYL)-1,2,3,3A,4,9B-HEXAHYDROCYCLOPENTA[C]CHROMEN-9-OL
Interactions	DRUGBANK	2-(4-hydroxyphenyl)benzo[b]thiophen-6-ol
Interactions	DRUGBANK	Tibolone
Interactions	DRUGBANK	Eugenol
Interactions	DRUGBANK	Synthetic Conjugated Estrogens, A
Interactions	DRUGBANK	Synthetic Conjugated Estrogens, B
Interactions	DRUGBANK	Polyestradiol phosphate
Interactions	DRUGBANK	Norethynodrel
Interactions	DRUGBANK	Octocrylene
Interactions	DRUGBANK	Homosalate
Interactions	DRUGBANK	Enzacamene
Interactions	DRUGBANK	Zeranol
Interactions	DRUGBANK	Ractopamine
Interactions	DRUGBANK	Gestrinone
Interactions	DRUGBANK	Estetrol
Interactions	DRUGBANK	Propyl Gallate
Interactions	DRUGBANK	Ormeloxifene
Interactions	DRUGBANK	2-Methoxy-6-{(E)-[(4-methylphenyl)imino]methyl}phenol
Interactions	DRUGBANK	Testosterone cypionate

Main Category	Category	Terms
Interactions	DRUGBANK	Testosterone enanthate
Interactions	DRUGBANK	Stanolone acetate
Interactions	DRUGBANK	Estradiol acetate
Interactions	DRUGBANK	Estradiol benzoate
Interactions	DRUGBANK	Estradiol cypionate
Interactions	DRUGBANK	Estradiol dienanthate
Interactions	DRUGBANK	Estradiol valerate
Interactions	DRUGBANK	Zinc acetate
Interactions	DRUGBANK	Zinc chloride
Interactions	DRUGBANK	Zinc sulfate, unspecified form
Interactions	DRUGBANK	Estriol tripropionate
Interactions	DRUGBANK	Fluoroestradiol F-18
Interactions	PUBCHEM	(3R,4S)-3-(4-(4-Fluorophenyl)-4-hydroxypiperidin-1-yl)-3,4-dihydro-2 chromene-4,7-diol
Interactions	PUBCHEM	Altrenogest
Interactions	PUBCHEM	1-Decyl-3-methylimidazolium hexafluorophosphate
Interactions	PUBCHEM	Amiprofos-methyl
Interactions	PUBCHEM	FR167356
Interactions	PUBCHEM	Pazopanib
Interactions	PUBCHEM	Selumetinib
Interactions	PUBCHEM	Pinane
Interactions	PUBCHEM	Tenovin-1
Interactions	PUBCHEM	Glycocholic Acid
Interactions	PUBCHEM	1,1'-Oxybis-2-propanol dibenzoate
Interactions	PUBCHEM	Rhein
Interactions	PUBCHEM	Nandrolone Cyclohexylpropionate
Interactions	PUBCHEM	Ssr161421
Interactions	PUBCHEM	Epitestosterone
Interactions	PUBCHEM	Aloe emodin
Interactions	PUBCHEM	Menadiol
Interactions	PUBCHEM	Emetine
Interactions	PUBCHEM	Estriol Tripropionate
Interactions	PUBCHEM	9-Phenanthrol

Main Category	Category	Terms
Interactions	PUBCHEM	PharmaGSID_47315
Interactions	PUBCHEM	Binapacryl
Interactions	PUBCHEM	Eucatropine hydrochloride
Interactions	PUBCHEM	1-Dodecyl-3-methylimidazolium hexafluorophosphate
Interactions	PUBCHEM	Alexidine dihydrochloride
Interactions	PUBCHEM	4,4'-(Octahydro-4,7-methano-5H-inden-5-ylidene)bis(phenol)
Interactions	PUBCHEM	PharmaGSID_48509
Interactions	PUBCHEM	3-Methylanthralin
Interactions	PUBCHEM	Isobornyl cyclohexanol
Interactions	PUBCHEM	Iminodibenyl
Interactions	PUBCHEM	Azoxybenzene
Interactions	PUBCHEM	3,4-Diaminotoluene
Interactions	PUBCHEM	PharmaGSID_48519
Interactions	PUBCHEM	4,6-Di-tert-butyl-3-methylphenol
Interactions	PUBCHEM	Chlorazanil
Interactions	PUBCHEM	Olivetol
Interactions	PUBCHEM	Thioglycolic acid anilide
Interactions	PUBCHEM	Fenamidone
Interactions	PUBCHEM	1-Hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide
Interactions	PUBCHEM	N-[(3R)-1-(2-Fluorophenyl)-4-oxo-3,4,6,7-tetrahydro[1,4]diazepino[6, hi]indol-3-yl]-1H-indole-2-carboxamide
Interactions	PUBCHEM	Ethyl 4-(5-chloro-2,3-dihydro-2-oxo-1H-benzimidazol-1-yl)-1-piperidinecarboxylate
Interactions	PUBCHEM	4-Amino-3-nitrophenyl thiocyanate
Interactions	PUBCHEM	2-[2-[4-(7-Methyloctyl)phenoxy]ethoxy]ethanol
Interactions	PUBCHEM	Fulvestrant
Interactions	PUBCHEM	Potassium silver cyanide
Interactions	PUBCHEM	Flurogestone Acetate
Interactions	PUBCHEM	Methylene Blue Trihydrate
Interactions	PUBCHEM	SB236057A
Interactions	PUBCHEM	tert-BUTYL CHLORIDE
Interactions	PUBCHEM	Vanoxerine dihydrochloride
Interactions	PUBCHEM	Chlorobenzilate

Main Category	Category	Terms
Interactions	PUBCHEM	1-Propene, 1-chloro-2-methyl-
Interactions	PUBCHEM	Tetraiodoethylene
Interactions	PUBCHEM	Pyrogallol
Interactions	PUBCHEM	Podophyllotoxin
Interactions	PUBCHEM	Mestanolone
Interactions	PUBCHEM	Androstenediol
Interactions	PUBCHEM	Dihydrotestosterone
Interactions	PUBCHEM	Flavone
Interactions	PUBCHEM	(3R,4R)-3,4-bis((3-hydroxyphenyl)methyl)oxolan-2-one
Interactions	PUBCHEM	Trimethylpentanediyl dibenzoate
Interactions	PUBCHEM	4,4'-(1,2-Diethylethylene)diphenyl dipropionate
Interactions	PUBCHEM	2-Ethylhexyl p-hydroxybenzoate
Interactions	PUBCHEM	N-(4-Acetyl-phenyl)-2-[1-(3-methoxy-phenyl)-1H-tetrazol-5-ylsulfany acetamide
Interactions	PUBCHEM	2-Chloroacetophenone
Interactions	PUBCHEM	Amyleine hydrochloride
Interactions	PUBCHEM	C.I. Basic Orange 2
Interactions	PUBCHEM	Indoxacarb
Interactions	PUBCHEM	3-(2,4,5-Trihydroxyphenyl)-DL-alanine
Interactions	PUBCHEM	(7-Hydroxy-2-p-tolyl-pyrazolo[1,5-a]pyrimidin-5-yl)-acetic acid ethyl
Interactions	PUBCHEM	Apomorphine Hydrochloride
Interactions	PUBCHEM	Dazomet
Interactions	PUBCHEM	1,3-Dichloroacetone
Interactions	PUBCHEM	2-Methyl-4,6-dinitrophenol
Interactions	PUBCHEM	Tamibarotene
Interactions	PUBCHEM	Trabectedin
Interactions	PUBCHEM	3-Methoxyaniline
Interactions	PUBCHEM	Domiphen Bromide
Interactions	PUBCHEM	Dibenzyl sulfide
Interactions	PUBCHEM	Dicyclohexylcarbodiimide
Interactions	PUBCHEM	Benzenamine, 4-methyl-, hydrochloride (1:1)
Interactions	PUBCHEM	3,3'-Dimethylbenzidine dihydrochloride
Interactions	PUBCHEM	Dipentaerythritol pentaacrylate

Main Category	Category	Terms
Interactions	PUBCHEM	Selenious Acid
Interactions	PUBCHEM	3-Methyl-2-butenoic acid
Interactions	PUBCHEM	Maleimide
Interactions	PUBCHEM	Bis(Chloromethyl) Ether
Interactions	PUBCHEM	Octodrine
Interactions	PUBCHEM	Fluoxastrobin
Interactions	PUBCHEM	Crystal Violet
Interactions	PUBCHEM	Amitriptyline Hydrochloride
Interactions	PUBCHEM	1-Butyl-4-methylpyridinium Chloride
Interactions	PUBCHEM	1-Naphthyl Isothiocyanate
Interactions	PUBCHEM	alpha-Terpinyl acetate
Interactions	PUBCHEM	Nandrolone Furylpropionate
Interactions	PUBCHEM	Anecortave Acetate
Interactions	PUBCHEM	3-Nitrophenol
Interactions	PUBCHEM	S-Metolachlor
Interactions	PUBCHEM	Methyl isothiocyanate
Interactions	PUBCHEM	Phosphonium, trihexyltetradecyl-, salt with 1,1,1-trifluoro-N-((trifluoromethyl)sulfonyl)methanesulfonamide (1:1)
Interactions	PUBCHEM	N-(5-Acetylamino-4-thiophen-2-yl-thiazol-2-yl)-2-methoxy-benzamic
Interactions	PUBCHEM	6-Chloro-3-[5-(methylamino)-1,3,4-thiadiazol-2-yl]chromen-2-one
Interactions	PUBCHEM	Ataluren
Interactions	PUBCHEM	Donormil
Interactions	PUBCHEM	1-Butyl-4-methylpyridinium hexafluorophosphate
Interactions	PUBCHEM	3-Methylpentan-2-One
Interactions	PUBCHEM	Formestane
Interactions	PUBCHEM	Picoxystrobin
Interactions	PUBCHEM	Chlorotrianisene
Interactions	PUBCHEM	Pararosaniline Hydrochloride
Interactions	PUBCHEM	Malachite Green
Interactions	PUBCHEM	Pirarubicin
Interactions	PUBCHEM	Congo red
Interactions	PUBCHEM	Trihexyltetradecylphosphonium chloride
Interactions	PUBCHEM	N-(4-chloro-2,5-dimethoxyphenyl)-2-[4-(1-tetrazolyl)phenoxy]acetar

Main Category	Category	Terms
Interactions	PUBCHEM	Methyl 3,3-dimethyl-4-pentenoate
Interactions	PUBCHEM	Allethrins
Interactions	PUBCHEM	Toluene 2,4-Diisocyanate
Interactions	PUBCHEM	1-Methyl-3-octylimidazolium tetrafluoroborate
Interactions	PUBCHEM	1-Decyl-3-methylimidazolium Tetrafluoroborate
Interactions	PUBCHEM	Terpinolene
Interactions	PUBCHEM	N-{4-[(4-methylpyrimidin-2-yl)sulfamoyl]phenyl}-2-(thiophen-2-yl)acetamide
Interactions	PUBCHEM	Chloroacetanilide
Interactions	PUBCHEM	Tetradecylthioacetic acid
Interactions	PUBCHEM	3-Hexanone
Interactions	PUBCHEM	(2-Ethyl-2-((2-((1-oxoallyl)oxy)ethoxy)methyl)-1,3-propanediyl)bis(o ethanediyl) diacrylate
Interactions	PUBCHEM	Diquat dibromide monohydrate
Interactions	PUBCHEM	Carfilzomib
Interactions	PUBCHEM	2-Tridecanone
Interactions	PUBCHEM	2H-1-Benzopyran-2-methanol, alpha,alpha'-(iminobis(methylene))b fluoro-3,4-dihydro-, hydrochloride (1:1), (alphaR,alpha'R,2R,2'S)-rel-
Interactions	PUBCHEM	[(2R)-3-[2-[[(2S)-2-[[(4R)-4-[[(2S)-2-[[(2R)-2-[(3R,4R,5S,6R)-3-acetami dihydroxy-6-(hydroxymethyl)oxan-4-yl]oxypropanoyl]amino]propanoyl]amino]-5-amino-5-oxopentanoyl]amino]propanoyl]amino]ethoxy-hydroxyphosphoryl]o hexadecanoyloxypropyl] hexadecanoate
Interactions	PUBCHEM	5Hpp-33
Interactions	PUBCHEM	dimethylazanium;N,N-dimethylcarbamodithioate
Interactions	PUBCHEM	4-Cumylphenol
Interactions	PUBCHEM	2-(1-Benzo[1,3]dioxol-5-yl-1H-tetrazol-5-ylsulfanyl)-N-(3,4-dimethyl-phenyl)-acetamide
Interactions	PUBCHEM	tert-Butyl 4-hydroxybenzoate
Interactions	PUBCHEM	1,8-Dinitronaphthalene
Interactions	PUBCHEM	9-Nitroanthracene
Interactions	PUBCHEM	Triphenylbismuth
Interactions	PUBCHEM	2-Hydroxyanthraquinone
Interactions	PUBCHEM	Butafenacil
Interactions	PUBCHEM	2-Nitrofluorene
Interactions	PUBCHEM	Pentabromophenol

Main Category	Category	Terms
Interactions	PUBCHEM	3,4,5-Trichlorophenol
Interactions	PUBCHEM	2,6-Dichloro-1,4-benzenediamine
Interactions	PUBCHEM	3,5-Diethyl-1-phenyl-2-propyl-1,2-dihydropyridine
Interactions	PUBCHEM	Guanidine, N,N'''-1,4-butanediylbis-, sulfate
Interactions	PUBCHEM	3-Methylfentanyl hydrochloride, cis-
Interactions	PUBCHEM	4-(2-Chlorophenyl)-2-(2-(4-isobutylphenyl)ethyl)-6,9-dimethyl-6H-thieno(3,2-f)(1,2,4)triazolo(4,3-a)(1,4)diazepine
Interactions	PUBCHEM	Clofarabine
Interactions	PUBCHEM	3,3'-Dichlorobenzidine dihydrochloride
Interactions	PUBCHEM	2-Aminoanthracene
Interactions	PUBCHEM	Benzoylhydrazine
Interactions	PUBCHEM	Iobenguane sulfate
Interactions	PUBCHEM	SB 206553 hydrochloride
Interactions	PUBCHEM	5,12-Naphthacenedione, 7,8,9,10-tetrahydro-6,8,11-trihydroxy-8-(2-hydroxyacetyl)-1-methoxy-10-[[2,3,6-trideoxy-3-[(2S)-2-methoxy-4-morpholinyl]-a-Llyxo-hexopyranosyl]oxy]-, (8S,10S)-
Interactions	PUBCHEM	Oxytocin acetate
Interactions	PUBCHEM	2-Pyrrolidone
Interactions	PUBCHEM	1-Hexyl-3-methylimidazolium Iodide
Interactions	PUBCHEM	(3aS,4R,9bR)-4-(pyridin-2-yl)-3a,4,5,9b-tetrahydro-3H-cyclopenta[c]quinoline-8-carboxylic acid
Interactions	PUBCHEM	1-Decyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide
Interactions	PUBCHEM	4-Nitroanthranilic acid
Interactions	PUBCHEM	3-Tolunitrile
Interactions	PUBCHEM	N-(3-cyano-4,5-diphenyl-2-furyl)-2-(2-pyrimidinylsulfanyl)acetamide
Interactions	PUBCHEM	Bis(4-hydroxyphenyl)methane
Interactions	PUBCHEM	7-(4-Bromophenyl)-4-(3-pyridinyl)isoxazolo[3,4-d]pyridazine
Interactions	PUBCHEM	Mercury, chloro(4-hydroxyphenyl)-
Interactions	PUBCHEM	1-Methyl-3-n-octylimidazolium Bis(trifluoromethanesulfonyl)imide
Interactions	PUBCHEM	Zoxamide
Interactions	PUBCHEM	(+)-Pentazocine succinate
Interactions	PUBCHEM	2-{[1-(3,4-dimethylphenyl)-1H-tetrazol-5-yl]thio}-N-(1-oxo-1,3-dihydr benzofuran-5-yl)acetamide
Interactions	PUBCHEM	Barbaloin
Interactions	PUBCHEM	4-Hydroxy-4'-chlorobenzophenone

Main Category	Category	Terms
Interactions	PUBCHEM	(3S,5R,8R,9S,10S,13S,14S,17S)-17-(1-hydroxyethyl)-10,13-dimethyl-2,3,4,5,6,7,8,9,11,12,14,15,16,17-tetradecahydro-1H-cyclopenta[a]phenanthren-3-ol
Interactions	PUBCHEM	Gefitinib
Interactions	PUBCHEM	Heptadecane
Interactions	PUBCHEM	Brilliant Green
Interactions	PUBCHEM	Berberine chloride
Interactions	PUBCHEM	Methyl 2-[(2-methyl-3-furoyl)amino]-4-(4-pyridinyl)-3-thiophenecarb
Interactions	PUBCHEM	2,3',4,4'-Tetrahydroxybenzophenone
Interactions	PUBCHEM	2-Toluidine hydrochloride
Interactions	PUBCHEM	Erythromycin
Interactions	PUBCHEM	2-{[5-(5-bromo-1-benzofuran-2-yl)-4-ethyl-4H-1,2,4-triazol-3-yl]thio}-fluorophenyl)acetamide
Interactions	PUBCHEM	N-[3-({[3-(2-quinoxalinyl)phenyl]amino}carbonyl)phenyl]-2-furamide
Interactions	PUBCHEM	Dibutyltin dichloride
Interactions	PUBCHEM	2-Ethylhexyl methacrylate
Interactions	PUBCHEM	Dodecanedioic Acid
Interactions	PUBCHEM	gamma-Hexalactone
Interactions	PUBCHEM	4-(2-Quinoxalinyl)phenyl 5-bromo-2-furoate
Interactions	PUBCHEM	N-[2-(naphthalen-2-yl)-1,3-benzoxazol-5-yl]thiophene-2-carboxamid
Interactions	PUBCHEM	24R,25-Dihydroxyvitamin D3
Interactions	PUBCHEM	Tolylfluanid
Interactions	PUBCHEM	2-methyl-N-[[3-[[(2-methylfuran-3-carbonyl)amino]methyl]phenyl]methyl]furan-3-carboxamide
Interactions	PUBCHEM	2-[(14Z)-9-[4-(dimethylamino)-3-hydroxy-6-methyloxan-2-yl]oxy-3-et hydroxy-2,8,12,16-tetramethyl-5,13-dioxo-4,17-dioxabicyclo[14.1.0]heptadec-14-en-10-yl]acetaldehyde
Interactions	PUBCHEM	Bensulide
Interactions	PUBCHEM	Ro 23-7637
Interactions	PUBCHEM	5-bromo-N-[2-(2-fluorophenyl)-1,3-benzoxazol-5-yl]nicotinamide
Interactions	PUBCHEM	Spiro[isobenzofuran-1(3H),9'-[9H]xanthen]-3-one, 3',6'-dihydroxy-2', tetraiodo-, sodium salt (1:2)
Interactions	PUBCHEM	Butyltrimethylammonium bis(trifluoromethylsulfonyl)imide
Interactions	PUBCHEM	Sulbactam
Interactions	PUBCHEM	N-cyclohexyl-2-[(8-methoxy-6-oxo-3-benzo[c][1]benzopyranyl)oxy]acetamide

Main Category	Category	Terms
Interactions	PUBCHEM	2,2-Bis(2-hydroxy-5-biphenylyl)propane
Interactions	PUBCHEM	Carbophenothion
Interactions	PUBCHEM	O,P'-Ddt
Interactions	PUBCHEM	Ulipristal Acetate
Interactions	PUBCHEM	ethyl 8-({[3-(2-furylmethyl)-4-oxo-3,4-dihydro-2-quinazolinyl]thio} methyl)-4H-1,3-benzodioxine-6-carboxylate
Interactions	PUBCHEM	Minopafant
Interactions	PUBCHEM	1,4-Benzenediamine, N-(1,3-dimethylbutyl)-N'-phenyl-
Interactions	PUBCHEM	Levonorgestrel
Interactions	PUBCHEM	(1,3-Dioxoisoindol-2-yl)methyl 3-methyl-1-phenylthieno[2,3-c]pyrazo carboxylate
Interactions	PUBCHEM	2-Hydroxyethyl acrylate
Interactions	PUBCHEM	1,10-Phenanthroline
Interactions	PUBCHEM	Hexythiazox
Interactions	PUBCHEM	4-Phenoxyphenol
Interactions	PUBCHEM	Phenformin Hydrochloride
Interactions	PUBCHEM	Pranazepide
Interactions	PUBCHEM	1-(Phenylazo)-2-naphthol
Interactions	PUBCHEM	Etonitazene
Interactions	PUBCHEM	1,1,1-Trichloropropanone
Interactions	PUBCHEM	1,1,3-Trichloroacetone
Interactions	PUBCHEM	Folic Acid
Interactions	PUBCHEM	Salinazid
Interactions	PUBCHEM	N-(2-nitrophenyl)-3-oxo-1,2,3,5-tetrahydropyrido[1,2-a]benzimidazo carboxamide
Interactions	PUBCHEM	4-[[2-[(6-methyl-4-oxo-1H-pyrimidin-2-yl)thio]-1-oxoethyl]amino]ben acid ethyl ester
Interactions	PUBCHEM	Tegaserod
Interactions	PUBCHEM	Nolatrexed Dihydrochloride
Interactions	PUBCHEM	Aprepitant
Interactions	PUBCHEM	N-(6-ethyl-1,3-benzothiazol-2-yl)-2-(morpholin-4-ylmethyl)-4-oxo-1H quinazoline-7-carboxamide
Interactions	PUBCHEM	N-(furan-2-ylmethylcarbamoyl)-2-[methyl-[(4-oxo-1H-quinazolin-2-yl)methyl]amino]acetamide
Interactions	PUBCHEM	2-(4-fluoroanilino)-6-[(1-phenyltetrazol-5-yl)sulfanylmethyl]-1H-pyri

Main Category	Category	Terms
		one
Interactions	PUBCHEM	ethyl (5E)-2-methyl-4-oxo-5-{[5-({[oxo(phenylamino)acetyl]amino} methyl)furan-2-yl]methylidene}-4,5-dihydro-1H-pyrrole-3-carboxylat
Interactions	PUBCHEM	Resistomycin
Interactions	PUBCHEM	N-Methyl-N'-nitro-N-nitrosoguanidine
Interactions	PUBCHEM	6-ethylsulfanyl-1-(4-methylphenyl)-2H-pyrazolo[3,4-d]pyrimidin-4-or
Interactions	PUBCHEM	5-[(4-Ethoxyphenyl)methyl]-2-(2-pyridylazamethylene)-1,3-thiazolid
Interactions	PUBCHEM	2-methyl-3-[[(Z)-(2-oxo-6,7,8,9-tetrahydrodibenzofuran-1-ylidene)methyl]amino]quinazolin-4-one
Interactions	PUBCHEM	methyl (5E)-5-({5-[({[(3,4-dimethylphenyl)amino](oxo)acetyl} amino)methyl]furan-2-yl}methylidene)-2-methyl-4-oxo-4,5-dihydro-1 pyrrole-3-carboxylate
Interactions	PUBCHEM	N-Butyl-N'-nitro-N-nitrosoguanidine
Interactions	PUBCHEM	N-Nitroso-N-phenylhydroxylamine, ammonium salt
Interactions	PUBCHEM	6-[2-(4-chlorophenyl)-2-oxoethyl]sulfanyl-1-phenyl-2H-pyrazolo[3,4-d]pyrimidin-4-one
Interactions	PUBCHEM	ethyl 4-oxo-5-[(1-{2-oxo-2-[(tetrahydro-2-furanylmethyl)amino]ethyl} indol-3-yl)methylene]-2-(4-toluidino)-4,5-dihydro-3-thiophenecarbox
Interactions	PUBCHEM	(5Z)-4-(4-ethoxyanilino)-5-[(4-methoxyphenyl)methylidene]-1,3-thia one
Interactions	PUBCHEM	Capreomycin IB
Interactions	PUBCHEM	Torkinib
Interactions	PUBCHEM	Dabigatran Etexilate
Interactions	PUBCHEM	3-(3-Oxo-2-benzo[f][1]benzopyranyl)-1-phenyl-4-pyrazolecarboxalde
Interactions	PUBCHEM	N-(3-Chlorophenyl)-N'-(4,5-dihydro-1-methyl-4-oxo-1H-imidazol-2-yl
Interactions	PUBCHEM	N-Nitrosopyrrolidine
Interactions	PUBCHEM	(7S,11S,12R,13S,14R,15R,16R,17S,18S,19Z,21Z)-26-[(E)-[(4-cyclopentylpiperazin-1-yl)imino]methyl]-2,15,17,27,29-pentahydroxy methoxy-3,7,12,14,16,18,22-heptamethyl-6,23-dioxo-8,30-dioxa-24-azatetracyclo[23.3.1.1^{4,7}.0^{5,28}]triaconta-1,3,5(28),9,19,21,25 octaen-13-yl acetate
Interactions	PUBCHEM	3-Methoxycatechol
Interactions	PUBCHEM	Dtxcid5010773
Interactions	PUBCHEM	2,3,5,6-Tetrachlorophenol
Interactions	PUBCHEM	N-Phenylmaleimide
Interactions	PUBCHEM	C8-18-Alkydimethylbenzyl ammonium chlorides
Interactions	PUBCHEM	Ethylestrenol

Main Category	Category	Terms
Interactions	PUBCHEM	Testolactone
Interactions	PUBCHEM	Dinobuton
Interactions	PUBCHEM	Estradiol Valerate
Interactions	PUBCHEM	Ethyl 2-(6-(ethylamino)-3-(ethylimino)-2,7-dimethyl-3H-xanthen-9-yl)benzoate hydrochloride (1:1)
Interactions	PUBCHEM	C.I. Direct Red 2
Interactions	PUBCHEM	Hexamethyldisilazane
Interactions	PUBCHEM	Dtxcid401326699
Interactions	PUBCHEM	SSR 241586 HCl
Interactions	PUBCHEM	Pyrrolnitrin
Interactions	PUBCHEM	Endosulfan Sulfate
Interactions	PUBCHEM	CI-959
Interactions	PUBCHEM	1,2,4-Trichlorobenzene
Interactions	PUBCHEM	Chromium(III) acetate
Interactions	PUBCHEM	2-Ethylhexyl dihydrogen phosphate
Interactions	PUBCHEM	Hexylparaben
Interactions	PUBCHEM	Heptyl p-hydroxybenzoate
Interactions	PUBCHEM	Dichlofluanid
Interactions	PUBCHEM	Bps-mpe4
Interactions	PUBCHEM	N-(4-ethylphenyl)-6-{[(1-phenyl-1H-tetrazol-5-yl)thio]methyl}-1,3,5-triazine-2,4-diamine
Interactions	PUBCHEM	Dodecyltrimethylammonium bromide
Interactions	PUBCHEM	Tetradecyltrimethylammonium bromide
Interactions	PUBCHEM	8-Hydroxyquinoline salicylate
Interactions	PUBCHEM	4-Nitropyridine N-oxide
Interactions	PUBCHEM	3-[3-(furan-2-ylmethylamino)imidazo[1,2-a]pyridin-2-yl]-6-methyl-1H-quinolin-2-one
Interactions	PUBCHEM	4-Hydroxybenzophenone
Interactions	PUBCHEM	3,3',4',5-Tetrachlorosalicylanilide
Interactions	PUBCHEM	Dodecyl Gallate
Interactions	PUBCHEM	Dibenzosuberone
Interactions	PUBCHEM	Octylparaben
Interactions	PUBCHEM	Benzylhydrazine dihydrochloride
Interactions	PUBCHEM	Furazabol

Main Category	Category	Terms
Interactions	PUBCHEM	3beta-Acetoxyandrost-5-en-17-one
Interactions	PUBCHEM	Ethidium Bromide
Interactions	PUBCHEM	Rabeprazole Sodium
Interactions	PUBCHEM	Testosterone phenylpropionate
Interactions	PUBCHEM	Docetaxel
Interactions	PUBCHEM	2,4,6-Tribromophenol
Interactions	PUBCHEM	Benzyldimethyl(2-(2-((4-(1,1,3,3-tetramethylbutyl)-M-tolyl)oxy)ethoxy)ethyl)ammonium chloride
Interactions	PUBCHEM	(4-Amino-3-methylthieno[2,3-c]isothiazol-5-yl)(4-bromophenyl)meti
Interactions	PUBCHEM	3,4',5-Tribromosalicylanilide
Interactions	PUBCHEM	5-Chlorosalicylanilide
Interactions	PUBCHEM	Arsenic Trioxide
Interactions	PUBCHEM	2,4-Dinitrophenol
Interactions	PUBCHEM	Dinoterb
Interactions	PUBCHEM	Manidipine dihydrochloride
Interactions	PUBCHEM	Tributyltin Chloride
Interactions	PUBCHEM	Tetrabutyltin
Interactions	PUBCHEM	Cilomilast
Interactions	PUBCHEM	Vatalanib
Interactions	PUBCHEM	3-[2-(6H-indolo[2,3-b]quinoxalin-6-yl)ethyl]-1,3-benzoxazol-2(3H)-on
Interactions	PUBCHEM	4,4'-Dimethoxyazoxybenzene
Interactions	PUBCHEM	Bisphenol A diglycidyl ether dimethacrylate
Interactions	PUBCHEM	2,4-Dinitrophenyl thiocyanate
Interactions	PUBCHEM	Daminozide
Interactions	PUBCHEM	Mercuric acetate
Interactions	PUBCHEM	Aclarubicin Hydrochloride
Interactions	PUBCHEM	N-((3S)-2,3-Dihydro-6-(phenylmethoxy)-3-benzofuranyl)-N-hydroxyu
Interactions	PUBCHEM	2-Aminofluorene
Interactions	PUBCHEM	Bazedoxifene Acetate
Interactions	PUBCHEM	1,9-Decadiene
Interactions	PUBCHEM	Capsaicin
Interactions	PUBCHEM	Zuclomiphene
Interactions	PUBCHEM	Fmoc-leucine

Main Category	Category	Terms
Interactions	PUBCHEM	Sodium aristolochate
Interactions	PUBCHEM	1-Phenyl-2-nitropropene
Interactions	PUBCHEM	Beveno
Interactions	PUBCHEM	2-Mercaptopyrimidine
Interactions	PUBCHEM	alpha-Hexylcinnamaldehyde
Interactions	PUBCHEM	Cinnamyl cinnamate
Interactions	PUBCHEM	2,6-Bis(2-thienylmethylene)cyclohexan-1-one
Interactions	PUBCHEM	Testosterone decanoate
Interactions	PUBCHEM	Nefopam Hydrochloride
Interactions	PUBCHEM	4-Hydroxyazobenzene
Interactions	PUBCHEM	Ioxynil
Interactions	PUBCHEM	Bromoxynil
Interactions	PUBCHEM	Bromoxynil octanoate
Interactions	PUBCHEM	Dtxcid8020208
Interactions	PUBCHEM	Mordant Orange 1
Interactions	PUBCHEM	2,3,7,8-Tetrachlorodibenzo-P-Dioxin
Interactions	PUBCHEM	2-Amino-6-methoxybenzothiazole
Interactions	PUBCHEM	Besonprodil
Interactions	PUBCHEM	Canertinib
Interactions	PUBCHEM	Doramapimod
Interactions	PUBCHEM	4-n-Octylphenol
Interactions	PUBCHEM	Lactitol
Interactions	PUBCHEM	Nitrofen
Interactions	PUBCHEM	Lercanidipine Hydrochloride
Interactions	PUBCHEM	Octabenzone
Interactions	PUBCHEM	N-[5-(4-chlorophenyl)-1,2,4-thiadiazol-3-yl]-2-[(1-phenyl-1H-tetrazol-yl)thio]acetamide
Interactions	PUBCHEM	Chlorothalonil
Interactions	PUBCHEM	Ethylpyridinium bromide
Interactions	PUBCHEM	1H-Indole-1-carboxamide, 2,3-dihydro-5-methyl-N-[6-[(2-methyl-3-pyridinyl)oxy]-3-pyridinyl]-6-(trifluoromethyl)-, hydrochloride (1:1)
Interactions	PUBCHEM	Pipofezine
Interactions	PUBCHEM	PharmaGSID_48518

Main Category	Category	Terms
Interactions	PUBCHEM	Dtxcid0027802
Interactions	PUBCHEM	17-Epi Pimecrolimus
Interactions	PUBCHEM	[4-[(8aR,9R)-5-[[(2R,4aR,6R,7R,8R,8aS)-7,8-dihydroxy-2-methyl-4,4a,6,7,8,8a-hexahydropyrano[3,2-d][1,3]dioxin-6-yl]oxy]-8-oxo-5a,6,8a,9-tetrahydro-5H-[2]benzofuro[5,6-f][1,3]benzodioxol-9-y dimethoxyphenyl] dihydrogen phosphate
Interactions	PUBCHEM	N-(2-chlorobenzyl)-1-phenylpropan-2-amine hydrochloride
Interactions	PUBCHEM	4-[methyl(Nitroso)amino]benzene-1,2-Diol
Interactions	PUBCHEM	147098-20-2 (Calcium)
Interactions	PUBCHEM	Chloroxuron
Interactions	PUBCHEM	[(3S,17S)-13-methyl-17-propanoyloxy-1,2,3,6,7,8,9,10,11,12,14,15,16 tetradecahydrocyclopenta[a]phenanthren-3-yl] propanoate
Interactions	PUBCHEM	4-n-Heptylphenol
Interactions	PUBCHEM	Fipexide hydrochloride
Interactions	PUBCHEM	1-Ethyl-2,3-dimethylimidazolium triflate
Interactions	PUBCHEM	Methyltricaprylylammonium bistriflamide
Interactions	PUBCHEM	(S)-trans-2-Amino-4-(2-aminoethoxy)-3-butenoic acid hydrochloride
Interactions	PUBCHEM	1-Dodecyl-3-methyl-1H-imidazol-3-ium iodide
Interactions	PUBCHEM	Quinine sulfate dihydrate
Interactions	PUBCHEM	Trihexyltetradecylphosphonium hexafluorophosphate
Interactions	PUBCHEM	1-Methyl-3-n-octylimidazolium Trifluoromethanesulfonate
Interactions	PUBCHEM	Trihexyltetradecylphosphonium dicyanamide
Interactions	PUBCHEM	Trihexyltetradecylphosphonium bromide
Interactions	PUBCHEM	alpha-Amylcinnamaldehyde, (E)-
Interactions	PUBCHEM	Methiocarb
Interactions	PUBCHEM	Diallyl Trisulfide
Interactions	PUBCHEM	Sclareol
Interactions	PUBCHEM	Pimozide
Interactions	PUBCHEM	Fasudil Hydrochloride
Interactions	PUBCHEM	Amitrole
Interactions	PUBCHEM	2-Amino-4-(p-nitrophenyl)thiazole
Interactions	PUBCHEM	Ethyl p-nitrophenyl benzenethiophosphonate
Interactions	PUBCHEM	Bromophos
Interactions	PUBCHEM	2-{Methyl[(2-oxo-1,2-dihydrobenzo[cd]indol-6-yl)sulfonyl]amino}ben acid

Main Category	Category	Terms
Interactions	PUBCHEM	Chloranocryl
Interactions	PUBCHEM	Isooctyl acrylate
Interactions	PUBCHEM	Azaribine
Interactions	PUBCHEM	L-l+/-lpha-Noracetylmethadol hydrochloride
Interactions	PUBCHEM	Phenylmercuric Acetate
Interactions	PUBCHEM	Dibutyltin diacetate
Interactions	PUBCHEM	Bis(tributyltin) oxide
Interactions	PUBCHEM	Tributyltin methacrylate
Interactions	PUBCHEM	Tributyltin benzoate
Interactions	PUBCHEM	Nitromersol
Interactions	PUBCHEM	Triphenylplumbyl acetate
Interactions	PUBCHEM	Bis(acetato-O)hydroxyaluminium
Interactions	PUBCHEM	2,7-Dihydroxy-1,3,2-benzodioxabismole-5-carboxylic acid
Interactions	PUBCHEM	Phenylmercuric lactate
Interactions	PUBCHEM	Tributyltin acrylate
Interactions	PUBCHEM	Thimerosal
Interactions	PUBCHEM	1-Piperidineethanol, alpha-(5-phenyl-3-isoxazolyl)-, 2-hydroxy-1,2,3-propanetricarboxylate (2:1)
Interactions	PUBCHEM	1,4-Naphthalenediamine
Interactions	PUBCHEM	1,5-Diaminonaphthalene
Interactions	PUBCHEM	(5-benzylfuran-3-yl)methyl (1S,3R)-2,2-dimethyl-3-[(E)-(2-oxothiolan ylidene)methyl]cyclopropane-1-carboxylate
Interactions	PUBCHEM	Ipratropium bromide hydrate
Interactions	PUBCHEM	Phenethyl Isothiocyanate
Interactions	PUBCHEM	3-Methylcholanthrene
Interactions	PUBCHEM	(+-)-Clopidogrel bisulfate
Interactions	PUBCHEM	MK-571 sodium
Interactions	PUBCHEM	Pipethanate Ethobromide
Interactions	PUBCHEM	6-benzylamino-9-(2-tetrahydropyranyl)-9H-purine
Interactions	PUBCHEM	p-Ethoxychrysoidine
Interactions	PUBCHEM	Fluorescein
Interactions	PUBCHEM	Urea, (3,4-dichlorophenyl)-
Interactions	PUBCHEM	Ergotaman-3',6',18-trione, 9,10-dihydro-12'-hydroxy-2',5'-bis(1-methylethyl)-, (5'alpha,10alpha)-, methanesulfonate (1:1)

Main Category	Category	Terms
Interactions	PUBCHEM	Fast Green FCF
Interactions	PUBCHEM	Didecyl dimethyl ammonium bromide
Interactions	PUBCHEM	Dexamethasone Sodium Phosphate
Interactions	PUBCHEM	Hydroxyprogesterone Caproate
Interactions	PUBCHEM	(2,4,6-Trioxo-1,3,5-triazinane-1,3,5-triyl)triethane-2,1-diyl trisprop-2-
Interactions	PUBCHEM	Captafol
Interactions	PUBCHEM	Drometrizole
Interactions	PUBCHEM	2-Dodecylphenol
Interactions	PUBCHEM	C.I. Basic Yellow 2
Interactions	PUBCHEM	1-Chlorononane
Interactions	PUBCHEM	Benzenamine, 4,4'-[1,3-phenylenebis(oxy)]bis-
Interactions	PUBCHEM	Solvent Yellow 56
Interactions	PUBCHEM	4-(Octyloxy)benzoic acid
Interactions	PUBCHEM	CID 172717
Interactions	PUBCHEM	Dtxcid9031486
Interactions	PUBCHEM	p-(Chloromercuri)benzoic acid
Interactions	PUBCHEM	4-Chloro-3-methylphenol
Interactions	PUBCHEM	Ethyl 8-bromo-2-(methoxymethyl)-3-methyl-4,5-dioxobenzo[e]indol carboxylate
Interactions	PUBCHEM	Imazalil sulfate
Interactions	PUBCHEM	7-Benzothiazolesulfonic acid, 2-(p-aminophenyl)-6-methyl-
Interactions	PUBCHEM	N-(4-hydroxyphenyl)-3,7-dimethyl-9-(2,6,6-trimethylcyclohexen-1-yl)nona-2,4,6,8-tetraenamide
Interactions	PUBCHEM	C.I. Direct Red 81 disodium salt
Interactions	PUBCHEM	1,2-Benzisothiazol-3(2H)-one
Interactions	PUBCHEM	4-Nonylphenol
Interactions	PUBCHEM	Azinphos-Ethyl
Interactions	PUBCHEM	4-Chloro-beta-(4-chlorophenyl)benzeneethanol
Interactions	PUBCHEM	4-Nitrohippuric acid
Interactions	PUBCHEM	Clofilium Tosylate
Interactions	PUBCHEM	Benzoic acid, 4-hydroxy-, dodecyl ester
Interactions	PUBCHEM	4,4'-Thiodiphenol
Interactions	PUBCHEM	3-Aminofluoranthene

Main Category	Category	Terms
Interactions	PUBCHEM	Enzastaurin
Interactions	PUBCHEM	Bifenazate
Interactions	PUBCHEM	Cypromid
Interactions	PUBCHEM	Triethyltin bromide
Interactions	PUBCHEM	Benzyldimethyltetradecylammonium chloride dihydrate
Interactions	PUBCHEM	Spirodiclofen
Interactions	PUBCHEM	Acetaldehyde
Interactions	PUBCHEM	4-Amino-m-cresol
Interactions	PUBCHEM	Isophorone diamine
Interactions	PUBCHEM	Sulazepam
Interactions	PUBCHEM	Dtxcid3025962
Interactions	PUBCHEM	Limonin
Interactions	PUBCHEM	2,4-Diisopropylphenol
Interactions	PUBCHEM	Fabesetron Hydrochloride
Interactions	PUBCHEM	5-Bromo-5-Nitro-1,3-Dioxane
Interactions	PUBCHEM	Mercury(+2) Cation Diiodide
Interactions	PUBCHEM	N,N'-1,3-Phenylenedimaleimide
Interactions	PUBCHEM	SB413217A
Interactions	PUBCHEM	Doxifluridine
Interactions	PUBCHEM	Tralopyril
Interactions	PUBCHEM	p-tert-Butylphenyl glycidyl ether
Interactions	PUBCHEM	Cyclohexyl isocyanate
Interactions	PUBCHEM	1-Hexadecyl-3-methylimidazolium tetrafluoroborate
Interactions	PUBCHEM	Didodecyldimethylammonium bromide
Interactions	PUBCHEM	Trioctyl trimellitate
Interactions	PUBCHEM	Dithianon
Interactions	PUBCHEM	Deflazacort
Interactions	PUBCHEM	CI-1002 hydrochloride
Interactions	PUBCHEM	Pentaerythritol triacrylate
Interactions	PUBCHEM	3-Phenoxytoluene
Interactions	PUBCHEM	8,12,16,16-Tetramethyl-3,11-dioxapentacyclo[10.7.1.01,15.04,20.05,10]icosa-5,7,9-trien-6-ol
Interactions	PUBCHEM	Hexestrol

Main Category	Category	Terms
Interactions	PUBCHEM	8-Hydroxyquinoline
Interactions	PUBCHEM	Diundecyl phthalate
Interactions	PUBCHEM	Etravirine
Interactions	PUBCHEM	Chlorophacinone
Interactions	PUBCHEM	Diloxanide furoate
Interactions	PUBCHEM	2-(3-Furan-2-ylmethyl-2,4-dioxo-3,4,6,7,8,9-hexahydro-2H,5H-10-thia diaza-benzo[a]azulen-1-yl)-N-phenyl-acetamide
Interactions	PUBCHEM	Estradiol, 17-propionate
Interactions	PUBCHEM	Nafenopin
Interactions	PUBCHEM	2,2-diphenyl-N-(4-pyrrolidin-1-ylphenyl)acetamide
Interactions	PUBCHEM	Sodium (2-pyridylthio)-N-oxide
Interactions	PUBCHEM	Docebenone
Interactions	PUBCHEM	N-(4,5-diphenyl-2-oxazolyl)-2-[(1-methyl-5-tetrazolyl)thio]acetamide
Interactions	PUBCHEM	FD&C Blue No. 1
Interactions	PUBCHEM	Fuberidazole
Interactions	PUBCHEM	3,6-Acridinediamine, hydrochloride (1:1)
Interactions	PUBCHEM	2-[[4-amino-5-(4-methoxyphenyl)-1,2,4-triazol-3-yl]thio]-N-(9-ethyl-3 carbazolyl)acetamide
Interactions	PUBCHEM	Acetochlor
Interactions	PUBCHEM	Nitroxoline
Interactions	PUBCHEM	p-Phenylene diisothiocyanate
Interactions	PUBCHEM	Diethylene glycol diacrylate
Interactions	PUBCHEM	Dtxcid2026363
Interactions	PUBCHEM	Dtxcid0028909
Interactions	PUBCHEM	Scopolamine hydrobromide trihydrate
Interactions	PUBCHEM	2,2-Bis(3,5-dibromo-4-(2-hydroxyethoxy)phenyl)propane
Interactions	PUBCHEM	Neopentyl glycol dibenzoate
Interactions	PUBCHEM	Isobutylparaben
Interactions	PUBCHEM	Cladribine
Interactions	PUBCHEM	Afloqualone
Interactions	PUBCHEM	2,4,5-Trimethoxybenzaldehyde
Interactions	PUBCHEM	Tetrabutylurea
Interactions	PUBCHEM	Triethanolamine stearate

Main Category	Category	Terms
Interactions	PUBCHEM	Tetradecyltrimethylammonium chloride
Interactions	PUBCHEM	Alachlor
Interactions	PUBCHEM	2-[[2-(4,4-diethyl-2,5-dioxoimidazolidin-1-yl)acetyl]amino]-5,6-dihyd cyclopenta[b]thiophene-3-carboxamide
Interactions	PUBCHEM	Albendazole
Interactions	PUBCHEM	3-Ethylcyclopentanepropanoic acid
Interactions	PUBCHEM	N-(3-cyanothiophen-2-yl)-2-[(4-methyl-5-phenyl-1,2,4-triazol-3-yl)sulfanyl]acetamide
Interactions	PUBCHEM	Dronedarone
Interactions	PUBCHEM	Lapatinib
Interactions	PUBCHEM	Fluorophene
Interactions	PUBCHEM	2,3,4,5-Tetrachlorophenol
Interactions	PUBCHEM	Estradiol Enanthate
Interactions	PUBCHEM	(3,5-Dimethyl-1,2-oxazol-4-yl)methyl 4-methylthieno[2,3-b]indole-2-carboxylate
Interactions	PUBCHEM	Cyclohexylazanium;chloride
Interactions	PUBCHEM	Alprazolam
Interactions	PUBCHEM	Trioctylmethylammonium chloride
Interactions	PUBCHEM	Althiazide
Interactions	PUBCHEM	1,5-diphenyl-N-(1,3-thiazol-2-yl)-1,2,4-triazole-3-carboxamide
Interactions	PUBCHEM	Ro 40-6055
Interactions	PUBCHEM	4-Chlorobenzotrichloride
Interactions	PUBCHEM	Fenhexamid
Interactions	PUBCHEM	Famoxadone
Interactions	PUBCHEM	Carboxin
Interactions	PUBCHEM	Oxycarboxin
Interactions	PUBCHEM	1-Hydroxypyrene
Interactions	PUBCHEM	N-(2-carbamoyl-1-benzofuran-3-yl)-1,3-benzothiazole-2-carboxamid
Interactions	PUBCHEM	Deferasirox
Interactions	PUBCHEM	6-[5-(4-Chlorophenyl)-2-methylfuran-3-yl]-3-(2-methylfuran-3-yl)-[1,2,4]triazolo[3,4-b][1,3,4]thiadiazole
Interactions	PUBCHEM	4-Amino-3-hydroxybutanoic acid
Interactions	PUBCHEM	(Z)-4-(2,5-dioxopyrrolidin-1-yl)-N-(3-ethyl-4,7-dimethoxybenzo[d]thiazol-2(3H)-ylidene)benzamide
Interactions	PUBCHEM	Isoquinolinium, 2-dodecyl-, p-toluenesulfonate

Main Category	Category	Terms
Interactions	PUBCHEM	Theophylline
Interactions	PUBCHEM	2-Aminothiazole
Interactions	PUBCHEM	Lasofoxifene
Interactions	PUBCHEM	Dihydrostreptomycin Sulfate
Interactions	PUBCHEM	1-Nitropyrene
Interactions	PUBCHEM	Amoxapine
Interactions	PUBCHEM	Betamethasone Dipropionate
Interactions	PUBCHEM	Chlorpyrifos oxon
Interactions	PUBCHEM	2-[[5-(4-aminophenyl)-4-ethyl-1,2,4-triazol-3-yl]sulfanyl]-N-dibenzofu ylacetamide
Interactions	PUBCHEM	Dtxcid8027872
Interactions	PUBCHEM	1-Hexylpyridinium iodide
Interactions	PUBCHEM	1,1-Bis(3-cyclohexyl-4-hydroxyphenyl)cyclohexane
Interactions	PUBCHEM	Anisindione
Interactions	PUBCHEM	Ethyl 5-acetyl-4-methyl-2-({[(5-phenyl-1,3,4-oxadiazol-2-yl)thio]acety amino)-3-thiophenecarboxylate
Interactions	PUBCHEM	N-[5-(3-methoxyphenyl)-1,3,4-thiadiazol-2-yl]-2-(2-thienyl)acetamide
Interactions	PUBCHEM	Demecolcine
Interactions	PUBCHEM	1-Decyl-3-methylimidazolium Bromide
Interactions	PUBCHEM	N-(2-phenyl-2H-1,2,3-benzotriazol-5-yl)-1,3-benzodioxole-5-carboxa
Interactions	PUBCHEM	2-Methoxy-3-dibenzofuranamine
Interactions	PUBCHEM	Chloropropylate
Interactions	PUBCHEM	Terbuthylazine
Interactions	PUBCHEM	1-Decyl-3-methylimidazolium triflate
Interactions	PUBCHEM	Estradiol Benzoate
Interactions	PUBCHEM	N-{4-[5-(4-methylphenyl)-1,3,4-oxadiazol-2-yl]phenyl}-2-furamide
Interactions	PUBCHEM	Equilin
Interactions	PUBCHEM	4-[(5-bromo-2-furyl)methylene]-2-(methylthio)-1,3-thiazol-5(4H)-one
Interactions	PUBCHEM	6-Bromo-2-{[4-(2-furoyl)-1-piperazinyl]carbonyl}pyrazolo[1,5-a]pyrim
Interactions	PUBCHEM	N-(3-fluorophenyl)-4-methylsulfanylbenzenesulfonamide
Interactions	PUBCHEM	Vatalanib dihydrochloride
Interactions	PUBCHEM	Cyclophosphamide Monohydrate
Interactions	PUBCHEM	2,2'-Dihydroxy-6,6'-dinaphthyldisulfide

Main Category	Category	Terms
Interactions	PUBCHEM	Astemizole
Interactions	PUBCHEM	2-Chloro-p-phenylenediamine sulfate
Interactions	PUBCHEM	Aurintricarboxylic Acid
Interactions	PUBCHEM	Azelaic Acid
Interactions	PUBCHEM	Methyltricaprylylammonium triflate
Interactions	PUBCHEM	Azinphos-Methyl
Interactions	PUBCHEM	Azobenzene
Interactions	PUBCHEM	Methylene bis(thiocyanate)
Interactions	PUBCHEM	Chlorinated paraffins (C12, 60% chlorine)
Interactions	PUBCHEM	Emamectin benzoate [ISO]
Interactions	PUBCHEM	3-{5-[(3-{2-[(3,4-Dimethylphenyl)amino]-2-oxoethyl}-2,4-dioxo-1,3-thiazolidin-5-ylidene)methyl]-2-furyl}-4-methylbenzoic acid
Interactions	PUBCHEM	Bisphenol A diglycidyl ether
Interactions	PUBCHEM	Bithionolate Sodium
Interactions	PUBCHEM	Norethisterone enanthate
Interactions	PUBCHEM	Nandrolone Phenylpropionate
Interactions	PUBCHEM	C.I. Acid Red 114
Interactions	PUBCHEM	Ammonium nitrate
Interactions	PUBCHEM	Pentyl 4-hydroxybenzoate
Interactions	PUBCHEM	N,N-dibenzyl-2-[N-(dimethylsulfamoyl)anilino]acetamide
Interactions	PUBCHEM	1-Butyl-3-methylpyridinium trifluoromethanesulfonate
Interactions	PUBCHEM	Testosterone Isocaproate
Interactions	PUBCHEM	1-Dodecyl-3-methylimidazolium trifluoromethanesulfonate
Interactions	PUBCHEM	1,1-Bis(4-hydroxyphenyl)cyclohexane
Interactions	PUBCHEM	Benserazide
Interactions	PUBCHEM	2,2,4-Trimethyl-1,3-pentanediol diisobutyrate
Interactions	PUBCHEM	Benzbromarone
Interactions	PUBCHEM	2,2'-Dithiobis[thiophene]
Interactions	PUBCHEM	Benzethonium
Interactions	PUBCHEM	Benzo[a]pyrene
Interactions	PUBCHEM	Benzocaine
Interactions	PUBCHEM	2-(Dimethylamino)-2-methylpropanol
Interactions	PUBCHEM	3-[Ethyl(phenyl)sulfamoyl]-4-methylbenzoic acid

Main Category	Category	Terms
Interactions	PUBCHEM	[2-(Furan-2-ylmethylamino)-2-oxoethyl] 3-phenoxypropanoate
Interactions	PUBCHEM	6-Amino-1,3-dimethyl-5-[(3-methyl-1-phenyl-5-thieno[2,3-c]pyrazoly oxomethyl]pyrimidine-2,4-dione
Interactions	PUBCHEM	Butyl Benzyl Phthalate
Interactions	PUBCHEM	Bepridil
Interactions	PUBCHEM	p-Dimethylaminophenyl thiocyanate
Interactions	PUBCHEM	[2-(Furan-2-ylmethylamino)-2-oxoethyl] 4-oxo-4-thiophen-2-ylbutan
Interactions	PUBCHEM	Didecyldimethylammonium Chloride
Interactions	PUBCHEM	[2-(furan-2-ylmethylamino)-2-oxoethyl] 3-(3,4-dihydro-1H-isoquinoli ylsulfonyl)benzoate
Interactions	PUBCHEM	[2-(Furan-2-ylmethylamino)-2-oxoethyl] 4,5-dimethoxy-2-(thiophene carbonylamino)benzoate
Interactions	PUBCHEM	Allylestrenol
Interactions	PUBCHEM	Beta-Naphthoflavone
Interactions	PUBCHEM	Tazobactam Sodium
Interactions	PUBCHEM	Beraprost Sodium
Interactions	PUBCHEM	Sodium pentachlorophenolate
Interactions	PUBCHEM	Sodium 5-(3-pyridinylmethyl)-2-benzofurancarboxylate
Interactions	PUBCHEM	Cerivastatin Sodium
Interactions	PUBCHEM	Sodium sulfanilate
Interactions	PUBCHEM	Sodium 2-(2-amino-3-benzoylphenyl)acetate
Interactions	PUBCHEM	Sodium Tetradecyl Sulfate
Interactions	PUBCHEM	Menadione Sodium Bisulfite
Interactions	PUBCHEM	Liothyronine Sodium
Interactions	PUBCHEM	Sodium Tridecyl Sulfate
Interactions	PUBCHEM	Sodium 4-octylbenzene sulfonate
Interactions	PUBCHEM	Dexamethasone Metasulfobenzoate Sodium
Interactions	PUBCHEM	Sodium Amidotrizoate
Interactions	PUBCHEM	Docusate Sodium
Interactions	PUBCHEM	Actinoquinol
Interactions	PUBCHEM	Potassium perfluorobutanesulfonate
Interactions	PUBCHEM	Hydroxymethanesulfinic acid monosodium salt dihydrate
Interactions	PUBCHEM	Sodium nonyl sulfate
Interactions	PUBCHEM	Methicillin Sodium

Main Category	Category	Terms
Interactions	PUBCHEM	Sodium abietate
Interactions	PUBCHEM	Phenol Red sodium salt
Interactions	PUBCHEM	Phenol, 4,4'-(1,1-dioxido-3H-2,1-benzoxathiol-3-ylidene)bis(2,6-dibr methyl-, sodium salt (1:1)
Interactions	PUBCHEM	Sodium prasterone sulfate dihydrate
Interactions	PUBCHEM	Estriol 3-glucuronide (sodium salt)
Interactions	PUBCHEM	Itriglumide sodium
Interactions	PUBCHEM	Benzododecinium Bromide
Interactions	PUBCHEM	Warfarin Potassium
Interactions	PUBCHEM	Potassium Ricinoleate
Interactions	PUBCHEM	2,5-Di-tert-butylhydroquinone
Interactions	PUBCHEM	1-(Furan-2-yl)-2-[[5-[2-(furan-2-yl)-2-oxoethyl]sulfanyl-1,3,4-thiadiazo yl]sulfanyl]ethanone
Interactions	PUBCHEM	Bicalutamide
Interactions	PUBCHEM	Bisacodyl
Interactions	PUBCHEM	N-(6-methylsulfonyl-1,3-benzothiazol-2-yl)-4-oxo-4-thiophen-2-ylbutanamide
Interactions	PUBCHEM	Selenium Dioxide
Interactions	PUBCHEM	2-Phenoxybenzoic acid [2-(2-furanylmethylamino)-2-oxoethyl] este
Interactions	PUBCHEM	Bithionol
Interactions	PUBCHEM	Clazolam
Interactions	PUBCHEM	4-(3-Dodecanyl)benzenesulfonate, calcium salt
Interactions	PUBCHEM	2-amino-N-(furan-2-ylmethyl)-1-[(4-methoxyphenyl)methyl]pyrrolo[3 b]quinoxaline-3-carboxamide
Interactions	PUBCHEM	Tribenzylamine
Interactions	PUBCHEM	Etocrylene
Interactions	PUBCHEM	N-(3-carbamoyl-5,6-dihydro-4H-cyclopenta[b]thiophen-2-yl)-1-benzofuran-2-carboxamide
Interactions	PUBCHEM	Benzoic Acid
Interactions	PUBCHEM	Tetrasodium Pyrophosphate
Interactions	PUBCHEM	Metoprine
Interactions	PUBCHEM	(6,7-dimethyl-2-oxochromen-4-yl)methyl 1H-indazole-3-carboxylate
Interactions	PUBCHEM	2-(4-Acetylanilino)-1-(4-phenylpiperazin-1-yl)ethanone
Interactions	PUBCHEM	Brotizolam
Interactions	PUBCHEM	Cadmium chloride, dihydrate

Main Category	Category	Terms
Interactions	PUBCHEM	Propham
Interactions	PUBCHEM	Buformin
Interactions	PUBCHEM	Triisopropanolamine
Interactions	PUBCHEM	Desmedipham
Interactions	PUBCHEM	Phenmedipham
Interactions	PUBCHEM	Glycol monostearate
Interactions	PUBCHEM	4-Nonylphenol diethoxylate
Interactions	PUBCHEM	[2-(2,2-dimethyl-3-oxo-4H-quinoxalin-1-yl)-2-oxoethyl] 4-[(3,5-dimeth oxazol-4-yl)methoxy]benzoate
Interactions	PUBCHEM	Buspirone
Interactions	PUBCHEM	Busulfan
Interactions	PUBCHEM	Dtxcid7027334
Interactions	PUBCHEM	Antimony trichloride
Interactions	PUBCHEM	Methenolone Enanthate
Interactions	PUBCHEM	Butyl 4-aminobenzoate
Interactions	PUBCHEM	Coenzyme A trilithium dihydrate
Interactions	PUBCHEM	2-(5-morpholin-4-ylsulfonylthiophen-2-yl)-N-(4,5,6,7-tetrahydro-1,3-benzothiazol-2-yl)acetamide
Interactions	PUBCHEM	1-Methyl-3-octadecylimidazolium hexafluorophosphate
Interactions	PUBCHEM	Mk-2206
Interactions	PUBCHEM	2-(2,3-dihydro-1,4-benzodioxin-6-yl)-N-(5-ethyl-6,7-dihydro-4H-[1,3]thiazolo[5,4-c]pyridin-2-yl)acetamide
Interactions	PUBCHEM	N-(furan-2-ylmethylcarbamoyl)-2-[(2-methylphenyl)methylsulfanyl]acetamide
Interactions	PUBCHEM	Ro3280
Interactions	PUBCHEM	Trenbolone
Interactions	PUBCHEM	2,2-Dibromo-3-nitrilopropionamide
Interactions	PUBCHEM	Ethyl 4-(dimethylamino)benzoate
Interactions	PUBCHEM	2-(6-ethyl-4-oxo-3-prop-2-enylthieno[2,3-d]pyrimidin-2-yl)sulfanyl-N-(furan-2-ylmethylcarbamoyl)acetamide
Interactions	PUBCHEM	Dialifor
Interactions	PUBCHEM	Pheniramine maleate2-[3-(dimethylammonio)-1-phenylpropyl]pyrid (2Z)-but-2-enedioate
Interactions	PUBCHEM	Mibolerone
Interactions	PUBCHEM	Caffeine

Main Category	Category	Terms
Interactions	PUBCHEM	Eugenyl phenylacetate
Interactions	PUBCHEM	Stanozolol
Interactions	PUBCHEM	1-[2,5-Dimethyl-1-(thiophen-2-ylmethyl)pyrrol-3-yl]-2-[[5-(2-fluorophenyl)-1,3,4-oxadiazol-2-yl]sulfanyl]ethanone
Interactions	PUBCHEM	2-[(4-benzyl-5-methyl-1,2,4-triazol-3-yl)sulfanyl]-N-(furan-2-ylmethylcarbamoyl)acetamide
Interactions	PUBCHEM	Camphor
Interactions	PUBCHEM	Candesartan
Interactions	PUBCHEM	Cannabinol
Interactions	PUBCHEM	2-[5-(3,4-dimethoxyphenyl)tetrazol-2-yl]-1-(7-ethyl-1H-indol-3-yl)eth
Interactions	PUBCHEM	2,2'-Dichlorobiphenyl
Interactions	PUBCHEM	1-{4H,5H,6H-cyclopenta[b]thiophene-2-carbonyl}-4-(2-methoxyphenyl)piperazine
Interactions	PUBCHEM	Methyl 4-(bromomethyl)benzoate
Interactions	PUBCHEM	Carbofuran
Interactions	PUBCHEM	Carboquone
Interactions	PUBCHEM	N-(furan-2-ylmethylcarbamoyl)-2-[[5-(2-methylanilino)-1,3,4-thiadiaz yl]sulfanyl]acetamide
Interactions	PUBCHEM	Estramustine Phosphate
Interactions	PUBCHEM	Estramustine
Interactions	PUBCHEM	3-methyl-N-[4-[2-[4-(1,3,4-oxadiazol-2-yl)phenoxy]acetyl]phenyl]butanamide
Interactions	PUBCHEM	Gold trichloride
Interactions	PUBCHEM	Carbonyl cyanide (m-chlorophenyl)hydrazone
Interactions	PUBCHEM	Pyrithione Zinc
Interactions	PUBCHEM	N-(furan-2-ylmethylcarbamoyl)-2-[1-(2-methoxy-5-methylphenyl)tet yl]sulfanylacetamide
Interactions	PUBCHEM	2-(5,6-dimethylthieno[2,3-d]pyrimidin-4-yl)sulfanyl-N-(furan-2-ylmethylcarbamoyl)acetamide
Interactions	PUBCHEM	Quinalphos
Interactions	PUBCHEM	Hydrocortisone Butyrate
Interactions	PUBCHEM	1,6-Hexanediol dibenzoate
Interactions	PUBCHEM	Tris(1,3-dichloro-2-propyl) phosphate
Interactions	PUBCHEM	5-Methyl-2,3-hexanedione
Interactions	PUBCHEM	3-Phenoxybenzyl alcohol

Main Category	Category	Terms
Interactions	PUBCHEM	Bisoxatin Acetate
Interactions	PUBCHEM	N-(furan-2-ylmethyl)-2-[[5-(4-methylanilino)-1,3,4-thiadiazol-2-yl]sulfanyl]acetamide
Interactions	PUBCHEM	Parbendazole
Interactions	PUBCHEM	Celecoxib
Interactions	PUBCHEM	Zinc Diethyldithiocarbamate
Interactions	PUBCHEM	Ferbam
Interactions	PUBCHEM	2-[[5-[4-(difluoromethoxy)phenyl]-1,3,4-oxadiazol-2-yl]sulfanyl]-N-(fu ylmethylcarbamoyl)acetamide
Interactions	PUBCHEM	Benserazide Hydrochloride
Interactions	PUBCHEM	Napropamide
Interactions	PUBCHEM	1-(2,6-Dichlorophenyl)indolin-2-one
Interactions	PUBCHEM	Chloroxine
Interactions	PUBCHEM	9-Aminoacridine hydrochloride monohydrate
Interactions	PUBCHEM	Thioguanine
Interactions	PUBCHEM	Ethyl cysteinate hydrochloride
Interactions	PUBCHEM	(3R,4R,5S,6R)-3-amino-6-(hydroxymethyl)oxane-2,4,5-triol hydrochl
Interactions	PUBCHEM	Ethylenethiourea
Interactions	PUBCHEM	4'-tert-Butyl-4-chlorobutyrophenone
Interactions	PUBCHEM	1,10-Phenanthroline monohydrochloride monohydrate
Interactions	PUBCHEM	Pregnenolone Acetate
Interactions	PUBCHEM	Tacrine Hydrochloride
Interactions	PUBCHEM	Thiosemicarbazide
Interactions	PUBCHEM	N-Methylhydrazinecarbothioamide
Interactions	PUBCHEM	Indigo carmine
Interactions	PUBCHEM	Copper monocyanide
Interactions	PUBCHEM	Polyethylene Glycol Cetyl Ether
Interactions	PUBCHEM	Digoxin
Interactions	PUBCHEM	Malachite Green Oxalate
Interactions	PUBCHEM	Quinacrine dihydrochloride dihydrate
Interactions	PUBCHEM	L-Cysteine, methyl ester, hydrochloride
Interactions	PUBCHEM	Tamoxifen Citrate
Interactions	PUBCHEM	Tamoxifen

Main Category	Category	Terms
Interactions	PUBCHEM	Tetraoctylammonium bromide
Interactions	PUBCHEM	1-Hexyl-3-methylimidazolium chloride
Interactions	PUBCHEM	1-Hexylpyridinium chloride
Interactions	PUBCHEM	1-Hexyl-3-methylimidazolium hexafluorophosphate
Interactions	PUBCHEM	1-Hexylpyridinium Hexafluorophosphate
Interactions	PUBCHEM	1-Hexyl-3-methylimidazolium tetrafluoroborate
Interactions	PUBCHEM	1-Butylpyridinium Tetrafluoroborate
Interactions	PUBCHEM	1-Octyl-3-methylimidazolium chloride
Interactions	PUBCHEM	1-Decyl-3-methylimidazolium chloride
Interactions	PUBCHEM	1-Dodecyl-3-methyl-1H-imidazol-3-ium Chloride
Interactions	PUBCHEM	1-Methyl-3-tetradecylimidazolium chloride
Interactions	PUBCHEM	1-Hexadecyl-3-methylimidazolium Chloride
Interactions	PUBCHEM	1-Hexyl-3-methylimidazolium bromide
Interactions	PUBCHEM	1-Hexylpyridinium bromide
Interactions	PUBCHEM	1-Methyl-3-octylimidazolium hexafluorophosphate
Interactions	PUBCHEM	1-Hexyl-3-methylimidazolium Trifluoromethanesulfonate
Interactions	PUBCHEM	1-Hexylpyridinium trifluoromethanesulfonate
Interactions	PUBCHEM	D-Fructose 1,6-bisphosphate trisodium
Interactions	PUBCHEM	4-Anisidine hydrochloride
Interactions	PUBCHEM	N,N'-Diethylthiourea
Interactions	PUBCHEM	3-Chloro-4-fluorobenzoyl Chloride
Interactions	PUBCHEM	tert-Butyl-3-aminobenzoate
Interactions	PUBCHEM	Trimethylolpropane triacrylate
Interactions	PUBCHEM	2,2',5,5'-Tetrachlorobenzidine
Interactions	PUBCHEM	Cilostamide
Interactions	PUBCHEM	Cilostazol
Interactions	PUBCHEM	2,3,4-Trichlorophenol
Interactions	PUBCHEM	1-Ethyl-3-methylimidazolium hexafluorophosphate
Interactions	PUBCHEM	Ethionamide
Interactions	PUBCHEM	1,3-Didecyl-2-methyl-1H-imidazolium chloride
Interactions	PUBCHEM	Ammonium perfluorooctanoate
Interactions	PUBCHEM	Gestrinone

Main Category	Category	Terms
Interactions	PUBCHEM	Clioquinol
Interactions	PUBCHEM	Levamisole Hydrochloride
Interactions	PUBCHEM	Clofazimine
Interactions	PUBCHEM	Clofibrate
Interactions	PUBCHEM	Clofoctol
Interactions	PUBCHEM	Clomiphene
Interactions	PUBCHEM	Clotrimazole
Interactions	PUBCHEM	5-Chloro-8-hydroxyquinoline
Interactions	PUBCHEM	6-[2-(2,4-dimethylphenyl)-2-oxoethyl]sulfanyl-4-(furan-2-yl)-2-oxo-3, dihydro-1H-pyridine-5-carbonitrile
Interactions	PUBCHEM	Danazol
Interactions	PUBCHEM	Benzene, (2-iodoethyl)-
Interactions	PUBCHEM	Tomatine
Interactions	PUBCHEM	Disperse Red 60
Interactions	PUBCHEM	4-cyclohex-3-en-1-yl-6-methoxy-8-nitro-3a,4,5,9b-tetrahydro-3H-cyclopenta[c]quinoline
Interactions	PUBCHEM	6-(morpholine-4-carbonyl)-3a,4,5,9b-tetrahydro-3H-cyclopenta[c]quinoline-4-carboxylic acid
Interactions	PUBCHEM	Clorgyline hydrochloride
Interactions	PUBCHEM	2-(Cyclohexylthio)-1H-isoindole-1,3(2H)-dione
Interactions	PUBCHEM	Benomyl
Interactions	PUBCHEM	P-Cresol
Interactions	PUBCHEM	Tetraethylene glycol diacrylate
Interactions	PUBCHEM	Iodofenphos
Interactions	PUBCHEM	Cyclofenil
Interactions	PUBCHEM	Catechol
Interactions	PUBCHEM	1-(2-Ethoxy-phenyl)-5,5-bis-(2-pyridin-4-yl-ethyl)-pyrimidine-2,4,6-tri
Interactions	PUBCHEM	Methabenzthiazuron
Interactions	PUBCHEM	Isoxathion
Interactions	PUBCHEM	N-(4-Dipropylsulfamoyl-phenyl)-2-thiophen-2-yl-acetamide
Interactions	PUBCHEM	N-[4-(4-Methyl-piperidine-1-sulfonyl)-phenyl]-2-thiophen-2-yl-acetar
Interactions	PUBCHEM	4-Hexyloxyphenol
Interactions	PUBCHEM	8-Methoxy-3-[2-(4-methylpiperidin-1-yl)-2-oxoethoxy]benzo[c]chrom one

Main Category	Category	Terms
Interactions	PUBCHEM	Oryzalin
Interactions	PUBCHEM	2-(5,6-Di-furan-2-yl-[1,2,4]triazin-3-ylsulfanyl)-N-o-tolyl-acetamide
Interactions	PUBCHEM	3,4,5-trimethoxy-N-[5-(3-methyl-2-thienyl)-1,3,4-thiadiazol-2-yl]benz
Interactions	PUBCHEM	1,8-Dihydroxyanthraquinone
Interactions	PUBCHEM	2-(2,4-dichlorophenoxy)-N-[2-(4-pyridinyl)-1,3-benzoxazol-5-yl]propanamide
Interactions	PUBCHEM	5-Methyl-7-(pentafluoroethyl)-4,5,6,7-tetrahydropyrazolo[1,5-a]pyrimidine-2-carboxylic acid
Interactions	PUBCHEM	Oxadiazon
Interactions	PUBCHEM	N-[1-(1-adamantyl)-1H-pyrazol-4-yl]-5-(2-furyl)-7-(trifluoromethyl)pyrazolo[1,5-a]pyrimidine-3-carboxamide
Interactions	PUBCHEM	Ethyl 1-[(6-bromopyrazolo[1,5-a]pyrimidin-2-yl)carbonyl]-3-piperidinecarboxylate
Interactions	PUBCHEM	Dephostatin
Interactions	PUBCHEM	Benzyl 6-(furan-2-yl)-3-methyl-4-oxo-1,5,6,7-tetrahydroindole-2-carb
Interactions	PUBCHEM	Triphenylbismuthdichloride
Interactions	PUBCHEM	6-[5-(2,5-dichlorophenyl)furan-2-yl]-3-propyl-7H-[1,2,4]triazolo[3,4-b][1,3,4]thiadiazine
Interactions	PUBCHEM	[1-(4-Chlorophenyl)-3-methylthieno[2,3-c]pyrazol-5-yl]-(4-pyridin-2-ylpiperazin-1-yl)methanone
Interactions	PUBCHEM	7-Bromo-2-[[5-(2,3-dihydro-1,4-benzodioxin-3-yl)-4-methyl-1,2,4-triaz yl]sulfanylmethyl]pyrido[1,2-a]pyrimidin-4-one
Interactions	PUBCHEM	1-(2-Fluorophenoxy)-4-methylphthalazine
Interactions	PUBCHEM	2-[[2-(4-ethyl-4-methyl-2,5-dioxo-1-imidazolidinyl)-1-oxoethyl]amino dihydro-4H-cyclopenta[b]thiophene-3-carboxamide
Interactions	PUBCHEM	N-(furan-2-ylmethylcarbamoyl)-2-[[5-[(3-methylphenoxy)methyl]-4-p enyl-1,2,4-triazol-3-yl]sulfanyl]acetamide
Interactions	PUBCHEM	ethyl 2,5-dimethyl-4-[2-[[5-(3,4,5-trimethoxyphenyl)-1,3,4-oxadiazol-yl]sulfanyl]acetyl]-1H-pyrrole-3-carboxylate
Interactions	PUBCHEM	5,6-dihydro-4H-cyclopenta[b]thiophene-2-carboxylic acid [2-[(1,1-di thiolanyl)-(2-methylpropyl)amino]-2-oxoethyl] ester
Interactions	PUBCHEM	N-(5-ethylsulfanyl-1,3,4-thiadiazol-2-yl)-2-(furan-2-carbonyl)-3,4-dihydro-1H-isoquinoline-3-carboxamide
Interactions	PUBCHEM	5-[4-(1-Benzofuran-2-yl)-1,3-thiazol-2-yl]-6-methylimidazo[2,1-b][1,3]thiazole
Interactions	PUBCHEM	N-(2,5-diethoxy-4-morpholin-4-ylphenyl)-2-(furan-2-yl)quinoline-4-carboxamide
Interactions	PUBCHEM	2-(furan-2-carbonyl)-N-[2-morpholin-4-yl-5-(trifluoromethyl)phenyl]-dihydro-1H-isoquinoline-3-carboxamide

Main Category	Category	Terms
Interactions	PUBCHEM	Zeranol
Interactions	PUBCHEM	N-[4-(4-methylphenyl)-1,3-thiazol-2-yl]-1,2-oxazole-5-carboxamide
Interactions	PUBCHEM	Chlordecone
Interactions	PUBCHEM	2-(6-Imidazo[2,1-b]thiazolyl)acetic acid 1,3-benzothiazol-2-ylmethy
Interactions	PUBCHEM	1-[2,5-Dimethyl-1-(thiophen-2-ylmethyl)pyrrol-3-yl]-2-[[5-(4-methoxyphenyl)-1,3,4-oxadiazol-2-yl]sulfanyl]ethanone
Interactions	PUBCHEM	2-(6,7-dihydro-5H-cyclopenta[f]benzofuran-3-yl)acetic acid (4-ketopyrimido[2,1-b][1,3]benzothiazol-2-yl)methyl ester
Interactions	PUBCHEM	Luliconazole
Interactions	PUBCHEM	Toremifene Citrate
Interactions	PUBCHEM	2-Amino-m-cresol
Interactions	PUBCHEM	Tetraoctylphosphonium bromide
Interactions	PUBCHEM	Diazepam
Interactions	PUBCHEM	2-amino-1-((4-fluorophenyl)methyl)-1H-benzimidazole
Interactions	PUBCHEM	Dibutyl Phthalate
Interactions	PUBCHEM	(7S)-7-(Hexadecanoyloxy)-4-hydroxy-N,N,N-trimethyl-10-oxo-3,5,9-t phosphapentacosan-1-aminium 4-oxide
Interactions	PUBCHEM	Fosfestrol
Interactions	PUBCHEM	Daunorubicin
Interactions	PUBCHEM	Tributylstannanyl
Interactions	PUBCHEM	Thiophanate
Interactions	PUBCHEM	Naproxol
Interactions	PUBCHEM	Fluazifop-p-butyl
Interactions	PUBCHEM	Zuclomiphene Citrate
Interactions	PUBCHEM	Gestodene
Interactions	PUBCHEM	Azoxystrobin
Interactions	PUBCHEM	1-Chloro-4-(2,2,2-trichloro-1-(4-chlorophenyl)ethyl)benzene
Interactions	PUBCHEM	Dichlorophen
Interactions	PUBCHEM	Dichlorvos
Interactions	PUBCHEM	Dasatinib
Interactions	PUBCHEM	Emetine dihydrochloride
Interactions	PUBCHEM	Diisopropyl Methylphosphonate
Interactions	PUBCHEM	Dilazep
Interactions	PUBCHEM	N-(4-Hydroxy-2-methyl-6-quinolinyl)acetamide

Main Category	Category	Terms
Interactions	PUBCHEM	Dimethadione
Interactions	PUBCHEM	Mefexamide hydrochloride
Interactions	PUBCHEM	4-(6-Nitro-1H-benzimidazol-2-yl)benzenamine
Interactions	PUBCHEM	Alclofenac
Interactions	PUBCHEM	Benzophenone
Interactions	PUBCHEM	Dipyrithione
Interactions	PUBCHEM	Bromocriptine Mesylate
Interactions	PUBCHEM	Bromocriptine
Interactions	PUBCHEM	Disulfiram
Interactions	PUBCHEM	Benzylhexadecyldimethylammonium chloride
Interactions	PUBCHEM	Benzyldimethylstearylammonium chloride
Interactions	PUBCHEM	4-Hydroxydiphenylamine
Interactions	PUBCHEM	Hydrazobenzene
Interactions	PUBCHEM	2-[5-(4-Methoxy-phenyl)-[1,3,4]oxadiazol-2-yl]-benzo[f]chromen-3-o
Interactions	PUBCHEM	Cetylpyridinium Chloride
Interactions	PUBCHEM	Cetyldimethylethylammonium bromide
Interactions	PUBCHEM	Tetradecanal
Interactions	PUBCHEM	Triamcinolone
Interactions	PUBCHEM	Oxyphenisatin
Interactions	PUBCHEM	Thiorphan
Interactions	PUBCHEM	Butonate
Interactions	PUBCHEM	Neopentyl glycol
Interactions	PUBCHEM	Linalyl benzoate
Interactions	PUBCHEM	3-Hydroxy-2-(4-quinolyl)inden-1-one
Interactions	PUBCHEM	Tris(2,3-dibromopropyl) Phosphate
Interactions	PUBCHEM	Tributyl Phosphate
Interactions	PUBCHEM	2,4,7,9-Tetramethyl-5-decyne-4,7-diol
Interactions	PUBCHEM	Methacrylonitrile
Interactions	PUBCHEM	3-[[(2E)-2-[4-(2-cyanoethylsulfanyl)-5-methyl-1,3-dithiol-2-ylidene]-5 methyl-1,3-dithiol-4-yl]sulfanyl]propanenitrile
Interactions	PUBCHEM	Fludrocortisone
Interactions	PUBCHEM	Sodium 3-nitrobenzenesulfonate
Interactions	PUBCHEM	Pregnanolone

Main Category	Category	Terms
Interactions	PUBCHEM	2,6-Di-tert-butyl-4-methylphenol
Interactions	PUBCHEM	Pyrene
Interactions	PUBCHEM	2-Methyl-1-nitroanthraquinone
Interactions	PUBCHEM	Rafoxanide
Interactions	PUBCHEM	Metipranolol
Interactions	PUBCHEM	N,N-Dimethyl-N-(2-phenoxyethyl)-1-dodecanaminium
Interactions	PUBCHEM	3-[(1-tert-butyltetrazol-5-yl)-[furan-2-ylmethyl(oxolan-2-ylmethyl)amino]methyl]-8-methyl-1H-quinolin-2-one
Interactions	PUBCHEM	N-(3-cyano-6-methyl-4,5,6,7-tetrahydro-1-benzothiophen-2-yl)-5-[(4-nitropyrazol-1-yl)methyl]furan-2-carboxamide
Interactions	PUBCHEM	8,9-dimethyl-6-nitro-4-(pyridin-2-yl)-3a,4,5,9b-tetrahydro-3H-cyclopenta[c]quinoline
Interactions	PUBCHEM	CID 3168140
Interactions	PUBCHEM	CID 3168141
Interactions	PUBCHEM	Doxorubicin
Interactions	PUBCHEM	Pirimiphos-Ethyl
Interactions	PUBCHEM	Triazophos
Interactions	PUBCHEM	Emodin
Interactions	PUBCHEM	Endosulfan
Interactions	PUBCHEM	5-{2-[Cyano(cyclopentylidene)methyl]-1,3-thiazol-4-yl}-6-methyl-2-(propylthio)nicotinonitrile
Interactions	PUBCHEM	5-bromo-N-[4-[5-(2-ethoxyphenyl)-1,2,4-oxadiazol-3-yl]phenyl]furan-carboxamide
Interactions	PUBCHEM	N-(5-chloro-2-methoxyphenyl)-4-methyl-1,3-oxazole-5-carboxamide
Interactions	PUBCHEM	ethyl (4Z)-4-{[(1,1-dioxidotetrahydrothien-3-yl)amino]methylene}-2-methyl-5-oxo-4,5-dihydronaphtho[1,2-b]furan-3-carboxylate
Interactions	PUBCHEM	N-(2-furylmethyl)-2-(5-methyl-2-furyl)quinoline-4-carboxamide
Interactions	PUBCHEM	N-(2-furylmethyl)-4,6-dihydrothieno[3,4-b]thiophene-2-carboxamide
Interactions	PUBCHEM	6-Methoxy-2-({4-[3-(trifluoromethyl)phenyl]piperazin-1-yl} carbonyl)furo[2,3-b]quinoline
Interactions	PUBCHEM	N-[2-[2-(cyclopropylamino)-2-oxoethyl]sulfanyl-1,3-benzothiazol-6-y methylimidazol-2-yl)sulfanylacetamide
Interactions	PUBCHEM	7-{4-[4-(4-chlorophenyl)piperazin-1-yl]-4-oxobutyl}-6-thioxo-6,7-dihydro[1,3]dioxolo[4,5-g]quinazolin-8(5H)-one
Interactions	PUBCHEM	Coumarin
Interactions	PUBCHEM	[2,3-Bis(furan-2-yl)quinoxalin-6-yl]-[4-(2-methoxyphenyl)piperazin-1 yl]methanone

Main Category	Category	Terms
Interactions	PUBCHEM	N-(3-imidazo[1,2-a]pyrimidin-2-ylphenyl)-1,3-benzodioxole-5-carbox
Interactions	PUBCHEM	7-chloro-N-(2-furylmethyl)furo[2,3-b]quinoline-2-carboxamide
Interactions	PUBCHEM	N-(1,3-benzothiazol-2-yl)-2-(4-oxochromeno[4,3-c]pyrazol-1-yl)aceta
Interactions	PUBCHEM	8-Ethoxy-3-(6-methylimidazo[1,2-a]pyridin-2-yl)chromen-2-one
Interactions	PUBCHEM	N-(4-methyl-3-oxo-1,4-benzothiazin-6-yl)-4-piperidin-1-ylpiperidine-1 carboxamide
Interactions	PUBCHEM	N-(1,3-benzodioxol-5-ylmethyl)-3-(4-methoxybenzyl)-2,4-dioxo-1,2,3 tetrahydroquinazoline-7-carboxamide
Interactions	PUBCHEM	N-[4-({(E)-[2-(4-methylphenyl)-5-oxo-1,3-oxazol-4(5H)-ylidene]methy amino)phenyl]acetamide
Interactions	PUBCHEM	Ethyl 4-((4-((6-fluorobenzo[d]thiazol-2-yl)carbamoyl)phenyl)sulfonyl)piperazine-1-carboxylate
Interactions	PUBCHEM	3-hydroxy-2-phenyl-2,3-dihydro-1,5-benzothiazepin-4(5H)-one
Interactions	PUBCHEM	2-[[3-[2-(3,4-dimethoxyphenyl)ethyl]-4-oxo-2-thieno[3,2-d]pyrimidiny N-(5-methyl-3-isoxazolyl)acetamide
Interactions	PUBCHEM	8-Ethoxy-3-[5-(furan-2-ylmethylamino)-1,3,4-thiadiazol-2-yl]chromen
Interactions	PUBCHEM	6-{[ethyl(phenyl)amino]sulfonyl}-N-(2-methoxyethyl)-4-oxo-1,4-dihydroquinoline-3-carboxamide
Interactions	PUBCHEM	N-(4-methoxybenzyl)-5-thien-2-ylisoxazole-3-carboxamide
Interactions	PUBCHEM	6-Amino-4-(2-ethoxyphenyl)-3-phenyl-4,7-dihydroisoxazolo[5,4-b]pyridine-5-carbonitrile
Interactions	PUBCHEM	4-methyl-N-[2-(2-methylindol-1-yl)ethyl]benzenesulfonamide
Interactions	PUBCHEM	9-(3-chlorophenyl)-2-(3,4-dimethoxyphenyl)-8-oxo-8,9-dihydro-7H-p carboxamide
Interactions	PUBCHEM	Cuminol
Interactions	PUBCHEM	Ethacrynic Acid
Interactions	PUBCHEM	Ethion
Interactions	PUBCHEM	2-Hexyldecanoic acid
Interactions	PUBCHEM	3-(1-Dodecen-1-yl)dihydro-2,5-furandione
Interactions	PUBCHEM	Ethylenediamine
Interactions	PUBCHEM	Etizolam
Interactions	PUBCHEM	Exalamide
Interactions	PUBCHEM	1-((3-(2-Chlorophenyl)-2-(4-fluorophenyl)-2-oxiranyl)methyl)-1H-1,2 triazole
Interactions	PUBCHEM	Cambendazole
Interactions	PUBCHEM	Frentizole

Main Category	Category	Terms
Interactions	PUBCHEM	Felodipine
Interactions	PUBCHEM	Fenbendazole
Interactions	PUBCHEM	Ethofumesate
Interactions	PUBCHEM	Fenipentol
Interactions	PUBCHEM	Fenofibrate
Interactions	PUBCHEM	Fenoldopam
Interactions	PUBCHEM	Fenthion
Interactions	PUBCHEM	2-n-Octyl-4-isothiazolin-3-one
Interactions	PUBCHEM	Fipronil
Interactions	PUBCHEM	Sodium Azide
Interactions	PUBCHEM	Penfluridol
Interactions	PUBCHEM	N-[4-(1,3-benzothiazol-2-yl)-1,3-thiazol-2-yl]-2,3-dihydro-1,4-benzodioxine-3-carboxamide
Interactions	PUBCHEM	Norflurazon
Interactions	PUBCHEM	1,8-Dinitro-4,5-dihydroxyanthraquinone
Interactions	PUBCHEM	5-Fluorouracil
Interactions	PUBCHEM	4-Nonylphenol, branched
Interactions	PUBCHEM	GW-7647
Interactions	PUBCHEM	Flurbiprofen
Interactions	PUBCHEM	Fluspirilene
Interactions	PUBCHEM	Flutamide
Interactions	PUBCHEM	Cetraxate hydrochloride
Interactions	PUBCHEM	2-[(4-Methyl-5-sulfanylidene-1,3,4-thiadiazol-2-yl)thio]acetic acid et ester
Interactions	PUBCHEM	Methfuroxam
Interactions	PUBCHEM	N-{2-[(4-methoxyphenyl)amino]-2-oxo-1-phenylethyl}-N-methyl-6-ox dihydropyridine-3-carboxamide
Interactions	PUBCHEM	Furosemide
Interactions	PUBCHEM	Bromuconazole
Interactions	PUBCHEM	Dinitramine
Interactions	PUBCHEM	Prodiamine
Interactions	PUBCHEM	(2-Amino-1,3-thiazol-4-yl)acetic acid
Interactions	PUBCHEM	2-Amino-5,6-dimethylbenzothiazole
Interactions	PUBCHEM	zinc;N,N-dimethylcarbamodithioate

Main Category	Category	Terms
Interactions	PUBCHEM	Gossypol
Interactions	PUBCHEM	Lauryldiethanolamine
Interactions	PUBCHEM	Zidovudine
Interactions	PUBCHEM	2-(4-Bromo-phenyl)-1,2,3,5,6,7-hexahydro-cyclopenta[4,5]thieno[2,3 d]pyrimidin-4-one
Interactions	PUBCHEM	Halazone
Interactions	PUBCHEM	Haloperidol
Interactions	PUBCHEM	Haloprogin
Interactions	PUBCHEM	Sodium 4-styrenesulfonate
Interactions	PUBCHEM	2-(4-Acetylphenoxy)-1-(1-piperidinyl)ethanone
Interactions	PUBCHEM	4-chloro-N-[2-[3-[2-[(5-methyl-1,2-oxazol-3-yl)amino]-2-oxoethyl]sulfanylindol-1-yl]ethyl]benzamide
Interactions	PUBCHEM	Niguldipine hydrochloride, (+/-)-
Interactions	PUBCHEM	Hexachlorophene
Interactions	PUBCHEM	Hexylresorcinol
Interactions	PUBCHEM	Pentabromodiphenyl ethers
Interactions	PUBCHEM	Hycanthone
Interactions	PUBCHEM	Alinidine
Interactions	PUBCHEM	4-Hexylaniline
Interactions	PUBCHEM	Etoposide
Interactions	PUBCHEM	Butralin
Interactions	PUBCHEM	Ibuprofen, (+-)-
Interactions	PUBCHEM	Prothiofos
Interactions	PUBCHEM	Indigotindisulfonic acid
Interactions	PUBCHEM	Diflubenzuron
Interactions	PUBCHEM	Indole-3-Carbinol
Interactions	PUBCHEM	Imazalil
Interactions	PUBCHEM	Indoprofen
Interactions	PUBCHEM	Iodamide
Interactions	PUBCHEM	5,7-Diiodo-8-hydroxyquinoline
Interactions	PUBCHEM	Iopanoic Acid
Interactions	PUBCHEM	Ioxilan
Interactions	PUBCHEM	Ipriflavone

Main Category	Category	Terms
Interactions	PUBCHEM	Dimepranol
Interactions	PUBCHEM	Ribavirin
Interactions	PUBCHEM	N-(3-bromo-1-diethoxyphosphoryl-4-oxonaphthalen-1-yl)benzenesulfonamide
Interactions	PUBCHEM	Isocarboxazid
Interactions	PUBCHEM	Isoconazole
Interactions	PUBCHEM	2-Imino-10-oxo-1-pyridin-3-ylmethyl-1,10-dihydro-2H-1,9,10a-triaza-anthracene-3-carboxylic acid 4-fluoro-benzylamide
Interactions	PUBCHEM	Amikacin
Interactions	PUBCHEM	Hortiacine
Interactions	PUBCHEM	4-[(4-hydroxyphenyl)hydrazono]-3-(4-nitrophenyl)-5-oxo-4,5-dihydro pyrazole-1-carbothioamide
Interactions	PUBCHEM	Octanoic Acid
Interactions	PUBCHEM	(3Z)-6-chloro-3-[5-(3,4-dimethoxyphenyl)-1-methylsulfonylpyrazolic ylidene]-4-phenylquinolin-2-one
Interactions	PUBCHEM	Cridanimod
Interactions	PUBCHEM	4-Hexyloxyaniline
Interactions	PUBCHEM	Pendimethalin
Interactions	PUBCHEM	2-Bromo-5-hydroxybenzaldehyde
Interactions	PUBCHEM	Bortezomib
Interactions	PUBCHEM	Profenofos
Interactions	PUBCHEM	Ciclopirox Olamine
Interactions	PUBCHEM	Leflunomide
Interactions	PUBCHEM	Oxyfluorfen
Interactions	PUBCHEM	Triadimefon
Interactions	PUBCHEM	Difenzoquat metilsulfate
Interactions	PUBCHEM	Lonidamine
Interactions	PUBCHEM	Loratadine
Interactions	PUBCHEM	Vecuronium Bromide
Interactions	PUBCHEM	Bardoxolone Methyl
Interactions	PUBCHEM	Thidiazuron
Interactions	PUBCHEM	2-Amino-4-phenylthiazole hydrobromide monohydrate
Interactions	PUBCHEM	4-Aminophenol
Interactions	PUBCHEM	Flamprop-isopropyl

Main Category	Category	Terms
Interactions	PUBCHEM	Memantine
Interactions	PUBCHEM	Menadione
Interactions	PUBCHEM	Mephenesin
Interactions	PUBCHEM	Sorafenib Tosylate
Interactions	PUBCHEM	5-Aminosalicylic Acid
Interactions	PUBCHEM	Oxfendazole
Interactions	PUBCHEM	methyl 3-[6-(1,3-benzodioxol-5-ylmethylamino)-6-oxohexyl]-2,4-diox quinazoline-7-carboxylate
Interactions	PUBCHEM	Methylmercury chloride
Interactions	PUBCHEM	Desogestrel
Interactions	PUBCHEM	Theofibrate
Interactions	PUBCHEM	Nicardipine Hydrochloride
Interactions	PUBCHEM	8-Methoxypsoralen
Interactions	PUBCHEM	Methoxychlor
Interactions	PUBCHEM	Tripropylene glycol diacrylate
Interactions	PUBCHEM	Methyl Parathion
Interactions	PUBCHEM	Methyl Salicylate
Interactions	PUBCHEM	Carbosulfan
Interactions	PUBCHEM	Alpha-methyldopa
Interactions	PUBCHEM	Methyl 2-[2-(2-chloroanilino)-2-oxoethyl]sulfanyl-4-oxo-3-(oxolan-2- ylmethyl)quinazoline-7-carboxylate
Interactions	PUBCHEM	Nitazoxanide
Interactions	PUBCHEM	3-[4-[4-(3-chlorophenyl)piperazin-1-yl]-4-oxobutyl]-2-sulfanylidene-1 quinazolin-4-one
Interactions	PUBCHEM	2-Cyclopentylacetic acid [2-[(3-chloro-9,10-dioxo-2-anthracenyl)am oxoethyl] ester
Interactions	PUBCHEM	Mitotane
Interactions	PUBCHEM	Moclobemide
Interactions	PUBCHEM	Dinaline
Interactions	PUBCHEM	Ethyl 2-(benzo[d]thiazole-6-carboxamido)-4-methylthiazole-5-carbo
Interactions	PUBCHEM	N,N-diethyl-m-toluamide
Interactions	PUBCHEM	Midazolam Hydrochloride
Interactions	PUBCHEM	N-(3-chloro-1-diethoxyphosphoryl-4-oxonaphthalen-1- yl)benzenesulfonamide
Interactions	PUBCHEM	Vindesine Sulfate

Main Category	Category	Terms
Interactions	PUBCHEM	Fenarimol
Interactions	PUBCHEM	Propiconazole
Interactions	PUBCHEM	Cefonicid Sodium
Interactions	PUBCHEM	2-Hydrazinyl-4-(4-nitrophenyl)thiazole
Interactions	PUBCHEM	Ouabain
Interactions	PUBCHEM	Acifluorfen sodium
Interactions	PUBCHEM	(-)-Camphene
Interactions	PUBCHEM	Nabumetone
Interactions	PUBCHEM	Quinidine
Interactions	PUBCHEM	Griseofulvin
Interactions	PUBCHEM	Digitoxin
Interactions	PUBCHEM	(1R,4S,4'Z,5'S,6R,6'S,8R,10E,13R,14E,20R,21R,24S)-21,24-dihydroxy methoxyimino-5',11,13,22-tetramethyl-6'-[(E)-4-methylpent-2-en-2-yl]spiro[3,7,19-trioxatetracyclo[15.6.1.14,8.020,24]pentacosa-10,14 tetraene-6,2'-oxane]-2-one
Interactions	PUBCHEM	Rocuronium Bromide
Interactions	PUBCHEM	Paromomycin Sulfate
Interactions	PUBCHEM	Linezolid
Interactions	PUBCHEM	Hydrocortisone Sodium Phosphate
Interactions	PUBCHEM	6-(2,3-Dihydro-1,4-benzodioxin-3-yl)-3-(phenoxymethyl)-[1,2,4]triazo b][1,3,4]thiadiazole
Interactions	PUBCHEM	3-Thiatetradecanoic Acid
Interactions	PUBCHEM	Naled
Interactions	PUBCHEM	6-[2-(4-Hydroxy-phenyl)-propyl]-pyridin-3-ol
Interactions	PUBCHEM	trans-6-[1-Ethyl-2-(4-hydroxy-phenyl)-propyl]-pyridin-3-ol
Interactions	PUBCHEM	cis-6-[1-Ethyl-2-(4-hydroxy-phenyl)-butyl]-pyridin-3-ol
Interactions	PUBCHEM	trans-6-[1-Ethyl-2-(4-hydroxy-phenyl)-butyl]-pyridin-3-ol
Interactions	PUBCHEM	6-[2-(4-Hydroxy-phenyl)-1-methyl-ethyl]-pyridin-3-ol
Interactions	PUBCHEM	6-[1-(4-Hydroxy-benzyl)-propyl]-pyridin-3-ol
Interactions	PUBCHEM	D-Phenylalaninamide, (3R)-3-(1,3-benzodioxol-5-yl)-N-((6-methoxy- naphthalenyl)sulfonyl)-beta-alanyl-4-(((2R,6S)-2,6-dimethyl-1- piperidinyl)methyl)-N-methyl-N-(1-methylethyl)-, hydrochloride (1:1
Interactions	PUBCHEM	Ochratoxin A
Interactions	PUBCHEM	Acriflavine
Interactions	PUBCHEM	L-gamma-Glutamyl-S-(2-(4-azidophenyl)-2-oxoethyl)-L-cysteinylgly

Main Category	Category	Terms
Interactions	PUBCHEM	Puromycin Hydrochloride
Interactions	PUBCHEM	Devazepide
Interactions	PUBCHEM	(4S,4aR,7S,7aR,12bS)-1,2,3,4,4a,7,7a,13-octahydro-4,12-methanobenzofuro[3,2-e]isoquinoline-7,9-diol
Interactions	PUBCHEM	1,2-Benzisothiazol-3(2H)-one, 1,1-dioxide, sodium salt, hydrate
Interactions	PUBCHEM	Carubicin
Interactions	PUBCHEM	Barnidipine Hydrochloride
Interactions	PUBCHEM	Epitiostanol
Interactions	PUBCHEM	Doxorubicin Hydrochloride
Interactions	PUBCHEM	Oxendolone
Interactions	PUBCHEM	Terguride
Interactions	PUBCHEM	Estramustine Phosphate Sodium
Interactions	PUBCHEM	Tibolone
Interactions	PUBCHEM	Sultamicillin tosylate
Interactions	PUBCHEM	Loteprednol Etabonate
Interactions	PUBCHEM	(2R,3R,4R,5R,6R)-5-((2R,3R,4R,5S,6R)-5-((2R,3R,4S,5S,6R)-3,4-dihyd methyl-5-((1S,4R,5S,6S)-4,5,6-trihydroxy-3-(hydroxymethyl)cyclohex enylamino)-tetrahydro-2H-pyran-2-yloxy)-3,4-dihydroxy-6-(hydroxym tetrahydro-2H-pyran-2-yloxy)-6-(hydroxymethyl)-tetrahydro-2H-pyra triol
Interactions	PUBCHEM	N-[(2S,3R,4S,6S)-6-[(2S,3S,4S,6R)-6-[(1'R,2R,3S,4'S,6S,8'R,10'E,12'S,13'S,14'E,16'E,20'R,21'R,24'S)-2-buta yl-21',24'-dihydroxy-3,11',13',22'-tetramethyl-2'-oxospiro[2,3-dihydropyran-6,6'-3,7,19-trioxatetracyclo[15.6.1.14,8.020,24]pentacosa-10,14,16,22-tetraene yl]oxy-4-methoxy-2-methyloxan-3-yl]oxy-4-methoxy-2-methyloxan-3 yl]acetamide
Interactions	PUBCHEM	Retinoic Acid
Interactions	PUBCHEM	Alpha-Cyclodextrin
Interactions	PUBCHEM	Fumaric Acid
Interactions	PUBCHEM	Resveratrol
Interactions	PUBCHEM	Retinol
Interactions	PUBCHEM	Foscarnet Sodium
Interactions	PUBCHEM	5-bromo-N-[[3-(2,4-dinitrophenoxy)benzylidene]amino]nicotinamide
Interactions	PUBCHEM	Mycophenolic Acid
Interactions	PUBCHEM	Lycopene
Interactions	PUBCHEM	Nicardipine

Main Category	Category	Terms
Interactions	PUBCHEM	Niclosamide
Interactions	PUBCHEM	[4-[(2-Hydroxybenzoyl)diazenyl]anilino] 4-methylbenzoate
Interactions	PUBCHEM	Diethylstilbestrol
Interactions	PUBCHEM	9-Cis-Retinoic Acid
Interactions	PUBCHEM	Nilutamide
Interactions	PUBCHEM	4-Hydroxytamoxifen
Interactions	PUBCHEM	Nisoldipine
Interactions	PUBCHEM	Nordihydroguaretic acid
Interactions	PUBCHEM	1-O-[(2S,3S)-4-methoxy-16,18-dioxa-10-azapentacyclo[11.7.0.02,6.06,10.015,19]icosa-1(20),4,13,15(19)-te yl] 4-O-methyl (2R)-2-hydroxy-2-(4-hydroxy-4-methylpentyl)butaned
Interactions	PUBCHEM	6-chloro-2-[2-(2,5-dimethyl-1H-pyrrol-1-yl)-3-thienyl]-5-methylpyrazo a]pyrimidin-7-ol
Interactions	PUBCHEM	(2R,4S)-Ketoconazole
Interactions	PUBCHEM	Dactinomycin
Interactions	PUBCHEM	Diofenolan
Interactions	PUBCHEM	(4E)-2-(1,3-benzodioxol-5-yl)-4-[(4-methylanilino)methylidene]isoquinoline-1,3-dione
Interactions	PUBCHEM	Azaserine
Interactions	PUBCHEM	Osalmid
Interactions	PUBCHEM	Melphalan
Interactions	PUBCHEM	Oxazepam
Interactions	PUBCHEM	2-(4-chlorophenoxy)-N-[2-methyl-5-(3-methyl-[1,2,4]triazolo[3,4-b][1,3,4]thiadiazol-6-yl)phenyl]acetamide
Interactions	PUBCHEM	1-Dodecyl-3-methylimidazolium tetrafluoroborate
Interactions	PUBCHEM	2-Hydroxy-4-methoxybenzophenone
Interactions	PUBCHEM	Tolcapone
Interactions	PUBCHEM	Bosutinib methanoate
Interactions	PUBCHEM	Posaconazole
Interactions	PUBCHEM	Dibekacin
Interactions	PUBCHEM	Copper dimethyldithiocarbamate
Interactions	PUBCHEM	Rubitecan
Interactions	PUBCHEM	Fenpropathrin
Interactions	PUBCHEM	Phenacemide
Interactions	PUBCHEM	Phenolphthalein

Main Category	Category	Terms
Interactions	PUBCHEM	Phenol Red
Interactions	PUBCHEM	Micafungin
Interactions	PUBCHEM	Phenylarsine Oxide
Interactions	PUBCHEM	Phloretin
Interactions	PUBCHEM	Phorate
Interactions	PUBCHEM	Forskolin
Interactions	PUBCHEM	Fenoxaprop-ethyl
Interactions	PUBCHEM	Phthalaldehyde
Interactions	PUBCHEM	Piperazine
Interactions	PUBCHEM	Piromidic Acid
Interactions	PUBCHEM	[2-(Benzylcarbamoylamino)-2-oxoethyl] 4,5-dimethoxy-2-(thiophene carbonylamino)benzoate
Interactions	PUBCHEM	Prazepam
Interactions	PUBCHEM	Praziquantel
Interactions	PUBCHEM	Primaquine
Interactions	PUBCHEM	Probenecid
Interactions	PUBCHEM	Proglumetacin
Interactions	PUBCHEM	Propachlor
Interactions	PUBCHEM	Propanil
Interactions	PUBCHEM	Propargite
Interactions	PUBCHEM	Propyl Gallate
Interactions	PUBCHEM	protoporphyrin IX
Interactions	PUBCHEM	Psilocin
Interactions	PUBCHEM	Lusutrombopag
Interactions	PUBCHEM	Pyrimethamine
Interactions	PUBCHEM	2-[[6-methyl-4-oxidanylidene-3-(2-phenylethyl)-6,7-dihydrothieno[3,2 d]pyrimidin-2-yl]sulfanyl]-N-(5-methyl-1,3,4-thiadiazol-2-yl)ethanam
Interactions	PUBCHEM	Isosulfan Blue
Interactions	PUBCHEM	Diclofenac sodium
Interactions	PUBCHEM	Raloxifene
Interactions	PUBCHEM	2-(Tert-butylimino)-5-phenyl-3-(propan-2-yl)-1,3,5-thiadiazinan-4-on
Interactions	PUBCHEM	Propyl pyrazole triol
Interactions	PUBCHEM	Pivoxil Sulbactam

Main Category	Category	Terms
Interactions	PUBCHEM	Riluzole
Interactions	PUBCHEM	Ritanserin
Interactions	PUBCHEM	N-(4-ethylphenyl)-5,6-dihydro-4H-cyclopenta[d][1,2]oxazole-3-carbo
Interactions	PUBCHEM	Avobenzone
Interactions	PUBCHEM	Mitoxantrone Hydrochloride
Interactions	PUBCHEM	4-Nitro-1,2-phenylenediamine
Interactions	PUBCHEM	2-Amino-3-(1,3-benzothiazol-2-yl)-1-cyclohexyl-6-methyl-4-nitropyrr d]pyridazin-7-one
Interactions	PUBCHEM	Sanguinarine
Interactions	PUBCHEM	Fomesafen
Interactions	PUBCHEM	Fenoxycarb
Interactions	PUBCHEM	Salsalate
Interactions	PUBCHEM	Potassium Perchlorate
Interactions	PUBCHEM	Hydroxyquinoline sulfate
Interactions	PUBCHEM	1,3,5-Triazine-2,4,6(1H,3H,5H)-trione, 1,3-dichloro-, sodium salt
Interactions	PUBCHEM	Sodium diethyldithiocarbamate trihydrate
Interactions	PUBCHEM	Semustine
Interactions	PUBCHEM	2-Acetylthiazole
Interactions	PUBCHEM	Metamizole Sodium
Interactions	PUBCHEM	Spiperone
Interactions	PUBCHEM	Isoamyl Cinnamate
Interactions	PUBCHEM	Benzyl Cinnamate
Interactions	PUBCHEM	Desmethoxyyangonin
Interactions	PUBCHEM	Quercetin
Interactions	PUBCHEM	Dinoprost
Interactions	PUBCHEM	Biochanin A
Interactions	PUBCHEM	Formononetin
Interactions	PUBCHEM	Apigenin
Interactions	PUBCHEM	Ergocalciferol
Interactions	PUBCHEM	Vitamin D3
Interactions	PUBCHEM	Rutin
Interactions	PUBCHEM	Kaempferol
Interactions	PUBCHEM	Genistein

Main Category	Category	Terms
Interactions	PUBCHEM	Budesonide
Interactions	PUBCHEM	Quetiapine Fumarate
Interactions	PUBCHEM	Oxymetholone
Interactions	PUBCHEM	Mycophenolate Mofetil
Interactions	PUBCHEM	Mivacurium Chloride
Interactions	PUBCHEM	(1S,2S,6R,14R,15R,16R)-5-(cyclopropylmethyl)-16-(2-hydroxy-3,3-dimethylbutan-2-yl)-15-methoxy-13-oxa-5-azahexacyclo[13.2.2.12,8.01,6.02,14.012,20]icosa-8(20),9,11-trien-ol;hydrochloride
Interactions	PUBCHEM	Paricalcitol
Interactions	PUBCHEM	Canthaxanthin
Interactions	PUBCHEM	Crocin
Interactions	PUBCHEM	(+)-Kavain
Interactions	PUBCHEM	Yangonin
Interactions	PUBCHEM	Zearalenone
Interactions	PUBCHEM	Chrysin
Interactions	PUBCHEM	Morin
Interactions	PUBCHEM	Myricetin
Interactions	PUBCHEM	Coumestrol
Interactions	PUBCHEM	Daidzein
Interactions	PUBCHEM	Matox
Interactions	PUBCHEM	Stearic Acid
Interactions	PUBCHEM	Cilnidipine
Interactions	PUBCHEM	Gamma-Oryzanol
Interactions	PUBCHEM	(2R)-3,4-Dihydro-2,5,7,8-tetramethyl-2-((4R,8R)-4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl (+-)-retinoate
Interactions	PUBCHEM	S-Hydroprene
Interactions	PUBCHEM	Zanapezil Fumarate
Interactions	PUBCHEM	Sofalcone
Interactions	PUBCHEM	(11beta)-11,17-Dihydroxy-21-(((2E,6E)-3,7,11-trimethyl-1-oxo-2,6,10 dodecatrien-1-yl)oxy)pregna-1,4-diene-3,20-dione
Interactions	PUBCHEM	Tranilast
Interactions	PUBCHEM	Menatetrenone
Interactions	PUBCHEM	Etretinate
Interactions	PUBCHEM	Isotretinoin

Main Category	Category	Terms
Interactions	PUBCHEM	Palmitelaidic acid
Interactions	PUBCHEM	2,4-Decadienal
Interactions	PUBCHEM	1alpha-Hydroxy-5,6-trans-vitamin D3
Interactions	PUBCHEM	(5E)-1alpha,25-dihydroxyvitamin D3 / (5E)-1alpha,25-dihydroxycholecalciferol
Interactions	PUBCHEM	(1,1'-Biphenyl)-4-amine, hydrochloride (1:1)
Interactions	PUBCHEM	3-(4-chlorophenyl)-N,N-dimethyl-3-pyridin-2-ylpropan-1-aminium (2 carboxyprop-2-enoate
Interactions	PUBCHEM	Cinnamyl Anthranilate
Interactions	PUBCHEM	Mercury(2+) dichloride
Interactions	PUBCHEM	Phenoxybenzamine Hydrochloride
Interactions	PUBCHEM	Quercetin Dihydrate
Interactions	PUBCHEM	Tetrachlorvinphos
Interactions	PUBCHEM	Toxaphene
Interactions	PUBCHEM	Zinc dibutyldithiocarbamate
Interactions	PUBCHEM	Acitretin
Interactions	PUBCHEM	Astromicin
Interactions	PUBCHEM	Chlormadinone
Interactions	PUBCHEM	Ogen
Interactions	PUBCHEM	Ethisterone
Interactions	PUBCHEM	Nalmefene
Interactions	PUBCHEM	(2S)-2-[(2S,4R,5R,6R)-4-[(2S,4R,5R,6R)-4-[(2S,4S,5R,6R)-4,5-dihydrox dimethyloxan-2-yl]oxy-5-hydroxy-6-methyloxan-2-yl]oxy-5-hydroxy-6 methyloxan-2-yl]oxy-3-[(1S,3S,4R)-3,4-dihydroxy-1-methoxy-2-oxope [(2S,4R,5S,6R)-4-[(2S,4R,5S,6R)-4,5-dihydroxy-6-methyloxan-2-yl]oxy hydroxy-6-methyloxan-2-yl]oxy-8,9-dihydroxy-7-methyl-3,4-dihydro-2 anthracen-1-one
Interactions	PUBCHEM	Proscillaridin
Interactions	PUBCHEM	Dtxcid603638
Interactions	PUBCHEM	4'-Hydroxytamoxifen
Interactions	PUBCHEM	6alpha-Hydroxyestradiol
Interactions	PUBCHEM	Fenretinide
Interactions	PUBCHEM	N-(4-chlorophenyl)-2-cyano-2-(1,3-dihydro-2H-benzimidazol-2-ylidene)acetamide
Interactions	PUBCHEM	7-[4-(furan-2-carbonyl)piperazine-1-carbonyl]-2-(4-methoxyphenyl)- isoindol-1-one

Main Category	Category	Terms
Interactions	PUBCHEM	2-methoxyethyl 6-(furan-2-yl)-3-methyl-4-oxo-4,5,6,7-tetrahydro-1H-indole-2-carboxylate
Interactions	PUBCHEM	3-(2-Ethoxyphenyl)-8-(2-methoxynaphthalen-1-yl)-6-oxo-2,4,7,8-tetrahydropyrido[2,1-b][1,3,5]thiadiazine-9-carbonitrile
Interactions	PUBCHEM	[(E)-but-2-enyl] 6-(furan-2-yl)-3-methyl-4-oxo-1,5,6,7-tetrahydroindol carboxylate
Interactions	PUBCHEM	2-[[4-(2,3-Dimethylphenyl)-1-piperazinyl]-oxomethyl]-1,5-dimethyl-4- pyrrolo[3,2-c]quinolinone
Interactions	PUBCHEM	N-(1,3-benzodioxol-5-yl)-1,5-dimethyl-4-oxopyrrolo[3,2-c]quinoline-2 carboxamide
Interactions	PUBCHEM	2-(4-methoxyphenyl)-3-oxo-1H-isoindole-4-carboxylic acid
Interactions	PUBCHEM	N-[(2-pyridin-3-yl-1,3-thiazol-4-yl)methyl]thiophene-2-sulfonamide
Interactions	PUBCHEM	6-(4-methoxyphenyl)-3-methyl-N-pyridin-4-ylimidazo[2,1-b][1,3]thiaz carboxamide
Interactions	PUBCHEM	5-chloro-N-[2-(2-pyridin-3-yl-1,3-thiazol-4-yl)ethyl]thiophene-2-sulfo
Interactions	PUBCHEM	N-[2-(3-methylimidazo[2,1-b][1,3]thiazol-6-yl)ethyl]-2,3-dihydro-1,4-benzodioxine-6-sulfonamide
Interactions	PUBCHEM	4H-1-Benzopyran-4-one, 7-hydroxy-2-methyl-3-phenoxy-
Interactions	PUBCHEM	Desonide
Interactions	PUBCHEM	Fluprostenol
Interactions	PUBCHEM	Iloprost
Interactions	PUBCHEM	Lacidipine
Interactions	PUBCHEM	Vorinostat
Interactions	PUBCHEM	Cinnamyl Alcohol
Interactions	PUBCHEM	3-Phenylpropyl cinnamate
Interactions	PUBCHEM	Furazolidone
Interactions	PUBCHEM	[2-(6-methylheptyl)-4,6-dinitrophenyl] (E)-but-2-enoate
Interactions	PUBCHEM	Gsk163929B
Interactions	PUBCHEM	Choline bis(trifluoromethylsulfonyl)imide
Interactions	PUBCHEM	Flupirtine
Interactions	PUBCHEM	Sulfamethazine
Interactions	PUBCHEM	AG-538
Interactions	PUBCHEM	Semaxanib
Interactions	PUBCHEM	ethyl (2R,3S)-3-amino-2-phenyl-2,3-dihydro-1-benzofuran-2-carboxy
Interactions	PUBCHEM	N-((1S,4R)-4-(3-(4-Fluorophenoxy)phenoxy)-2-cyclopenten-1-yl)-N-hydroxyurea

Main Category	Category	Terms
Interactions	PUBCHEM	MK-274
Interactions	PUBCHEM	Dtxcid6027349
Interactions	PUBCHEM	1-Ethyl-3-methyl-1H-imidazol-3-ium diethyl phosphate
Interactions	PUBCHEM	1-Dodecyl-3-methylimidazolium Bis(trifluoromethanesulfonyl)imid
Interactions	PUBCHEM	Diethylmethylsulfonium bis(trifluoromethylsulfonyl)imide
Interactions	PUBCHEM	Nifuroxazide
Interactions	PUBCHEM	AC1Nsuzb
Interactions	PUBCHEM	Sulfisoxazole
Interactions	PUBCHEM	5-(4-methylphenyl)-4-{[3-(2-nitrophenyl)-2-propen-1-ylidene]amino}-4H-1,2,4-triazole-3-thiol
Interactions	PUBCHEM	(E)-4-Hydroxytamoxifen
Interactions	PUBCHEM	Molsidomine
Interactions	PUBCHEM	(2-Ethylcrotonoyl)urea
Interactions	PUBCHEM	Cinnamyl propionate
Interactions	PUBCHEM	Linalyl cinnamate
Interactions	PUBCHEM	trans-Cefuroxime
Interactions	PUBCHEM	(R,R)-Fenvalerate
Interactions	PUBCHEM	Benazepril Hydrochloride
Interactions	PUBCHEM	(2-Dodecenyl)succinic anhydride
Interactions	PUBCHEM	11-Hexadecenal, (11Z)-
Interactions	PUBCHEM	Emulphor
Interactions	PUBCHEM	Phenethyl Cinnamate
Interactions	PUBCHEM	((Z)-Hex-3-Enyl) 2-Hydroxybenzoate
Interactions	PUBCHEM	Oleic acid diethanolamide
Interactions	PUBCHEM	alpha-Chlorfenvinphos
Interactions	PUBCHEM	1,4-Dimethylamino-4-nitrostilbene
Interactions	PUBCHEM	Polyoxyethylene dioleate
Interactions	PUBCHEM	Gatifloxacin
Interactions	PUBCHEM	Tazarotene
Interactions	PUBCHEM	Tcpobop
Interactions	PUBCHEM	Tegafur
Interactions	PUBCHEM	Trimebutine Maleate
Interactions	PUBCHEM	Vinblastine Sulfate

Main Category	Category	Terms
Interactions	PUBCHEM	6-Chloro-7-hydroxy-4-methyl-8-[(4-methylpiperidin-1-yl)methyl]chro one
Interactions	PUBCHEM	Temazepam
Interactions	PUBCHEM	Temephos
Interactions	PUBCHEM	4-methyl-N'-[(5-nitro-2-thienyl)methylene]benzohydrazide
Interactions	PUBCHEM	N-(2-furylmethyl)-7-hydroxy-2-oxo-2H-chromene-3-carboxamide
Interactions	PUBCHEM	4-Nitrobenzaldehyde
Interactions	PUBCHEM	Amorolfine
Interactions	PUBCHEM	Thiabendazole
Interactions	PUBCHEM	4-Phenyl-N-(2-phenylethyl)-2-thiazolamine
Interactions	PUBCHEM	Simvastatin
Interactions	PUBCHEM	Nitrofurazone
Interactions	PUBCHEM	Thioridazine
Interactions	PUBCHEM	Thiram
Interactions	PUBCHEM	Carbocyanine
Interactions	PUBCHEM	5-Methyl-2-phenyl-hex-2-enal, (2Z)-
Interactions	PUBCHEM	L-Ascorbic Acid
Interactions	PUBCHEM	Dicoumarol
Interactions	PUBCHEM	Meloxicam
Interactions	PUBCHEM	Pyrvinium Pamoate
Interactions	PUBCHEM	2-Naphthacenecarboxamide, 4-(dimethylamino)-1,4,4a,5,5a,6,11,12 octahydro-3,5,6,10,12,12a-hexahydroxy-6-methyl-1,11-dioxo-, hydrochloride (1:1), (4S,4aR,5S,5aR,6S,12aS)-
Interactions	PUBCHEM	Chlortetracycline hydrochloride
Interactions	PUBCHEM	Rolitetracycline
Interactions	PUBCHEM	Minocycline hydrochloride
Interactions	PUBCHEM	Ticlopidine
Interactions	PUBCHEM	Tilorone
Interactions	PUBCHEM	Tinidazole
Interactions	PUBCHEM	Tinoridine
Interactions	PUBCHEM	Pregabalin
Interactions	PUBCHEM	Raloxifene Hydrochloride
Interactions	PUBCHEM	Nefazodone Hydrochloride
Interactions	PUBCHEM	Ilmofosine

Main Category	Category	Terms
Interactions	PUBCHEM	Tolazamide
Interactions	PUBCHEM	Mifepristone
Interactions	PUBCHEM	Mannuronic acid lactone
Interactions	PUBCHEM	Itraconazole
Interactions	PUBCHEM	Iofetamine hydrochloride
Interactions	PUBCHEM	Triacetin
Interactions	PUBCHEM	Triamterene
Interactions	PUBCHEM	Triazolam
Interactions	PUBCHEM	Triclosan
Interactions	PUBCHEM	Trifluoperazine
Interactions	PUBCHEM	Triton X 100
Interactions	PUBCHEM	Troglitazone
Interactions	PUBCHEM	Sodium Dimethyldithiocarbamate
Interactions	PUBCHEM	Rjx9mkq44K
Interactions	PUBCHEM	1-(3-Phenyl-1,2,4-triazin-5-yl)-1-butanone (4-chlorophenyl)hydrazon
Interactions	PUBCHEM	Roxatidine Acetate Hydrochloride
Interactions	PUBCHEM	(1S,4S,7Z,10S,16E,21R)-7-Ethylidene-4,21-bis(1-methylethyl)-2-oxa-dithia-5,8,20,23-tetraazabicyclo(8.7.6)tricos-16-ene-3,6,9,19,22-pen
Interactions	PUBCHEM	4-[(5,7-Dimethyl-[1,2,4]triazolo[1,5-a]pyrimidin-2-yl)sulfanylmethyl]-7 hydroxychromen-2-one
Interactions	PUBCHEM	Iproniazid sulfate
Interactions	PUBCHEM	Ethaverine Hydrochloride
Interactions	PUBCHEM	(S)-(+)-2-Methylbutyl P-[(P-methoxybenzylidene)amino]cinnamate
Interactions	PUBCHEM	3,3'-Diethylthiacarbocyanine iodide
Interactions	PUBCHEM	ethyl 7-amino-6-cyano-8-(2-methoxybenzylidene)-3-(2-methoxyphe oxo-3,4-dihydro-8H-pyrrolo[2,1-c][1,4]thiazine-1-carboxylate
Interactions	PUBCHEM	Donepezil Hydrochloride
Interactions	PUBCHEM	[(1S,4S)-4-[2-amino-6-(cyclopropylamino)purin-9-yl]cyclopent-2-en-yl]methanol
Interactions	PUBCHEM	Mitomycin C
Interactions	PUBCHEM	Cortisol
Interactions	PUBCHEM	Estriol
Interactions	PUBCHEM	Estradiol
Interactions	PUBCHEM	(+)-Lysergic acid diethylamide

Main Category	Category	Terms
Interactions	PUBCHEM	8-Methoxy-3-(5-p-tolyl-[1,3,4]oxadiazol-2-ylmethylsulfanyl)-5H-[1,2,4]triazino[5,6-b]indole
Interactions	PUBCHEM	Reserpine
Interactions	PUBCHEM	(8Z)-2-amino-4-(furan-3-yl)-8-(furan-3-ylmethylidene)-5,6,7,8-tetrahydro-4H-chromene-3-carbonitrile
Interactions	PUBCHEM	Benzoic acid, 4-amino-, 4-acetoxy-2,2,6,6-tetramethyl-1-piperidinyl e
Interactions	PUBCHEM	Cibenzoline Succinate
Interactions	PUBCHEM	(5Z)-2-(2-furyl)-5-[(5-methyl-2-furyl)methylene][1,3]thiazolo[3,2-b][1,2,4]triazol-6(5H)-one
Interactions	PUBCHEM	2-Aminophenol
Interactions	PUBCHEM	Tiratricol
Interactions	PUBCHEM	Dienestrol diacetate
Interactions	PUBCHEM	3-[(3,4-Dimethylphenoxy)methyl]-2-furoic acid
Interactions	PUBCHEM	N-(4-Methoxyphenyl)Acetamide
Interactions	PUBCHEM	Norethisterone Acetate
Interactions	PUBCHEM	Spironolactone
Interactions	PUBCHEM	Prednisolone Acetate
Interactions	PUBCHEM	Tetrachlorovinphos
Interactions	PUBCHEM	Lynestrenol
Interactions	PUBCHEM	Prednisone
Interactions	PUBCHEM	Nonyl 4-Hydroxybenzoate
Interactions	PUBCHEM	Estrone
Interactions	PUBCHEM	Methylprednisolone Acetate
Interactions	PUBCHEM	Dehydroepiandrosterone
Interactions	PUBCHEM	DIBENZ(a,h)ANTHRACENE
Interactions	PUBCHEM	2-Acetylaminofluorene
Interactions	PUBCHEM	Liothyronine
Interactions	PUBCHEM	N-Nitrosodiethylamine
Interactions	PUBCHEM	Metol
Interactions	PUBCHEM	N,N-Dimethyl-4-[2-(3-methylphenyl)diazenyl]benzenamine
Interactions	PUBCHEM	Nitrogen mustard hydrochloride
Interactions	PUBCHEM	Deoxycorticosterone acetate
Interactions	PUBCHEM	BENZ(a)ANTHRACENE
Interactions	PUBCHEM	alpha-ethyl-2-oxo-1-Pyrrolidineacetamide

Main Category	Category	Terms
Interactions	PUBCHEM	Hexadecyltrimethylammonium Bromide
Interactions	PUBCHEM	Tepoxalin
Interactions	PUBCHEM	Vincristine
Interactions	PUBCHEM	Physostigmine
Interactions	PUBCHEM	Ethinylestradiol
Interactions	PUBCHEM	Chlordane, technical
Interactions	PUBCHEM	Progesterone
Interactions	PUBCHEM	Testosterone Propionate
Interactions	PUBCHEM	7,12-Dimethylbenz(a)anthracene
Interactions	PUBCHEM	Apomorphine
Interactions	PUBCHEM	Methyltestosterone
Interactions	PUBCHEM	Testosterone
Interactions	PUBCHEM	Sertindole
Interactions	PUBCHEM	Promethazine Hydrochloride
Interactions	PUBCHEM	Adapalene
Interactions	PUBCHEM	(4S)-4-(((2S)-2-(((4R)-2-((1R,2R)-1-amino-2-methylbutyl)-4,5-dihydro thiazole-4-carbonyl)amino)-4-methylpentanoyl)amino)-5-(((2R,3S)- (((3R,6R,9R,12R,15R,18R,21R)-3-(2-amino-2-oxoethyl)-18-(3-aminopropyl)-12-benzyl-15-((2R)-butan-2-yl)-6-(carboxymethyl)-9-(imidazol-5-ylmethyl)-2,5,8,11,14,17,20-heptaoxo-1,4,7,10,13,16,19-heptazacyclopentacos-21-yl)amino)-3-methyl-1-oxopentan-2-yl)am oxopentanoic acid
Interactions	PUBCHEM	Tylosin Tartrate
Interactions	PUBCHEM	2-{[(2E)-2-(1-Azabicyclo[2.2.2]oct-3-ylidene)-2-fluoroethyl]oxy}-9H-carbazole
Interactions	PUBCHEM	3-Aminopropan-1-ol;1-hydroxy-4-methyl-6-(2,4,4-trimethylpentyl)py one
Interactions	PUBCHEM	N-[(3S,4S)-3-methyl-1-(2-phenylethyl)piperidin-4-yl]-N-phenylpropanamide;hydrochloride
Interactions	PUBCHEM	(-)-Metazocine fumarate (1:1)
Interactions	PUBCHEM	Dtxcid5028530
Interactions	PUBCHEM	Mecobalamin
Interactions	PUBCHEM	Potassium antimony(III) tartrate hydrate
Interactions	PUBCHEM	Glycyrrhizic acid, monosodium salt
Interactions	PUBCHEM	Dibromopropamidine dihydrochloride
Interactions	PUBCHEM	Dtxcid6028995

Main Category	Category	Terms
Interactions	PUBCHEM	1-Hexadecyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imid
Interactions	PUBCHEM	1-Methyl-3-tetradecylimidazolium bis(trifluoromethylsulfonyl)imide
Interactions	PUBCHEM	1-Methyl-3-propylimidazolium hexafluorophosphate
Interactions	PUBCHEM	1-Butyl-1-methylpiperidinium trifluoromethanesulfonate
Interactions	PUBCHEM	Tributyltetradecylphosphonium dodecylbenzenesulfonate
Interactions	PUBCHEM	N-[(S)-[(3S)-1-Azabicyclo[2.2.2]octan-3-yl](phenyl)methyl]-2,6-dichl (trifluoromethyl)benzamide-hydrogen chloride (1:1)
Interactions	PUBCHEM	SR146131 trifluoroacetate (1:1)
Interactions	PUBCHEM	Ave6324
Interactions	PUBCHEM	Ssr126768
Interactions	PUBCHEM	Dtxcid2027387
Interactions	PUBCHEM	Dtxcid5022065
Interactions	PUBCHEM	3-(3-((8-Methoxyoctyl)oxy)propyl)-1,1,1,3,5,5,5-heptamethyltrisiloxa
Interactions	PUBCHEM	Indium(III) chloride hydrate
Interactions	PUBCHEM	PharmaGSID_48166
Interactions	PUBCHEM	PharmaGSID_48507
Interactions	PUBCHEM	PharmaGSID_48514
Interactions	PUBCHEM	PharmaGSID_48172
Interactions	PUBCHEM	Exemestane
Interactions	PUBCHEM	Cephalothin
Interactions	PUBCHEM	Triphenylethylene
Interactions	PUBCHEM	2,3,4,6-Tetrachlorophenol
Interactions	PUBCHEM	Amlodipine Besylate
Interactions	PUBCHEM	4-Aminoazobenzene
Interactions	PUBCHEM	C.I. Solvent Yellow 2
Interactions	PUBCHEM	Pioglitazone Hydrochloride
Interactions	PUBCHEM	Cidofovir
Interactions	PUBCHEM	Marbofloxacin
Interactions	PUBCHEM	Topotecan Hydrochloride
Interactions	PUBCHEM	Tenidap
Interactions	PUBCHEM	Gadoteridol
Interactions	PUBCHEM	Gemcitabine Hydrochloride
Interactions	PUBCHEM	Pecazine

Main Category	Category	Terms
Interactions	PUBCHEM	Aripiprazole
Interactions	PUBCHEM	Monatepil
Interactions	PUBCHEM	Bisphenol E
Interactions	PUBCHEM	Atorvastatin Calcium
Interactions	PUBCHEM	Papaverine Hydrochloride
Interactions	PUBCHEM	Saquinavir Mesylate
Interactions	PUBCHEM	Tetramethylammonium hydroxide
Interactions	PUBCHEM	Methylene Blue
Interactions	PUBCHEM	2-Methylundecanal
Interactions	PUBCHEM	8-Hydroxyquinoline citrate
Interactions	PUBCHEM	Phenelzine Sulfate
Interactions	PUBCHEM	Trichloroacetamide
Interactions	PUBCHEM	1,1,1,2-Tetrabromoethane
Interactions	PUBCHEM	Tert-Amyl methyl ether
Interactions	PUBCHEM	Octyl Gallate
Interactions	PUBCHEM	Androstenedione
Interactions	PUBCHEM	L-Phenylalanine
Interactions	PUBCHEM	Cobaltous sulfate heptahydrate
Interactions	PUBCHEM	D-Lactic Acid
Interactions	PUBCHEM	Deoxycorticosterone
Interactions	PUBCHEM	Colchicine
Interactions	PUBCHEM	Cycloheximide
Interactions	PUBCHEM	4,4'-(Propane-1,2-diyl)diphenol
Interactions	PUBCHEM	3-Iodo-2-propynyl butylcarbamate
Interactions	PUBCHEM	9H-Carbazol-3-amine, 9-ethyl-, hydrochloride (1:1)
Interactions	PUBCHEM	Flumetralin
Interactions	PUBCHEM	beta-NAPHTHYL ANTHRANILATE
Interactions	PUBCHEM	Lactofen
Interactions	PUBCHEM	Benoxacor
Interactions	PUBCHEM	Norethindrone
Interactions	PUBCHEM	o-Dianisidine dihydrochloride
Interactions	PUBCHEM	Norethynodrel

Main Category	Category	Terms
Interactions	PUBCHEM	Scarlet red
Interactions	PUBCHEM	Tetradecylpyridinium bromide
Interactions	PUBCHEM	Benzylaminopurine
Interactions	PUBCHEM	17-Hydroxyprogesterone
Interactions	PUBCHEM	Chlorpromazine Hydrochloride
Interactions	PUBCHEM	Bis(TRICHLOROMETHYL) SULFONE
Interactions	PUBCHEM	Octrizole
Interactions	PUBCHEM	2-(1,5-Bis(5-nitrofuran-2-yl)penta-1,4-dien-3-ylidene)hydrazinecarboximidamide
Interactions	PUBCHEM	Cytarabine Hydrochloride
Interactions	PUBCHEM	tert-Octylphenol
Interactions	PUBCHEM	Tinuvin
Interactions	PUBCHEM	Pentaerythritol tetraacrylate
Interactions	PUBCHEM	Dioctyldimethylammonium chloride
Interactions	PUBCHEM	Phenacyl bromide
Interactions	PUBCHEM	1,1,2,2-Tetrakis(4-glycidoxyphenyl)ethane
Interactions	PUBCHEM	Dinitrofluorobenzene
Interactions	PUBCHEM	Sodium Persulfate
Interactions	PUBCHEM	Diiodomethyl p-tolyl sulfone
Interactions	PUBCHEM	Daunorubicin Hydrochloride
Interactions	PUBCHEM	Dimethyloctadecyl(3-(trimethoxysilyl)propyl)ammonium chloride
Interactions	PUBCHEM	S-Bioallethrin
Interactions	PUBCHEM	Reconcile
Interactions	PUBCHEM	Propranolol Hydrochloride
Interactions	PUBCHEM	Mestranol
Interactions	PUBCHEM	Methaqualone
Interactions	PUBCHEM	Alizarin
Interactions	PUBCHEM	P,P'-ddd
Interactions	PUBCHEM	Trovafloxacin mesylate
Interactions	PUBCHEM	Verapamil Hydrochloride
Interactions	PUBCHEM	Methandrostenolone
Interactions	PUBCHEM	Chlorquinaldol
Interactions	PUBCHEM	N,N'-Diphenyl-P-Phenylenediamine

Main Category	Category	Terms
Interactions	PUBCHEM	Ivermectin B1a
Interactions	PUBCHEM	Diphenyl phosphonate
Interactions	PUBCHEM	Triphenyltin oxide
Interactions	PUBCHEM	Myclobutanil
Interactions	PUBCHEM	phenyl N-(phenylsulfonyl)benzenecarbimidothioate
Interactions	PUBCHEM	Idarubicin Hydrochloride
Interactions	PUBCHEM	Bekanamycin Sulfate
Interactions	PUBCHEM	Cefpiramide
Interactions	PUBCHEM	Iodoform
Interactions	PUBCHEM	Cinnamaldehyde
Interactions	PUBCHEM	Geraniol
Interactions	PUBCHEM	Benzylideneacetone
Interactions	PUBCHEM	Fumaronitrile
Interactions	PUBCHEM	Retinal
Interactions	PUBCHEM	Piperine
Interactions	PUBCHEM	Retinol, acetate
Interactions	PUBCHEM	trans-Stilbene
Interactions	PUBCHEM	Acetone Cyanohydrin
Interactions	PUBCHEM	4-Styrylpyridine
Interactions	PUBCHEM	Efavirenz
Interactions	PUBCHEM	Pyraclostrobin
Interactions	PUBCHEM	Diphenoxylate Hydrochloride
Interactions	PUBCHEM	Benzothiazolium, 3-ethyl-2-(5-(3-ethyl-2(3H)-benzothiazolylidene)-1 pentadienyl)-, iodide
Interactions	PUBCHEM	Stilonium Iodide
Interactions	PUBCHEM	Pyrantel Tartrate
Interactions	PUBCHEM	avermectin B1a
Interactions	PUBCHEM	Flupirtine Maleate
Interactions	PUBCHEM	13-cis-Retinal
Interactions	PUBCHEM	(5Z)-5-(4-Chlorobenzylidene)-2,2-dimethyl-1-(1H-1,2,4-triazol-1-ylmethyl)cyclopentanol
Interactions	PUBCHEM	Diniconazole
Interactions	PUBCHEM	Ricinoleic Acid

Main Category	Category	Terms
Interactions	PUBCHEM	Triamcinolone Acetonide
Interactions	PUBCHEM	Beta-Zearalenol
Interactions	PUBCHEM	Lumefantrine
Interactions	PUBCHEM	Flavin Mononucleotide
Interactions	PUBCHEM	E-Guggulsterone
Interactions	PUBCHEM	(-)-Valerenic acid
Interactions	PUBCHEM	Everolimus
Interactions	PUBCHEM	Calmidazolium Chloride
Interactions	PUBCHEM	2-Methyl-2-nitro-1-propanol
Interactions	PUBCHEM	Fluoxymesterone
Interactions	PUBCHEM	2-[3-[3-[3-(1-Imidazolyl)propyl]-4-oxo-2-quinazolinyl]propyl]isoindole dione
Interactions	PUBCHEM	Guggulsterone
Interactions	PUBCHEM	Axitinib
Interactions	PUBCHEM	Bromocresol Green
Interactions	PUBCHEM	Sunitinib Malate
Interactions	PUBCHEM	N-[1-[(4-methylphenyl)methyl]-2-benzimidazolyl]-2-furancarboxami
Interactions	PUBCHEM	2-amino-4-(7-methoxy-1,3-benzodioxol-5-yl)-7-methyl-5-oxo-4H,5H- pyrano[4,3-b]pyran-3-carbonitrile
Interactions	PUBCHEM	Antioxidant 1010
Interactions	PUBCHEM	1-(2-Methyl-2,3-dihydro-indol-1-yl)-2-{5-[4-(pyrrolidine-1-sulfonyl)-ph [1,3,4]oxadiazol-2-ylsulfanyl}-ethanone
Interactions	PUBCHEM	1-(2,5-Dimethoxy-phenyl)-2-[5-(3,4-dimethoxy-phenyl)-[1,3,4]oxadia ylsulfanyl]-ethanone
Interactions	PUBCHEM	6-(4-Chloro-phenyl)-3-(5-furan-2-yl-[1,2,4]oxadiazol-3-yl)-2-methoxy- pyridine
Interactions	PUBCHEM	N-(1,3-benzodioxol-5-ylmethyl)-N-[(6-ethyl-2-oxo-1H-quinolin-3-yl)methyl]furan-2-carboxamide
Interactions	PUBCHEM	Diphenylcyclopropenone
Interactions	PUBCHEM	Tanespimycin
Interactions	PUBCHEM	2,2':5',2"-Terthiophene
Interactions	PUBCHEM	(5Z)-2-(2-furyl)-5-(2-methoxybenzylidene)[1,3]thiazolo[3,2-b] [1,2,4]triazol-6(5H)-one
Interactions	PUBCHEM	(4-Benzyl-4H-thieno[3,2-b]pyrrol-5-yl)-(2,3-dihydro-indol-1-yl)-metha
Interactions	PUBCHEM	N-(5-methyl-1,3,4-thiadiazol-2-yl)-2-(thiophen-2-yl)acetamide

Main Category	Category	Terms
Interactions	PUBCHEM	o-Cresyl phosphate
Interactions	PUBCHEM	Monocresyl diphenyl phosphate
Interactions	PUBCHEM	2-[[5-(2,3-dimethyl-1H-indol-5-yl)-1,3,4-oxadiazol-2-yl]sulfanyl]-1-(2-fluorophenyl)ethanone
Interactions	PUBCHEM	Ephedrine Hydrochloride
Interactions	PUBCHEM	Mefloquine Hydrochloride
Interactions	PUBCHEM	Dopamine Hydrochloride
Interactions	PUBCHEM	Glutathione Disulfide
Interactions	PUBCHEM	Triethyl Phosphate
Interactions	PUBCHEM	(5E,7E,22E)-9,10-Secoergosta-5,7,10(19),22-tetraen-3beta-OL
Interactions	PUBCHEM	Triparanol
Interactions	PUBCHEM	Dtxcid7010707
Interactions	PUBCHEM	Cadmium acetate, dihydrate
Interactions	PUBCHEM	2,3,4,5,6,8-Hexachlorodecane
Interactions	PUBCHEM	beta-D-Galactopyranoside, (2alpha,3beta,5alpha,15beta,25R)-2,15-dihydroxyspirostan-3-yl O-beta-D-glucopyranosyl-(1-3)-O-beta-D-galactopyranosyl-(1-2)-O-(beta-D-xylopyranosyl-(1-3))-O-beta-D-glucopyranosyl-(1-4)-
Interactions	PUBCHEM	Triciribine
Interactions	PUBCHEM	Norgestimate
Interactions	PUBCHEM	Isophorone
Interactions	PUBCHEM	Tetraiodothyroacetic acid
Interactions	PUBCHEM	3-[(1-cyclopentyltetrazol-5-yl)-(2,3-dihydroindol-1-yl)methyl]-6,7-dimethoxy-1H-quinolin-2-one
Interactions	PUBCHEM	Broquinaldol
Interactions	PUBCHEM	N-[3-[[2-[1-(3,4-dimethylphenyl)tetrazol-5-yl]sulfanylacetyl]amino]phenyl]furan-2-carboxamide
Interactions	PUBCHEM	Mezlocillin
Interactions	PUBCHEM	Sodium saccharin
Interactions	PUBCHEM	Trilostane
Interactions	PUBCHEM	Methylatropine bromide
Interactions	PUBCHEM	Flutropium bromide
Interactions	PUBCHEM	Nonaethylene Glycol Monododecyl Ether
Interactions	PUBCHEM	beta-Artesunate
Interactions	PUBCHEM	Benidipine Hydrochloride

Main Category	Category	Terms
Interactions	PUBCHEM	Dimethamphetamine
Interactions	PUBCHEM	Diethylstilbestrol Dipropionate
Interactions	PUBCHEM	Suloctidil (USAN/INN)
Interactions	PUBCHEM	Propylthiouracil
Interactions	PUBCHEM	3-(2-aminoethylamino)-7-benzyl-1-(benzylamino)-6,8-dihydro-5H-2, naphthyridine-4-carbonitrile
Interactions	PUBCHEM	Pifexole
Interactions	PUBCHEM	Oxaceprol
Interactions	PUBCHEM	Acrylamide
Interactions	PUBCHEM	N-(1-butyl-3-cyanopyrrolo[3,2-b]quinoxalin-2-yl)propanamide
Interactions	PUBCHEM	15,15-Dimethyl-16-oxa-19-thia-3,4,5,6,8,9,10,11-octazapentacyclo[10.7.0.02,6.07,11.013,18]nonadeca-1(12),2,4,7,9 hexaene
Interactions	PUBCHEM	Zaldaride
Interactions	PUBCHEM	Prulifloxacin
Interactions	PUBCHEM	4-Dodecylphenol
Interactions	PUBCHEM	CID 6603610
Interactions	PUBCHEM	Dantrolene Sodium
Interactions	PUBCHEM	Blue FPG-A
Interactions	PUBCHEM	Sodium Nitroprusside
Interactions	PUBCHEM	Nitrofurantoin
Interactions	PUBCHEM	Gabexate Mesilate
Interactions	PUBCHEM	Thioridazine Hydrochloride
Interactions	PUBCHEM	N-(3-cyano-5,6-dihydro-4H-cyclopenta[b]thiophen-2-yl)-2-(3-cyano- pyridin-3-ylpyridin-2-yl)sulfanylacetamide
Interactions	PUBCHEM	2,5-Di-t-pentylhydroquinone
Interactions	PUBCHEM	5-Chloro-2-[6-(5-chloro-1,3-dioxo-3a,4,7,7a-tetrahydroisoindol-2-yl)hexyl]-3a,4,7,7a-tetrahydroisoindole-1,3-dione
Interactions	PUBCHEM	Bisphenol B
Interactions	PUBCHEM	Tetrabromobisphenol A
Interactions	PUBCHEM	1-(3-Amino-7-methoxy-1-pyrazolo[3,4-b]quinolinyl)-2-(4-chlorophenyl)ethanone
Interactions	PUBCHEM	N-[5-(4-chlorophenyl)-7-(4-methoxyphenyl)-3,7-dihydro[1,2,4]triazol a]pyrimidin-2-yl]acetamide
Interactions	PUBCHEM	Tetrachlorobisphenol A

Main Category	Category	Terms
Interactions	PUBCHEM	Bisphenol C
Interactions	PUBCHEM	Sulphenone
Interactions	PUBCHEM	N,N-diethyl-4-[3-(2-methoxyphenyl)-[1,2,4]triazolo[3,4-b][1,3,4]thiadi yl]aniline
Interactions	PUBCHEM	Bisphenol A
Interactions	PUBCHEM	4,4'-Dichlorodiphenyl sulfone
Interactions	PUBCHEM	Bisphenol S
Interactions	PUBCHEM	Cumene Hydroperoxide
Interactions	PUBCHEM	N-(4,6-dimethylpyrimidin-2-yl)-2-[(4-methyl-5-sulfanylidene-1,3,4-thiadiazol-2-yl)sulfanyl]acetamide
Interactions	PUBCHEM	N-Cyclohexyl-4-methylbenzenesulfonamide
Interactions	PUBCHEM	2-Phenoxyethyl methacrylate
Interactions	PUBCHEM	Trenbolone Acetate
Interactions	PUBCHEM	6-Ketoestradiol
Interactions	PUBCHEM	Dicumyl peroxide
Interactions	PUBCHEM	Estradiol Phosphate
Interactions	PUBCHEM	4-tert-Pentylphenol
Interactions	PUBCHEM	3-[2-Methoxy-6-(4-methoxyphenyl)-3-pyridyl]-5-(2-thienyl)-1,2,4-oxa
Interactions	PUBCHEM	Tyramine hydrochloride
Interactions	PUBCHEM	2-[[4-(4-methoxyphenyl)-11-methyl-3-oxo-8-thia-4,6,11-triazatricyclo[7.4.0.02,7]trideca-1(9),2(7),5-trien-5-yl]sulfanyl]-N-(1,3 thiazol-2-yl)acetamide
Interactions	PUBCHEM	Hexaconazole
Interactions	PUBCHEM	(1R,9S)-11-[4-(4-methoxyphthalazin-1-yl)benzoyl]-7,11-diazatricyclo[7.3.1.02,7]trideca-2,4-dien-6-one
Interactions	PUBCHEM	2-[3-(Azepan-1-ylsulfonyl)phenyl]imidazo[1,2-a]pyrimidine
Interactions	PUBCHEM	p,p'-Oxybis(benzenesulfonyl hydrazide)
Interactions	PUBCHEM	(3-Acetyl-4-oxidoquinoxalin-4-ium-2-yl)methyl furan-2-carboxylate
Interactions	PUBCHEM	1-Imidazol-1-yl-3-propylpyrido[1,2-a]benzimidazole-4-carbonitrile
Interactions	PUBCHEM	4-(2-Aminopropan-2-yl)-1-methylcyclohexan-1-amine
Interactions	PUBCHEM	(3aR,6aS)-3-(2-chlorophenyl)-5-(2,5-dimethoxyphenyl)-3a,6a-dihydropyrrolo[3,4-d]isoxazole-4,6-dione
Interactions	PUBCHEM	2-[2-(5-bromofuran-2-yl)-3H-benzimidazol-5-yl]isoindole-1,3-dione
Interactions	PUBCHEM	6-(2,4-dichlorophenyl)-3-(5-methyl-1H-pyrazol-3-yl)-[1,2,4]triazolo[3, [1,3,4]thiadiazole

Main Category	Category	Terms
Interactions	PUBCHEM	N-(3,4-dihydro-1H-[1,4]thiazino[4,3-a]benzimidazol-8-yl)furan-2-carboxamide
Interactions	PUBCHEM	p-tert-Butylphenyl salicylate
Interactions	PUBCHEM	8-(5-Methylfuran-2-yl)-3-(3-methylphenyl)-6-oxo-2,4,7,8-tetrahydropyrido[2,1-b][1,3,5]thiadiazine-9-carbonitrile
Interactions	PUBCHEM	(5E)-N-(2,5-dimethoxyphenyl)-5-(4-methyl-6-oxocyclohexa-2,4-dien-ylidene)-1,2-dihydropyrazole-3-carboxamide
Interactions	PUBCHEM	Musk ketone
Interactions	PUBCHEM	2-Amino-1-naphthalenesulfonic acid
Interactions	PUBCHEM	2-[(3-Phenyl-4,5-dihydro-1,2-oxazol-5-yl)methoxy]benzamide
Interactions	PUBCHEM	2,6-Dimethylnitrobenzene
Interactions	PUBCHEM	Dienestrol
Interactions	PUBCHEM	Moskene
Interactions	PUBCHEM	1-Amino-2-methylanthraquinone
Interactions	PUBCHEM	(3aR,7aS)-3a,7a-dimethylhexahydro-4,7-epoxy-2-benzofuran-1,3-dio
Interactions	PUBCHEM	Xylenol Blue
Interactions	PUBCHEM	4,4'-Bis(4-hydroxyphenyl)valeric acid
Interactions	PUBCHEM	Diphacinone
Interactions	PUBCHEM	1,4-Dihydroxy-2-naphthoic acid
Interactions	PUBCHEM	Acenaphthenequinone
Interactions	PUBCHEM	2,2'-Dibenzoylaminodiphenyl disulfide
Interactions	PUBCHEM	Napthol asol
Interactions	PUBCHEM	3-Hydroxy-3'-nitro-2-naphthanilide
Interactions	PUBCHEM	Quinoline Yellow
Interactions	PUBCHEM	Pindone
Interactions	PUBCHEM	1,5-Dihydroxynaphthalene
Interactions	PUBCHEM	Musk ambrette
Interactions	PUBCHEM	1-(1H-1,2,3-benzotriazol-1-yl)-4-(4-fluorophenyl)phthalazine
Interactions	PUBCHEM	4,4,4-Trifluoro-1-phenyl-1,3-butanedione
Interactions	PUBCHEM	Rotenone
Interactions	PUBCHEM	Phenylthiourea
Interactions	PUBCHEM	4,7-Phenanthroline-5,6-dione
Interactions	PUBCHEM	Dibenzthione
Interactions	PUBCHEM	2-Ethylanthraquinone

Main Category	Category	Terms
Interactions	PUBCHEM	2-Methylanthraquinone
Interactions	PUBCHEM	Perfluorononanoic acid
Interactions	PUBCHEM	Diisobutyl phthalate
Interactions	PUBCHEM	Diquat Dibromide
Interactions	PUBCHEM	Chlorohydroxy benzophenone
Interactions	PUBCHEM	4,4'-Butylidenebis(3-methyl-6-tert-butylphenol)
Interactions	PUBCHEM	8-(3-(4-Fluorophenoxy)propyl)-1-phenyl-1,3,8-triazaspiro(4.5)decan
Interactions	PUBCHEM	N-Phenyl-o-phenylenediamine
Interactions	PUBCHEM	Dyclonine Hydrochloride
Interactions	PUBCHEM	N-Nitrosodiphenylamine
Interactions	PUBCHEM	(3-Quinoxalin-2-ylphenyl) furan-2-carboxylate
Interactions	PUBCHEM	Androstenone
Interactions	PUBCHEM	Scopolamine hydrochloride
Interactions	PUBCHEM	Clomipramine Hydrochloride
Interactions	PUBCHEM	Nitroscanate
Interactions	PUBCHEM	alpha-Estradiol
Interactions	PUBCHEM	Gold(I) potassium cyanide
Interactions	PUBCHEM	Econazole Nitrate
Interactions	PUBCHEM	Setoperone
Interactions	PUBCHEM	Fenspiride Hydrochloride
Interactions	PUBCHEM	Calcium Levulinate
Interactions	PUBCHEM	7-Amino-4-hydroxy-2-naphthalenesulfonic acid
Interactions	PUBCHEM	N-[(E)-(3-hydroxy-4-methoxyphenyl)methylideneamino]-4,6-diphenylpyrimidine-2-carboxamide
Interactions	PUBCHEM	2-amino-N-(2-methoxyethyl)-1-(2-thenylideneamino)pyrrolo[3,2-b]quinoxaline-3-carboxamide
Interactions	PUBCHEM	Artemisinin
Interactions	PUBCHEM	5-(4-methylphenyl)-4-{[(E,2E)-3-(2-nitrophenyl)-2-propenylidene]amino}-4H-1,2,4-triazol-3-yl hydrosulfide
Interactions	PUBCHEM	Clostebol
Interactions	PUBCHEM	Oxabolone Cipionate
Interactions	PUBCHEM	4,4'-Dihydroxybenzophenone
Interactions	PUBCHEM	Etonogestrel
Interactions	PUBCHEM	Eticlopride hydrochloride

Main Category	Category	Terms
Interactions	PUBCHEM	Dapiprazole Hydrochloride
Interactions	PUBCHEM	Resibufogenin
Interactions	PUBCHEM	Ciclesonide
Interactions	PUBCHEM	Mitiglinide Calcium Hydrate
Interactions	PUBCHEM	Temsirolimus
Interactions	PUBCHEM	Ci-1040
Interactions	PUBCHEM	Almitrine Mesylate
Interactions	PUBCHEM	Aclatonium Napadisilate
Interactions	PUBCHEM	2,2'-Methylenebis(4-ethyl-6-tert-butylphenol)
Interactions	PUBCHEM	Butylated hydroxymethylphenol
Interactions	PUBCHEM	Benzene, 1,1'-(1,2-ethanediylbis(thio))bis-
Interactions	PUBCHEM	4-Nitro-3-trifluoromethylphenol
Interactions	PUBCHEM	Dinoseb
Interactions	PUBCHEM	Quinine Ethylcarbonate
Interactions	PUBCHEM	2-(Furan-2-yl)quinoline-4-carboxylic acid
Interactions	PUBCHEM	2,6-Di-tert-butyl-4-nitrophenol
Interactions	PUBCHEM	Diphenyl isophthalate
Interactions	PUBCHEM	Perfluorooctanesulfonamide
Interactions	PUBCHEM	4-Methyl-2-nitroaniline
Interactions	PUBCHEM	2-Acetylfluorene
Interactions	PUBCHEM	4-Nitro-m-xylene
Interactions	PUBCHEM	2-Ethylphenol
Interactions	PUBCHEM	Mercufenol Chloride
Interactions	PUBCHEM	1-Chloro-2,4-Dinitrobenzene
Interactions	PUBCHEM	1-Chloronaphthalene
Interactions	PUBCHEM	Rosacetol
Interactions	PUBCHEM	Diphenylthiourea
Interactions	PUBCHEM	Ethanolamine
Interactions	PUBCHEM	Phenyl-1-naphthylamine
Interactions	PUBCHEM	Anthrone
Interactions	PUBCHEM	Xanthone
Interactions	PUBCHEM	Pivalophenone

Main Category	Category	Terms
Interactions	PUBCHEM	4,4'-Bis(dimethylamino)benzophenone
Interactions	PUBCHEM	Triphenylborane
Interactions	PUBCHEM	Chlorotriethylgermane
Interactions	PUBCHEM	(4-Chloro-phenyl)-(4-pyridin-4-yl-thiazol-2-yl)-amine
Interactions	PUBCHEM	(2S,3S)-2,3-disulfanylbutanedioic acid
Interactions	PUBCHEM	3-(Diethylamino)phenol
Interactions	PUBCHEM	Benzoguanamine
Interactions	PUBCHEM	Benzyltriphenylphosphonium chloride
Interactions	PUBCHEM	3,3'-Dichlorobenzidine
Interactions	PUBCHEM	M-Terphenyl
Interactions	PUBCHEM	3-Methoxy-9,10-dimethyl-[1]benzofuro[6,5-c]isochromen-5-one
Interactions	PUBCHEM	Alizarin Yellow A
Interactions	PUBCHEM	Tolonium chloride
Interactions	PUBCHEM	5-Nitro-2-furanmethanediol diacetate
Interactions	PUBCHEM	3,3-Diphenylacrolein
Interactions	PUBCHEM	4-Aminobiphenyl
Interactions	PUBCHEM	4-Hydroxybiphenyl
Interactions	PUBCHEM	3-Hydroxy-2-naphthoic acid
Interactions	PUBCHEM	Phenothiazine
Interactions	PUBCHEM	5-(4-fluorophenyl)-N-(4-methoxy-1,3-benzothiazol-2-yl)-1,3,4-oxadia amine
Interactions	PUBCHEM	4,4'-Dihydroxybiphenyl
Interactions	PUBCHEM	p-Terphenyl
Interactions	PUBCHEM	5-Methyl-8-hydroxyquinoline
Interactions	PUBCHEM	N,N-Diethyl-p-phenylenediamine
Interactions	PUBCHEM	Methyleugenol
Interactions	PUBCHEM	Romazarit
Interactions	PUBCHEM	Masoprocol
Interactions	PUBCHEM	2-Ethoxyethyl methacrylate
Interactions	PUBCHEM	Liarozole hydrochloride
Interactions	PUBCHEM	Maprotiline Hydrochloride
Interactions	PUBCHEM	Protoporphyrin disodium
Interactions	PUBCHEM	Hydroxyzine monohydrochloride

Main Category	Category	Terms
Interactions	PUBCHEM	Simetride
Interactions	PUBCHEM	2-Hydroxy-4-methoxy-4'-methylbenzophenone
Interactions	PUBCHEM	Benzoylacetone
Interactions	PUBCHEM	Benmoxin
Interactions	PUBCHEM	Batracylin
Interactions	PUBCHEM	Propylparaben
Interactions	PUBCHEM	Alfuzosin Hydrochloride
Interactions	PUBCHEM	Buparvaquone
Interactions	PUBCHEM	Isobutamben
Interactions	PUBCHEM	Benzyl 4-Hydroxybenzoate
Interactions	PUBCHEM	Chlormidazole
Interactions	PUBCHEM	Alloclamide
Interactions	PUBCHEM	Butylparaben
Interactions	PUBCHEM	Myralact
Interactions	PUBCHEM	Dipentamethylenethiuram disulfide
Interactions	PUBCHEM	Amino-6-ethoxybenzothiazole, 2-
Interactions	PUBCHEM	Phenethyl Benzoate
Interactions	PUBCHEM	Efaroxan
Interactions	PUBCHEM	o-Phenetidine
Interactions	PUBCHEM	Tioxidazole
Interactions	PUBCHEM	O-Phenylenediamine
Interactions	PUBCHEM	2-Chlorophenol
Interactions	PUBCHEM	3,4-Dimethylaniline
Interactions	PUBCHEM	Methylhydroquinone
Interactions	PUBCHEM	3,4-Dichlorophenol
Interactions	PUBCHEM	2,5-Dimethylaniline
Interactions	PUBCHEM	4-Chloro-o-phenylenediamine
Interactions	PUBCHEM	2-Amino-4-chlorophenol
Interactions	PUBCHEM	1,2,4,5-Tetrachlorobenzene
Interactions	PUBCHEM	Dicyclohexylthiourea
Interactions	PUBCHEM	Hexachlorocyclohexane
Interactions	PUBCHEM	2-Methylbutanal

Main Category	Category	Terms
Interactions	PUBCHEM	1,2,3-Trichloropropane
Interactions	PUBCHEM	Thioglycerol
Interactions	PUBCHEM	Santonox
Interactions	PUBCHEM	2-tert-Butyl-4-ethylphenol
Interactions	PUBCHEM	2,4-Di-tert-butylphenol
Interactions	PUBCHEM	4-Bromo-2-[3-(2-chlorophenyl)acryloyl]-7-(ethylamino)-2,4,6-cycloheptatrien-1-one
Interactions	PUBCHEM	Genite
Interactions	PUBCHEM	Fenticlor
Interactions	PUBCHEM	4-methoxy-3-nitro-N-phenylbenzamide
Interactions	PUBCHEM	Amperozide Hydrochloride
Interactions	PUBCHEM	Eprazinone Hydrochloride
Interactions	PUBCHEM	2-Methoxy-4-nitroaniline
Interactions	PUBCHEM	o-Aminoazotoluene
Interactions	PUBCHEM	Antimony Potassium Tartrate
Interactions	PUBCHEM	Tetramethylthiuram monosulfide
Interactions	PUBCHEM	4-[4-[(3aS,4R,9bR)-6,8-dimethyl-3a,4,5,9b-tetrahydro-3H-cyclopenta[c]quinolin-4-yl]phenyl]sulfonylmorpholine
Interactions	PUBCHEM	Prochloraz
Interactions	PUBCHEM	Clofentezine
Interactions	PUBCHEM	Flusilazole
Interactions	PUBCHEM	Crystal Violet lactone
Interactions	PUBCHEM	4-tert-Butylcatechol
Interactions	PUBCHEM	9-Anthracenemethanol
Interactions	PUBCHEM	2,4,4'-Trihydroxybenzophenone
Interactions	PUBCHEM	bisphenol AF
Interactions	PUBCHEM	(1R,9S,12S,15R,16Z,18R,19R,21R,23S,24E,26Z,30S,32S,35R)-1,18-dihydroxy-12-[(2R)-1-[(1S,3R,4R)-4-hydroxy-3-methoxycyclohexyl]pr yl]-19,30-dimethoxy-15,17,21,23,29,35-hexamethyl-11,36-dioxa-4-azatricyclo[30.3.1.04,9]hexatriaconta-16,24,26,28-tetraene-2,3,10,1 pentone
Interactions	PUBCHEM	Arsanilic Acid
Interactions	PUBCHEM	4-Ethylbenzenesulfonic acid
Interactions	PUBCHEM	9-Bromoanthracene
Interactions	PUBCHEM	2,6-Dichloro-4-nitroaniline

Main Category	Category	Terms
Interactions	PUBCHEM	Perfluorooctanesulfonic acid
Interactions	PUBCHEM	1,3-Dinitrobenzene
Interactions	PUBCHEM	Methyl 4-methylbenzoate
Interactions	PUBCHEM	Dimethyl-p-toluidine
Interactions	PUBCHEM	N,N-dimethyl-p-phenylenediamine
Interactions	PUBCHEM	4-Nitroaniline
Interactions	PUBCHEM	4'-Methoxyacetophenone
Interactions	PUBCHEM	4-(Dimethylamino)Benzaldehyde
Interactions	PUBCHEM	N-Methyl-4-nitroaniline
Interactions	PUBCHEM	Methyl 3-phenanthryl ketone
Interactions	PUBCHEM	1,1,1,2,2,3,3,4,4-Nonafluoro-6-iodohexane
Interactions	PUBCHEM	N,N,N',N'-tetramethyl-p-phenylenediamine
Interactions	PUBCHEM	4-Nitrophenetol
Interactions	PUBCHEM	3-(4-(4-Chlorophenyl)cyclohexyl)-4-hydroxynaphthalene-1,2-dione
Interactions	PUBCHEM	Irinotecan Hydrochloride
Interactions	PUBCHEM	Benzyl mercaptan
Interactions	PUBCHEM	Phenylmercuric Chloride
Interactions	PUBCHEM	N-Phenylhydroxylamine
Interactions	PUBCHEM	2-((3,5-Diiodo-4-oxo-2,5-cyclohexadien-1-ylidene)(4-hydroxy-3,5-diiodophenyl)methyl)benzoic acid
Interactions	PUBCHEM	Ethylene glycol diacrylate
Interactions	PUBCHEM	3,3'-Dichloro-4,4'-Diaminodiphenylmethane
Interactions	PUBCHEM	Basic Blue 7
Interactions	PUBCHEM	3-Hydroxy-N-phenylaniline
Interactions	PUBCHEM	Triclocarban
Interactions	PUBCHEM	2-(2-Furanyl)-4H-naphtho(1,2-b)pyran-4-one
Interactions	PUBCHEM	Pyridostigmine Bromide
Interactions	PUBCHEM	2-Hydroxyfluorene
Interactions	PUBCHEM	2,4'-Dihydroxydiphenylmethane
Interactions	PUBCHEM	Tris(isopropylphenyl) phosphate
Interactions	PUBCHEM	4-Chloro-8-(diethylamino)pyrano[3,2-c]chromene-2,5-dione
Interactions	PUBCHEM	4-Benzylphenol
Interactions	PUBCHEM	4,4'-Dimethoxydiphenylamine

Main Category	Category	Terms
Interactions	PUBCHEM	Diphenan
Interactions	PUBCHEM	N-isopropyl-N'-phenyl-p-phenylenediamine
Interactions	PUBCHEM	4,4'-Diaminodiphenyl Ether
Interactions	PUBCHEM	Resorcinol diglycidyl ether
Interactions	PUBCHEM	N,N'-Di-sec-butyl-p-phenylenediamine
Interactions	PUBCHEM	Acetoacetanilide
Interactions	PUBCHEM	Diphenylguanidine
Interactions	PUBCHEM	Carbanilide
Interactions	PUBCHEM	2,4-Dicumylphenol
Interactions	PUBCHEM	2-(4-Morpholino)thiobenzothiazole
Interactions	PUBCHEM	Boric Acid
Interactions	PUBCHEM	1,1,1-Trichloro-2,2-bis(4-hydroxyphenyl)ethane
Interactions	PUBCHEM	2-Ethylhexyl Acetate
Interactions	PUBCHEM	Monobenzone
Interactions	PUBCHEM	Bis(2-ethylhexyl) azelate
Interactions	PUBCHEM	9,9-Bis(4-hydroxyphenyl)fluorene
Interactions	PUBCHEM	8-Methoxy-3-(7-methylimidazo[1,2-a]pyridin-2-yl)chromen-2-one
Interactions	PUBCHEM	4-Chlorophenyl Isocyanate
Interactions	PUBCHEM	Nelfinavir mesylate hydrate
Interactions	PUBCHEM	Benzonatate
Interactions	PUBCHEM	Benzenepropanal
Interactions	PUBCHEM	1,2-Diphenoxyethane
Interactions	PUBCHEM	3,3,4,4-Tetrachlorotetrahydrothiophene 1,1-dioxide
Interactions	PUBCHEM	P-Anisidine
Interactions	PUBCHEM	1,4-Benzoquinone dioxime
Interactions	PUBCHEM	2-Amino-5-(5-nitro-2-furyl)-1,3,4-oxadiazole
Interactions	PUBCHEM	4-(2,5-dimethyl-1H-pyrrol-1-yl)-3-methylbenzoic acid
Interactions	PUBCHEM	3,3'-Thiobis(methyl propionate)
Interactions	PUBCHEM	2-Chloroethyldiisopropylammonium chloride
Interactions	PUBCHEM	4-Bromoaniline
Interactions	PUBCHEM	4-Chloroaniline
Interactions	PUBCHEM	1,2-Epoxybutane

Main Category	Category	Terms
Interactions	PUBCHEM	1-Bromopropane
Interactions	PUBCHEM	Atraric Acid
Interactions	PUBCHEM	Allyl chloride
Interactions	PUBCHEM	Chloroacetonitrile
Interactions	PUBCHEM	Glyoxal
Interactions	PUBCHEM	Dibutyl phosphate
Interactions	PUBCHEM	1,3-Butanediol
Interactions	PUBCHEM	1-Methoxy-2-propanol
Interactions	PUBCHEM	1-Nitropropane
Interactions	PUBCHEM	M-Xylene
Interactions	PUBCHEM	2-[(8-quinolinylamino)methylene]-1-benzothiophen-3(2H)-one
Interactions	PUBCHEM	Isobutyl ketone
Interactions	PUBCHEM	2-Isopropylthioxanthone
Interactions	PUBCHEM	2-Chloro-4-biphenylol
Interactions	PUBCHEM	4,4'-Isopropylidenedi-2,6-xylol
Interactions	PUBCHEM	2-[4-(Diethylamino)-2-hydroxybenzoyl]benzoic acid
Interactions	PUBCHEM	Indole
Interactions	PUBCHEM	Pentanoic Acid
Interactions	PUBCHEM	Tetraconazole
Interactions	PUBCHEM	Propanediaminetetraacetic acid
Interactions	PUBCHEM	Isopropyl Myristate
Interactions	PUBCHEM	1,2-Dimethoxyethane
Interactions	PUBCHEM	2-(Ethylamino)ethanol
Interactions	PUBCHEM	3'-Amino-4'-methoxyacetanilide
Interactions	PUBCHEM	1,5-Pentanediol
Interactions	PUBCHEM	4,4'-(4-Methylpentane-2,2-diyl)diphenol
Interactions	PUBCHEM	1-Octene
Interactions	PUBCHEM	Dodecyltrimethylammonium chloride
Interactions	PUBCHEM	Hexadecyltrimethylammonium chloride
Interactions	PUBCHEM	Stearyltrimethylammonium chloride
Interactions	PUBCHEM	Heptyl acetate
Interactions	PUBCHEM	Decyl acetate

Main Category	Category	Terms
Interactions	PUBCHEM	Diethylene glycol diethyl ether
Interactions	PUBCHEM	Undecanal
Interactions	PUBCHEM	1-Chlorododecane
Interactions	PUBCHEM	Bexarotene
Interactions	PUBCHEM	2,7-Naphthalenedisulfonic acid, 5-(acetylamino)-4-hydroxy-3-[2-(2-methoxyphenyl)diazenyl]-, sodium salt (1:2)
Interactions	PUBCHEM	Estradiol Dipropionate
Interactions	PUBCHEM	Benzenamine, 4-(1-methyl-1-phenylethyl)-N-[4-(1-methyl-1-phenylethyl)phenyl]-
Interactions	PUBCHEM	Acetylphenylhydrazine
Interactions	PUBCHEM	Tribromoacetaldehyde
Interactions	PUBCHEM	Oxyphenisatin Acetate
Interactions	PUBCHEM	10-Chloro-9-anthraldehyde
Interactions	PUBCHEM	Bromophenol Blue
Interactions	PUBCHEM	Triphenyl Phosphate
Interactions	PUBCHEM	3-Hydroxybenzophenone
Interactions	PUBCHEM	Tetradifon
Interactions	PUBCHEM	4,4'-(Isopropylidenebis(4,1-phenyleneoxy))dianiline
Interactions	PUBCHEM	2,2'-Diallyl-4,4'-sulfonyldiphenol
Interactions	PUBCHEM	2-Aminoanthraquinone
Interactions	PUBCHEM	Dichlone
Interactions	PUBCHEM	Di(2-ethylhexyl) phthalate
Interactions	PUBCHEM	Bis(2-butoxyethyl) phthalate
Interactions	PUBCHEM	di-N-Octyl Phthalate
Interactions	PUBCHEM	1,3-Dichloro-5,5-dimethylhydantoin
Interactions	PUBCHEM	N-pyridin-2-yl-2-thiophen-2-ylacetamide
Interactions	PUBCHEM	Benzyl Salicylate
Interactions	PUBCHEM	2-Ethylhexyl salicylate
Interactions	PUBCHEM	4,4'-Methylenebis[2,6-diethylaniline]
Interactions	PUBCHEM	Chloranil
Interactions	PUBCHEM	Tetramethrin
Interactions	PUBCHEM	2,2'-Methylenebis(4-methyl-6-tert-butylphenol)
Interactions	PUBCHEM	o-Tolidine

Main Category	Category	Terms
Interactions	PUBCHEM	Veratraldehyde
Interactions	PUBCHEM	Chlorophene
Interactions	PUBCHEM	Dibenzoylmethane
Interactions	PUBCHEM	Ethylparaben
Interactions	PUBCHEM	Dimethyl terephthalate
Interactions	PUBCHEM	2,4-Di-tert-pentylphenol
Interactions	PUBCHEM	1,1-Bis(4-hydroxyphenyl)-2,2-dichloroethylene
Interactions	PUBCHEM	Benzethonium Chloride
Interactions	PUBCHEM	2-Amino-5-Nitrothiazole
Interactions	PUBCHEM	1-Chloro-3-nitrobenzene
Interactions	PUBCHEM	Quinolinium, 1-(phenylmethyl)-, chloride (1:1)
Interactions	PUBCHEM	Sulfan Blue
Interactions	PUBCHEM	1,4-Naphthoquinone
Interactions	PUBCHEM	Butyl 3-mercaptopropionate
Interactions	PUBCHEM	Dinex
Interactions	PUBCHEM	Pamoic Acid
Interactions	PUBCHEM	Deserpidine
Interactions	PUBCHEM	N-Methyl-D-glucamine
Interactions	PUBCHEM	Benzophenone-6
Interactions	PUBCHEM	2,2',4,4'-Tetrahydroxybenzophenone
Interactions	PUBCHEM	2,4-Dihydroxybenzophenone
Interactions	PUBCHEM	N-(1-Methylheptyl)-N'-phenyl-p-phenylenediamine
Interactions	PUBCHEM	Captan
Interactions	PUBCHEM	Folpet
Interactions	PUBCHEM	Phenyl 4-aminosalicylate
Interactions	PUBCHEM	Tebuconazole
Interactions	PUBCHEM	Fenbuconazole
Interactions	PUBCHEM	Phenethyl anthranilate
Interactions	PUBCHEM	2-Benzyl-2-(dimethylamino)-1-(4-(morpholin-4-yl)phenyl)butan-1-or
Interactions	PUBCHEM	Difenoconazole
Interactions	PUBCHEM	Metconazole
Interactions	PUBCHEM	Ipconazole

Main Category	Category	Terms
Interactions	PUBCHEM	Bromofenoxim
Interactions	PUBCHEM	PharmaGSID_48505
Interactions	PUBCHEM	Menthyl Anthranilate
Interactions	PUBCHEM	Tebufenpyrad
Interactions	PUBCHEM	Cyprodinil
Interactions	PUBCHEM	Sulfentrazone
Interactions	PUBCHEM	Fludioxonil
Interactions	PUBCHEM	Lipoamide
Interactions	PUBCHEM	4-Methylbenzophenone
Interactions	PUBCHEM	2-Butyl p-hydroxybenzoate
Interactions	PUBCHEM	2-Naphthol
Interactions	PUBCHEM	Tetrahydromyrcenol
Interactions	PUBCHEM	N-phenyl-2-naphthylamine
Interactions	PUBCHEM	2-(Thiophene-2-carbonylamino)-4,5,6,7,8,9-hexahydrocycloocta[b]thiophene-3-carboxamide
Interactions	PUBCHEM	Diazoaminobenzene
Interactions	PUBCHEM	Phenazopyridine hydrochloride
Interactions	PUBCHEM	Butyl Benzoate
Interactions	PUBCHEM	Diamthazole
Interactions	PUBCHEM	N-(3-acetylphenyl)-2-thiophen-2-ylacetamide
Interactions	PUBCHEM	Ziram
Interactions	PUBCHEM	Phenylparaben
Interactions	PUBCHEM	Tetrachloropicolinonitrile
Interactions	PUBCHEM	N,N-dimethyl-4-nitrosoaniline
Interactions	PUBCHEM	Benzododecinium Chloride
Interactions	PUBCHEM	Benzyldimethyltetradecylammonium chloride
Interactions	PUBCHEM	2-Propen-1-amine, N-ethyl-2-methyl-
Interactions	PUBCHEM	2-phenyl-N-(2-pyridin-3-yl-1,3-benzoxazol-5-yl)acetamide
Interactions	PUBCHEM	4,4'-Thiodianiline
Interactions	PUBCHEM	Benzyl acetate
Interactions	PUBCHEM	4'-Chloroacetyl(acetanilide)
Interactions	PUBCHEM	Aramite
Interactions	PUBCHEM	Pentamidine Isethionate

Main Category	Category	Terms
Interactions	PUBCHEM	4-Tert-octylphenol
Interactions	PUBCHEM	Cetylpyridinium Bromide
Interactions	PUBCHEM	Dibutyl succinate
Interactions	PUBCHEM	Citronellyl propionate
Interactions	PUBCHEM	Acetylethanolamine
Interactions	PUBCHEM	Isoamyl 4-N,N-dimethylaminobenzoate
Interactions	PUBCHEM	Magnesium Acetate
Interactions	PUBCHEM	2-(2-Chlorophenyl)-5-[(4-ethoxyphenyl)amino]-1,3-oxazole-4-carbor
Interactions	PUBCHEM	2,2,4-Trimethyl-1,3-pentanediol
Interactions	PUBCHEM	Pregnenolone
Interactions	PUBCHEM	Melatonin
Interactions	PUBCHEM	Diphenhydramine Hydrochloride
Interactions	PUBCHEM	Quinestrol
Interactions	PUBCHEM	2,2-Dimethoxy-2-phenylacetophenone
Interactions	PUBCHEM	4-Nitrosodiphenylamine
Interactions	PUBCHEM	Tiagabine Hydrochloride
Interactions	PUBCHEM	Isocetyl Stearate
Interactions	PUBCHEM	Galaxolide
Interactions	PUBCHEM	Fluoranthene
Interactions	PUBCHEM	Benzo(K)Fluoranthene
Interactions	PUBCHEM	Liral
Interactions	PUBCHEM	Hydroxyflutamide
Interactions	PUBCHEM	Benzo(B)Triphenylene
Interactions	PUBCHEM	Tolclofos-methyl
Interactions	PUBCHEM	Nuarimol
Interactions	PUBCHEM	Pencycuron
Interactions	PUBCHEM	Penconazole
Interactions	PUBCHEM	Triflumizole
Interactions	PUBCHEM	Chlorfluazuron
Interactions	PUBCHEM	Fluazinam
Interactions	PUBCHEM	Dimethenamid
Interactions	PUBCHEM	Pyridaben

Main Category	Category	Terms
Interactions	PUBCHEM	Flufenoxuron
Interactions	PUBCHEM	Difethialone
Interactions	PUBCHEM	Chlorfenapyr
Interactions	PUBCHEM	cis-N-(1-(2-(2-Thienyl)ethyl)-3-methyl-4-piperidyl)-N-phenylpropana hydrochloride
Interactions	PUBCHEM	2,3,4,4'-Tetrahydroxybenzophenone
Interactions	PUBCHEM	L-2-Aminoadipic Acid
Interactions	PUBCHEM	Hibon
Interactions	PUBCHEM	Maltotriose
Interactions	PUBCHEM	Phenoxathiin
Interactions	PUBCHEM	Benzofuran
Interactions	PUBCHEM	Tebutam
Interactions	PUBCHEM	2-Methyl-1-(4-(methylthio)phenyl)-2-morpholino-1-propanone
Interactions	PUBCHEM	Ethynodiol Diacetate
Interactions	PUBCHEM	Lopinavir
Interactions	PUBCHEM	Bis(2-ethylhexyl) phosphate
Interactions	PUBCHEM	Tetrabromophenolphthalein ethyl ester
Interactions	PUBCHEM	5alpha-Androstan-3beta-ol
Interactions	PUBCHEM	Tolperisone Hydrochloride
Interactions	PUBCHEM	Fosfomycin calcium
Interactions	PUBCHEM	1,1,1-Tris(4-hydroxyphenyl)ethane
Interactions	PUBCHEM	1,3,5-Tri(4-tert-butyl-2,6-dimethyl-3-hydroxybenzyl)-1,3,5-triazine-2,4,6(1H,3H,5H)-trione
Interactions	PUBCHEM	Forchlorfenuron
Interactions	PUBCHEM	Hydralazine Hydrochloride
Interactions	PUBCHEM	N,N'-9H-Fluorene-2,7-diylbis(acetamide)
Interactions	PUBCHEM	5,6-Dioxo-3,4-dihydro-2H,5H,6H-11-oxa-1-thia-4a,12-diaza-naphtha carbaldehyde
Interactions	PUBCHEM	2,6-Diiodo-4-nitrophenol
Interactions	PUBCHEM	Paraoxon
Interactions	PUBCHEM	Estradiol Cypionate
Interactions	PUBCHEM	2,4-Dichloro-1,3-dinitro-5-(trifluoromethyl)benzene
Interactions	PUBCHEM	1-(Bromomethyl)-3-phenoxybenzene
Interactions	PUBCHEM	4'-Fluoro(1,1'-biphenyl)-4-amine

Main Category	Category	Terms
Interactions	PUBCHEM	Hydrogenated bisphenol a
Interactions	PUBCHEM	4,4'-Dinitrocarbanilide
Interactions	PUBCHEM	Bde 47
Interactions	PUBCHEM	Azane;ruthenium(4+);bis(ruthenium(5+));hexachloride;dihydrate
Interactions	PUBCHEM	Argipressin monoacetate
Interactions	PUBCHEM	(6E)-2-ethyl-6-[[5-(2-furanylmethylthio)-2-furanyl]methylidene]-5-imi [1,3,4]thiadiazolo[3,2-a]pyrimidin-7-one
Interactions	PUBCHEM	N"-(4-chloro-3-nitro-benzoyl)-3-nitro-4-(tetrahydrofurfurylthio)benzohydrazide
Interactions	PUBCHEM	10-Methyl-1-(4-methylphenyl)-2,5-bis(oxidanylidene)dipyrido[3,4-c: f]pyrimidine-3-carbonitrile
Interactions	PUBCHEM	5-(3-methoxyphenyl)-N-(5-methyl-2-phenyl-3-pyrazolyl)-3-isoxazolecarboxamide
Interactions	PUBCHEM	5-{[(4-Methylphenyl)sulfonyl]methyl}furan-2-carboxylic acid
Interactions	PUBCHEM	5-chloro-N-[[2-(3-pyridyl)thiazol-4-yl]methyl]thiophene-2-sulfonamid
Interactions	PUBCHEM	N-{4-[4-(morpholin-4-ylsulfonyl)phenyl]-1,3-thiazol-2-yl}benzamide- dioxane (1:1)
Interactions	PUBCHEM	Chlorhexidine
Interactions	PUBCHEM	Perfluorooctanoic acid
Interactions	PUBCHEM	Perfluorodecanoic acid
Interactions	PUBCHEM	Oxiconazole Nitrate
Interactions	PUBCHEM	Chlorhexidine Acetate
Interactions	PUBCHEM	Mecloqualone
Interactions	PUBCHEM	Carumonam Sodium
Interactions	PUBCHEM	Norelgestromin (USAN:INN:BAN)
Interactions	PUBCHEM	2-[(2,5-Dimethoxybenzylidene)amino]-4,5-bis(2-furyl)-3-furonitrile
Interactions	PUBCHEM	2-amino-N-(2-methylpropyl)-1-(pyridin-2-ylmethylideneamino)pyrro b]quinoxaline-3-carboxamide
Interactions	PUBCHEM	3,5-Dioxa-6-aza-4-phosphaoct-6-ene-8-nitrile, 4-ethoxy-7-phenyl-, 4-
Interactions	PUBCHEM	2-[4-[(E)-N-hydroxy-C-methylcarbonimidoyl]phenoxy]-1-piperidin-1-ylethanone
Interactions	PUBCHEM	Chlorhexidine Hydrochloride
Interactions	PUBCHEM	9,10-dihydrobenzo[a]pyren-7(8H)-one
Interactions	PUBCHEM	Fenpyroximate
Interactions	PUBCHEM	Gimatecan

Main Category	Category	Terms
Interactions	PUBCHEM	Methyl (2Z)-(methoxyimino)(2-((((E)-(1-(3-(trifluoromethyl)phenyl)ethylidene)amino)oxy)methyl)phenyl)aceta
Interactions	PUBCHEM	4-methoxy-N-[1-(4-phenyl-3a,4,5,9b-tetrahydro-3H-cyclopenta[c]qui yl)ethylideneamino]benzamide
Interactions	PUBCHEM	Ethanimidothioic acid, 2-(dimethylamino)-N-(((methylamino)carbonyl)oxy)-2-oxo-, methyl ester
Interactions	PUBCHEM	6-(4-chlorophenyl)imidazo(2,1-b)(1,3)thiazole-5-carbaldehyde O-(3, dichlorobenzyl)oxime
Interactions	PUBCHEM	3-Hydroxyfluorene
Interactions	PUBCHEM	2-[[4-[(2E)-2-[(5-chloro-3-methyl-1-phenyl-4-pyrazolyl)methylidene]hydrazinyl]-3-nitrophenyl]sulfonylamino]ben acid
Interactions	PUBCHEM	N-[(1E)-3-Chloro-4-oxocyclohexa-2,5-dien-1-ylidene]-2,4,6-trimethylbenzene-1-sulfonamide
Interactions	PUBCHEM	4-Chloro-2-(3-chloro-4-fluorophenyl)-5-[methyl-(3-pyridinylmethylideneamino)amino]-3-pyridazinone
Interactions	PUBCHEM	Nandrolone Decanoate
Interactions	PUBCHEM	2-(2,2,2-Trifluoroacetamido)fluorene
Interactions	PUBCHEM	Dieldrin
Interactions	PUBCHEM	Curcumin
Interactions	PUBCHEM	Halocarban
Interactions	PUBCHEM	2-Methyl-4-(3-phenyl[1,2,4]triazolo[3,4-b][1,3,4]thiadiazol-6-yl)quino
Interactions	PUBCHEM	2-(4-chlorophenyl)sulfanyl-N-(2-ethyltetrazol-5-yl)acetamide
Interactions	PUBCHEM	Triphenyltin fluoride
Interactions	PUBCHEM	Ergotamine Tartrate
Interactions	PUBCHEM	2-Nitro-alpha,alpha,alpha-trifluorotoluene
Interactions	PUBCHEM	CP-401387
Interactions	PUBCHEM	Ave5638
Interactions	PUBCHEM	Calcipotriol, (5E)-
Interactions	PUBCHEM	Estradiol Acetate
Interactions	PUBCHEM	2-(4-(7-hydroxy-2,3-dihydro-1H-inden-4-yloxy)-3,5-dimethylphenylar oxoacetic acid
Interactions	PUBCHEM	Abiraterone Acetate
Interactions	PUBCHEM	4-Methyl-1-phenylpyrazolidin-3-one
Interactions	PUBCHEM	CP-457920
Interactions	PUBCHEM	Sincalide

Main Category	Category	Terms
Interactions	PUBCHEM	Prallethrin
Interactions	PUBCHEM	3,6-Acridinediamine, sulfate (2:1)
Interactions	PUBCHEM	Surinabant
Interactions	PUBCHEM	C14 Linker; DECA-14; Quinolinium
Interactions	PUBCHEM	Palmitic Acid
Interactions	PUBCHEM	1-(2-(3,4-dichlorophenoxy)-5-fluorophenyl)-N-methylmethanamine
Interactions	PUBCHEM	Cyazofamid
Interactions	PUBCHEM	CP-544439
Interactions	PUBCHEM	Cyproterone Acetate
Interactions	PUBCHEM	N-(2-(1,6,7,8-Tetrahydro-2H-indeno(5,4-b)furan-8-yl)ethyl)propanam
Interactions	PUBCHEM	Hmr-1426
Interactions	PUBCHEM	Oxalato(trans-l-1,2-cyclohexanediamine)platinum(II)
Interactions	PUBCHEM	Methanaminium, N-(4-((4-(dimethylamino)phenyl)phenylmethylene cyclohexadien-1-ylidene)-N-methyl-, ethanedioate (1:1)
Interactions	PUBCHEM	Tri-n-butyl tetradecyl phosphonium chloride
Interactions	PUBCHEM	Pyrrolidine, 1-(2-(4-(3,4-dihydro-7-methoxy-2,2-dimethyl-3-phenyl-2H benzopyran-4-yl)phenoxy)ethyl)-, cis-
Interactions	PUBCHEM	Nelivaptan
Interactions	PUBCHEM	Nandrolone
Interactions	PUBCHEM	UK-373911
Interactions	PUBCHEM	Formamide, N-((1-((2-(diethylamino)ethyl)amino)-7-methoxy-9-oxo thioxanthen-4-yl)methyl)-
Interactions	PUBCHEM	Flavopiridol hydrochloride
Interactions	PUBCHEM	Xanthinol Nicotinate
Interactions	PUBCHEM	N-[4-chloro-2-[[2-[(2-methoxyphenyl)carbamoyl]phenyl]carbamoyl]phenyl]furan-2-carbo
Interactions	PUBCHEM	Sorbitan mono-(9Z)-9-octadecenoate
Interactions	PUBCHEM	Alloisoleucine, DL-
Interactions	PUBCHEM	Pentachlorophenol
Interactions	PUBCHEM	2H-Pyran-4-carboxamide, tetrahydro-4-(3-((4-(2-methyl-1H-imidazo yl)phenyl)thio)phenyl)-, methanesulfonate (1:1)
Interactions	PUBCHEM	Oxeladin Citrate
Interactions	PUBCHEM	2-Hydroxyethyl (2R)-2,5,7,8-tetramethyl-2-((4R,8R)-4,8,12-trimethyltridecyl)-3,4-dihydro-2H-chromen-6-yl butanedioate
Interactions	PUBCHEM	1,2,4-Triazolo(4,3-a)pyridine, 6-(4-(2,5-difluorophenyl)-5-oxazolyl)-3

Main Category	Category	Terms
		methylethyl)-
Interactions	PUBCHEM	4-Methylcatechol
Interactions	PUBCHEM	3-((Dimethylamino)methyl)-4-(3-methyl-4-(methylthio)phenoxy)benzenesulfonamide
Interactions	PUBCHEM	PD 0343701
Interactions	PUBCHEM	1-Bromo-4-fluorobenzene
Interactions	UP_KW_LIGAND	Lipid-binding
Interactions	UP_KW_LIGAND	Metal-binding
Interactions	UP_KW_LIGAND	Steroid-binding
Interactions	UP_KW_LIGAND	Zinc
Literature	GENERIF_SUMMARY	The ligand-binding domain of estrogen receptor alpha has been expressed, purified, and crystallized to yield high amounts of solub protein, in order to solve the structure in its native form without renaturation or modification steps.
Literature	GENERIF_SUMMARY	ERa but not ERbeta is present in human preadipocytes
Literature	GENERIF_SUMMARY	Association of a T262C transition in exon 1 of estrogen-receptor-a gene with skeletal responsiveness to estrogen in post-menopausa women.
Literature	GENERIF_SUMMARY	To determine whether receptor-induced changes in DNA structure related to transactivation, we compared the abilities of ER alpha ar beta to activate transcription and induce distortion and bending in
Literature	GENERIF_SUMMARY	Estrogen receptors are found in brain areas involved in regulation c intake. The anorexic effects of estrogen are accentuated by stress that variation in the estrogen receptors may contribute to the gene susceptibility to AN in females.
Literature	GENERIF_SUMMARY	Reduction of coactivator expression by antisense oligodeoxynucle inhibits ERalpha transcriptional activity and MCF-7 proliferation
Literature	GENERIF_SUMMARY	mutations targeted to predicted helix in the extreme carboxyl-term region alter its response to estradiol and 4-hydroxytamoxifen
Literature	GENERIF_SUMMARY	expression of estrogen receptor alpha and estrogen receptor beta studied in leiomyomas and homologous myometrium from women proliferative phase of the menstrual cycle and from women treated gonadotropin-releasing hormone analogue
Literature	GENERIF_SUMMARY	Estrogen receptors play a role in the activation of amino acid trans system A by estrogen.
Literature	GENERIF_SUMMARY	patients that progressed to breast cancer showed significantly hig alpha expression in their HUT (hyperplasia of usual type)foci comp with controls
Literature	GENERIF_SUMMARY	There was strong linkage disequilibrium between the three loci and significant sex difference was observed in allele (for (TA)(n), PvuII) genotype (for XbaI) frequencies.
Literature	GENERIF_SUMMARY	Results indicate a weak and transient activation of estrogen recep estradiol stimulation that may be enhanced by growth factors.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	RTA, a negative coregulator for ERalpha was isolated
Literature	GENERIF_SUMMARY	Coronary artery wall atherosclerosis in relation to the estrogen rec gene polymorphism
Literature	GENERIF_SUMMARY	role in blood pressure
Literature	GENERIF_SUMMARY	Mutations of tyrosine 537 in the human estrogen receptor-alpha selectively alter the receptor's affinity for estradiol and the kinetics interaction
Literature	GENERIF_SUMMARY	these results show that although estrogen can up-regulate endoge PR gene expression in osteoblasts via both ER isoforms, ER-alpha predominant inducer
Literature	GENERIF_SUMMARY	Postmenopausal women with coronary disease who have the ER-a IVS1-401 C/C genotype, or several other closely related genotypes an augmented response of HDL cholesterol to hormone-replaceme therapy.
Literature	GENERIF_SUMMARY	Results show that DNA-102 produced a 36-fold increase in the leve total ER alpha mRNA and a 12-fold increase in the level of mRNA f isoform transcribed from the upstream F promoter, and induced a differentiated osteoblastic phenotype.
Literature	GENERIF_SUMMARY	the extent of AF-2-dependent cofactor recruitment by ERalpha or E is affected both by ER ligands and estrogen-responsive element sequences
Literature	GENERIF_SUMMARY	Association of polymorphisms of the oestrogen receptor alpha ge the age of menarche
Literature	GENERIF_SUMMARY	Interaction of transcriptional intermediary factor 2 nuclear recepto peptides with the coactivator binding site of estrogen receptor alp
Literature	GENERIF_SUMMARY	crystal structures of the ER alpha ligand binding domain (LBD) bou both THC and a fragment of the transcriptional coactivator GRIP1, ER beta LBD bound to THC
Literature	GENERIF_SUMMARY	MAP kinase mediates growth factor-induced nuclear translocation estrogen receptor alpha.
Literature	GENERIF_SUMMARY	Identification of ten exon deleted ERbeta mRNAs in human ovary, b uterus and bone tissues
Literature	GENERIF_SUMMARY	possesses a novel large-scale chromatin unfolding activity when t or recruited to DNA
Literature	GENERIF_SUMMARY	demonstrated that ERalpha but not ERbeta directly interacts with calmodulin
Literature	GENERIF_SUMMARY	effect on cyclin D1 expression
Literature	GENERIF_SUMMARY	genetic variants in panic disorder
Literature	GENERIF_SUMMARY	in addition to the alternative cis-splicing, the hER alpha gene is als subject to natural trans-splicing
Literature	GENERIF_SUMMARY	estrogen via estrogen receptor 1 directly modulates differentiation bipotential stromal cells into the osteoblast and adipocyte lineage causing a lineage shift toward the osteoblast

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	contribution of genetic polymorphism of oestrogen and androgen receptor (AR) genes in male infertility
Literature	GENERIF_SUMMARY	ADA3-containing complexes associate with estrogen receptor alph
Literature	GENERIF_SUMMARY	PRMT2 is a novel ERalpha coactivator.
Literature	GENERIF_SUMMARY	review of connections and regulation of the human estrogen recep alpha
Literature	GENERIF_SUMMARY	tentative findings provide evidence that genetic variation in ESR1 m modify coronary reactivity and LDL oxidation and reflect difference early pathogenesis of coronary dysfunction in these healthy young
Literature	GENERIF_SUMMARY	Expression of c-erbB-2 in node negative breast cancer does not co with estrogen receptor status, predictors of hormone responsivene PCNA expression.
Literature	GENERIF_SUMMARY	The constitutively activated MAP kinase cascade in endometrial ca cells enhances ERalpha transcriptional activity via the AF1 domain activation pathway may be involved in the stimulatory effect of tam on endometrial cancer.
Literature	GENERIF_SUMMARY	Structural regions (BOX-1 & -2)required for the ability of ERalpha to transcription synergistically from tandem ERE sequences are also for the interaction of the receptor with the co-regulatory proteins.
Literature	GENERIF_SUMMARY	Formation of an hER alpha-COUP-TFI complex enhances hER alpha through Ser118 phosphorylation by MAPK.
Literature	GENERIF_SUMMARY	These results suggest that the estrogen receptor alpha gene is a susceptibility locus for bone mass, especially for the femoral neck elderly Japanese women.
Literature	GENERIF_SUMMARY	We outline additional exons in the genes' 5'- and 3'-untranslated re new polymorphic ER alpha microsatellite and a nested gene.
Literature	GENERIF_SUMMARY	MCF-7 cell line stably expressing GFP-tagged human estrogen rec alpha under control of the tetracycline-on system to determine the of GFP-ER expression on cell proliferation and expression of endog ER and hormone-responsive genes
Literature	GENERIF_SUMMARY	decrease in estrogen receptor alpha and progesterone receptor A a concentration in the early phase of first stage labor may play a role cervical dilation at term
Literature	GENERIF_SUMMARY	Activation of ER-alpha by estrogen results in tumor progression by stimulating cell growth and invasiveness via acceleration of the expression of matrix metalloproteinases.
Literature	GENERIF_SUMMARY	ERalpha is phosphorylated at Serine 118 in vivo
Literature	GENERIF_SUMMARY	ER-alpha and its mRNA were detected in both the cytosolic and nu compartments of skin fibroblasts.
Literature	GENERIF_SUMMARY	Data show that ER alpha and not ER beta silences caveolin-1/-2 expression in an epigenetic fashion in neuronal cells.
Literature	GENERIF_SUMMARY	leucine 372 may be an important part of the interaction surface on estrogen receptor alpha that is responsible for corepressor binding
Literature	GENERIF_SUMMARY	differences in the abilities of estrogen receptors alpha and beta to

Main Category	Category	Terms
		integrate activation functions through functional interactions betw amino and carboxyl termini are critical for the transcriptional stren ER subtypes
Literature	GENERIF_SUMMARY	Upstream open reading frames regulate the translation of the mult mRNA variants
Literature	GENERIF_SUMMARY	both transactivation and down-regulation of estrogen receptor alpl adrenal androgens increase with aromatase overexpression in transfected MCF7 cells
Literature	GENERIF_SUMMARY	Lack of association between ESR1 gene polymorphism and Parkin disease with dementia.
Literature	GENERIF_SUMMARY	Polymorphic variation of the ER-alpha gene appears to have little e on the pathogenesis of osteoporosis in men.
Literature	GENERIF_SUMMARY	identification and characterization of a new tissue-specific promot is involved in the expression of ER-alpha in the epididymis and in te
Literature	GENERIF_SUMMARY	ERalpha and ERbeta make unique contributions to xenoestrogen a target cells
Literature	GENERIF_SUMMARY	present data indicate that polymorphism of the estrogen receptor gene and multidrug resistance 3 gene mutation are unlikely to play significant role in obstetric cholestasis in affected Finnish women
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha is present in the human growth plate in ch and adolescence
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms of estrogen receptor alpha may l role in pathogenesis of human renal cell carcinoma possible genet in carcinogenesis
Literature	GENERIF_SUMMARY	Association between the estrogen receptor alpha gene polymorph XbaI and PvuII and circulating levels of androstenedione was demonstrated, with the highest levels of the hormone being found xx and pp genotypes, as compared to XX or PP genotypes.
Literature	GENERIF_SUMMARY	expression of either ERalpha or beta lowers both basal and stimula PPARgamma-mediated transactivation, demonstrating that signal talk exists bidirectionally between PPARgamma and ER in breast c cells
Literature	GENERIF_SUMMARY	Polymorphic in endometrial cancer
Literature	GENERIF_SUMMARY	Although 17beta-estradiol and hypoxia induce proteasome-depend degradation of ERalpha, their effects on transactivation are differe this may have implications for clinical treatment of mammary tum
Literature	GENERIF_SUMMARY	present data indicate that polymorphism of the estrogen receptor and multidrug resistance 3 genes 1712delT mutation are unlikely t any significant role in obstetric cholestasis in affected Finnish wor
Literature	GENERIF_SUMMARY	ER alpha can physically and functionally couple to the lipid kinase phosphatidylinositol 3-OH kinase (PI3K). This interaction leads to activation of PI3K signaling cascade to Ser/Thr kinase Akt.
Literature	GENERIF_SUMMARY	These results demonstrate a novel pathogenic mechanism whereb mutations in BRCA1, via their interaction with ER-alpha, could prom tumorigenesis through the hormonal regulation of mammary epith cell proliferation and impaired VEGF

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Transcriptional synergism between a p160 coactivator and this pro depends on the coactivator subtype, the type of estrogen response element, and the promoter context.
Literature	GENERIF_SUMMARY	studies suggest that the cooperative interaction of the estrogen re with Fos and Jun proteins helps confer estrogen responsiveness to endogenous progesterone receptor gene
Literature	GENERIF_SUMMARY	the binding of different ligands to ESRA influences the A/D-F doma interaction and results in specific functional outcomes
Literature	GENERIF_SUMMARY	Estrogen receptor alpha gene polymorphisms Pvu II and Xba I influ association between leptin receptor gene polymorphism (Gln223A bone mineral density in young men.
Literature	GENERIF_SUMMARY	estrogen receptor alpha-mediated transcription is regulated by a d interaction with protein phosphatase 2A
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	data indicate that coordinated expression of ERalpha and ERbeta r be necessary for normal estrogen action in myometrium and that estrogen appears a dominant regulator of both receptors in the myometrium
Literature	GENERIF_SUMMARY	Estrogen receptor alpha binds to different selective estrogen recep modulators based on the sequence of ERalpha
Literature	GENERIF_SUMMARY	our results indicate an important role for Mediator, as well as its functional interplay with p300/CBP-SRC, in the enhancement of ER dependent transcription with chromatin templates
Literature	GENERIF_SUMMARY	estrogen receptor alpha has a critical amino acid (Asp-538) in helix
Literature	GENERIF_SUMMARY	A novel coregulatory protein, template-activating factor-Ibeta (TAF is identified, which binds to ERalpha in vitro when the receptor is n complexed with an estrogen response element.[TAF-Ibeta]
Literature	GENERIF_SUMMARY	A truncated estrogen receptor alpha 46-kDa isoform in vascular endothelial cells mediates activation of nitric oxide synthase by es
Literature	GENERIF_SUMMARY	Ligands differentially modulate the protein interactions of the hum estrogen receptors alpha and beta
Literature	GENERIF_SUMMARY	a single residue in the ligand-binding domain determines the stereoselectivity of ERalpha
Literature	GENERIF_SUMMARY	The estrogenic down-regulation of PTP gamma expression in hum breast is associated with ER alpha.
Literature	GENERIF_SUMMARY	Data show that liganded ERalpha is a much more potent transcript activator than ERbeta with chromatin templates, but not with nake
Literature	GENERIF_SUMMARY	dinucleotide repeat polymorphism of the ER alpha gene may contr specific components of personality.
Literature	GENERIF_SUMMARY	Mechanistic differences in the activation of this protein's and ER b dependent gene expression by cAMP signaling pathway(s).
Literature	GENERIF_SUMMARY	Interaction domains of Smad4 and ERalpha are mapped and show essential for transcriptional repression of ERalpha by Smad4.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	F domain interactions with nuclear cofactors are required for ERalp Sp1 action
Literature	GENERIF_SUMMARY	Estrogen receptor alpha has a role in rapid, nontranscriptional resp to estrogen in endothelial cells through interacting with phosphatidylinositol 3-kinase.
Literature	GENERIF_SUMMARY	Estrogen receptor and progesterone receptor concentrations and E transcription levels were not statistically different between ethnic backgrounds.A causative role for these receptors in the ethnic vari leiomyoma biology seems unlikely
Literature	GENERIF_SUMMARY	Resveratrol acts as an estrogen receptor (ER) agonist in breast car cells stably transfected with ER alpha.
Literature	GENERIF_SUMMARY	This protein is expressed and synthesized in the human internal th artery.
Literature	GENERIF_SUMMARY	Association study of oestrogen receptor alpha gene polymorphism suicidal behaviours in major depressive disorder in females.
Literature	GENERIF_SUMMARY	Inhibitory AhR-ERalpha cross talk is linked to a new pathway for de ERalpha in which TCDD initially induces formation of a nuclear AhR complex which coordinately recruits ERalpha & the proteasome co resulting in degradation of both receptors.
Literature	GENERIF_SUMMARY	2 domains of PRB, ERID-I and -II, mediate a direct interaction with t ligand-binding domain of ERalpha.
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	The considerable increase in estrogen production during pregnanc play a role, via the ER-alpha, in the stimulation of cytotrophoblast differentiation & promote function in normal placentae.
Literature	GENERIF_SUMMARY	analysis of isoforms of estrogen receptor alpha
Literature	GENERIF_SUMMARY	Cyclic turnover of hERalpha permits continuous responses to char the concentration of estradiol
Literature	GENERIF_SUMMARY	Due to alternative splicing, the 46-kDa endothelial cell protein (ER4 amino-terminal truncated product of full-length ER alpha (ER66).
Literature	GENERIF_SUMMARY	study of the equilibrium dissociation constant for homodimerizatio ERalpha
Literature	GENERIF_SUMMARY	ERalpha co-activator caveolin is negatively regulated by the steroid receptor itself
Literature	GENERIF_SUMMARY	estrogen-induced estrogen receptor degradation serves to control physiological responses in human vascular endothelium by down-regulating estrogen receptor alpha and estrogen receptor beta thro different proteasomal uptake mechanisms
Literature	GENERIF_SUMMARY	estrogens may inhibit thymoma growth via estrogen receptor alpha estrogen receptor alpha immunoreactivity may act as a prognostic in human thymoma
Literature	GENERIF_SUMMARY	there are at least ten isoforms of ER alpha mRNA containing differ untranslated regions, exons (A), B, C, D, T1-T2, T1, F-E2, F, E1-E2 an expressed in human, and that these are involved in tissue specific expression of the gene.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Mutation of Leu-536 in estrogen receptor-alpha alters the coupling between ligand binding, transcription activation, and receptor conformation.
Literature	GENERIF_SUMMARY	During trophoblast differentiation, ER-alpha is associated with a le differentiated state. Asymmetric segregation of ER-alpha may play in asymmetric division of estrogen-dependent cells.
Literature	GENERIF_SUMMARY	p53-independent activation of the hdm2-P2 promoter through mult transcription factor response elements results in elevated hdm2 expression in estrogen receptor alpha-positive breast cancer cells.
Literature	GENERIF_SUMMARY	Polymorphisms have been determined in the coding region of the ERalpha gene. It is hypothesized that these polymorphisms are associated with renal cell carcinoma.
Literature	GENERIF_SUMMARY	estrogen receptor-mediated estrogen signalling is modulated by a regulatory-like function of activated AhR/Arnt, giving rise to advers oestrogen-related actions of dioxin-type environmental contamina
Literature	GENERIF_SUMMARY	estrogen (E2)-depleted medium in MCF-7 breast cancer cell line re hypersensitivity to E2 coinciding with elevated levels of estrogen re (ER) alpha phosphorylated on Ser118 and MAPK
Literature	GENERIF_SUMMARY	results indicate that the naturally occurring variant ERDeltaE7 is a dominant negative receptor that, when expressed at high levels rel wild-type ERs, might have profound effects on several estrogen-dependent functions
Literature	GENERIF_SUMMARY	ER-alpha gene may have minor effects on bone density variation in Chinese population.
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha regulates the degradation of insulin recep substrates 1 and 2 in breast cancer cells
Literature	GENERIF_SUMMARY	ESR1 dinucleotide repeat is not associated with premature myoca infarction or with hyperlipidemia in caucasian women.
Literature	GENERIF_SUMMARY	In women there was an allele dose effect of haplotype ""px"" and a number of minisatellite repeats with decreased lumbar spine bone mineral density (BMD) and decreased vertebral bone area."
Literature	GENERIF_SUMMARY	Estrogen receptor alpha regulates expression of the orphan recept small heterodimer partner
Literature	GENERIF_SUMMARY	polymorphisms in the ERalpha gene are associated with radiograp of the knee, and in particular with osteophytosis, in both elderly me elderly women.
Literature	GENERIF_SUMMARY	results suggest that, in MCF-7 breast cancer cells, estrogen recept turnover, localization, and activity are maintained by an equilibrium between various phosphorylation pathways, which are differently modulated by ER ligands and protein kinases
Literature	GENERIF_SUMMARY	Data indicate that serines 104, 106, and 118 are important for ligar dependent ER dimerization, and that MAP kinase mediated phosphorylation may be important for ER function, in yeast model systems.
Literature	GENERIF_SUMMARY	malignant ovarian epithelial cells demonstrated multiple alteration expression of estrogen receptor-alpha, estrogen receptor-beta, progesterone receptor A, and progesterone receptor B

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	SNCG is required for efficient ER-alpha signaling in brest cancer.
Literature	GENERIF_SUMMARY	estrogen receptor alpha binds to thymine-DNA glycosylase and potentiates transcription of estrogen-regulated genes
Literature	GENERIF_SUMMARY	Truncated estrogen receptor product-1 stimulates estrogen recept alpha transcriptional activity by titration of repressor proteins
Literature	GENERIF_SUMMARY	polymorphism in codon 10 of ERalpha may be a risk factor for pros cancer.
Literature	GENERIF_SUMMARY	Oestrogen-activated ERbeta presents a lower transcriptional activi compared with ERalpha, probably due to structural differences in t regions of the receptors. Genistein and daidzein act as weak agon the oestrogenic activity via ERalpha.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha was seen in the sebaceous gland.
Literature	GENERIF_SUMMARY	The methylation status of estrogen receptor-alpha is cell specific i cancer cell lines.
Literature	GENERIF_SUMMARY	interaction between estrogen receptor 1 alpha and vitamin D recep gene polymorphisms leads to increased risk of osteoporotic vertel fractures in women, largely independent of bone mineral density
Literature	GENERIF_SUMMARY	ERK7 is a novel regulator of estrogen responsiveness through its c of ER alpha turnover
Literature	GENERIF_SUMMARY	ER-alpha and vitamin D receptor polymorphisms are related to bon mineral density in pre- and post-menopausal women
Literature	GENERIF_SUMMARY	Activation of the mitogen activated protein kinase (MAPK) induces independent coactivation of ERalpha by SRA
Literature	GENERIF_SUMMARY	ESR1 polymorphism maybe associated with risk of endometrial neoplasms.
Literature	GENERIF_SUMMARY	adrenomedullin may be involved in both normal and malignant tiss growth and because of the correlations with estrogen receptor alp beta status, there is evidence that ovarian cancer is endocrine-rela
Literature	GENERIF_SUMMARY	estradiol treatment results in binding to membrane ER-alpha and interaction with a heterodimer containing ErbB2, leading to tyrosin phosphorylation. This results in the activation of PI 3-K and Akt. Ak turn, may interact with nuclear ER-alpha
Literature	GENERIF_SUMMARY	up-regulates the MIS promoter activity under low concentrations o
Literature	GENERIF_SUMMARY	Results of this study indicate that genetic polymorphisms in the ER gene may play a role in the etiology of breast cancer.
Literature	GENERIF_SUMMARY	estrogen receptor alpha Px haplotype and collagen IA1 s allele ma involved in causing the phenotypic expression of higher circulating of parathyroid hormone and higher bone turnover, which may lead loss
Literature	GENERIF_SUMMARY	ESR1 polymorhism is significantly associated with familial prostat carcinoma risk.
Literature	GENERIF_SUMMARY	ERalpha immunocytochemical expression in AD patients did not di from that in elderly controls.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Subgroup of women with vulvar vestibulitis syndrome who exhibit abnormal estrogen receptor-alpha expression. May be helpful in explaining why some women are resistant to medical treatment
Literature	GENERIF_SUMMARY	Functional and clinical implications of observed ER alpha distributi the hypothalamus are discussed in relation to reproduction, autono function, mood, cognition, and neuroprotection in health and disea
Literature	GENERIF_SUMMARY	Findings show that in endometrial hyperplasia and low-grade endometrioid endometrial carcinoma, coordinate increase in cdc2 phosphorylated ER-alpha occurs.
Literature	GENERIF_SUMMARY	The Px haplotype of the ER alpha gene is associated with low BMD in females, as the phenotype is gender dependent (the association not observed in males.
Literature	GENERIF_SUMMARY	Women homozygous for the T-allele of the PvuII polymorphism of estrogen receptor-alpha gene who are treated with coronary stent implantation have a higher risk of in-stent restenosis than men.
Literature	GENERIF_SUMMARY	data provide an alternative mechanism of LDL receptor gene expre by non-classical estradiol- and tamoxifen-stimulated induction thro ER-alpha/Sp1 complex
Literature	GENERIF_SUMMARY	Our results reveal an association between a common estrogen rec genotype, ESR1 c.454-397CC, and increased odds of myocardial infarction.
Literature	GENERIF_SUMMARY	ER ligand and estrogen response element (ERE) sequence work to to regulate in vivo binding of ER to estrogen-responsive promoters
Literature	GENERIF_SUMMARY	upregulation of VEGF-A expression by 2-Methoxyestradiol is media through ER-alpha
Literature	GENERIF_SUMMARY	qualitative and quantitative analysis and comparison of mRNA ER- and its isoforms concentration profile in endometrial adenocarcino and normal endometrium
Literature	GENERIF_SUMMARY	The findings of key roles for histone deacetylases and nucleosome remodeling complexes in limiting transcription implies that transcr activation is a cyclical process that requires both activating and repressive epigenetic processes.
Literature	GENERIF_SUMMARY	Thus, our findings indicated that TAS-108 acts as a novel estrogen antagonist that recruits co-repressors to ERs without AF-1 activati prevention of DNA binding.
Literature	GENERIF_SUMMARY	AIB1-Delta3 causes a significant increase in the efficacy of 17beta estradiol at both estrogen receptor-alpha (ER-alpha) and ER-beta in ovarian, breast and endometrial cancer cell lines
Literature	GENERIF_SUMMARY	In node-negative primary breasst tumors, IGF-IR positively correlat ERbeta but not with ERalpha, tumor size or grade.
Literature	GENERIF_SUMMARY	a role for genetic variations in the estrogen receptor alpha gene in determining adult stature in women was demonstrated
Literature	GENERIF_SUMMARY	differential recruitment of steroid receptor coactivator-1 and silenc mediator for retinoid and thyroid receptors by estrogen receptor-al and beta in breast cancer may be central to the response of the tu endocrine treatment

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	transcriptional repression by tamoxifen-bound ER at E-regulated ge promoters involves a dynamic interplay of multiple distinct chroma modifying/remodeling complexes
Literature	GENERIF_SUMMARY	ERalpha and ERbeta are located within the MCF7 cell mitochondria matrix
Literature	GENERIF_SUMMARY	ERalpha and ERbeta interaction with their cognate target DNAs rec the C-terminal extension and HMGB-1/-2
Literature	GENERIF_SUMMARY	Tranhcriptional activation of endothelial nitric oxide synthase via e receptor (ER)alpha and ERbeta in human pregnant myometrium.
Literature	GENERIF_SUMMARY	both ER-alpha and ER-beta mRNA are expressed in human mature adipocytes with a large predominance of ER-alpha mRNA
Literature	GENERIF_SUMMARY	Overexpression of ESRA suppresses gap junctional intercellular communication in endometrial carcinoma cells.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha genotype associations were statistically significant in explaining rate of perimenopausal bone loss and bon turnover.
Literature	GENERIF_SUMMARY	Shc and IGF-1R serve as key elements in the translocation of ERalp the cell membrane and in the facilitation of ERalpha-mediated rapi action
Literature	GENERIF_SUMMARY	ER-alpha has a role in CpG island methylation near the TGA initiatio codon of the ER-alpha gene in gastric cancer cell lines
Literature	GENERIF_SUMMARY	HMGA1 cooperates with either the N- or the C-terminal transcriptio activation domain of the estrogen receptor to stimulate estrogen response element binding and promoter transcriptional activity
Literature	GENERIF_SUMMARY	Estrogen receptor alpha has a role in human breast cancer [review]
Literature	GENERIF_SUMMARY	estrogen receptor-alpha may have a role in male breast cancer development [review]
Literature	GENERIF_SUMMARY	Polymorphisms in ER-alpha gene can affect breast cancer suscept and may be related to other protein expression.
Literature	GENERIF_SUMMARY	direct functional activation of ERalpha is caused by amplification o estrogen signal induced by leptin
Literature	GENERIF_SUMMARY	Data describe the influence of allelic variations of the vitamin D rec and estrogen receptor genes on bone mineral density.
Literature	GENERIF_SUMMARY	ER alpha interacts with the p55 regulatory subunit of PI3K, wherea beta interacts with Akt1 in spermatozoa
Literature	GENERIF_SUMMARY	Molecular determinants of Era in the selectivity of 16 alpha-iodoes were studied.
Literature	GENERIF_SUMMARY	human estrogen receptor-alpha functions are modulated by s-palmitoylation
Literature	GENERIF_SUMMARY	Of the approximately 100 regulated genes we identified, some wer stimulated by E2 equally through ERalpha and ERbeta, whereas oth were selectively stimulated via ERalpha or ERbeta
Literature	GENERIF_SUMMARY	No evidence of the ESR1 missense mutation A908G was found in

Main Category	Category	Terms
		carcinomas, epithelial hyperplasias or in breast cancer cell lines st
Literature	GENERIF_SUMMARY	ER-/Progesterone receptor- breasst tumors in age > 50. ER+/PR+ s was associated with higher frequency of methylation.
Literature	GENERIF_SUMMARY	nuclear factor-kappaB has a role in mitogen-activated protein kinas induced estrogen receptor-alpha down-regulation in breast cancer
Literature	GENERIF_SUMMARY	T(4), like E(2), causes phosphorylation by MAPK of nuclear ERalph serine-118 in MCF-7 cells and promotes cell proliferation through t by a MAPK-dependent pathway.
Literature	GENERIF_SUMMARY	Data identify two peptides that are highly selective for tamoxifen-b estrogen receptor (ER) alpha or beta, and suggest that these pepti recognize a novel tamoxifen-induced binding surface that is distin the agonist-induced AF-2 surface.
Literature	GENERIF_SUMMARY	SAFB1 represses ERalpha activity via indirect association with hist deacetylation and interaction with the basal transcription machine
Literature	GENERIF_SUMMARY	ER-alpha genotype may modulate relationship between BMD or rat bone loss and estrogen levels in men. Bone mass in men with the alleles may be more susceptible to consequences of estrogen defi than in men with the xx or pp genotypes.
Literature	GENERIF_SUMMARY	Effect of estrogen was caused through the estrogen receptor subt alpha.
Literature	GENERIF_SUMMARY	Shift of ERalpha from nucleus to cytoplasm in elderly females with absence of Alzheimer's disease(AD). AD showed more nuclear ER and much stronger AD neuropathology.
Literature	GENERIF_SUMMARY	the transcriptional activity of ERalpha relies on its AF-1 transactiva function in most estrogen target cell types
Literature	GENERIF_SUMMARY	Male and female AD patients had lower ER alpha and beta densitie compared to non-AD patients.
Literature	GENERIF_SUMMARY	role in expression of progesterone receptor B-isoform
Literature	GENERIF_SUMMARY	When present in pituitary tumors, estrogen receptor-alpha exhibits concentration, and is more common in nonfunctioning and invasiv adenomas, but absent in ACTH-secreting ones.
Literature	GENERIF_SUMMARY	ER alpha mRNA was increased at 6 days but not at 14 days whilst mRNA expression was only increased at 14 days with 17 beta estr treatment
Literature	GENERIF_SUMMARY	genetic variation in hormone receptors, in particular the ESR1 gene play a role in migraine.
Literature	GENERIF_SUMMARY	ERalpha has a role in tamoxifen induction of maspin by recruiting cofactors to the maspin promoter region
Literature	GENERIF_SUMMARY	FOXP1 is a potential ER coregulator in human breast carcinoma ar also independently regulate additional important pathways that co the progression of breast cancer
Literature	GENERIF_SUMMARY	estrogen receptor-alpha regulates metastatic tumor antigen 3 path
Literature	GENERIF_SUMMARY	Data suggest that the exon 3-deleted estrogen receptor (ERDeltaE3 inhibits wild-type ER activity and negatively regulates proliferation

Main Category	Category	Terms
		MCF-7 cells.
Literature	GENERIF_SUMMARY	Data show that estrogen receptor (ER)beta was degraded in the pr of 17beta-estradiol, and in contrast to ERalpha, ERbeta degradatior not abolished by inhibitors of the proteasome-mediated protein degradation pathway.
Literature	GENERIF_SUMMARY	estrogen receptor redistribution to the cytoplasm and its interactio HER2 are important downstream effects of HER2 overexpression
Literature	GENERIF_SUMMARY	Novel single-nucleotide polymorphism provides a tool for human g studies of diseases in the African population.
Literature	GENERIF_SUMMARY	Keratinocytes express ER-alpha. Estradiol up-regulates ER-alpha in keratinocytes & induces their proliferation. The binding of estradiol in keratinocytes has a calculated Bmax of 14.2 fmol/mg of protein
Literature	GENERIF_SUMMARY	Postmenopausal women who carry ESR1 haplotype 1 (c.454-397 T and c.454-351 A allele) have an increased risk of myocardial infarc and ischemic heart disease, independent of known cardiovascular factors.
Literature	GENERIF_SUMMARY	ERalpha activated the PAI-1 promoter through a proximal ERE (-427 possibly additional EREs located within the PAI-1 promoter.
Literature	GENERIF_SUMMARY	estrogen receptor alpha is inhibited by melatonin via a calmodulin pathway
Literature	GENERIF_SUMMARY	ER alpha exists as dimers, a structural requirement that supports r signal transduction and affects cell physiology.
Literature	GENERIF_SUMMARY	ERalpha and ERbeta genes show CpG methylation in breast neopla and CpG methylation of the ERbeta gene seems to play a role in ce proliferation
Literature	GENERIF_SUMMARY	Era-positive breast cancer cells were identified by gene expression profiling of an intrinsic set of estrogen regulated genes.
Literature	GENERIF_SUMMARY	methylation of the CpG sites inside ERalpha gene promoter F here analyzed may contribute to ERalpha transcriptional control, influen the tissue specific expression of the gene
Literature	GENERIF_SUMMARY	Neither the PvuII nor the XbaI polymorphism or haplotypes thereof associated with the risk of all-cause dementia or Alzheimer's disea Nondemented women who carried the PvuII p allele or haplotype 'p smaller amygdalar volumes on MRI.
Literature	GENERIF_SUMMARY	Data show that CCND1 promoter activation by estrogens in humar cancer cells is mediated by recruitment of a c-Jun/c-Fos/estrogen receptor alpha/progesterone receptor complex to the tetradecanoy phorbol acetate-responsive element of the gene.
Literature	GENERIF_SUMMARY	AIB1 plays a dual role in regulating ERalpha activity, one in recruitin transcription factors including other coactivators involved in gene activation and the other in regulating ERalpha protein degradation mediated by the ubiquitin-proteosome machinery.
Literature	GENERIF_SUMMARY	there is cross-talk between Wnt and estrogen signaling pathways v functional interaction between beta-catenin and ERalpha
Literature	GENERIF_SUMMARY	These results suggest that ERalpha is the functionally dominant pa the ERalpha/beta heterodimer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	nuclear IRS-1 interacts with ERalpha and that this interaction migh influence ERalpha transcriptional activity
Literature	GENERIF_SUMMARY	In the presence of estradiol ERalpha induced CRH-binding protein promoter activity in a dose-dependent manner.
Literature	GENERIF_SUMMARY	ERalpha transcriptional activity is regulated by XBP-1 through modification of large-scale chromatin unfolding
Literature	GENERIF_SUMMARY	the agonist activity of tamoxifen-bound ERalpha is enhanced by SF which may contribute to tamoxifen resistance
Literature	GENERIF_SUMMARY	The present study shows that Pvull-Xbal ERalpha gene polymorphi associated with BMD during childhood.
Literature	GENERIF_SUMMARY	methylation of ESR1 is both age-dependent and tumor differentiati dependent and age-dependent methylation of ESR1 may represent mechanism linking aging and prostate cancer
Literature	GENERIF_SUMMARY	results indicate that the mitochondrial ERs can interact with mtERE suggest that they may be directly involved in E2 induction of mtDN transcription
Literature	GENERIF_SUMMARY	The genetic variation at the ESR1 locus influences susceptibility to persistent HBV infection in a Chinese population.
Literature	GENERIF_SUMMARY	ESR1 may be a possible risk factor for female or generalized type c vitiligo patients.
Literature	GENERIF_SUMMARY	17beta estradiol promotes Hela cell survival through ER alpha non- genomic signaling and cell death through ER beta non-genomic sig
Literature	GENERIF_SUMMARY	Fibronectin and type IV collagen activate ERalpha AF-1 by the c-Sro pathway in breast cancer
Literature	GENERIF_SUMMARY	DRIP205 coactivation of estrogen receptor alpha (ERalpha) involve multiple domains of both proteins
Literature	GENERIF_SUMMARY	Uterine leiomyomas, myometrium, and endometrium display distin patterns of estrogen receptor alpha and beta expression.
Literature	GENERIF_SUMMARY	estrogen receptor alpha has a role in regulating an association bet exercise and pubertal bone mineral density
Literature	GENERIF_SUMMARY	results demonstrate that transcriptional adaptor ADA3 protein dire binds to human estrogen receptor alpha and beta and enhances th transcription of estrogen receptor-responsive genes
Literature	GENERIF_SUMMARY	ERalpha palmitoylation has a role in the receptor localization to the membrane and in the regulation of the E2-induced cell proliferation
Literature	GENERIF_SUMMARY	expression of AP-1-dependent genes is inhibited by the cJun domi negative; the AP-1 blockade does not affect mRNA ERalpha expres estrogen induction of estrogen-responsive element activity
Literature	GENERIF_SUMMARY	The relative binding affinities (RBA) of typical ligands with the alph subtype of the human estrogen receptor ligand-binding domain (hE LBD)are calculating their binding energies.
Literature	GENERIF_SUMMARY	The ESR1 gene exerts differential genetic effects on bone mineral and fracture risk.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the context of the lactoferrin gene influences the ERalpha-mediate transcriptional activity.
Literature	GENERIF_SUMMARY	the ERE sequence impacts estradiol-and 4-hydroxytamoxifen-occu ERalpha interaction with coregulators as measured by transcriptio activity in mammalian cells
Literature	GENERIF_SUMMARY	The expression of Delta5 variation in uterine carcinoma may provi mechanism by which tumors could maintain expression of E2-resp genes in the absence of E2.
Literature	GENERIF_SUMMARY	Studies demonstrate the existence of a variant ER-alpha(var2) prot localized in the nuclei in breast cancer cell lines. Specific binding c 17beta-E2 by these cells suggest the ER-alpha(var2) may act as ac receptor.
Literature	GENERIF_SUMMARY	Coronary artery cross-sections were immunostained for Androgen receptor, estrogesn rseceptor alpha, and Estrogen receptor beta ar measured semiquantitatively in each arterial wall.
Literature	GENERIF_SUMMARY	No evidence was found for a relationship between common varian the ERalpha gene and the CYP17 gene with age at natural menopa
Literature	GENERIF_SUMMARY	differential expression of AR, ERalpha and ERbeta in dermal papilla follicles
Literature	GENERIF_SUMMARY	estrogen receptor alpha and beta signalling have roles in the narin induced apoptotic cascade in cancer cells
Literature	GENERIF_SUMMARY	ESRA signaling complex at the cell membrane was analyzed.
Literature	GENERIF_SUMMARY	striatin is the molecular anchor that localizes ERalpha to the meml and organizes the ERalpha-eNOS membrane signaling complex
Literature	GENERIF_SUMMARY	Estrogen receptor alpha mRNA levels were reduced in the basome nucleus in major depressive disorder and bipolar disorder.
Literature	GENERIF_SUMMARY	PELP1 functionally interacts with both ERalpha and ERbeta and en their transcriptional responses.
Literature	GENERIF_SUMMARY	High levels of ER-alpha in endometrial cancer may inhibit cancer g by modulating angiogenic factors, thereby limiting the blood suppl growing tumor.
Literature	GENERIF_SUMMARY	ER alpha and Ki67 might be involved in distinct pathological and molecular features during breast cancer development
Literature	GENERIF_SUMMARY	abundance and specific localization of estrogen receptors alpha a in human term myometrium seems to be independent of its contra activity which may point to the specific role of those receptors in la pregnancy myometrium
Literature	GENERIF_SUMMARY	ESRA has a role in modulating IGF-I receptor signaling and functio breast cancer
Literature	GENERIF_SUMMARY	Loss of estrogen receptor alpha is associated with prostatic tumo
Literature	GENERIF_SUMMARY	found in nuclei of peripheral cells of sebaceous glands while secre epithelium of apocrine glands of axilla were negative
Literature	GENERIF_SUMMARY	Estrogen receptor alpha mRNA and protein in kidney alters with ag shows sex differences.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	cultured and freshly isolated endometrial endothelial cells each ex ER-beta mRNA but not ER-alpha.
Literature	GENERIF_SUMMARY	ER may play a role in the rapid effects of resveratrol in EC and som atheroprotective effects of resveratrol may be mediated through ra activation of ER signaling in EC
Literature	GENERIF_SUMMARY	PvuII polymorphism and the ER-alpha haplotype may be associate the bone mineral density at several femur sites in postmenopausa Korean women
Literature	GENERIF_SUMMARY	Coactivation of ERalpha by DRIP150 in ZR-75 cells is NR box-indep and requires a novel sequence with putative alpha-helical structure
Literature	GENERIF_SUMMARY	ERalpha interacts with Sp1 transcription factor in breast cancer ce
Literature	GENERIF_SUMMARY	The affinity, specificity and stability of complexes formed between estrogen receptor alpha and estrogen response elements are exan and an alternate mechanism proposed for the various effectors inv in the fine-tuning of gene expression.
Literature	GENERIF_SUMMARY	levels of mER-alpha play a role in the temporal coordination of phosphorylation/dephosphorylation events for the ERKs in breast cells
Literature	GENERIF_SUMMARY	molecular mechanisms underlying ligand-dependent signaling in th AP-1 pathway
Literature	GENERIF_SUMMARY	Expressed in primary biliary cirrhosis(PBC) but not in normal subje Low ER-alpha in PBC and disappearance in advanced histological s suggests that estrogenic deficiency could favour evolution toward ductopenia.
Literature	GENERIF_SUMMARY	association of ERalpha gene PvuII polymorphism with PCOS deter
Literature	GENERIF_SUMMARY	binding to calmodulin is required for normal transcriptional functio ERalpha
Literature	GENERIF_SUMMARY	Myasthenia gravis (MG) patients showed increased expression of estrogen receptor(ER) alpha on thymocytes and both ER alpha and on T cells from peripheral blood mononuclear cells
Literature	GENERIF_SUMMARY	a partial BRCA1
Literature	GENERIF_SUMMARY	Positive immunohistochemical staining for ERalpha of an an adenocarcinoma in a gastric biopsy is diagnostic for metastatic br carcinoma.
Literature	GENERIF_SUMMARY	This study suggested that BCAS2 might play an important role in b cancer development by increasing the estrogen receptor's functior
Literature	GENERIF_SUMMARY	Transfection of wild-type or specifically mutated human ERalpha in sheep cells expressing tranfected COX-1 shows that the AF-1 regio within the A-B domains of ER is important for Sp1 activation. Simil domains within AF-1 may bind Sp1 & AP-2.
Literature	GENERIF_SUMMARY	Data suggest that the ER-alpha gene polymorphisms and ER status have an interactive effect on breast cancer survival.
Literature	GENERIF_SUMMARY	the subnuclear positioning of SMRT is influenced by the ligand-bou ERalpha, and this activity is dependent on the ratio of the co-expre ERalpha and SMRT

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Expression of mRNA for estrogen receptors-alpha and -beta in brea tumors before and after 3-month neoadjuvant hormone therapy wi antiestrogen tamoxifen and/or aromatase inhibitors.
Literature	GENERIF_SUMMARY	The activation of ER gene expression by 5-aza-2'-deoxycytidine inv the release of the repressor complex involving various methyl-bind proteins, DNMTs, and HDAC1.
Literature	GENERIF_SUMMARY	ER alpha* 14 TA repeats and the CYP17* A1 allele are associated v increased risk of endometriosis
Literature	GENERIF_SUMMARY	Alternative splice variants more ofter seen in Alzheimer disease th normal mammillary nucleus mrna.
Literature	GENERIF_SUMMARY	expression of ESRA, bcl-2 and c-myc gene expression in fibroaden and adjacent normal breast is related to nodule size, hormonal and reproductive features
Literature	GENERIF_SUMMARY	ARA70 is a coactivator for estrogen receptor alpha (ERalpha) and r represent a functional link between ERalpha/androgen receptor (A modulating their cross-talk in models of estrogen signaling in MCF HeLa cells
Literature	GENERIF_SUMMARY	Expression of estrogen receptors (ERs)-alpha and -beta analyzed i human fetal ovaries at various gestational ages.
Literature	GENERIF_SUMMARY	regulation of phosphorylation of ER-alpha and EGFR may play critic roles in EGF-induced transcriptional activation of WISP-2 gene in b tumor cells
Literature	GENERIF_SUMMARY	loss of estrogen receptor alpha correlates with increasing maligna transformation of endometrial carcinoma
Literature	GENERIF_SUMMARY	N-CoR and SMRT play an active role in preventing tamoxifen from stimulating proliferation in breast cancer cells through repression subset of target genes involved in ERalpha function and cell prolife
Literature	GENERIF_SUMMARY	ERalpha/beta heterodimers exert unique effects on estrogen- and tamoxifen-dependent gene expression in human U2OS osteosarco cells
Literature	GENERIF_SUMMARY	Activation of protein kinase C delta by phorbol myristate acetate re in activation and nuclear translocation of ERalpha.
Literature	GENERIF_SUMMARY	ER alpha-AHR-ARNT protein-protein interactions mediate estradiol- dependent transrepression of dioxin-inducible gene transcription
Literature	GENERIF_SUMMARY	The Xba I gene polymorphism of ERalpha gene was associated wi breast cancer risk, more significantly in postmenopausal women.
Literature	GENERIF_SUMMARY	Polymorphisms in combination with lifestyle factors may be assoc with breast cancer risk in older Caucasian women.
Literature	GENERIF_SUMMARY	analysis of estrogen receptor alpha membrane localization and fu [review]
Literature	GENERIF_SUMMARY	Estrogen receptor (ER) alpha and beta mRNA levels increased fron first to the second trimester and then decreased until normal term delivery.
Literature	GENERIF_SUMMARY	hsp90 ligands induced an estsrasdiol and MG132-inhibited decrea both estrogen receptor-alpha and -betas cell content.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Valproic acid invokes reversible promoter shutoff of estrogen rece alpha.
Literature	GENERIF_SUMMARY	Transient transfection of either all-trans-retinoic acid (ATRA) recep alpha or estrogen receptor alpha expression vectors increased cell retinoic acid binding protein II expression in MDA-MB-231 cells
Literature	GENERIF_SUMMARY	Estrogen receptor alphaeffects on SULT2A1 were greater than the stimulation seen in response to steroidogenic factor 1 (SF1).
Literature	GENERIF_SUMMARY	demonstrate that Smad4 induces apoptosis by regulating Bim spli an initial intrinsic signal in ERalpha-positive cells
Literature	GENERIF_SUMMARY	The association of cryptorchidism with homozygosity for the spec ESR1 haplotype suggests the relevance of genetic susceptibility to estrogenic environmental endocrine disruptors
Literature	GENERIF_SUMMARY	SF3a p120 potentiated hERalpha-mediated RNA splicing, and nota potentiation of RNA splicing by SF3a p120 depended on hER Ser-1 phosphorylation
Literature	GENERIF_SUMMARY	Our studies suggested that pRb2/p130-complexes bind to the ER-a promoter and could be involved in the transcriptional regulation of alpha gene by altering chromatin structure and DNA methylation p
Literature	GENERIF_SUMMARY	Estrogen receptor alpha codon 594 genotype may influence the development of systemic lupus erythematosus at a younger age, a as a certain disease clinical pattern.
Literature	GENERIF_SUMMARY	increased estrogen down regulated gene 1 expression results in in of cyclin T1 recruitment and ERalpha DNA binding
Literature	GENERIF_SUMMARY	Haplotypes may incrast thse risk of anxiety disorder, but not a dep disorder.
Literature	GENERIF_SUMMARY	MAPK and phosphatidylinositol 3-kinase cascades are involved in nuclear redistribution of ERalpha by EGF and IGF-I, respectively, an the activation function-2 domain of ERalpha may be needed for the nuclear redistribution of ERalpha
Literature	GENERIF_SUMMARY	ERRalpha mRNA increased 2 h post-exercise in skeletal muscle
Literature	GENERIF_SUMMARY	Sumoylation is a new mechanism modulating ERalpha transcriptio activity and ERalpha dependent cellular response.
Literature	GENERIF_SUMMARY	AHR-mediated transcription
Literature	GENERIF_SUMMARY	There is differential regulation of estrogen-inducible proteolysis an transcription by the Era N terminus.
Literature	GENERIF_SUMMARY	Protein phosphatase 2A regulates estrogen receptor alpha (ER) expression through modulation of ER mRNA stability.
Literature	GENERIF_SUMMARY	Common ESR1 polymorphisms have a significant effect on lipopro metabolism in women.
Literature	GENERIF_SUMMARY	The association of the estrogen receptor (ER) with the complete nonrepetitive sequence of human chromosomes 21 and 22 by con chromatin immunoprecipitation (ChIP) with tiled microarrays has b mapped.
Literature	GENERIF_SUMMARY	CHIP promotes ERalpha degradation and attenuates receptor-med

Main Category	Category	Terms
		gene transcription.
Literature	GENERIF_SUMMARY	Age-dependent methylation of the ESR1 Cpg island leads rapidly to silencing of the gene within the cells, and hence the colonic crypt v which it occurs.
Literature	GENERIF_SUMMARY	expression of estrogen receptors alpha and beta do not vary signif with cancer grade, but expression of the androgen receptor is incre estrogen receptor negative tissue as well as in grade 2 and grade 3 tumours
Literature	GENERIF_SUMMARY	HDAC4 regulates ERalpha activity as a N-terminal coregulator.
Literature	GENERIF_SUMMARY	ERalpha directly regulates human suppressor of cytokine signaling promoter activity in human breast cancer cells
Literature	GENERIF_SUMMARY	role for genetic variations in the ESR1 gene in determining post-menopausal E(2) levels in women
Literature	GENERIF_SUMMARY	GSK-3 interacts with and phosphorylates ESRalpha and is involved regulation of receptor activity
Literature	GENERIF_SUMMARY	This study suggests that Bst UI polymorphism in ER-alpha gene ma absent or rare in Chinese Han population. Pvu II polymorphism pos influences the loss of trabecular bone mass in old men.
Literature	GENERIF_SUMMARY	interplay between WT1 and ERalpha in control of IGF-IR gene trans
Literature	GENERIF_SUMMARY	ablation of forkhead box A1 expression in MCF-7 cells suppressed ERalpha binding to the prototypic TFF1 promoter
Literature	GENERIF_SUMMARY	LRH-1 is transcriptionally regulated by the estrogen receptor alpha reinforcing the hypothesis that LRH-1 could exert potential oncoge effects in breast cancer formation.
Literature	GENERIF_SUMMARY	Data suggest that mutation of the estrogen receptor beta but not a an independent risk marker for premature coronary artery disease.
Literature	GENERIF_SUMMARY	Activation of ERalpha by different phytoestrogens in a breast canc context; a screening bioassay tool to evaluate the estrogenic activi extracts of herbs and foods
Literature	GENERIF_SUMMARY	Bangladeshi natural plant extracts modulate expression of estroge receptor alpha gene in breast cancer cells treated with transcriptio decoy
Literature	GENERIF_SUMMARY	the differential localization of AR and ER isoforms in human sperm reveals distinct roles of these receptors in the physiology of sperm
Literature	GENERIF_SUMMARY	the ERalpha TA repeat polymorphism has a significant effect on th estrogen responsiveness of lumbar spine bone mineral density aft year of hormone replacement therapy in Korean women
Literature	GENERIF_SUMMARY	The activation function-1 domain of estrogen receptor alpha in ute stromal cells is not required for human uterine epithelial response estrogen.
Literature	GENERIF_SUMMARY	ERalpha46 antagonizes the proliferative action of ERalpha66 in MC cells in part by inhibiting ERalpha66 AF-1 activity
Literature	GENERIF_SUMMARY	There was no evidence to suggest that the TA repeat polymorphism ERalpha was associated with venous ulceration.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The ESR1 c.454-397C and c.454-351G haplotype and longer (TA)n are associated with the extent of coronary artery disease(CAD) in y subjects, independent of the known CAD risk factors.
Literature	GENERIF_SUMMARY	Our study does not support an association of the ER-alpha gene w bone mineral density nor body mass index in the Chinese populatio
Literature	GENERIF_SUMMARY	Her-2 expression was inversely related to estrogen and progestero receptor expression in breast cancer.
Literature	GENERIF_SUMMARY	Phosphorylation of ER-alpha Ser167 is helpful in selecting patients may benefit from endocrine therapy and is a prognostic marker in metastatic breast cancer.
Literature	GENERIF_SUMMARY	Patients with stage I/II endometriosis show a higher incidence of estrogen receptor alpha alleles with fewer (TA)n repeats (12-15 rep compared with controls.
Literature	GENERIF_SUMMARY	At elevated levels, ERalpha activated, but could not repress, known estrogen-responsive genes.
Literature	GENERIF_SUMMARY	the ESR1 c.454-397CC genotype may have a role in stroke
Literature	GENERIF_SUMMARY	oestrogen receptor alpha gene is linked and/or associated with ag menarche in different ethnic groups
Literature	GENERIF_SUMMARY	pp and XX genotype of estrogen receptor gene may play a certain maintaining bone mineral density.
Literature	GENERIF_SUMMARY	397T/C and -351A/G polymorphisms of ESR1 or haplotypes based these polymorphisms are not associated with myocardial infarctio white population.
Literature	GENERIF_SUMMARY	In patients with endometriosis, epithelial and stromal cells from endometriotic lesions both express significantly higher levels of m metalloproteinase 1 and lower levels of ER alpha than correspondi in uterine endometrium.
Literature	GENERIF_SUMMARY	Results support a relevant role for the estrogenic pathway, notably ESR1 gene, in human male reproductive function and advocate a c trait model for male infertility.
Literature	GENERIF_SUMMARY	The results of this study suggest that the PvuII polymorphism in th alpha, or another mutation in linkage disequilibrium with PvuII, in combination with high E2 levels increases breast cancer risk in postmenopausal women
Literature	GENERIF_SUMMARY	Results underline the importance of the A/B domain of estrogen re alpha in both the differentiating and the neuroprotective effects of estrogens.
Literature	GENERIF_SUMMARY	Data suggest that piceatannol and myricetin may contribute to the estrogenicity of red wine since they transactivate endogenous estr receptors alpha, and compete with 17beta-estradiol for binding to estrogen receptors.
Literature	GENERIF_SUMMARY	NFAT3 may play a role in estrogen receptor signaling in breast can cells
Literature	GENERIF_SUMMARY	Data show that estrogen receptor alpha (ERa) reduces levels of peroxisome proliferator-activated receptor alpha (PPARa) in breast cell lines, and that reduction of ERa by sodium butyrate results in a

Main Category	Category	Terms
		increase in PPARa expression.
Literature	GENERIF_SUMMARY	The study concludes that genetic variations at ER alpha gene locus perhaps, are associated with BMD in Indian women and may influe some determinant of bone metabolism resulting in accelerated bo with age.
Literature	GENERIF_SUMMARY	Data suggested suggested Arg548 may play an important role in E function in precocious puberty.
Literature	GENERIF_SUMMARY	14-3-3 sigma expression inversely correlated with estrogen recept alpha, progesterone receptor and estrogen-responsive finger prote positively correlated with myometrial invasion and lymph node metastasis in endometrial adenocarcinoma.
Literature	GENERIF_SUMMARY	The largest so far to screen for the ER-alpha A908G mutation in br cancer confirms the presence of the mutant in invasive breast tum
Literature	GENERIF_SUMMARY	ERbeta influences ERalpha protein levels
Literature	GENERIF_SUMMARY	Epithelial ovarian carcinoma is mediated by activation of ERalpha-mediated transcription.
Literature	GENERIF_SUMMARY	More than two-thirds of invasive breast cancers showed a high de methylation of the ER-beta promoter.
Literature	GENERIF_SUMMARY	The replacement of Asp-351 with lysine (D351K) or leucine (D351 completely abolished estrogen-dependent transactivation without affecting estrogen-binding, DNA-binding and homodimerization ac in ERalpha.
Literature	GENERIF_SUMMARY	ESR1 G325C polymorphism is associated with migraine in spain population.
Literature	GENERIF_SUMMARY	The (TA)n repeat polymorphism situated in the promoter region of ERalpha gene has a significant effect on sperm count.
Literature	GENERIF_SUMMARY	study proposes a new mechanism by which estrogen receptors an retinoic acid receptors regulate BRCA1-mediated DNA repair by me CBP
Literature	GENERIF_SUMMARY	MUC1 binds directly to the ERalpha DNA binding domain and stabi ERalpha by blocking its ubiquitination and degradation.
Literature	GENERIF_SUMMARY	suppressing p53 function is an important component in the pro-proliferative role of ERalpha
Literature	GENERIF_SUMMARY	Results show that TAK-778 enhances osteoblast differentiation th an ERalpha-dependent pathway by binding to this receptor and not increasing the expression of ER.
Literature	GENERIF_SUMMARY	ER1 haplotype does not have a strong effect on the estrogen-induc changes in haemostasis and inflammation risk markers for arteria venous thrombosis
Literature	GENERIF_SUMMARY	Data demonstrate that estrogen receptor alpha is a TNFalpha-indu coactivator that becomes a repressor in the presence of estradiol recruiting GRIP1.
Literature	GENERIF_SUMMARY	The expression of estrogen receptor-alpha had no effect on surviv
Literature	GENERIF_SUMMARY	ERalpha PvuII pp and ERalpha XbaI xx polymorphisms were assoc

Main Category	Category	Terms
		with a nonsignificant elevated risk; associations seemed to be stro among women with a family history of breast cancer.
Literature	GENERIF_SUMMARY	Acetylation is a modulator of the ligand-dependent gene regulatory activity of ERalpha; conserved lysine residues in ERalpha (Lys266 a Lys268) are the primary targets of p300-mediated acetylation.
Literature	GENERIF_SUMMARY	Women affected by endometriosis with ERalpha polymorphic allele if heterozygous, have worse prognosis. ERalpha gene polymorphis may be included among genetic risk factors for endometriosis.
Literature	GENERIF_SUMMARY	a polymorphism in the estrogen-receptor-signaling pathway increa breast cancer risk
Literature	GENERIF_SUMMARY	No evidence for estrogen receptor 1 gene polymorphisms in intron exon 4 with migraine susceptibility.
Literature	GENERIF_SUMMARY	ESR1 gene variants are associated with bronchial hyperresponsive and a more rapid lung function decline especially in female subjec
Literature	GENERIF_SUMMARY	Our data suggest a novel ER subtype-specific neuroprotective acti repressing calpain expression and calpain activity under condition massive Ca2+ influx.
Literature	GENERIF_SUMMARY	Expresssion of estrogen receptor-alpha was reduced in cultured ce compared with whole tissue.
Literature	GENERIF_SUMMARY	The association of 2 ESR1 gene polymorphisms (the XbaI and Pvu with cognitive impairment on the Orientation-Memory-Concentrati in postmenopausal Danish women showed a marginal effect of th polymorphism.
Literature	GENERIF_SUMMARY	analysis of estrogen receptor alpha signaling in non-tumorigenic h mammary epithelial cells
Literature	GENERIF_SUMMARY	vinca alkaloids decrease ERalpha protein levels in the human brea cancer cell line MCF-7
Literature	GENERIF_SUMMARY	Sp3 and Sp4 cooperatively interact with ERalpha to activate VEGF
Literature	GENERIF_SUMMARY	Carriage of the ESR1 IVS1-397 T/C (PvuII), COMT G158A, and the 34T-->C SNPs is not associated with the susceptibility to uterine leiomyoma in a Caucasian population.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha interacts with Galpha13 to drive actin rem and endothelial cell migration.
Literature	GENERIF_SUMMARY	ERalpha-mediated repression of hepatic transporters and alteratio bile acid biosynthesis may contribute to development of EE2-induc hepatotoxicity
Literature	GENERIF_SUMMARY	Estrogen-occupied estrogen receptor represses cyclin G2 gene expression and recruits a repressor complex at the cyclin G2 prom
Literature	GENERIF_SUMMARY	The results indicate that estrogen plus overexpressed ERalpha ind LoVo cell apoptosis might mediate through the increase of hTNF-a gene expression, which in turn activate caspase-8, -9 and caspase- lead to the DNA fragmentation and apoptosis.
Literature	GENERIF_SUMMARY	overexpressed hER-beta but not hER-alpha may induce caspase-8- mediated apoptosis by increasing the hTNF-alpha gene expressio ligand-dependent manner in poorly differentiated HA22T cells

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	BARX2 and estrogen receptor alpha may coordinately regulate cell growth, survival and invasion pathways that are critical to breast c progression.
Literature	GENERIF_SUMMARY	classical estrogen receptors mediate rapid signals induced by estr these cells
Literature	GENERIF_SUMMARY	metastasis associated 1 family, member 2 is a repressor of ERalph activity and that it could represent a new therapeutic target of ERa action in human breast tumors
Literature	GENERIF_SUMMARY	ER/PR/AR-negative cells transfected with AR were killed by DHEAS treatment, providing evidence that AR is responsible for this effect provides the first AR-targeted hormonal therapy for ER breast canc
Literature	GENERIF_SUMMARY	ERbeta selectivity of OBCP ligands can be attributed to a combinat their interactions with Met-336 in ERbeta and Met-421 in ERalpha
Literature	GENERIF_SUMMARY	aryl hydrocarbon receptor-ERalpha/Sp1 transcription factor cross t due, in part, to enhanced association of aryl hydrocarbon receptor ERalpha
Literature	GENERIF_SUMMARY	a 1.8-fold increase in ERalpha mRNA and protein in end-stage hum dilated cardiomyopathy (DCM, n=41), as compared with controls (n
Literature	GENERIF_SUMMARY	estrogen receptor alpha expression is downregulated by polyamine analogs in human breast cancer cells
Literature	GENERIF_SUMMARY	the DLC1-KIBRA interaction is essential for ER transactivation in br cancer cells
Literature	GENERIF_SUMMARY	PDI plays a critical role in estrogen responsiveness by functioning molecular chaperone and assisting the receptor in differentially reg target gene expression
Literature	GENERIF_SUMMARY	upregulation of aromatase combined with downregulation of ERalp provides a basis for selective action of estradiol produced locally b autocrine or paracrine mechanisms
Literature	GENERIF_SUMMARY	To test bioactivity of estrogens, human ESRA ws expressed in a genetically modified Saccharomyces cerevisiae strain.
Literature	GENERIF_SUMMARY	Cav-1-deficient mammary acini displayed increased ERalpha levels enhanced sensitivity toward estrogen-stimulated growth, with spec regulation of cyclin D1
Literature	GENERIF_SUMMARY	Data show that estrogen receptor (ER)-alpha36 is a membrane-bas alternatively spliced variant of hER-alpha66 that acts as a dominan negative effector of both estrogen-dependent and -independent sig through hER-alpha66 and ER-beta.
Literature	GENERIF_SUMMARY	estrogen receptor-beta/NFkappaB crosstalk mediates ICI 182,780- regulated gene expression in DU145 prostate cancer cells
Literature	GENERIF_SUMMARY	The estrogen receptor alpha variant lacking exon 5 inhibits estroge receptor transactivation in hepatoma cells.
Literature	GENERIF_SUMMARY	There is no significant influence of ER Px haplotype on bone miner density in Chinese women.
Literature	GENERIF_SUMMARY	The relationship between venous thromboembolism ESR1 polymo was consistent with a dominant model in women and a recessive

Main Category	Category	Terms
		in men.
Literature	GENERIF_SUMMARY	ESR1 polymorphism is associated with susceptibility to hepatocell carcinoma with hepatitis B virus carriers.
Literature	GENERIF_SUMMARY	BAF57 is a critical regulator of estrogen receptor function in breas cancer cells
Literature	GENERIF_SUMMARY	The crystal structure of the ligand-binding domain (LBD) associate hydroxytamoxifen (HT) was determined to 2.2 A and revealed two molecules of HT bound to the protein.
Literature	GENERIF_SUMMARY	PvuII and XbaI polymorphisms in the ER-alpha gene were associat the risk of developing microsatellite instability in colorectal cancer patients. The BsmI polymorphism in the VDR gene was linked to th of developing colorectal cancer
Literature	GENERIF_SUMMARY	Longer Tandem Repeats in estrogen receptor alpha is associated w breast cancer
Literature	GENERIF_SUMMARY	Estrogen receptor alpha and APOEepsilon4 polymorphisms interac increase risk for sporadic AD in Italian females.
Literature	GENERIF_SUMMARY	The prevalence of ER, PR and HER2/neu positivity in breast cancer Jordanian women is lower than that of the western populations an to other populations such as the Chinese and the minor ethnic gro Northern America (African Americans).
Literature	GENERIF_SUMMARY	specific polymorphisms in the CYP1B1, ESR1, and ESR2 genes ma role in progression of benign breast disease to breast cancer amo Caucasian women
Literature	GENERIF_SUMMARY	FoxM1 and FoxO3a cooperate to regulate ERalpha gene transcripti
Literature	GENERIF_SUMMARY	ERbeta had diverse effects on gene expression, enhancing or counteracting ERalpha regulation for distinct subsets of estrogen t genes
Literature	GENERIF_SUMMARY	findings suggest that the rate of bone resorption was influenced by ERalpha genotypes, even though these biochemical differences we evident in bone mass indices
Literature	GENERIF_SUMMARY	Both ER alpha and beta could mediate estrogen action in spermatogenetic cells. Presence of short ER-beta form in abnorm cells and its disappearance in mature sperm, support estrogen modulation via different ERs during sperm maturation.
Literature	GENERIF_SUMMARY	both ERs were significantly overexpressed in macrophages of won with endometriosis compared with controls
Literature	GENERIF_SUMMARY	there is an E2-dependent, ER isoform-specific transcriptional activa the RBBP1 gene, which in part, is explained by the differential activ ER AF1 and enhancer element binding
Literature	GENERIF_SUMMARY	ERalpha activation switched smooth muscle cells to a dedifferenti phenotype and may contribute to plaque instability.
Literature	GENERIF_SUMMARY	A range of ERbeta isoform expression occurs in ERalpha-negative tumours.
Literature	GENERIF_SUMMARY	Among the glassy cell carcinoma (GCC) cases, estrogen receptor (progesterone receptor (PR) and Her2/neu were positive in 2 (18.1%

Main Category	Category	Terms
		(9.1%) and 5 (45.4%) cases, respectively.
Literature	GENERIF_SUMMARY	Contribution of an autocrine ERBB4/estrogen receptor signaling pa to tumor growth and therapeutic response should be considered w managing patients with ER-positive breast cancer.
Literature	GENERIF_SUMMARY	the allele frequencies of the exon 1 and the intron 1 of the ESR1 ge not differ between uterine leiomyoma cases and controls (P = 0.32 0.175, respectively)
Literature	GENERIF_SUMMARY	A study evaluating potential associations between single nucleotic polymorphism variants of the estrogen receptor genes ESR1 and E and circulating estradiol concentrations in women of 4 races/ethni is presented.
Literature	GENERIF_SUMMARY	The relationship between ESR1 and ESR2 polymorphisms and bon mineral density in women of 4 races/ethnicities who were premen or in early perimenopause is reported.
Literature	GENERIF_SUMMARY	Increases in levels of phosphorylation at serine may play a key role emergence of endocrine resistance in breast cancer.
Literature	GENERIF_SUMMARY	LMP2-associated proteasome is recruited to the entire sequence c target genes, implicating a role for the proteasome in both transcri initiation and elongation.
Literature	GENERIF_SUMMARY	ERalpha-truncation mutants indicated that helix-12 and interaction co-regulators influenced large-scale chromatin modeling.
Literature	GENERIF_SUMMARY	polymorphims of the ER-alpha gene, age, years since menopause a educational level are associated with bone density in postmenopa Vietnamese women
Literature	GENERIF_SUMMARY	The association between Akt activation and PR negative expressio observed even in the ER-positive cases
Literature	GENERIF_SUMMARY	mammalian PHB2 has roles in mitochondria and its nuclear transl is estrogen receptor-dependent
Literature	GENERIF_SUMMARY	analysis of estrogen receptor-alpha ligand-selective interdomain conformations
Literature	GENERIF_SUMMARY	Nuclear ER alpha is enhanced during aging in women, indicating availability of locally synthesized estrogens that may up-regulate E alpha, by which neuronal metabolism can be augmented in the hippocampus after menopause.
Literature	GENERIF_SUMMARY	Estrogens operate in leiomyoma smooth muscle cells through diff receptors, alpha and beta.
Literature	GENERIF_SUMMARY	Study in a large series of Chinese patients does not show any asso of PvuII and XbaI single nucleotide polymorphisms of ESR alpha w increased risk of having adolescent idiopathic scoliosis.
Literature	GENERIF_SUMMARY	role for hematopoietic PBX-interaction protein-microtubule comple regulating 17beta-estradiol-ERalpha responses in mammalian cells discovered an inherent role of microtubules in the action of nuclea receptor
Literature	GENERIF_SUMMARY	Altered expression of ERalpha and ERbeta in tumors compared wit corresponding normal gastric tissues was more common in poorly differentiated adenocarcinomas and related to malignant propertie

Main Category	Category	Terms
		as lymph node metastasis.
Literature	GENERIF_SUMMARY	the PvuII polymorphism of EDR-alpha had a very weak association femoral neck bone mineral density (BMD) whereas XbaI polymorph was unlikely to be a predictor of femoral neck or spine BMD in Chi women
Literature	GENERIF_SUMMARY	Novel polymorphism in the promoter of a cardiac mRNA splice iso ERalpha is associated with left ventricular hypertrophy.
Literature	GENERIF_SUMMARY	Gains in area-adjusted bone mineral content in late pubertal girls a strongly associated with ESR1 polymorphisms
Literature	GENERIF_SUMMARY	ER alpha down-regulation in the mid-secretory phase, leading to de uterine receptivity.
Literature	GENERIF_SUMMARY	ERalpha-351 XbaI*G- and -397 PvuII*C-related genotypes/alleles w correlated with higher susceptibilities of endometriosis or leiomyo
Literature	GENERIF_SUMMARY	expression of mammalian GINS is regulated by 17beta-Estradiol-stimulated estrogen receptor alpha, and PSF3 acts as a gene resp to transcription factor E2F1
Literature	GENERIF_SUMMARY	The present study shows that 17beta-estradiol inhibits the transcri and apoptotic activities of the APPct complex by a process involvi interaction of estrogen receptor alpha (ERalpha) with Fe65.
Literature	GENERIF_SUMMARY	Binding and dissociation of full-length steroid receptor coactivator (SRC1a) from full-length estrogen receptor alpha or beta.
Literature	GENERIF_SUMMARY	In an analysis of breast cancer-specific survival up to 5 years, high high p53, and negative ER as well as aneuploidy of the tumour wer significant prognostic factors.
Literature	GENERIF_SUMMARY	The expression levels of ER subtypes and AR may be important fo regulation of follicular mesenchymal cells in human scalp.
Literature	GENERIF_SUMMARY	estrogen receptor alpha and beta gene polymorphisms may have a deep vein thrombosis
Literature	GENERIF_SUMMARY	Regions within the F domain of the hERalpha selectively modulate activity and its interactions with other proteins.
Literature	GENERIF_SUMMARY	Data indicate that in the T47D cancer cell line VACM-1 inhibits gro attenuating estrogen-dependent/estrogen receptor alpha signaling responses.
Literature	GENERIF_SUMMARY	provide only minor support for the role of the androgen receptor a gene in the etiology of breast cancer
Literature	GENERIF_SUMMARY	estrogen receptor expression is decreased in primary tumors and node metastases of human ductal breast cancer after chemothera treatment
Literature	GENERIF_SUMMARY	Reduced ERalpha mRNA levels observed in adipose tissue from ob compared to non-obese women support a role for oestrogen signa ERalpha, in control of body weight.
Literature	GENERIF_SUMMARY	Genetic variation in genes involved in steroid biosynthesis, metabo and signal transduction is reported in Greek patients with cholelith
Literature	GENERIF_SUMMARY	Data suggest that cytoplasmic mutant p53 increases Bcl-2 expres

Main Category	Category	Terms
		estrogen receptor-positive breast cancer cells.
Literature	GENERIF_SUMMARY	ERalpha associated with the ninth intron of the endogenous BARD in MCF-7 cells.
Literature	GENERIF_SUMMARY	results support the hypothesis that homozygosity for the specific E 'AGATA' haplotype may increase the susceptibility to the developm male genital abnormalities in response to estrogenic environmenta endocrine disruptors
Literature	GENERIF_SUMMARY	Estrogen Receptor-alpha (ER-alpha) in vaginal wall endothelial cell smooth muscle cells, and fibrocytes was significantly lower in pre- Menopausal women with Stress Urinary Incontence
Literature	GENERIF_SUMMARY	ICI182,780 and Ral locally alter the ERalpha ligand binding structur specific hydrophobic residues of H12 and decrease its transcriptio activity through tighter association with an insoluble nuclear struct
Literature	GENERIF_SUMMARY	Posttranscriptional regulation of ERalpha by a micro-RNA in the co breast cancer.
Literature	GENERIF_SUMMARY	There was a highly significant positive association between p27(K and estrogen receptor/progesterone receptor (ER/PR) status and v p27(KIP-1) and cyclin D1 expression. A trend between cyclin A and status was also identified.
Literature	GENERIF_SUMMARY	IGF-1 induces phosphorylation and an increase of the AP1 comple which is phosphorylated and binds to the pS2/TFF1 promoter, allo recruitment of the chromatin remodeling factor Brg1 followed by b of ERalpha via its interaction with c-Jun
Literature	GENERIF_SUMMARY	The specific XbaI and PvuII polymorphisms of the Er gene are asso with low bMD at all bMD measurement sites in the bulgarian fema population. they might become useful genetic markers in osteopor risk assessment in this specific population.
Literature	GENERIF_SUMMARY	Persons with one or two copies of ESR1 PvuII-XbaI haplotype 1 ha increased risk of late AMD (aging-macula disorder), especially of th form.
Literature	GENERIF_SUMMARY	The relationship between ESR1 intron variants and type 2 diabetes stage renal disease was studied.
Literature	GENERIF_SUMMARY	ERalpha and ERbeta are inhibitory modifiers of Apc-dependent colo tumorigenesis.
Literature	GENERIF_SUMMARY	FXR is expressed in breast cancer interacts with ER
Literature	GENERIF_SUMMARY	The ESR1 PvuII and XbaI gene polymorphisms seem to affect lipop metabolism in healthy subjects of peak reproductive age.
Literature	GENERIF_SUMMARY	F promoter of ERalpha activity may be tightly regulated by a dynam interplay between these two Runx2 sites, with a predominance of negative effect of the Runx2 (a) site.
Literature	GENERIF_SUMMARY	ERalpha splice variants and their age-related changes are brain reg specific
Literature	GENERIF_SUMMARY	exonic, but not intronic polymorphisms of estrogen receptor alpha might intensify the cholesterol lowering effect of raloxifene
Literature	GENERIF_SUMMARY	the ERalpha cofactor RIZ gene has a prominent effect on bone ma

Main Category	Category	Terms
		density(BMD), and the P704 genotype modulates the impact of est on BMD
Literature	GENERIF_SUMMARY	Common ESR1 polymorphisms are significantly associated with ag related changes in left ventricular structure.
Literature	GENERIF_SUMMARY	There is a significant correlation of Pvu II polymorphism of ER gen dry eye syndrome in postmenopausal women.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha mRNA copy numbers are not significantly different in tissues that are graded as positive by immunohistoche and ligand binding assays.
Literature	GENERIF_SUMMARY	temporomandibular joint(TMJ) osteoarthritis(OA) patients with PX haplotype had higher risk of moderate or severe pain compared to without the haplotype; suggests ERalpha polymorphism may be associated with pain susceptibility in females with TMJ OA
Literature	GENERIF_SUMMARY	ESRA is a putative substrate for the BRCA1 ubiquitin ligase.
Literature	GENERIF_SUMMARY	ERalpha mRNA and protein were undetectable in adrenal zona reti zona glomerulosa and medulla during development
Literature	GENERIF_SUMMARY	Fat mass in 11-yr-old children was related to the rs7757956 ESR1 polymorphism
Literature	GENERIF_SUMMARY	There was an interaction between lean tissue mass (LTM) and ESR polymorphisms with respect to their effect on femoral neck bone d in postmenopausal women. Those with TC/CC and AG/GG pheno had larger effects on LTM.
Literature	GENERIF_SUMMARY	Alzheimer's disease risk was found affected by interaction betwee estrogen receptor-alpha and butyrylcholinesterase alleles.
Literature	GENERIF_SUMMARY	This is the first report demonstrating estrogen receptors and progesterone receptors in the smooth muscle cells in smooth mus metaplasia outside the endometriotic foci of rectovaginal endome
Literature	GENERIF_SUMMARY	The regulation of STYK1/NOK is achieved independent of ERalpha suggests further investigation to the relevance of this kinase in bre cancer progression.
Literature	GENERIF_SUMMARY	ESR1 amplification may be a common mechanism in proliferative b disease and a very early genetic alteration in a large subset of brea cancers.
Literature	GENERIF_SUMMARY	the C/C genotype of the ER alpha Pvu II polymorphism was associ with a modestly reduced risk of prevalent and incident hip osteoart in elderly Caucasian women
Literature	GENERIF_SUMMARY	ERalpha may be required for estradiol modulation of affective and behavior, respectively.
Literature	GENERIF_SUMMARY	Female, but not male end-stage renal disease patients with the ERa Ser10Ser CC genotype had lower prevalence of protein energy was lower serum triglyceride, higher serum albumin and lower hsCRP le
Literature	GENERIF_SUMMARY	ESR1 polymorphisms do not play a role in breast cancer risk in Cau postmenopausal women.
Literature	GENERIF_SUMMARY	Dominant expression of estrogen receptors in penile tissues of chi with hypospadias may be the postnatal finding of disrupted estrog

Main Category	Category	Terms
		receptor interaction during intrauterine development of external ge
Literature	GENERIF_SUMMARY	Estrogen receptor alpha (ERalpha) was significantly downregulated ovarian tissue of patients with ovarian endometriosis
Literature	GENERIF_SUMMARY	Data show that the induction of the gene regulated in breast cance (GREB1) is mediated by binding of estrogen receptor alpha to three consensus estrogen response elements spread over approximatel of upstream flanking sequences.
Literature	GENERIF_SUMMARY	biological function for DBC-1 in the modulation of ERalpha express and hormone-independent breast cancer cell survival
Literature	GENERIF_SUMMARY	The ratio ERalpha/ERbeta constitutes one decisive parameters to orientate the transcriptional mechanism of a target gene in the pre of agonist as well as of antagonist ligands.
Literature	GENERIF_SUMMARY	did not observe an association between ESR1 PvuII and XbaI gene polymorphisms and ischemic stroke or any subtype of ischemic st
Literature	GENERIF_SUMMARY	Results show a relationship between the presence of the P allele a higher values of cortical thickness in male subjects, indicating for ERalpha a role in the control of tibial bone geometry.
Literature	GENERIF_SUMMARY	There were significant phytoestrogen interactions with polymorphi ESR1 and NR1I2 genes affecting estrone levels; we conclude that phytoestrogens modulate sex hormone and SHBG levels in postmenopausal women.
Literature	GENERIF_SUMMARY	The homozygous T-A haplotype carriers of estrogen receptor alpha (ESR1) PvuII and XbaI polymorphisms showed an increased risk o preeclampsia.
Literature	GENERIF_SUMMARY	AHR (aryl hydrocarbon receptor) agonists induce ERalpha promote occupancy at AHR target genes through indirect activation of oest receptor alpha(ERalpha), and support a role for ERalpha in AHR transactivation
Literature	GENERIF_SUMMARY	progesterone and glucocorticoid, but not estrogen receptor mRNA altered in breast cancer stroma
Literature	GENERIF_SUMMARY	ESR1 contributes to type 2 diabetes and Cardiovascular Diseases pleiotropic effects, leading to insulin resistance, a poor lipid profile obesity
Literature	GENERIF_SUMMARY	Microarrays performed on 17beta-estradiol or 4OH-tamoxifen (4HT treated Hs578T ERalpha and ERbeta cells revealed distinct ligand receptor-dependent patterns of gene regulation, while the inductio ERbetacx did not alter gene expression patterns.
Literature	GENERIF_SUMMARY	Results suggest that a relationship between arterial stiffness and or 30T/C polymorphisms in ER-alpha is different between sexes in humans.
Literature	GENERIF_SUMMARY	estrogen receptor alpha A908G mutation has a role in breast canc invasiveness
Literature	GENERIF_SUMMARY	Expression of ERalpha correlates with cytokine production in wom endometriosis.
Literature	GENERIF_SUMMARY	Oral contraceptives may interact with the ESR1 A908G mutant rece drive the development of some breast tumors.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	These results are consistent with previous observations indicating of ESR1 genotypes in human infertility and give insight into the cor interactions between genotypes and reproductive behaviours in hu populations.
Literature	GENERIF_SUMMARY	The mean intima thicknesses in the three genotype groups were 428+/-298 microm (TT), 494+/-371 microm (CT) and 636+/-436 m (CC). On average, women with the CC genotype had a thicker coron intima compared with that of women with the TT genotype
Literature	GENERIF_SUMMARY	polymorphism of genes involving estrogen-metabolizing pathway, estrogen- biosynthesizing pathway and estrogen receptor pathway play an important role in the etiology of BRCA1/2 negative breast c with hereditary predisposing factors.
Literature	GENERIF_SUMMARY	U2-OS cells were used to study direct interactions between fluores fusion proteins of ERalpha and the NF-kappaB subunits p50 and p
Literature	GENERIF_SUMMARY	Data indicate that SMRT, in conjunction with gene-specific and cell dependent factors, is required for positively regulating agonist-dep ERalpha transcriptional activity.
Literature	GENERIF_SUMMARY	ERalpha interactions with insulin receptor substrate 1 (IRS-1) are d and involve the ERalpha activation function(AF)-1/DNA-binding do
Literature	GENERIF_SUMMARY	IL6 and ESR1 polymorphisms may have a role in breast cancer risk associated with active and passive smoking
Literature	GENERIF_SUMMARY	Estrogen receptor polymorphisms may be associated with increas of developing dysplasia of the hip.
Literature	GENERIF_SUMMARY	Per2, a core clock gene, links the circadian cycle to the ERalpha sig network.
Literature	GENERIF_SUMMARY	Correlation was observed between promoter methylation and loss protein expression confirming our hypothesis that promoter methy is an important mechanism for transcriptional silencing of these g breast cancer
Literature	GENERIF_SUMMARY	Four SNPs in intron 4 of ESR1 showed significantly different genot allele distributions between premenstrual dysphoric disorder patie control subjects.
Literature	GENERIF_SUMMARY	the ERs participate in the transduction of mechanical forces into p survival signaling in bone cells, albeit in a ligand-independent man
Literature	GENERIF_SUMMARY	GSK-3 is required for E2-induced ERalpha phosphorylation at Ser-1 full transcriptional activity of the receptor upon E2 stimulation.
Literature	GENERIF_SUMMARY	Expression of the NERKI receptor resulted in the suppression of m levels for AP, col I, OCN, and IGF-I. While E2 increased mRNA levels OCN, col I, and IGF-I in ER-alpha cells, E2 inhibits col I mRNA levels NERKI cells.
Literature	GENERIF_SUMMARY	There may be distinct groups of genes down-regulated by ERalpha role for ERK phosphorylation at serine 118 in activation function-1 regulating expression of the set of genes down-regulated by 4-hydroxytamoxifen.
Literature	GENERIF_SUMMARY	GATA-3 as a critical component of the master cell-type-specific transcriptional network including ER alpha and FoxA1 that dictates phenotype of hormone-dependent breast cancer

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Increased expression of estrogen receptor (ER)-beta in ER-alpha-p breast cancer cells antagonizes a defined group of ER-alpha/estro stimulated genes that are involved in cell cycle regulation and DNA replication.
Literature	GENERIF_SUMMARY	Tight nuclear binding of the estrogen-occupied estrogen receptor a predominantly mediated by estrogen receptor alpha-DNA interactio
Literature	GENERIF_SUMMARY	ERR alpha can be deactivated by the synthetic molecule XCT790. T compound also induces a proteasome degradation of ERR alpha
Literature	GENERIF_SUMMARY	ERalpha and ERbeta do not express in the levator ani muscle of wo
Literature	GENERIF_SUMMARY	Positive expression of ER alpha in addition to negative expression estrogen receptor beta proved to be an unfavorable independent prognostic indicator insquamous cell carcinoma of the esophagus
Literature	GENERIF_SUMMARY	ERbeta was 2.5-fold statistically significantly over expressed in postmenopausal leiomyomas compared to patient matched myom (p=0.038), whereas ERalpha and PGR were not significantly differe expressed.
Literature	GENERIF_SUMMARY	CYP2A6 is directly induced by estradiol in an ERalpha-dependent m implying a biological role of CYP2A6 in estrogen-
Literature	GENERIF_SUMMARY	Aberrant methylation of multiple genes (E-cadherin, estrogen recep RB1 , p16, p15, p14, and MGMT) is involved in gastric carcinogenes
Literature	GENERIF_SUMMARY	Physiological postnatal vasculogenesis involves cyclic, E2-regulate bioactivity of bone marrow-derived endothelial progenitor cells, predominantly through the ER alpha.
Literature	GENERIF_SUMMARY	there is a progressive loss of ERalpha cofactor recruitment to the estrogen-dependent pS2 gene and reduced requirement for cofact expression in acquired endocrine resistant breast cancer
Literature	GENERIF_SUMMARY	estrogen-stimulated GREB1 transcription may involve coordinated binding to all three distal consensus ERE motifs
Literature	GENERIF_SUMMARY	The collective data indicate that peripheral estrogen receptor alpha dependent mechanisms determine the shape of in vivo growth hor secretory bursts and that such mechanisms operate with secretag selectivity.
Literature	GENERIF_SUMMARY	ability of estrogen receptor alpha to regulate NS5B was suggested serve as a potential novel target for anti-hepatitis C virus therapeut
Literature	GENERIF_SUMMARY	Polymorphisms at the ESR1 gene are associated with premature o failure.
Literature	GENERIF_SUMMARY	The present study examined whether the estrogenic action of imidazolines and piperazines is associated with a down regulation ERalpha, as reported for conventional agonists.
Literature	GENERIF_SUMMARY	identify KLF9 as a transcriptional repressor of estrogen receptor al signaling
Literature	GENERIF_SUMMARY	cross-talk among ER-alpha, Snail, and the TGF-beta pathway appea control critical phenotypic properties of breast cancer cells
Literature	GENERIF_SUMMARY	These results demonstrate that ER DNA binding is not absolutely r for estrogen to induce breast cancer cell growth.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Beta-Catenin was co-immunoprecipitated with ERalpha, indicating these two proteins form a new signaling complex and transcriptior
Literature	GENERIF_SUMMARY	These findings suggest that PvuII and XbaI polymorphisms may af age at onset of the first stroke and the probability of developing cerebrovascular disease.
Literature	GENERIF_SUMMARY	Ligands caused a robust phenotype, leading to an almost total immobilization of ERalpha, whereas ERbeta retained its mobility.
Literature	GENERIF_SUMMARY	Most adult granulosa cell tumors and Sertoli-Leydig cell tumors sh overlapping patterns of expression of ERalpha and PR with each o a subset of cases in each tumor category exhibits unique ERalpha, immunoprofiles
Literature	GENERIF_SUMMARY	Several SNPs in the ESR1 and ESR2 genes were associated with ri developing cognitive impairment. These findings suggest that estr receptor genetic variants may play a role in cognitive aging.
Literature	GENERIF_SUMMARY	genetic variation in ER gene may influence blood levels of VCAM-1 women after the menopause
Literature	GENERIF_SUMMARY	Estrogen receptor alpha and beta gene polymorphisms are not risk factors for recurrent miscarriage in a Brazilian population.
Literature	GENERIF_SUMMARY	an association for breast cancer risk between short (SS) alleles fo repeat variants of the ESR2 and AR genes was found in women of descent.
Literature	GENERIF_SUMMARY	estrogen receptor-alpha phosphorylation at Ser167 is indicative of disease-free and overall survival in breast cancer patients
Literature	GENERIF_SUMMARY	In addition to the activity of RhoGDI alpha in the cytoplasm, it also influences ER alpha signaling in the nucleus.
Literature	GENERIF_SUMMARY	Our results point to an antiproliferative function of the unliganded alpha through its physical interactions with p21(WAF1) that may a explain the favorable prognosis of ER alpha-positive breast cancer
Literature	GENERIF_SUMMARY	A role for estrogenically regulated LRP16 as an ERalpha coactivato providing a positive feedback regulatory loop for ERalpha signal transduction.
Literature	GENERIF_SUMMARY	study suggests that age, serum vitamin B-12 concentrations, and t presence of benign or neoplastic lesions are independent determir ER promoter methylation in colorectal mucosa
Literature	GENERIF_SUMMARY	upregulation of ER-alpha and downregulation of ER-beta may repre two discrete molecular events in lobular neoplasia pathogenesis.
Literature	GENERIF_SUMMARY	Data show that the inhibition of NFkappaB by ER is mediated by CI is recruited to the NFkappaB response element of the MCP-1 and I promoters and displaces CBP.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha and BRCA1 are specifically targeted for methylation in sporadic breast cancers
Literature	GENERIF_SUMMARY	ER-alpha gene Pvu II and Xba I polymorphism seemed not potentia influence on BMD and bone turnover markers in the pubertal girls i Beijing, suggesting that it might not be the major genetic factor re this population.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The effect of chrysin and 7-O-beta-D-glucopyranosylchrysin on est receptor alpha transcription in MCF-7 cells is reported.
Literature	GENERIF_SUMMARY	ER and PR are present in transverse carpal ligament and flexor ten synovitis, suggesting a role for sex steroid hormones in the pathog of carpal tunnel syndrome.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha and estrogen receptor beta act differently through a non-estrogen response element-mediated pathway and play different roles in medullary thyroid carcinoma tumor growth a progression.
Literature	GENERIF_SUMMARY	the G2014A polymorphism is associated with the presence of osteoporosis while the TA dinucleotide repeat is not
Literature	GENERIF_SUMMARY	Short-term exposure of breast cancer cells to 1 nM TCDD suppres ERalpha protein expression, while chronic exposure for more than results in the partial re-expression of the receptor.
Literature	GENERIF_SUMMARY	The c.454-397T/T genotype associates with first-ever cerebral hemorrhage(CH), particularly in combination with hypertension. Alterations in ER alpha-mediated signaling may be involved in pathophysiology of CH, with putative impact on primary preventior
Literature	GENERIF_SUMMARY	The complementary abilities of ER alpha and NM23-H1 together to influence gene expression, cell migration, and apoptosis could be factors in helping to determine tumor cell fate
Literature	GENERIF_SUMMARY	Therefore, a loss of cell-cell junctions, a key process that occurs du the epithelial-mesenchymal transition, should have a broad impact alpha transcriptional functions.
Literature	GENERIF_SUMMARY	Strongly positive estrogen receptor is associated with leiomyomas uterus.
Literature	GENERIF_SUMMARY	BTF3 interacts with ERalpha that has been activated either by 17be estradiol (ligand-dependent activation) or by epidermal growth fac (ligand-independent activation).
Literature	GENERIF_SUMMARY	activation of PKCtheta inhibits the FOXO3a/ERalpha/p27(Kip1) axi normally maintains an epithelial cell phenotype and induces c-Rel genes, thereby promoting proliferation, survival, and more invasive cancer.
Literature	GENERIF_SUMMARY	Report show that ESR1 expression was appreciably higher in malig breast cancer tissues compared to benign tissues.
Literature	GENERIF_SUMMARY	ERbeta/ERalpha mRNA was significantly elevated in endometriosis Expression of PR-C mRNA relative to PR-A and PR-B mRNA was significantly increased in endometriosis.
Literature	GENERIF_SUMMARY	ESR1 gene alleles are unlikely to be a major cause of obesity in wo
Literature	GENERIF_SUMMARY	In our community-based study of elderly, a polymorphism in the es receptor alpha gene was associated with endothelium-dependent vasodilation in resistance, but not conduit arteries
Literature	GENERIF_SUMMARY	may play an important role in the function of human periodontal lig fibroblasts.
Literature	GENERIF_SUMMARY	Pituitary and/or peripheral estrogen-receptor alpha regulates follic stimulating hormone secretion, whereas central estrogenic pathwa

Main Category	Category	Terms
		direct growth hormone and prolactin secretion in postmenopausal women.
Literature	GENERIF_SUMMARY	this follow-up study confirms previous findings that the ER-alpha P polymorphism (Pp) modulates the association between exercise a bone mass
Literature	GENERIF_SUMMARY	Data suggest that as cells adapt to chronic estrogen stimulation, e receptor alpha is stabilized by a decreased rate of proteolysis and accumulation of proteasome-resistant, phosphorylated form of rec
Literature	GENERIF_SUMMARY	The ligand-binding domain is produced in Escherichia coli as a clea thioredoxin (Trx) fusion in order to improve solubility and the struct determined.
Literature	GENERIF_SUMMARY	Adrenal cortical carcinomas are character- ized by an imbalance b estrogen receptor alpha and estrogen receptor beta levels and incr aromatase expression.
Literature	GENERIF_SUMMARY	loss of p21's cyclin-dependent kinase inhibitory property results in hyperphosphorylation of estrogen receptor-alpha, with subsequent increased gene expression of estrogen receptor-regulated genes
Literature	GENERIF_SUMMARY	our data suggest that the four analyzed genes do not play an impo role in susceptibility to schizophrenia.
Literature	GENERIF_SUMMARY	Immunohistochemical expression of ERalpha seems to decrease i ovaries of women after menopause.
Literature	GENERIF_SUMMARY	Distribution of ERalpha PvuII and XbaI genotypes in population of postmenopausal women, is evaluated.
Literature	GENERIF_SUMMARY	Estrogen receptor-beta (ERbeta) expression, but not Estrogen rece alpha, was significantly higher in normal ovarian tissues compared malignant ovarian tissues
Literature	GENERIF_SUMMARY	Determine estrogen receptor alpha and beta in uterine fibroids
Literature	GENERIF_SUMMARY	17beta-estradiol induces ERRalpha gene expression in MCF-7 cells through active recruitment of co-activators and release of co-repre when ERRalpha and AP1 bind and ERalpha is tethered to the multip hormone-response element
Literature	GENERIF_SUMMARY	Intron 1 and exon 1 alpha estrogen receptor gene polymorphisms not associated with endometriosis in the population studied.
Literature	GENERIF_SUMMARY	The positive and significant correlations between the number of CD cells and positivity of the primary tumour for estrogen receptor and progesterone receptor suggest a possible role for CD1a as a progn marker for breast cancer.
Literature	GENERIF_SUMMARY	Results show that disease stage, tumor grade, SES, race/ethnicity, negative ER and PR status, rather than negative HER2 status, were factors for survival.
Literature	GENERIF_SUMMARY	Polymorphisms in the ESR1 gene are unrelated to open-angle glau but ESR2 polymorphisms seem to lead to increased risk of OAG in
Literature	GENERIF_SUMMARY	BRCA1 exerts its tissue-specific function through the regulation of progesterone receptor and estrogen receptor-alpha.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha, rather than estrogen receptor beta, plays

Main Category	Category	Terms
		role in regulating autoimmunity in (NZB x NZW) mice, and suggest the first time that estrogen receptor alpha promotes lupus.
Literature	GENERIF_SUMMARY	the ERalpha signalling pathway is disrupted by PPARgamma agoni
Literature	GENERIF_SUMMARY	In the present review we discuss recent findings showing that, in a to the canonical estrogen receptor-alpha (ERalpha), the level of var ERalpha splice variants is changed in the human brain in aging and Alzheimer's disease [review]
Literature	GENERIF_SUMMARY	mechanisms of resistance of breast cancer cells to endocrine ther due to the crosstalk between the ER and the HER growth factor rec signaling pathways
Literature	GENERIF_SUMMARY	Hsp22 induction represents a new aspect of the estrogenic respon potential significance for the biology of estrogen receptor-positive cancer cells.
Literature	GENERIF_SUMMARY	Different modulation of ERalpha palmitoylation exerted by differen ligands represents a pivotal mechanism that drives cancer cell to proliferation or apoptosis.
Literature	GENERIF_SUMMARY	Thus, the perilipin gene expression is regulated by a transcriptiona network controlling energy metabolism, substantiating the functio importance of perilipin in the maintenance of body energy balance
Literature	GENERIF_SUMMARY	Results support a model in which ERalpha/AP-1 complexes modul promoter activity under conditions of 17-beta-estradiol stimulation
Literature	GENERIF_SUMMARY	In conclusion, 17beta-estradiol suppressed OPG production by hur breast cancer cell lines in a dose-dependent and specific manner, indicating that the RANKL/OPG cytokine system is an estrogen-responsive target in breast cancer.
Literature	GENERIF_SUMMARY	MNAR plays an important role in ERa activation of Src/MAPK and Akt signaling pathways [review]
Literature	GENERIF_SUMMARY	p53 has a role in preventing centrosome amplification, ERalpha phenotypic heterogeneity and metastasis in breast cancer
Literature	GENERIF_SUMMARY	Our results do not support a strong association between common variants in the ESR1 and EGF genes and breast cancer risk, tumour characteristics or survival.
Literature	GENERIF_SUMMARY	Data show a relationship between wild-type ER-alpha and level of cognitive impairment in Alzheimer's, and also suggest the possibil variant isoforms of ER-alpha may be present in frontal cortex of pa with and without AD.
Literature	GENERIF_SUMMARY	a statistically significant relationship between the estrogen recepto alpha c.454-397T>C polymorphism and coronary artery disease we found independent of known CAD risk factors in a Turkish populati
Literature	GENERIF_SUMMARY	The data demonstrate that, in contrast to mice, no alterations in th expression of androgen and estrogen hormone receptors were evi after surgery in patients.
Literature	GENERIF_SUMMARY	The PvuII T allele may be a germline risk factor for familial forms o breast cancer and is associated with a specific subset of immunohistochemical tumor phenotype.
Literature	GENERIF_SUMMARY	evaluated variants across the entire ESR1 gene for association in A

Main Category	Category	Terms
		Americans with type 2 diabetes and end stage renal disease
Literature	GENERIF_SUMMARY	Results reveal that estrogen receeptor alpha signaling is depender protein acetylation and epigenetic regulation.
Literature	GENERIF_SUMMARY	We have not been able to replicate the previously reported associa between variations in the ESR1 gene and risk of stroke
Literature	GENERIF_SUMMARY	Higher levels of ERalpha mRNA is associated with bladder transiti carcinoma
Literature	GENERIF_SUMMARY	HER2 and estrogen receptor alpha expression depends upon nucle localization of Y-box binding protein-1 in human breast cancers
Literature	GENERIF_SUMMARY	depletion of HDAC2 results in simultaneous depletion of ER and P potentiates the effects of antihormonal therapy in ER-positive cells
Literature	GENERIF_SUMMARY	ESR1 knock-in anmals are sensitive to diethylstilbestrol-induced in due to upregulaterated utrerine estrogen receptor beta levels.
Literature	GENERIF_SUMMARY	The XbaI/X allele in ERalpha gene is associated to smaller gray ma volumes of the cerebral and cerebellar cortex. This allele might inc the susceptibility for senile neurodegenerative conditions, being associated to smaller cerebral reserve.
Literature	GENERIF_SUMMARY	the T allele of rs2881766 could be a useful genetic marker of PIH. A-T haplotype of rs2881766-rs1643821-rs988328 and the G-A hap of rs2881766-rs1643821 appear to be resistance markers of pregn induced hypertension.
Literature	GENERIF_SUMMARY	The multifunctional estrogen receptor-alpha F domain.
Literature	GENERIF_SUMMARY	estrogen, growth factors and cAMP elicit surprisingly distinct ERal dependent transcriptional responses in human MCF7 breast cance
Literature	GENERIF_SUMMARY	the ERNE-145 genotype alters transcription and is associated with increased HDL
Literature	GENERIF_SUMMARY	the MAPK stimulation of ERalpha activity involves the phosphoryla not only of S118 but also of S104 and S106; MAPK-mediated hyperphosphorylation of ERalpha at these sites may contribute to resistance to tamoxifen in breast cancer.
Literature	GENERIF_SUMMARY	These findings suggest that the G allele containing variants of ESR and the G allele containing variants of ESR2 2681-4A>G may decre risk of hypospadias, whereas the ESR1 C-A haplotype may increas risk.
Literature	GENERIF_SUMMARY	methylated ESR1 and 14-3-3-sigma may have roles in developmen breast cancer
Literature	GENERIF_SUMMARY	Results show that estrogen receptor alpha genotype may be assoc with pelvic organ prolapse risk.
Literature	GENERIF_SUMMARY	K302/303 of ERalpha protect ERalpha from basal degradation and necessary for efficient E2- and ICI-induced turnover in breast canc
Literature	GENERIF_SUMMARY	ERa gene polymorphism is associated with hypomenorrhea with u etiology.
Literature	GENERIF_SUMMARY	luminal ER(+)PR(+) breast tumors contain a minor ER(-)PR(-)CK5(+ population that has the capacity to generate the majority of

Main Category	Category	Terms
		ER(+)PR(+)CK18(+)CK5(-) cells
Literature	GENERIF_SUMMARY	Adenocarcinoma of gallbladder is negative for both ER(alpha) and progesterone receptor.
Literature	GENERIF_SUMMARY	analysis of how ligand dissociation from estrogen receptor is med receptor dimerization
Literature	GENERIF_SUMMARY	characterization of estrogen receptor expression profile & regulatio ER-alpha during differentiation into osteoclasts; results provide be understanding of role of ER-alpha regulation during bone resorptio mediated by osteoclasts
Literature	GENERIF_SUMMARY	constructed and analysed a multi-step model of hER-induced gene transcription that explains the multiple forms of a simple dose-res curve in an in vitro transcription system
Literature	GENERIF_SUMMARY	role for estrogen receptor activity in the development of Alzheimer disease in women with Down syndrome
Literature	GENERIF_SUMMARY	This is the first report of the essential role of SPOP in ERalpha ubiquitination by the Cullin3-based E3 ubiquitin ligase complex.
Literature	GENERIF_SUMMARY	Suggest that epistatic interactions of ESR1 and ESR2 alleles could associated with advancing age at menarche among Spanish wom
Literature	GENERIF_SUMMARY	The variation in the ESR1 gene is associated with schizophrenia ar mechanism of this association may involve alternative gene regula and transcript processing.
Literature	GENERIF_SUMMARY	The available data indicate a deleterious action of the ESR1 deletio both male and couple fertility.
Literature	GENERIF_SUMMARY	results suggest a novel B cell-mediated pathophysiological mecha for the etiology of osteoporosis, which is that downregulation of E and MAPK3 in B cells regulates secretion of factors leading to alte osteoclastogenesis or osteoblastogenesis
Literature	GENERIF_SUMMARY	providing a plausible explanation for distinct PGC-1alpha binding t ERRalpha monomers and homodimers
Literature	GENERIF_SUMMARY	ERalpha protein exists in neurones in mammalian prefrontal cortex
Literature	GENERIF_SUMMARY	We have discovered common sequence variants that are consiste associated with bone mineral density and with low-trauma fracture three populations of European descent.
Literature	GENERIF_SUMMARY	female-specific association between ESR1 SNPs and neurobehavi phenotypes
Literature	GENERIF_SUMMARY	In youths, evidence of association between rs9322331 & rs934079 apoC-II was stronger in males than in females. These findings sug that ESR1 variation plays an age- and sex-dependent role in determ plasma lipid & apolipoprotein levels.
Literature	GENERIF_SUMMARY	The dose of receptor critically mediates estradiol's ability to regula growth factors and survival kinases.
Literature	GENERIF_SUMMARY	analysis of a human estrogen receptor alpha mutation with differe responsiveness to nonsteroidal ligands
Literature	GENERIF_SUMMARY	Crucial role of ER in the regulation of the CA12 gene in breast canc

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	SNP12 in ESR1 is not associated with the occurrence of cryptorchi but is associated with the severity of cryptorchidism.
Literature	GENERIF_SUMMARY	ERalpha (estrogen receptor alpha) expression was present in ten c endometrioid adenocarcinomas but in only five of ten serous carci
Literature	GENERIF_SUMMARY	Results show that estrogen receptor alpha is directly methylated a 302 (K302) by the SET7 methyltransferase.
Literature	GENERIF_SUMMARY	disruption of ERalpha-p53 interaction in vivo resulting in restoratio functional p53 is a cellular response to radiation
Literature	GENERIF_SUMMARY	TIEG is rapidly induced in response to estrogen in osteoblasts by E but not ERalpha
Literature	GENERIF_SUMMARY	the Ret receptor tyrosine kinase pathway functionally interacts wit ERalpha pathway in breast cancer
Literature	GENERIF_SUMMARY	estrogen receptor-alpha is activated by the anion nitrite
Literature	GENERIF_SUMMARY	Data show that the TNF alpha may partly regulate the growth of M breast cancer cells through the down-regulation of ER alpha expre which is primarily mediated by a PI3K/Akt signaling.
Literature	GENERIF_SUMMARY	In North Indian population, ERalpha and CYP19 genes may be play role in the risk of prostate cancer.
Literature	GENERIF_SUMMARY	Estrogen receptor and HER2/neu status affect epigenetic differenc tumor-related genes in primary breast tumors
Literature	GENERIF_SUMMARY	ER alpha is involved in activation of ERK/mitogen activated protein by genistein by its early association with IGF-IR, leading to hyper-responsiveness of leiomyoma cells, confirming that ER signaling is enhanced by activation of ERK/MAPK.
Literature	GENERIF_SUMMARY	Functional interaction between estrogen receptor(ER)s and adenor polyposis coli (APC) shows that APC may physically associate wit alpha in a ligand-dependent manner and enhance estrogen-depend transactivation of the estrogen response element.
Literature	GENERIF_SUMMARY	Recruitment of coactivator glucocorticoid receptor interacting prot an estrogen receptor transcription complex is regulated by the 3',5 adenosine 5'-monophosphate-dependent protein kinase.
Literature	GENERIF_SUMMARY	Homozygous ER-alpha disruption markedly affects bone growth, m content, and structure but not periosteal circumference. ER-alpha heterozygosity appears to not impair spine aBMD.
Literature	GENERIF_SUMMARY	All tested ligands repressed the MMP1 promoter through the L540 mutant with identical efficacy.
Literature	GENERIF_SUMMARY	ERalpha concentrations were low in normal adrenal gland, in adren tissues adjacent to adrenocortical ademoma, in carcinoma, and lo adrenocortical adenoma
Literature	GENERIF_SUMMARY	Findings suggest that Vav3 overexpression may aberrantly enhanc ERalpha-mediated signaling axis and play a role in breast cancer development and/or progression.
Literature	GENERIF_SUMMARY	a combinatorial eNOS/ERalpha complex at the hTERT estrogen res element site and that active eNOS and ligand-activated ERs coope regulating hTERT expression in the endothelium.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	GATA3 is hypothesized to be integral to the ERalpha pathway
Literature	GENERIF_SUMMARY	Jab1 is a target of EGFR signaling in ERalpha- cell lines and breast
Literature	GENERIF_SUMMARY	The co-occupancy of ERalpha and Sp1 on TFF1 promoter remains similar level with and without estrogen, while that of ERalpha and S increased in the presence of estrogen
Literature	GENERIF_SUMMARY	Tamoxifen treatment of breast cancer cells reduced the expressio alpha and increased the expression of SMRT.
Literature	GENERIF_SUMMARY	Phosphorylation of ERalpha Ser118 and ERalpha Ser167 affects su in ER-positive breast cancer and could be helpful in distinguishing patients who are likely to benefit from endocrine therapy alone fro who are not.
Literature	GENERIF_SUMMARY	These studies demonstrate positive cross talk between ER, c-Src, E and STAT5b in ER+ breast cancer cells.
Literature	GENERIF_SUMMARY	ER alpha plays a protective role in modulating the induction of DNA damage by E 2 in human breast cancer cells.
Literature	GENERIF_SUMMARY	The mean number of ERA-positive stromal cells was lower in here vs sporadic cancer.
Literature	GENERIF_SUMMARY	ERAP75 functions as a novel coactivator that can modulate ER alp function in the prostate stromal cells.
Literature	GENERIF_SUMMARY	Data suggest that the aryl hydrocarbon receptor repressor might p role of co-repressor on the transcriptional activity of the estrogen r alpha homodimer.
Literature	GENERIF_SUMMARY	The results of immunoblotting experiments revealed that both est receptor alpha (ESR1) and beta (ESR2) proteins are expressed in n but not in the cytoplasm of peripheral blood mononuclear cells.
Literature	GENERIF_SUMMARY	Aberrant methylation of ESR1 promoter is associated with acute m leukemia
Literature	GENERIF_SUMMARY	Data show that RbAp46 and RbAp48 interact with estrogen recept (ER)alpha at endogenous, estrogen-responsive genes and alter expression of endogenous, ERalpha-activated and -repressed gene MCF-7 breast cancer cells.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha decreases the DNA binding of NF-kappaB subunits p65 and p50.
Literature	GENERIF_SUMMARY	PvuII gene polymorphisms affect CAD susceptibility in man only. T PvuII CC is a risk factor for CAD and it is associated with angiogra CAD severity.
Literature	GENERIF_SUMMARY	The effect of Etrogen receptor alpha on slug repression is depende the overall level of ERalpha, confirming that slug is an estrogen rec alpha reponsive gene.
Literature	GENERIF_SUMMARY	miR-206 could be a novel candidate for endocrine therapy that targ only ERalpha in breast cancer.
Literature	GENERIF_SUMMARY	Cross-talk between notch and the estrogen receptor in breast canc suggests novel therapeutic approaches.
Literature	GENERIF_SUMMARY	ER- breast cancer cells become dependent on Notch-survivin signa

Main Category	Category	Terms
		their maintenance, in vivo.
Literature	GENERIF_SUMMARY	In summary, our results confirm the stimulation of ER alpha and ER on CRH expression and demonstrate the important roles of the ER sites and CRE for the action of ER alpha and ER beta.
Literature	GENERIF_SUMMARY	the effects of ring B unsaturated estrogens are mainly mediated vi ERbeta and that the presence of both ER subtypes further enhance activity
Literature	GENERIF_SUMMARY	A phosphorylated form of the membrane estrogen receptor-alpha found to characterize the invasive breast cancer state
Literature	GENERIF_SUMMARY	Variation within the promoter region of ESR1 is associated with the of hypertrophy in male hypertrophic cardiomyopathy patients.
Literature	GENERIF_SUMMARY	A significant association was found between necessity of hormone replacement therapy and rs1256049, rs1271572, and D14S1026 in and rs2228480 in ERalpha.
Literature	GENERIF_SUMMARY	Genetic polymorphisms of ER-alpha are associated with breast ca
Literature	GENERIF_SUMMARY	estrogen/ER-responsive breast cancer genes linked to cell growth invasion pathways are associated with loss of PR and earlier disea specific mortality
Literature	GENERIF_SUMMARY	cyclin D1b protein is aberrantly regulated and could contribute to therapeutic failure in the context of ER-positive breast cancer
Literature	GENERIF_SUMMARY	estrogen receptor 1, vitamin C receptors SLC23A1 and SLC23A2, a matrix metalloproteinase MMP3 and MMP9 are associated with susceptibility to lymphoma
Literature	GENERIF_SUMMARY	Effects of estrogen-like compounds on cell proliferation are depen the actual ERalpha/ERbeta expression levels in these cells or tissu
Literature	GENERIF_SUMMARY	Estradiol-induced AKT-dependent phosphorylation of FKHR drives association with ERalpha, thereby triggering complex export from nucleus necessary for initiation of DNA synthesis and S phase ent
Literature	GENERIF_SUMMARY	Hyermethylation of CpG islands in the ER-alpha gene promoter reg found in high frequency in atherosclerosis patients.
Literature	GENERIF_SUMMARY	ERalpha plays important roles in conjugated linoleic acid induced apoptosis in human breast tissues
Literature	GENERIF_SUMMARY	Results identify single nucleotide polymorphisms that are significa associated with hip fractures in a Chinese population.
Literature	GENERIF_SUMMARY	ESR1 single nucleotide polymorphisms could have a mild influence scores in healthy people mainly but also in suicide attempters.
Literature	GENERIF_SUMMARY	The X allele of ERalpha is associated with endometriosis in a Chin population.
Literature	GENERIF_SUMMARY	erk1 E2, a sex hormone, stimulates MUC5B gene overexpression b interaction with estrogen receptor alpha (ERalpha) and by acting th extracellular signal-regulated kinase 1/2 (ERK1/2)-mitogen-activat protein kinase (MAPK).
Literature	GENERIF_SUMMARY	Estrogen receptor gene amplification occurs rarely in ovarian canc

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ERalpha46 may be involved in the development and/or progressior colorectal cancer via mediating growth inhibition and apoptosis of cells in the presence of 17beta-oestradiol.
Literature	GENERIF_SUMMARY	This study clearly implicates WT1 as a mediator of antiestrogen resistance in breast cancer through down-regulation of ERalpha expression.
Literature	GENERIF_SUMMARY	ESR1 promoter hypermethylation does not predict atypia in RPFNA persistent atypia after 12 months of tamoxifen chemoprevention.
Literature	GENERIF_SUMMARY	Significant difference in estrogen receptor expression between hig and non-high-grade tumors with bone metastasis suggests that di panels of molecular markers should be used to predict bone meta in these 2 groups of tumors.
Literature	GENERIF_SUMMARY	These findings indicate that the activation of hTERT expression an telomerase activity by estradiol in human mesenchymal stem cells depends on ERalpha, but not on ERbeta.
Literature	GENERIF_SUMMARY	ESR1 amplification might be one mechanism for ER over-expressio endometrial carcinoma, and suggest an early role for ESR1 amplifi in the development of a significant fraction of endometrial carcino
Literature	GENERIF_SUMMARY	Differential patterns of allelic loss in estrogen receptor-positive inf lobular and ductal breast cancer.
Literature	GENERIF_SUMMARY	Altered levels of estrogen receptor (ER) subtypes resulting in an in ERalpha
Literature	GENERIF_SUMMARY	distal ER binding sites are bona fide transcriptional enhancers that involved in long range chromosomal interaction, transcription com formation, and distinct structural modifications of chromatin acros genomic spans
Literature	GENERIF_SUMMARY	In breast cancer cases with tumors positive for ER, the proportion positive for ER-alpha was similar post-menopausally and pre-menopausally
Literature	GENERIF_SUMMARY	The data support that hNOS2A transcription is regulated positively ESR2 and negatively by ESR1 in VSMC, supporting differential acti these two estrogen receptors on a physiologically relevant gene in
Literature	GENERIF_SUMMARY	249 mature micro-RNA (miRNA) transcripts were quantified in estr receptor-positive (ER(+)) primary breast tumors of patients with ly node-negative (LNN) disease to identify miRNAs associated with metastatic capability.
Literature	GENERIF_SUMMARY	Data show that the interleukin-6 -634C/G and estrogen receptor-al PvuII polymorphism were significantly associated with bone miner density accrual and that they have an interactional effect on BMD in Chinese adolescent girls.
Literature	GENERIF_SUMMARY	The F-domain of estrogen receptor-alpha inhibits ligand induced re dimerization.
Literature	GENERIF_SUMMARY	Different from ERbeta, p53 interacts with HDAC1 and CtBP1 and fo inhibiting transcriptional complex that could compete for binding t sites with ERalpha transcriptional complex and inhibit BRCA2 transcription more significantly
Literature	GENERIF_SUMMARY	ER-alpha transcripts are expressed in human skin dermis, but bare

Main Category	Category	Terms
		detectable in epidermis or appendages. No site difference betwee arm, or hip. ER-alpha mRNA levels are approximately 10-fold highe those of ER-beta in human skin.
Literature	GENERIF_SUMMARY	Estrogen alpha and progesterone receptor may have a role in preve progression of breast cancer
Literature	GENERIF_SUMMARY	Epigenetic alteration of the ERalpha gene may play an important ro the pathogenesis of breast cancer
Literature	GENERIF_SUMMARY	ESR1 amplification is not present in our panel of 274 tumors (168 positive cases). Deletion and breaks occurred in the same region, t the 3 of the ESR1 locus, suggesting that fragility of this region may inactivation of the gene.
Literature	GENERIF_SUMMARY	model for the cell type-specific ERalpha occupancy of epigenetical marked cis-regulatory regions of target genes
Literature	GENERIF_SUMMARY	A comparison of ERs in tissues taken from gastric tumors and adja gastric mucosa revealed a loss of ERbeta and a gain of ERalpha in tumor cells
Literature	GENERIF_SUMMARY	ERalpha and ERbeta were generally expressed in the same nuclei i myofibres and in endothelial cells of the capillaries with no sex diff
Literature	GENERIF_SUMMARY	PR and ERalpha may play a role in the pathogenesis of the maturat arrest and Sertoli cell-only phenotype in patients with infertility
Literature	GENERIF_SUMMARY	These results suggest a unique role of AKT in modulating estroger signaling in ERalpha-positive breast cancers and highlights how extracellular signal activated kinases can change the landscape of transcription factor binding to the genome.
Literature	GENERIF_SUMMARY	The results suggest that the Pvull polymorphism of ER-alpha gene as Xbal polymorphism may not be linked with appearance and development of breast cancer.
Literature	GENERIF_SUMMARY	Vitamin D receptor allele B and estrogen receptor allele X are susc alleles for chronic periodontitis.
Literature	GENERIF_SUMMARY	Short-term oral contraceptives yielded lower ERalpha and PR conte Postmenopausal estrogen/progestogen combined therapy yielded PR content than estrogen-only therapy.
Literature	GENERIF_SUMMARY	ERalpha is involved in the transcriptional regulation of HSD17B8 ge response to E2 through its interaction with C/EBPbeta.
Literature	GENERIF_SUMMARY	ERalpha-phospho-Ser118 may be important in leiomyoma growth a possibly phosphorylated by phospho-p44/42 MAPK.
Literature	GENERIF_SUMMARY	The association between type 2 diabetes and ESR1 variants may b because of other unidentified ESR1 polymorphisms that regulate g homeostasis.
Literature	GENERIF_SUMMARY	Reactive oxygen species can induce post-translational modificatio ERalpha at serine 118 and serine 167, and may lead to ERalpha do regulation in human breast cancer cells
Literature	GENERIF_SUMMARY	Overexpression of ER, Ki-67 and cyclinD1 is associated with carcinogenesis of breast cancer.
Literature	GENERIF_SUMMARY	The influence of ERalpha polymorphism should be taken into acco

Main Category	Category	Terms
		among the negative factors influencing the development of osteop especially in the case of prolonged lack or minimal physical activit
Literature	GENERIF_SUMMARY	Two estrogen response element sequences near the PCNA gene a responsible for its estrogen-enhanced expression.
Literature	GENERIF_SUMMARY	Evaluation of p16 and ESR1 promoter methylation in blood using re PCR appears to be very useful for lung cancer diagnosis
Literature	GENERIF_SUMMARY	The different ligand regulations of KRT13 are due to ligand-differen recruitment of ER and coactivators.
Literature	GENERIF_SUMMARY	Findings suggest that the GGGA repeat from ESR1 and the CAG rep from AR may be associated with risk of late onset prostate cancer
Literature	GENERIF_SUMMARY	These results point at the importance of the cellular ERalpha/ERbe for the ultimate effect of (phyto)estrogens on cell proliferation.
Literature	GENERIF_SUMMARY	STEAP1 response to E2 in vitro suggest an effect mediated by the membrane-bound ERalpha (mbERalpha).
Literature	GENERIF_SUMMARY	A systematic study of posttranslational modifications of the estrog receptor isolated from the MCF-7 human breast cancer cell line is reported.
Literature	GENERIF_SUMMARY	intronic single nucleotide polymorphism is associated with endom risk cancer
Literature	GENERIF_SUMMARY	Estrogen receptor alpha is expressed in cultured and circulating endothelial progenitor cells during the menstrual cycle.
Literature	GENERIF_SUMMARY	Review focuses on the critical aspect of the insulin/insulin-like gro factor I (IGF-I) network involved in crosstalk with ERalpha and impl in breast cancer development and progression.
Literature	GENERIF_SUMMARY	decreased BRCA1 levels modify ERalpha-mediated transcription an regulation of cell proliferation in part by altering ERalpha-coregulat association.
Literature	GENERIF_SUMMARY	High levels of ERbeta suppress ERalpha expression and response estradiol in endometrial and endometriotic stromal cells via bindin classic and nonclassic DNA motifs in alternatively used ERalpha promoters.
Literature	GENERIF_SUMMARY	the estrogen receptor 1IVS1-397T allele carriers exhibited an enha inflammatory response in vivo and in vitro
Literature	GENERIF_SUMMARY	We investigated whether ESR1 promoter C driven (ESR1_C) gene expression may shed light on endocrine responsiveness in breast c
Literature	GENERIF_SUMMARY	the expression of SXR in breast cancer cells may alter the ER signa which may play crucial role for growth and differentiation of breast cells
Literature	GENERIF_SUMMARY	ER alpha mRNA is positively correlated with IGF-I mRNA, which imp that estrogen upregulates the gene encoding IGF-I through ER alph leiomyoma.
Literature	GENERIF_SUMMARY	In premenopausal women, increased number of ESR1 PvuII and Xb alleles was associated with higher baseline hot flash scores comp with those who had other haplotypes (P = .0026).

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Women with a CHEK2 mutation face a fourfold increase in the risk estrogen receptor-positive breast cancer
Literature	GENERIF_SUMMARY	miR-18a prevents translation of ERalpha, potentially blocking the protective effects of estrogen and promoting the development of hepatocellular carcinoma in women.
Literature	GENERIF_SUMMARY	the +261G>C polymorphism might influence lipoprotein and lipid le premenopausal women, but these effects seem to be modulated b smoking, whereas in men ESR1 polymorphisms were not associat high lipoprotein levels.
Literature	GENERIF_SUMMARY	No association between genetic variants in ESR1 and ESR2 and th metabolic syndrome was observed among these Caucasian wome
Literature	GENERIF_SUMMARY	A significant correlation between ON promoter hypermethylation a loss of ESR1 mRNA expression was detected in ovarian carcinoma and tisuses.
Literature	GENERIF_SUMMARY	Estrogen recpetor alpha gene may contribute to disease susceptib and to certain clinical manifestations of systemic sclerosis patient
Literature	GENERIF_SUMMARY	Results demonstrated that in ER+ breast cancer, the HER2 recepto commonly activated, but its low prevalence in ER+ tumors does no render it a useful prognostic parameter in tamoxifen-treated patier
Literature	GENERIF_SUMMARY	GAS interacts with oestrogen receptor-alpha (ERalpha) and partici both oestrogen receptor-regulated gene transcription and oestroge stimulated G1/S cell-cycle transition
Literature	GENERIF_SUMMARY	Pueraria mirifica has a beneficial effect on lipid metabolism in postmenopausal women, which may result from the activation of g transcription through selective binding of phytoestrogens to ERalp ERbeta.
Literature	GENERIF_SUMMARY	Presence of estrogen and/or progesterone expression in neuroenc tumors involving the lung should not exclude a primary pulmonary neoplasm.
Literature	GENERIF_SUMMARY	recombinant TF is capable of down-regulating ERalpha expression breast cancer cells, resulting in decreases in estrogen-mediated ce proliferation and increased invasiveness
Literature	GENERIF_SUMMARY	Single nucleotide polymorphism in ESR1 is associated with venous thromboembolism during tamoxifen therapy in breast cancer.
Literature	GENERIF_SUMMARY	These results provide evidence that hypoxia may stimulate yet unk factor(s), which can further stimulate ERalpha signal transduction pathways.
Literature	GENERIF_SUMMARY	A reduction in ERalpha transcriptional activity measured on a synth reporter gene was observed in SRC-2 or SRC-3 but not SRC-1 deple MCF-7 cells.
Literature	GENERIF_SUMMARY	Importance of ERE-BP as an attenuator of normal ERalpha signalin vivo and is a novel target for modulation by selective estrogen rece modulators.
Literature	GENERIF_SUMMARY	There was a down-regulation of ERalpha and aromatase in the stro obese prostate cancer patients compared to patients with normal
Literature	GENERIF_SUMMARY	regulates estrogen receptor alpha (ERalpha) by phosphorylating it

Main Category	Category	Terms
		serine 167, leading to transcriptional activation of ERalpha.
Literature	GENERIF_SUMMARY	Estradiol downregulation of the tumor suppressor gene BTG2 requ estrogen receptor-alpha and the REA corepressor
Literature	GENERIF_SUMMARY	These data identified a new negative regulatory protein of ERalpha provided a new pathway by which CDK11(p58) negatively regulate
Literature	GENERIF_SUMMARY	Single nucleotide polymorphism rs3020314, tagging a region of ES intron 4, is associated with an increase in breast cancer susceptibi with a dominant mode of action in European populations.
Literature	GENERIF_SUMMARY	CpG-island methylation of the ER promoter is associated with micrometastases for locoregional recurrence in colorectal cancer
Literature	GENERIF_SUMMARY	Results provide evidence for a mechanism by which overexpresse progesterone receptor-B is able to actively repress estrogen recept alpha gene expression.
Literature	GENERIF_SUMMARY	a polymorphism in the ERalpha negative transcriptional element w results in increased transcription in vitro is associated with substa augmentation of endothelial-dependent vasorelaxation
Literature	GENERIF_SUMMARY	ERalpha acting as a Trojan horse concentrating 4-OHEN in the nuc accelerate the rate of ROS generation and thereby amplify DNA da The Trojan horse mechanism may be of general importance beyon estrogen genotoxins.
Literature	GENERIF_SUMMARY	Data demonstrate that a non-aromatizable androgen metabolite, dihydrotestosterone (DHT), stimulated cell proliferation in vitro of b estrogen receptor-alpha (ER-alpha)-positive MCF-7 cells and ER-alp negative MDA-MB-231 human breast cancer cells.
Literature	GENERIF_SUMMARY	The disruption of the ERalpha negative transcriptional regulation e by this newly identified polymorphism may influence estrogen-sen phenotypes in vivo
Literature	GENERIF_SUMMARY	The association of certain allelic combinations of ERalpha gene th adenine (TA) polymorphism and aromatase gene TTTA polymorph bone mineral density (BMD) in young men, was assessed.
Literature	GENERIF_SUMMARY	Our study does not support a major contribution of the estrogen re gene and progesterone receptor gene to the pathogenesis of migra
Literature	GENERIF_SUMMARY	Data show that both AF-1 and AF-2 domains of estrogen receptor are required for regulation of MMP-13 promoter activity.
Literature	GENERIF_SUMMARY	Estrogen increases transcription in a rapid but transient manner at estrogen-repressed genes but that this is followed by recruitment corepressor CtBP1, a p300-interacting partner that plays an essent in the repressive process.
Literature	GENERIF_SUMMARY	up-regulation of Gli1 in breast cancer cells may be one of the mechanisms responsible for developing estrogen independence a process may be regulated through down-regulation of expression transactivation of ERalpha
Literature	GENERIF_SUMMARY	Results are consistent with the hypothesis that variation in ESR1 a ESR2, but not CYP19A1, is associated with an increased arterial st
Literature	GENERIF_SUMMARY	ER beta, AIB1, and TIF2 appear to be involved in colorectal tumorig and might have prognostic significance.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the association of the ESRalpha Pvu II and Xb I polymorphism with preeclampsia in a Southwest Chinese population
Literature	GENERIF_SUMMARY	The TA haplotype defined by PvuII and XbaI was associated with a increased risk of reproductive loss. No association was found for t ERbeta gene.
Literature	GENERIF_SUMMARY	Growth factor-induced resistance to tamoxifen is associated with a mutation of estrogen receptor alpha and its phosphorylation at ser
Literature	GENERIF_SUMMARY	ERalpha expression was slightly lower in pregnant subjects but the differences were not statistically significant.
Literature	GENERIF_SUMMARY	Data indicate that the level of HCCR-1 in breast cancer tissues is correlated with the HER2 overexpression, p53 mutation, and ER/PR status, and determination of HCCR-1 levels as options for HER2 te promising although it needs further evaluation.
Literature	GENERIF_SUMMARY	ER and PR showed inverse association with HER-2/neu while posit association was seen with lymph node metastases in breast carci patients in Pakistan
Literature	GENERIF_SUMMARY	Expression of estrogen (ER) and progesterone (PR) receptors in th smooth muscle component (SMC) of deep infiltrating endometrios
Literature	GENERIF_SUMMARY	Estrogen receptor (ER) gene polymorphism was associated with th postoperative result of an rotational acetabular osteotomy, while n association was observed between the acetabular dysplasia with v D receptor (VDR) and ER gene polymorphisms.
Literature	GENERIF_SUMMARY	Presence of A and T alleles (of the XbaI and PvuII polymorphisms, respectively) were associated with higher incidence of carotid rest
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	SRC-1 is a strong independent predictor of reduced disease free su whereas the interactions of the p160 proteins with estrogen recept alpha can predict the response of patients to endocrine treatment.
Literature	GENERIF_SUMMARY	does not provide evidence in favor of ESR1 PvuII and XbaI being associated with preeclampsia but with increased blood pressure le normal pregnant women, in a southwest Chinese population
Literature	GENERIF_SUMMARY	BRCA1/1a/1b fine tunes the dynamic complex interplay between S dependent/independent activities of Ubc9 on E2-induced ERalpha activation/repression and degradation in breast cancer cells
Literature	GENERIF_SUMMARY	cross-talk between A2AAR and ERalpha that is involved in ethanol This could open new perspectives for the therapy of estrogen-depe breast cancer.
Literature	GENERIF_SUMMARY	Estrogen/ERalpha exerts a positive feedback role, which promotes permissive histone modifications that are associated with inductio hCYP19 gene transcription.
Literature	GENERIF_SUMMARY	Methylation status of ESR1 in adenomas is not correlated with pat age, but is associated with K-ras mutation, suggesting that methyla ESR1 in tumors functions differently from that in normal colon mu
Literature	GENERIF_SUMMARY	No significant difference in expression of either ERalpha or ERbeta found between pure lobular carcinoma in situ (LCIS) of the breast, associated with invasive breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Study did not observe an association between ESR1 PvuII and Xba polymorphisms with CAD in the risk of CAD in an Iranian populatio
Literature	GENERIF_SUMMARY	ESR1 variants do not seem to affect myometrial invasion or endom cancer survival.
Literature	GENERIF_SUMMARY	ESR1 amplification may be helpful in selecting breast cancer patie may potentially benefit from endocrine therapy.
Literature	GENERIF_SUMMARY	Genomic responses from the estrogen-responsive elements-deper signaling pathway are required for estrogen-ERalpha to induce alte in cellular responses.
Literature	GENERIF_SUMMARY	study suggests that the protective association of enterolactone is reasonably robust across the investigated polymorphisms in the e receptor alpha and beta genes
Literature	GENERIF_SUMMARY	ER-alpha suppresses ER-alpha 36 promoter activity in an estrogen- independent manner, which can be released by ER-alpha 36 itself
Literature	GENERIF_SUMMARY	BCAR1 is essential for the rapid estrogen effect on osteoclast differentiation, through estrogen receptor alpha and possibly Traf6
Literature	GENERIF_SUMMARY	An association between menopausal status - but not BMI - and the tumoral expression of steroid sulfatase and ERalpha.
Literature	GENERIF_SUMMARY	Data show that p53 regulates ER expression through transcriptiona control of the ER promoter, accounting for their concordant expres human breast cancer.
Literature	GENERIF_SUMMARY	selective activation of a subset of the estrogen receptor alpha (ER cistrome in MCF7 breast cancer cells
Literature	GENERIF_SUMMARY	The absence of alterations in ER-alpha protein levels could be a characteristic behavior of fibroadenomas, unlike breast cancer.
Literature	GENERIF_SUMMARY	Results place LKB1 in a coactivator role for ERalpha signaling, broa the scientific scope of this tumor suppressor kinase and laying the groundwork for the use of LKB1 as a target for the development of therapies against breast cancer.
Literature	GENERIF_SUMMARY	Our results indicate a permanent repression of PAI-1 by ERalpha a suggest a ligand-independent crosstalk between ERalpha and TGF signaling in breast cancer cells.
Literature	GENERIF_SUMMARY	ESR1 haplotypes that included the variant C allele were associated reduced risk of obesity
Literature	GENERIF_SUMMARY	Both prolactin and estradiol resulted in prolonged ERalpha serine-1 phosphorylation and decreased the amount of ERalpha, but by diff mechanisms
Literature	GENERIF_SUMMARY	the ERalpha assay reflects summated activity of compounds in se
Literature	GENERIF_SUMMARY	ERalpha is crucial for long-distance regulation of CTSD expression involving a looping mechanism.
Literature	GENERIF_SUMMARY	estradiol affects the expression of MMP-9 and TIMP-1 and syndec but not of syndecan-4, through ERalpha
Literature	GENERIF_SUMMARY	these results suggest a novel role for the ERalpha that plays the du of ligand-independent activator and ligand-dependent repressor of

Main Category	Category	Terms
		cadherin in breast cancer cells.
Literature	GENERIF_SUMMARY	OTUB1 negatively regulates transcription mediated by ERalpha in transient reporter gene assays and transcription mediated by endo ERalpha in Ishikawa endometrial cancer cells
Literature	GENERIF_SUMMARY	Polymorphisms within ESR1 and ESR2 are associated with age-sp effects on various skeletal traits in healthy fertile women.
Literature	GENERIF_SUMMARY	A functional polymorphism (rs2234693) in estrogen receptor alpha a role in the pathophysiology of Japanese methamphetamine- indu psychosis.
Literature	GENERIF_SUMMARY	A cooperative role for ERalpha and c-Myc in estrogen regulation of and the ability of c-Myc to partially mimic estrogen regulation of angiogenesis.
Literature	GENERIF_SUMMARY	A novel mechanism of estrogen resistance of ovarian cancer by wl estrogen-repressed signaling pathway antagonizes estrogen-activa signaling transduction.
Literature	GENERIF_SUMMARY	Estrogen receptor polymorphism, estrogen content and idiopathic scoliosis in human
Literature	GENERIF_SUMMARY	polymorphism in the alpha estrogen receptor may increase the risk women developing temporomandibular joint disorder
Literature	GENERIF_SUMMARY	Data show that microRNA-22 represses estrogen receptor alpha expression most strongly and by directly targeting the ER alpha mR untranslated region.
Literature	GENERIF_SUMMARY	The distinct expression of ER and PgR in ductal breast cancers, dependent on evaluated zone of the tumour, confirms its heteroge character and exerts an effect on the type of applied treatment.
Literature	GENERIF_SUMMARY	ESR1 may not play a major role in the pathophysiology of schizoph and mood disorders.
Literature	GENERIF_SUMMARY	These findings suggest that the ERalpha estrogen receptor might b involved in regulating intracellular estrogen levels by modulating 17 HSD10 activity.
Literature	GENERIF_SUMMARY	DYX1C1 interacts with both ERs in the presence of 17beta-estradio neurons.
Literature	GENERIF_SUMMARY	This cross-sectional study showed that both equol production and ERalpha polymorphism are closely associated with each other in re to BMD.
Literature	GENERIF_SUMMARY	Data demonstrate that raloxifene exerted direct protective effects human osteoblasts in both estrogen receptor alpha/beta depende independent manners.
Literature	GENERIF_SUMMARY	PDIp can also serve as an effective modulator of the cellular levels biological actions of endogenous estrogens in the pancreas where estrogen receptors alpha and beta and PDIp are co-present.
Literature	GENERIF_SUMMARY	Report protective role of estrogen receptor-alpha on lower chlorina PCB congener-induced DNA damage and repair in human tumoral cells.
Literature	GENERIF_SUMMARY	the induction of Blimp1 represents a novel mechanism whereby th

Main Category	Category	Terms
		NF-kappaB subunit mediates repression, specifically of ERalpha, th promoting a more migratory phenotype in breast cancer cells
Literature	GENERIF_SUMMARY	Results indicate that Wip1 up-regulation is important in the pathog of p53(+) and ER(+) breast cancer through the inactivation of p53 dephosphorylation and the amplification of subsequent estrogenic effects through the E(2)-ERalpha-Wip1 pathway.
Literature	GENERIF_SUMMARY	Expression of ER, progesterone receptor, and HER2 proteins and th index appear to distinguish luminal A from luminal B breast cancer subtypes
Literature	GENERIF_SUMMARY	data suggest that the ESR1 (rs2234693 and rs9340799) and the ES (rs1255998 and rs944050) polymorphisms may be associated wit increased risk of developing endometrial cancer.
Literature	GENERIF_SUMMARY	estrogen receptor splice variant and ErbB4 may have roles in gene susceptibility pathways in schizophrenia
Literature	GENERIF_SUMMARY	ESR1 PvuII and XbaI gene polymorphisms did not seem to have an on lipoprotein metabolism or on inflammatory variables such as C
Literature	GENERIF_SUMMARY	Transcriptional profiling demonstrated that Bcl3 activates genes in in diverse pathways including a subset involved in cellular energy metabolism known to be regulated by PGC-1alpha, ERRalpha, and second nuclear receptor, PPARalpha.
Literature	GENERIF_SUMMARY	Data show that the combination of increasing chromatin accessibi inducing TFAP2C provides a more robust activation of the ERalpha in ERalpha-negative breast cancer cells.
Literature	GENERIF_SUMMARY	AF2 domain contributes to dioxin-induced recruitment of ERalpha target genes, but that both the AF1 and AF2 domains are required ERalpha-dependent increases in AHR activity.
Literature	GENERIF_SUMMARY	The three markers (ER, Vim and CEA) and their respective panel expressions showed statistically significant (p
Literature	GENERIF_SUMMARY	ERalpha is rarely expressed while ERbeta and PELP1/MNAR are in in colorectal tumorigenesis.
Literature	GENERIF_SUMMARY	The K303R ERalpha mutation might be a new predictive marker of response to aromatase inhibitors in mutation-positive breast tumo
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in ESR1, CETP and LPL had sign effects on baseline plasma levels of TG-rich and HDL subpopulatio
Literature	GENERIF_SUMMARY	Severe obesity in Han Chinese was associated with 3 novel comm SNPs for ESR1 and PPARgamma. These 2 genes collectively resul >5-fold risk of severe obesity.
Literature	GENERIF_SUMMARY	Studies discovered a novel molecular network between p53, KLF4 ERalpha.
Literature	GENERIF_SUMMARY	Methylated state of this set of genes may be more specific to the l rather than the early stage of NSCLC.
Literature	GENERIF_SUMMARY	ER alpha expression may influence vascular endothelial function in women by affecting protein levels and activation of eNOS.
Literature	GENERIF_SUMMARY	The results presented here identify SMYD3 as a new coactivator fo estrogen receptor-mediated transcription, providing a possible link

Main Category	Category	Terms
		between SMYD3 overexpression and breast cancer.
Literature	GENERIF_SUMMARY	The SNP T594T (ESR1), but not the D6S440 (ESR1)and D17S189 (THRA)is associated with thyroid cancer risk.
Literature	GENERIF_SUMMARY	The different subtypes of malignant ovarian germ cell tumours all ERalpha, ERbeta, and SNURF.
Literature	GENERIF_SUMMARY	M543V shows more ERalpha transcription with antiestrogens vs E Relative hydrophobicity of AA side chains on the inner face of Heli needed for active, agonist conformation with bound E2. The N-terr AF-1 plays a role in AP-1/ERalpha cross-talk.
Literature	GENERIF_SUMMARY	This result suggested that ERalpha plays a significant role in leukaemogenesis
Literature	GENERIF_SUMMARY	17beta-estradiol (E2) can rapidly enhance aromatase enzymatic ac through an increase of aromatase protein phosphorylation in breas cancer cell lines.
Literature	GENERIF_SUMMARY	Nuclear localization of P-ser118-ER alpha provides one explanatio sex-dependent differences in E(2)-genomic responses in lung adenocarcinoma cell lines.
Literature	GENERIF_SUMMARY	Study found that RARalpha/RARgamma exhibit extensive colocaliz of their genomic binding regions with ERalpha in the vicinity of ger are antagonistically regulated by estrogen and RA.
Literature	GENERIF_SUMMARY	results allowed the identification of mutation-resistant positions in estrogen receptors
Literature	GENERIF_SUMMARY	demonstrate not only that dioxin induces the recruitment of ERalph AHR target genes but that AHR is recruited to estrogen-responsive regions in a gene-manner, suggesting that AHR utilizes these mechanisms to modulate estrogen-dependent signaling
Literature	GENERIF_SUMMARY	The analyses indicate that both gene activation and repression by estrogen-bound ERalpha rely on an intact DNA-binding domain in v
Literature	GENERIF_SUMMARY	ER, PR and Her2 have roles in progression of breast cancer
Literature	GENERIF_SUMMARY	Association of ER-alpha gene polymorphism with metabolic phenc in Chinese Hans.
Literature	GENERIF_SUMMARY	Nuclear factor-kappaB regulates estrogen receptor-alpha transcrip the human heart
Literature	GENERIF_SUMMARY	ESR1 gene polymorphisms may be associated with the AAO in AD. present results provided information for possible associations bet certain polymorphisms of ESR1 gene and the risk of AD.
Literature	GENERIF_SUMMARY	NAG7 plays a potential role in promoting nasopharyngeal carcinon invasion by regulation of ERalpha and the H-ras/p-c-Raf and JNK2/ MMP1 signaling pathways.
Literature	GENERIF_SUMMARY	Studies indicate that estrogen receptor-positive tumors with increa proliferation, evaluated by Ki-67, may be selectively responsive to aromatase inhibitor therapy.
Literature	GENERIF_SUMMARY	Data demonstrate in caveolae of human cortex and hippocampus presence of pl-VDAC and ERalpha, in a complex with scaffolding caveolin-1 which likely provides ERalpha stability at the plasma

Main Category	Category	Terms
		membrane.
Literature	GENERIF_SUMMARY	Data show that DACH1 expression, which is lost in poor prognosis breast cancer, functions as an endogenous inhibitor of ERalpha fu
Literature	GENERIF_SUMMARY	Confirm the lack of association between the ESR1 rs223469 and c heart disease.
Literature	GENERIF_SUMMARY	These data support a model of repression wherein actions of ERal and Sin3A at the proximal promoter can overcome activating signa distal or proximal sites and ultimately decrease gene expression.
Literature	GENERIF_SUMMARY	Studied the association of EGFR and ESR1 SNPs with breast cance and prognosis in Tunisian patients. The ESR1 2014G allele showe significant association with breast cancer risk (P = .025).
Literature	GENERIF_SUMMARY	selected genetic polymorphisms of ESR1 and ESR2, overall, were r associated with prostate cancer risk. However, a variation in risk b and phytoestrogen intake was implicated
Literature	GENERIF_SUMMARY	The positive expression rate of ER decreased significantly from DC invasive ductal breast carcinoma. A significant positive correlation observed between ER and PgR receptor expression in DCIS and ID
Literature	GENERIF_SUMMARY	ESR1 PvuII polymorphism is not associated with angiographically significant CAD. ESR1 BtgI polymorphism is strongly associated w presence of normal coronary angiograms in women, which sugges protective effect.
Literature	GENERIF_SUMMARY	In univariate analyses, Tau- and ER-positive status were both asso with better DFS and OS (P
Literature	GENERIF_SUMMARY	ESR1 PvuII polymorphism is a significant risk factor for migraine particularly in women and migraine with aura patients, but ESR 325 polymorphism is not associated with migraine susceptibility
Literature	GENERIF_SUMMARY	Patterns and extent of ER and CK5 staining, when used together, a valuable adjunct stains to differentiate usual duct hyperplasia from atypical proliferations within papillary lesions on breast
Literature	GENERIF_SUMMARY	Data show that in multivariate analyses incorporating tumor size, t grade, lymph node status, ER, PR and HER2 status, low STARD10 expression was an independent predictor of death from breast car
Literature	GENERIF_SUMMARY	significant association between magnesium and ESR1 polymorphi
Literature	GENERIF_SUMMARY	Both antiestrogen sensitive and antiestrogen resistant breast canc showed the ability to switch between ERalpha and ErbB signaling depending on the applied treatment, thus supporting combined the targeting both receptor systems.
Literature	GENERIF_SUMMARY	identify the subset of microRNAs (miRNAs) modulated by ERalpha include upregulation of miRNAs derived from the processing of the paralogous primary transcripts (pri-) mir-17-92 and mir-106a-363
Literature	GENERIF_SUMMARY	Estrogen receptor alpha and beta expression distinguishes a subs NSCLC that has defined clinicopathologic and genetic features. In adenocarcinoma, estrogen receptor alpha expression correlates w EGFR mutations.
Literature	GENERIF_SUMMARY	Results suggest that PKA can negatively regulate ERalpha, at least through FoxH1.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	SH2D4A inhibited cell proliferation by suppression of the ERalpha/ gamma/PKC signaling pathway.
Literature	GENERIF_SUMMARY	A crucial role for p72 in ERalpha co-activation and oestrogen-depe cell growth.
Literature	GENERIF_SUMMARY	Studies show that the intrinsic molecular signature of five markers PR, HER-2, CK 5/6, and EGFR, demonstrated specificity of 100% an sensitivity of 75%, compared with classification by gene expressio profiling.
Literature	GENERIF_SUMMARY	certain olfactory sensory neurons naturally co-express olfactory re and ESR1
Literature	GENERIF_SUMMARY	TLR9 signal pathway, by activating NF-kappaB, negatively regulate ERalpha-mediated transactivation of breast cancer
Literature	GENERIF_SUMMARY	lower expression in menopausal women
Literature	GENERIF_SUMMARY	ERalpha C allele confers approximately 2-fold increased risk of cau both endometriosis and fibroids, indicating that it can be considere marker in Asian Indian Women.
Literature	GENERIF_SUMMARY	we described a tumor suppression function of miR-145 in breast c cell lines, and we linked miR-145 to TP53 and ER-alpha
Literature	GENERIF_SUMMARY	ER-alpha mRNA expression is associated with good patient outcor death from breast cancer.
Literature	GENERIF_SUMMARY	Polymorphisms in ESR1_rs910416 and AR_(CAG)n are associated postmenopausal breast cancer risk associated with menopausal hormone therapy.
Literature	GENERIF_SUMMARY	The conformational dynamics revealed by the use of hydrogen-deu exchange mass spectrometry for ERbeta ligand binding domain (L together with those for ERalpha LBD shed light on ER ligand intera and offer new structural insights.
Literature	GENERIF_SUMMARY	G protein-coupled receptor 30 expression is up-regulated by EGF a alpha in estrogen receptor alpha-positive cancer cells
Literature	GENERIF_SUMMARY	Data show that variant isoforms of estrogen receptor related beta have differential effects on ER alpha-dependent gene expression, a possibly on endometrial cell function.
Literature	GENERIF_SUMMARY	ERbeta1 and two ERbeta variants (ERbeta2, ERbeta5) were expres endometrial cancers and the proteins were immunolocalised to th of cells in both epithelial and stromal compartments.
Literature	GENERIF_SUMMARY	These results suggest that potentially functional ESR1 polymorphi may play a low penetrance role in breast cancer susceptibility
Literature	GENERIF_SUMMARY	results suggest that the PvuII polymorphism of ER-alpha gene as v XbaI polymorphism may not be linked with appearance and develo of endometrial cancer.
Literature	GENERIF_SUMMARY	The aim of the current study was to assess the expression levels c and phosphorylated Src kinase in human breast cancers and to es if these are linked to oestrogen receptor, progesterone receptor, an status or patient survival.
Literature	GENERIF_SUMMARY	describes in detail the BRET(2) assay as it is used to examine the

Main Category	Category	Terms
		interaction between the nuclear receptor ERalpha and the transcrip coactivator SRC-1
Literature	GENERIF_SUMMARY	Data show that valproic acid, inducing estroegn receptor alpha and FoxA1, confers to MDA-MB 231 cells an estrogen-sensitive ""phenc restoring their sensitivity to antiestrogen therapy."
Literature	GENERIF_SUMMARY	Testosterone induces ERK and Akt phosphorylation via the membr initiated signaling pathways mediated by ER-alpha36, suggesting a possible involvement of ER-alpha 36 in testosterone carcinogenes
Literature	GENERIF_SUMMARY	PvuII (rs2234693) and XbaI (rs9340799) restriction fragment polymorphisms of the oestrogen receptor alpha gene probably do a role in the pathomechanism of myasthenia gravis in Caucasian v
Literature	GENERIF_SUMMARY	ERalpha and Foxl2 interact and that Foxl2 selectively suppresses ERalpha-mediated transcription of AP1-regulated genes.
Literature	GENERIF_SUMMARY	Catechol-o-methyltransferase expression and 2-methoxyestradiol a microtubule dynamics and modify steroid receptors ERalpha and P signaling in leiomyoma cells
Literature	GENERIF_SUMMARY	ERalpha expression positively correlated with PR expression in endometrial cancer.
Literature	GENERIF_SUMMARY	A possible role of ESR1 variants on HA is suggested.
Literature	GENERIF_SUMMARY	estrogen receptor-alpha transcriptional activity is repressed by the megakaryoblastic leukemia 1 signaling pathway
Literature	GENERIF_SUMMARY	The presence of ERalpha in the majority of tumors diagnosed durin pregnancy indicates that the poorer outcome of these cases could related to the estrogen-mediated growth stimulus.
Literature	GENERIF_SUMMARY	abnormal expression of hormone receptor contribute to endometri polyps formation
Literature	GENERIF_SUMMARY	Data support a model in which Notch-1 can activate the transcripti ERalpha-target genes via IKKalpha-dependent cooperative chroma recruitment of Notch-CSL-MAML1 and ERalpha, which promotes th recruitment of p300.
Literature	GENERIF_SUMMARY	an important role for the K303R ERalpha mutation in hormonal reg of tumor growth and estrogen-regulated promoter dynamics in hur breast cancer
Literature	GENERIF_SUMMARY	G protein pathway suppressor 2 (GPS2) is a transcriptional corepre important for estrogen receptor alpha-mediated transcriptional reg
Literature	GENERIF_SUMMARY	Interactions of MinK and e-NOS gene polymorphisms appear to be inconsistent predictors of atrial fibrillation propensity, but long alle ESR1 promoter TA repeat may be a promising marker.
Literature	GENERIF_SUMMARY	Polymorphisms in the Estrogen Receptor Alpha gene are associate premature ovarian failure.
Literature	GENERIF_SUMMARY	Polymorphisms at the ESR1 gene are associated with infertility in t Turkish infertile women population.
Literature	GENERIF_SUMMARY	detected ERalpha binding to STAT3 and JAK2 (Janus kinase 2), res in enhanced JAK2 activity upstream of STAT3 in response to leptin might lead to an increased ERalpha-dependent cell viability

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	No relationship of Akt phosphorylation with HER2, ER-alpha or ER-l expression in breast cancer was found
Literature	GENERIF_SUMMARY	Results show no significant association between hand osteoarthri the investigated polymorphisms of ESR1 or ESR2 despite publishe reports of association and a priori hypotheses implicating their po roles.
Literature	GENERIF_SUMMARY	Notably the ratio of ERalpha-positive reactive astrocytes to total re astrocytes increased and significantly correlated with the severity cognitive impairment following HIV infection.
Literature	GENERIF_SUMMARY	estrogen receptor (ER), progesterone receptor (PR), and HER2 hav in breast cancer
Literature	GENERIF_SUMMARY	Data suggest post-translational modification (i.e., acetylation versu ubiquitination) of ESR1 is regulated by BRCA1 (breast cancer gene product). Thus, ESR1 activity is regulated by BRCA1.
Literature	GENERIF_SUMMARY	comprehensive mapping of the chromatin interaction network bou oestrogen receptor alpha (ER-alpha) in the human genome
Literature	GENERIF_SUMMARY	Studies reveal dynamic interplay between ESR1 and ESR2 in their selection of chromatin binding sites and reveal expansion of bindi exclusively for ESR1 or for ESR2 when both ESR are present.
Literature	GENERIF_SUMMARY	no relationship between ERalpha polymorphism and the stage of osteoporosis related to the spinal body mass density value
Literature	GENERIF_SUMMARY	Role of genetic polymorphism of estrogen receptor-alpha gene an prostate cancer in north Indian population
Literature	GENERIF_SUMMARY	Some common SNPs in the ESR1 could be important in modulatin carotid atherosclerosis and thereby CVD susceptibility in Taiwanes women.
Literature	GENERIF_SUMMARY	only a subpopulation of ER alpha was available to form complexes the interacting domains of cofactors in the cell
Literature	GENERIF_SUMMARY	Our results suggest that ESR1 might have a potential role in hyperlipidemia risk, independent of age, estradiol level, body mass and lifestyle in Chinese Han postmenopausal women.
Literature	GENERIF_SUMMARY	These findings provide further evidence of the importance of medi the growth factor signaling in ER regulation, introducing the Akt2/F axis as a pursuable target in therapy for ER-positive breast cancer.
Literature	GENERIF_SUMMARY	Clomiphene citrate may inhibit E2-induced endometrial epithelial c proliferation and ERE transactivation by inhibiting the recruitment c SRC-1 to ER alpha.
Literature	GENERIF_SUMMARY	No association was found between other ESR1 polymorphisms an levels. According to the results, the ESR1 PvuII TT genotype was associated with increased levels of HDL after 1 year of HRT.
Literature	GENERIF_SUMMARY	ERalpha expression and Bax/Bcl-2 ratio were similar in fresh and cryopreserved ovarian tissue
Literature	GENERIF_SUMMARY	Promoter hypermethylation was observed in PRB 67%, ERalpha 64 RASSF1A 63%, p16INK4A 51%, RARbeta2 22%, GSTP1 25% and BR 27% of the breast cancers, respectively.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Our findings suggest that the combined genetic polymorphisms of and ESR1P325P are associated with breast cancer in Korean wom
Literature	GENERIF_SUMMARY	ERalpha mRNA expression was increased in left ventricular specin from coronary disease and dilated cardiomyoapthy patients vs cor This might reflect a compensatory mechanism to counteract the d in ventricular function.
Literature	GENERIF_SUMMARY	Results reveal a novel link between proteasome activity and expres ERalpha in breast cancer and uncover distinct roles of the chymotr like activity of the proteasome in the regulation of the ERalpha pat
Literature	GENERIF_SUMMARY	Results from MCF-7 cells suggest that ERalpha interacts with 8 wi spaced regions of progesterone receptor (PR) gene (from 311 kb upstream to 4 kb downstream of the PR-B transcription start site).
Literature	GENERIF_SUMMARY	rs2077647 in ESR-alpha is associated with risk for prostate cancer
Literature	GENERIF_SUMMARY	The evaluated single nucleotide polymorphism in estrogen recepto not seem to influence bone marrow density in tamoxifen-treated su with breast cancer.
Literature	GENERIF_SUMMARY	Aberrations in the interaction between the ER and NFkappaB signa pathways may have a negative impact on implantation contributing pathologies such as early pregnancy loss and infertility
Literature	GENERIF_SUMMARY	the cyclin D1b represses breast cancer cell growth by antagonizing action of cyclin D1a on ERalpha-mediated transcription
Literature	GENERIF_SUMMARY	ER alpha polymorphisms were found to be associated with sperm and concentration, supporting the significance of this gene in spermatogenesis and semen quality
Literature	GENERIF_SUMMARY	ESR1 gene is quantitative trait loci (QTL) underlying fat mass varia young Chinese men.
Literature	GENERIF_SUMMARY	Toker cells and mammary Paget cells differ substantially in their expression of ER, c-erbB-2 and Ki-67.
Literature	GENERIF_SUMMARY	Morphology and ERalpha protein and gene expression in two breas cancer cell lines after three different fixatives.
Literature	GENERIF_SUMMARY	he role of oestrogen receptor {alpha} in human thyroid cancer.
Literature	GENERIF_SUMMARY	K18 attenuates estrogen receptor alpha-mediated signaling by sequestering LRP16 in cytoplasm.
Literature	GENERIF_SUMMARY	The growth of ER-negative SKBr3 breast cancer cells was inhibited ERalpha expression
Literature	GENERIF_SUMMARY	These results indicate that the GG genotype of the G2014A polymo is associated with the risk of failure in the mifepristone-induced ab
Literature	GENERIF_SUMMARY	17beta-estradiol enhances alpha(5) integrin subunit gene expressi through ERalpha-Sp1 interaction and reduces cell motility and inva ERalpha-positive breast cancer cells.
Literature	GENERIF_SUMMARY	a single nucleotide polymorphism downstream of ESR1, a gene ca spectrin repeat containing, nuclear envelope 1, was associated wit invasive ovarian cancer risk
Literature	GENERIF_SUMMARY	ESR1 PvuII and XbaI haplotype is associated with BP variation and

Main Category	Category	Terms
		reduction in cardiac autonomic nervous activity in young Japanese which may be precursors of future pathological episodes of cardiovascular diseases
Literature	GENERIF_SUMMARY	BRCA1 and BARD1 are required for estrogen receptor alpha ubiqui and degradation, and repression of either one leads to ERalpha accumulation, suggesting a feedback loop between BRCA1-BARD1 estrogen receptor alpha
Literature	GENERIF_SUMMARY	The age at menarche was not related to ERalpha gene polymorphis height is
Literature	GENERIF_SUMMARY	The 4th passage cell cultures from Fallopian tube muscularis exhi interstitial Cajal-like cells morphology showed the presence of ER- and/or PR-A by immunofluorescence.
Literature	GENERIF_SUMMARY	ERs recruit SMRT through an unusual mode of interaction involvin multiple contact surfaces.
Literature	GENERIF_SUMMARY	The human IP gene is directly regulated by estrogen that largely oc through an ERalpha-dependent transcriptional mechanism.
Literature	GENERIF_SUMMARY	Reporter gene assays revealed that ERalpha and ERbeta increased CYP2B6-regulated gene expression through a functional ERE locat -1669 to -1657 in the upstream regulatory region of CYP2B6.
Literature	GENERIF_SUMMARY	Chlamydial protease CT441 interacts with SRAP1 co-activator of e receptor alpha and partially alleviates its co-activation activity.
Literature	GENERIF_SUMMARY	ERalpha could in some cases actually lead to reactivation of the m toward at least some response elements.
Literature	GENERIF_SUMMARY	MYCN amplification may disrupt estrogen signaling sensitivity in p sympathetic cells through deregulation of ESR1.
Literature	GENERIF_SUMMARY	Retinoic acid receptor-alpha is an essential component of the estr receptor-alpha (ER)- complex, potentially by maintaining ER-cofact interactions, and different nuclear receptors can cooperate for transcriptional activity in breast cancer cells
Literature	GENERIF_SUMMARY	GPR30 antagonizes growth of ERalpha-positive breast cancer and represent a new target to combat this disease.
Literature	GENERIF_SUMMARY	The ERalpha pathway promotes cell proliferation by activating the pathway in a ligand-dependent manner through Shh induction of E positive gastric cancer.
Literature	GENERIF_SUMMARY	ESR1 gene including PvuII and XbaI polymorphisms may modify th of idiopathic premature ovarian failure (POF) but not idiopathic ea menopause (EM) risk.
Literature	GENERIF_SUMMARY	Contributions of estrogen receptor-alpha and estrogen receptor-s regulation of behavior.
Literature	GENERIF_SUMMARY	Data show there was a strong inverse correlation between slug an ERalpha and E-cadherin immunoreactivity in breast cancer cases.
Literature	GENERIF_SUMMARY	Findings suggest that RBCK1 regulates cell cycle progression and proliferation of ERalpha-positive breast cancer cells by supporting transcription of ERalpha and cyclin B1.
Literature	GENERIF_SUMMARY	Down-regulation of ER-alpha36 is associated with carcinogenesis

Main Category	Category	Terms
		progression of breast cancer.
Literature	GENERIF_SUMMARY	Study shows a simultaneous involvement of epigenetic regulation both CXCL12 and ESR1 genes in Brazilian women. Methylation sta both genes was significantly correlated with histologically advance disease, the presence of metastases and death.
Literature	GENERIF_SUMMARY	The association of some polymorphisms in the region of exon 4/ir of ESR1 and hip fractures was significant in females.
Literature	GENERIF_SUMMARY	Data show that estradiol, acting through estrogen receptor alpha, u regulates COX-1 and prostacyclin synthase expression, thus direct prostanoid balance toward increased prostacyclin production.
Literature	GENERIF_SUMMARY	analysis of transcriptional activation of DNA-dependent protein kin catalytic subunit gene expression by oestrogen receptor-alpha
Literature	GENERIF_SUMMARY	The ESR1 gene is associated with low bone mass and low estradio in all our study subjects. It is likely that the allele exerts its influenc the bone in early adulthood leading to an increased risk of osteopo later in life.
Literature	GENERIF_SUMMARY	characterized geneticc association between ER-alpha SNP and kne osteoarthritis in the Thai
Literature	GENERIF_SUMMARY	Data indicate that estradiol may be a key factor controlling the turn adult pancreatic beta cells by regulating the proliferation and differentiation of adult human islet-derived precursor cells through oestrogen receptor alpha.
Literature	GENERIF_SUMMARY	I3C is a novel anticancer agent in human cancers that coexpress E GATA3, and AhR, a combination found in a large percentage of bre cancers but not in other critical ERalpha target tissues essential to health
Literature	GENERIF_SUMMARY	Both transcript and protein products of the SRA1 gene co-modulat transcriptional activity of steroid receptors.
Literature	GENERIF_SUMMARY	Our study suggests that the minor allele -397C of the ESR1 gene co risk of coronary heart disease among Finnish men, both in homozy state and as part of a haplotype with the -351A allele.
Literature	GENERIF_SUMMARY	The data of this study demonstrated affective state-independent personality traits that distinguish patients with Premenstrual Dysp Disorder from controls and support ESR-1 polymorphic variants in regulation of non-reproductive behaviors.
Literature	GENERIF_SUMMARY	variants in estrogen receptor genes could play a role in the etiology biliary tract cancers, especially bile duct cancer in men.
Literature	GENERIF_SUMMARY	ER, PgR and E-cadherin are expressed in canine mammary gland columnar cell lesions; cytokeratin 34betaE-12, HER-2 and P53 are n expressed
Literature	GENERIF_SUMMARY	The effect of lapatinib on ERalpha signaling varied markedly deper on the nature of the HER2/ERalpha cross-talk
Literature	GENERIF_SUMMARY	Our results suggest that patients with tumors expressing Pak1 and pERalpha(ser305) in combination are a group in which tamoxifen treatment is insufficient
Literature	GENERIF_SUMMARY	microRNA-22 is an estrogen receptor alpha suppressor that is

Main Category	Category	Terms
		downregulated in estrogen receptor alpha-positive human breast c cell lines and clinical samples
Literature	GENERIF_SUMMARY	Results demonstrated that previously reported activities of GPR30 response to estrogen were through its ability to induce ER-alpha36 expression.
Literature	GENERIF_SUMMARY	Anacardic acid inhibits estrogen receptor alpha-DNA binding and r target gene transcription and breast cancer cell proliferation.
Literature	GENERIF_SUMMARY	The PP+Pp genotype of the ESR1 gene was more frequent than pp genotypes in subjects with dysmenorrhea than in subjects without dysmenorrhea.
Literature	GENERIF_SUMMARY	Report an estradiol-inducible adenovirus-mediated gene transfer s for nuclear delivery of a p16INK4A-estrogen receptor carboxy term fusion protein.
Literature	GENERIF_SUMMARY	Down-regulation of PROS1 gene expression by 17beta-estradiol via estrogen receptor alpha (ERalpha)-Sp1 interaction recruiting recep interacting protein 140 and the corepressor-HDAC3 complex.
Literature	GENERIF_SUMMARY	Genetic variation in ESR1 is associated with serum estradiol levels body fat content regulation in men.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha from MCF-7 breast carcinoma cell lysates immobilized inside a hollow-core photonic crystal fiber was detect using anti-ER primary antibody with Alexa Fluor 488 or 555 labeled anti-rabbit IgG as the secondary antibody.
Literature	GENERIF_SUMMARY	PKA-induced phosphorylation of ERalpha at serine 305 and high PA levels is associated with sensitivity to tamoxifen in ER-positive bre cancer.
Literature	GENERIF_SUMMARY	Single nucleotide polymorphsims in ESR1 do not associate with hypospadias in a Dutch study.
Literature	GENERIF_SUMMARY	Collectively, our findings suggest that tamoxifen resistance is a res increased ERalpha/Sp-1 interaction, which enhances the expressio E2F1 to promote tamoxifen resistance.
Literature	GENERIF_SUMMARY	Upregulated ESR1 is associated with breast cancer.
Literature	GENERIF_SUMMARY	Data show that, upon estrogen stimulation, DNA-PK forms a comp ERalpha in a breast cancer cell line (MELN).
Literature	GENERIF_SUMMARY	subcellular localization of activated AKT is dependent on TCL1B expression in estrogen receptor- and progesterone receptor-expres breast cancer
Literature	GENERIF_SUMMARY	CpG islands aberrant hypermethylation of estrogen receptor is ass with leukemia.
Literature	GENERIF_SUMMARY	MBD2 and ERalpha drive opposite effects on pS2 expression, whic associated with specific steady state levels of histone H3 acetylati methylation marks
Literature	GENERIF_SUMMARY	p21Waf1/Cip1 gene expression is induced in estrogen responsive breast cancer cells and Sp1 is involved in the recruitment of ERa to p21Waf1/Cip1 promoter.
Literature	GENERIF_SUMMARY	Data suggest that estrogen receptor (ER)alpha46 appears to enhar

Main Category	Category	Terms
		endocrine responses by inhibiting selected ERalpha66 responses.
Literature	GENERIF_SUMMARY	Results support a role of estrogen and aromatase in pathogenesis systemic lupus erythematosus and suggest that genetic variants i estrogen receptor alpha and beta genes might influence susceptib
Literature	GENERIF_SUMMARY	Results describe the expression of estrogen receptor alpha, estrog receptor beta, progesterone receptor and cKIT in a number of desr tumors and their role in determining treatment options.
Literature	GENERIF_SUMMARY	The HindIII polymorphism of the osteocalcin gene, but not the Pvu polymorphism of the estrogen receptor alpha gene or their potenti interaction, was associated with body mass index in premenopaus Chinese women.
Literature	GENERIF_SUMMARY	ER+/PR- tumors more frequently showed HER2 (human epidermal factor receptor) overexpression and represent a distinct subset in breast cancer patients.
Literature	GENERIF_SUMMARY	The actin network plays a role in nuclear ERalpha actions in breast cells.
Literature	GENERIF_SUMMARY	The binding of steroid receptor coactivator-1 (SRC-1) to the AF-1 d of ERalpha is essential but not sufficient to facilitate synergy betw AF-1 and AF-2 domains, which is required for a full agonistic respo estradiol.
Literature	GENERIF_SUMMARY	The positive correlations in positivity rate between the PRL-R and E expressions are found only in CerbB-2 positive patients with breas cancer.
Literature	GENERIF_SUMMARY	The interaction of estrogen receptor alpha polymorphisms with ca supplementation on bone density in Chinese girls at puberty is rep
Literature	GENERIF_SUMMARY	We investigated methylation patterns in the promoter regions of A ATM, BRCA1, CDH3, CDKN2A, CXCR4, ESR1, FBXW7, FOXC1, GSTP HMLH1, PPP2R2B, and PTEN75 in well-described pre-treatment sa from locally advanced breast cancer
Literature	GENERIF_SUMMARY	Positive estrogen receptor alpha is associated with gastric cancer.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha controls a gene network in luminal-like bre cancer cells comprising multiple transcription factors and microR
Literature	GENERIF_SUMMARY	Hsp90 is essential for the proper dimerization of ERalpha/alpha ar ERalpha/beta, it is not required for the proper dimerization of ERbe beta.
Literature	GENERIF_SUMMARY	The results suggest that the PvuII and XbaI RFLPs of the ER1 gene be genetic factors that affect QUS at the calcaneus
Literature	GENERIF_SUMMARY	phosphorylation of CARM1 by PKA at a single serine is necessary sufficient for direct binding to the unliganded hormone-binding dor ERalpha, and the interaction is necessary for cAMP activation of E
Literature	GENERIF_SUMMARY	E2-mediated thyroid cancer cell proliferation involves ERalpha and transcriptional and non-genomic signaling events
Literature	GENERIF_SUMMARY	Hemin inhibits cyclin D1 and IGF-1 expression via STAT5b under hy in ERalpha-negative MDA-MB 231 breast cancer cells
Literature	GENERIF_SUMMARY	This study suggested that strongly in favor of an association betw

Main Category	Category	Terms
		ESR1 polymorphisms and anorexia nervosa.
Literature	GENERIF_SUMMARY	Study show that miR-27a indirectly regulates E2-responsiveness in cells through suppression of ZBTB10, thereby enhancing expressio ERalpha.
Literature	GENERIF_SUMMARY	Genotyped six SNPs in 163 Caucasian cancer patients and 114 he controls using hybridization probes; performed Armitage;s associa trend-test to evaluate the risk. Linkage disequilibrium (LD) was ass for each pair of markers.
Literature	GENERIF_SUMMARY	Data suggest that the negative regulatory loop involving miR-221-2 ERalpha may confer proliferative advantage and migratory activity breast cancer cells and promote the transition from ER-positive to negative tumors.
Literature	GENERIF_SUMMARY	This work indicates that the sequence of the tri-nucleotide spacer random at receptor-bound genomic loci, influences estrogen recep alpha-DNA-binding affinity, and modulates transactivation potentia receptor-ligand-DNA complex.
Literature	GENERIF_SUMMARY	down-regulation of caveolin-1 protein expression leads to deregula estrogen receptor alpha (ERalpha) signaling and consequently earl transformation in mammary epithelia
Literature	GENERIF_SUMMARY	Common genetic variations of the aromatase and ER genes are associated with the risk of severe OA of the large joints of the lowe in a sex-specific manner.
Literature	GENERIF_SUMMARY	Our data show that alternative splicing of ERalpha mRNA is dimini the AD brain and more prominently in AD female cases.
Literature	GENERIF_SUMMARY	role of polymorphism in pathological process (Review)
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in ESR1 is associated with breas cancer.
Literature	GENERIF_SUMMARY	This paper briefly reviews the assessment of ERalpha, PgR, and HE breast cancer, emphasizing recent progress and persistent contro
Literature	GENERIF_SUMMARY	These results identify a conserved and novel function for Hsp27 w potential as a target for interrupting signaling from estrogen recep alpha and androgen receptors to tumor biology in hormone-respon cancers.
Literature	GENERIF_SUMMARY	These results suggest that PELP1 is a reader of H3 methylation m and has a crucial role in modulating the histone code at the ERalph target genes.
Literature	GENERIF_SUMMARY	ER-a overall expression was associated strongly to ESR1 gene cop in breast carcinoma and not in colon or hepatocellular carcinomas
Literature	GENERIF_SUMMARY	The level of ESR1 expression, determined by real-time PCR, directl correlated with the optical redox ratio in breast cancer.
Literature	GENERIF_SUMMARY	ESR1 promoter appeared to be unmethylated in patients with meta breast cancer.
Literature	GENERIF_SUMMARY	ERalpha may play a role in endometrial cancer and the 975C > G-E polymorphism may be considered an independent, early diagnosti marker in this disease.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	that methylation of ESR-1 promoter is higher in ulcerative colitis (UC)patients compared with age and sex matched controls
Literature	GENERIF_SUMMARY	ER-alpha36 expression may be used as a predictive marker for lym node metastasis of gastric cancer.
Literature	GENERIF_SUMMARY	Estrogens stimulate dynamic endothelial protein S-nitrosylation via mechanisms linked to specific ERs possibly on the plasma membr and endogenous nitric oxide.
Literature	GENERIF_SUMMARY	results indicate that both ERalpha and ERbeta were involved in the process of osteogenic differentiation of periodontal ligament stem
Literature	GENERIF_SUMMARY	The T29C genotype polymorphism of ESR1 is associated with the response to IFN-alpha in patients with chronic hepatitis B, and mig significant marker for predicting the initial response to IFN-alpha, a in this study population.
Literature	GENERIF_SUMMARY	E2 stimulation induced recruitment of ERalpha and loss of RNAPII proximal promoter of Kiss1.
Literature	GENERIF_SUMMARY	Our data showed intracellular and surface membrane expression o ERalpha in peripheral blood lymphocytes.
Literature	GENERIF_SUMMARY	confirmed a role for Runx1 in mediating ER alpha genomic recruitm and regulation of tethering genes
Literature	GENERIF_SUMMARY	The HER4 intracellular domain is a physiologically important ERalp coactivator and cooperates with ERalpha to potentiate PgR expres the normal and malignant breast.
Literature	GENERIF_SUMMARY	This pilot study supports our hypothesis that ESR1 amplification is associated with a poorer outcome following adjuvant treatment wi tamoxifen in ER positive early breast cancer.
Literature	GENERIF_SUMMARY	Data show that the concordance rates of histological type, estroge receptor and progesterone receptor status, and HER2 were 84.4%, 92.9%, and 89.3%, respectively between CNB and surgical specime
Literature	GENERIF_SUMMARY	There were no statistically significant differences in the overall associations between the ER-alpha exon I CT genotypes and uterin leiomyoma (P = 0.47).
Literature	GENERIF_SUMMARY	Our findings provide support for diverse roles of ESR1 polymorphis determining susceptibility in different stages of breast cancer.
Literature	GENERIF_SUMMARY	Higher levels of ER-alpha36 expression are required for preserving mass in postmenopausal and menopausal women.
Literature	GENERIF_SUMMARY	Results indicate that loss of ESR1 has a major role in mediating ce cancer invasion and progression.
Literature	GENERIF_SUMMARY	The objective of this study was to evaluate the presence of estroge receptors (ER) alpha and beta in various thyroid lesions and to ass practicality of this test.
Literature	GENERIF_SUMMARY	Genetic variants in ESR1, ESR2, and HSD17B1 genes could modify susceptibility to endometriosis and might influence the fertility sta endometriosis patients.
Literature	GENERIF_SUMMARY	phytoestrogen extracts isolated from flax have effects on estradio production and ER/PR expression in MCF7 breast cancer cells

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the binding of the DNA-binding domain of ERalpha to a 21-mer dsD oligo containing the estrogen response element motif was analyse
Literature	GENERIF_SUMMARY	Results indicate that ER-alpha36 is a new member of the ER-alpha that mediates membrane-initiated estrogen signaling and that syn gamma can replace the function of heat shock protein 90.
Literature	GENERIF_SUMMARY	The analysis of both estrogen receptors might be used as a marke identify high-risk patients only in a subset of patients with endome adenocarcinoma.
Literature	GENERIF_SUMMARY	The -351A/G ESR1 polymorphism might be associated with myoca infarction, but not with extreme longevity.
Literature	GENERIF_SUMMARY	The study shows association between the XbaI and PvuII polymor of the ESR1 gene and, RsaI and AluI polymorphisms of the ESR2 g male fertility status.
Literature	GENERIF_SUMMARY	Our data demonstrate that dual targeting of EphA2 and ER is a pro approach for delaying resistance to Tamoxifen.
Literature	GENERIF_SUMMARY	Results suggest that Hakai is a novel corepressor of ERalpha and play a negative role in the development and progression of breast cancers.
Literature	GENERIF_SUMMARY	Our data suggest that XbaI and possibly PvuII polymorphisms of t gene influence the impact of raloxifene treatment on endothelial fu
Literature	GENERIF_SUMMARY	Polymorphisms of ESR-1 were not associated with any subtype of incontinence.
Literature	GENERIF_SUMMARY	PRDX1 was associated with the cyclooxygenase (COX)-2 upstream promoter region at sites occupied by NF-kappaB in ER- but not in E breast cancer cells.
Literature	GENERIF_SUMMARY	Although certain symptoms of temporomandibular disorder (TMD found to have a relatively higher prevalence of one form or anothe estrogen receptor allele, no haplotype was confirmed to be a signi marker of TMD risk.
Literature	GENERIF_SUMMARY	SNPs associated with breast cancer differ for Asian, European, an African ancestry groups
Literature	GENERIF_SUMMARY	Report a decrease in ERalpha expression in breast cancer patients responding to preoperative systemic chemotherapy.
Literature	GENERIF_SUMMARY	These results elucidate that ER regulates an evolutionarily conserv apoptosis pathway
Literature	GENERIF_SUMMARY	vitamin D receptor and estrogen receptor 1 gene polymorphisms h effect on bone mineral density in postmenopausal women
Literature	GENERIF_SUMMARY	postmenopausal women with a high frequency of PvuII polymorph genotype are more predisposed to ischemic stroke
Literature	GENERIF_SUMMARY	evaluated and summarized results of studies involving rs2234693 rs9340799 and clustered risk factors for development of cardiova disease(CVD); studies provide some support for relationship betw polymorphisms in ESR1 and risk factors for CVD
Literature	GENERIF_SUMMARY	New insights into the possible effect of some lifestyle-related fact the aberrant methylation drift of ERalpha and RARbeta2 genes in b

Main Category	Category	Terms
		cancer.
Literature	GENERIF_SUMMARY	miR-30b is selectively downregulated in prefrontal cortex of femal not male subjects with schizophrenia and this is associated with sequence variations in the Esr1 estrogen receptor
Literature	GENERIF_SUMMARY	The mRNAs of SSTR subtypes 1, 2, 3, and 5, dopamine receptor an ERalpha were measured by in 59 non-functioning adenomas and 5 functioning adenomas.
Literature	GENERIF_SUMMARY	epistasis between ESR1 and CYP19A1 may be involved in the regu of folliculogenesis
Literature	GENERIF_SUMMARY	multicellular spheroids of MCF-7 breast cancer cells presented a paradoxical regulation of estrogen receptor alpha, with a decrease protein expression and a marked increase in mRNA steady-state le
Literature	GENERIF_SUMMARY	ERalpha stable transfection inhibits proliferation and migration cap of MDA-MB-231 cells and decreases expression of COX-2 and VEG
Literature	GENERIF_SUMMARY	ESR1 PP (odds ratio = 2.2) and CYP19 genotypes carrying the T all (odds ratio = 1.9) were significantly associated with longevity.
Literature	GENERIF_SUMMARY	ESR1 genotype was not associated with any changes in serum lipi concentrations in patients receiving treatment with tamoxifen.
Literature	GENERIF_SUMMARY	Estrogen receptor/progesterone receptor status were independent predictive factors of pathologic complete response to neoadjuvant docetaxel plus epirubicin combination chemotherapy in locally adv breast cancer.
Literature	GENERIF_SUMMARY	Variants in estrogen receptor alpha gene are associated with phenotypical expression of obsessive-compulsive disorder.
Literature	GENERIF_SUMMARY	Study provides the first evidence of an association between polymorphisms in the ESR gene with DF in high-fluoride-exposed populations.
Literature	GENERIF_SUMMARY	Esr1 bind to and activate the HOXC4 promoter to potentiate HoxC4 mediated AID induction, immunoglobulin class switch and somatic hypermutation.
Literature	GENERIF_SUMMARY	Estrogen receptor/progesterone receptor status (and nm23-H1) ar independent predictive factors of pathologic complete response to neoadjuvant docetaxel plus epirubicin combination chemotherapy patients with locally advanced breast cancer.
Literature	GENERIF_SUMMARY	Studies indicate that breast carcinomas Oncotype DX Recurrence S (RS) was correlated with six pathobiologic features
Literature	GENERIF_SUMMARY	Recruitment of FAK and N-WASP is necessary for cell migration an invasion induced by 17beta-estradiol in breast cancer cells.
Literature	GENERIF_SUMMARY	The data indicate that the studied SNPs on ERS1 gene do not influ the menstrual cycle timing and parity but there is a strong relations between the xx ERS1 SNP genotype and the incidence of miscarria the post-menopausal population analyzed.
Literature	GENERIF_SUMMARY	we were unable to demonstrate any association between ESR1 pro polymorphism and nonsyndromic cryptorchidism in Italian and Spa study populations

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The EGF-induced ERalpha cistrome specifically regulates genes fo overexpressed in ERBB2-positive human breast cancers
Literature	GENERIF_SUMMARY	Data show that DFS was statistically different between the two gro
Literature	GENERIF_SUMMARY	estrogen receptor, progesterone receptor, and HER-2 may have a ro progression in breast cancer despite response to doxorubicin/ cyclophosphamide therapy
Literature	GENERIF_SUMMARY	Estrogen receptor alpha36 mediates nongenomic estrogen signali through the epidermal growth factor receptor/Src/extracellular sig regulated kinase signaling pathway in estrogen receptor-negative b cancer cells.
Literature	GENERIF_SUMMARY	Raloxifene effects on prostate cancer cell growth control in vitro ar specific, depending on ERbeta or ERbeta/ERalpha relative expressi levels.
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha plays a central role in mediating alpha9-nA gene up-regulation in response to either nicotine or estradiol stimu
Literature	GENERIF_SUMMARY	estrogen-occupied ERalpha activates and interacts with extracellul ERK2, a downstream effector in the MAPK pathway, resulting in ER ERalpha colocalization at chromatin binding sites across the geno breast cancer cells
Literature	GENERIF_SUMMARY	the antimetastatic activity of raloxifene in mammary cancer may b related to expression of cytoplasmic ERalpha
Literature	GENERIF_SUMMARY	additive model suggest that the ESR-1 594 G>A (pooled OR 1.37; 9 1.02-1.83) and ESR-1 325 C>G (pooled OR 1.16; 95% CI 1.03-1.32) variants are associated with any migraine.
Literature	GENERIF_SUMMARY	identification of new splicing variants of estrogen receptor alpha in human endometrium
Literature	GENERIF_SUMMARY	Data show an association between ESR1 polymorphism and juven and an association between ESR2 polymorphism and adult SLE.
Literature	GENERIF_SUMMARY	ERalpha correlated positively with progesterone receptor expressio (normal controls), and maternal BMI at delivery (fetal growth restri
Literature	GENERIF_SUMMARY	ESR1 amplification as detected by multiplex ligation-dependent pro amplification is rare in breast cancer.
Literature	GENERIF_SUMMARY	Data show that inhibiting miR-375 in ERalpha-positive MCF-7 cells resulted in reduced ERalpha activation and cell proliferation.
Literature	GENERIF_SUMMARY	The loss of periodic expression of ERalpha in the endometrial-myo interface of adenomyosis may be associated with an abnormal reg of estrogen in adenomyosis.
Literature	GENERIF_SUMMARY	ER, PR and Her-2 are independent prognostic factors for locoregio recurrence and survival in node-positive breast cancer patients tre mastectomy.
Literature	GENERIF_SUMMARY	HBO1 was an important downstream molecule of ERalpha
Literature	GENERIF_SUMMARY	proteasome-dependent degradation of ERalpha is enhanced by p-f (1,2,3,6,-tetrahydro-1,3-dimethyl-2-oxo-6-thionpurin-8-ylthio) (TPSF) breast cancer cells

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	in ER negative basal and HER2+ breast cancer, gene expression m reflecting T-cell helper-1 (Th1) and T-cell helper-2 (Th2) mediated i responses play antagonistic roles as major risk factors for distant metastasis
Literature	GENERIF_SUMMARY	ERa gene polymorphism could affect the shift of T lymphocyte sub and the expression of the related cytokines in primary cholestasis cirrhosis patients.
Literature	GENERIF_SUMMARY	COX-2 activation is associated with Akt phosphorylation and poor in ER-negative, HER2-positive breast cancer
Literature	GENERIF_SUMMARY	Ressults suggest a possible involvement of ER-alpha36 in E2-depe growth-promoting effects in endometrial cancer cells.
Literature	GENERIF_SUMMARY	We have demonstrated that ERalpha expression in vestibular schwannoma is equivalent to that of the great auricular nerve. Hov the phosphorylated form of the receptor is up-regulated compared the great auricular nerve.
Literature	GENERIF_SUMMARY	High expression of ERalpha and the estrogen-induced gene EIG121 predicts shorter overall survival in patients with high-grade serous carcinoma.
Literature	GENERIF_SUMMARY	Differential recruitment of nuclear receptor coregulators in ligand-dependent transcriptional repression by estrogen receptor-alpha.
Literature	GENERIF_SUMMARY	MST and hSAV act as novel co-regulators of ERalpha and may play important role in breast cancer pathogenesis
Literature	GENERIF_SUMMARY	The estrogen receptor polymorphisms did not modify the increase of coronary heart disease, stroke, or venous thromboembolism du hormone therapy.
Literature	GENERIF_SUMMARY	CDB-4124 has an effect on progesterone and estrogen receptors, a may suppress the development of precancerous lesions
Literature	GENERIF_SUMMARY	ERalpha enhances endothelial cell motility through the dynamic co actin arrangement and the formation of focal adhesion complexes
Literature	GENERIF_SUMMARY	This small pilot study demonstrates the ubiquitous nature of ER sp variants in these tissue sites and suggests that ERalphaDelta5 ma involved in progression of prostate adenocarcinoma
Literature	GENERIF_SUMMARY	The authors identi fi ed one novel non-coding exon and several spl events that generate exon 1-containing and exon 1- lacking human alpha F isoforms in a wide variety of human tissues.
Literature	GENERIF_SUMMARY	An ER post-translational modification code may exist, reflecting th balance of ligand-mediated & cross-talk signal transduction pathw where integration of diverse signals occurs regulating breast canc growth & survival. Review.
Literature	GENERIF_SUMMARY	analysis of ER in breast cancer and false negative reactions by immunohistochemistry
Literature	GENERIF_SUMMARY	Studies indicate that aUPD is a common and non-random molecul feature of breast cancer that is most prominent in HER2/ER/PR tri negative cases.
Literature	GENERIF_SUMMARY	ER-alpha gene polymorphisms may be a susceptible indicator for E variation of lumbar spine L2-L4 and Ward in Chinese pre- and post-

Main Category	Category	Terms
		menopausal women patients with chronic periodontitis.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha pathway is involved in leptin-induced ovar cancer cell growth.
Literature	GENERIF_SUMMARY	data provide new insights on the protein-protein interaction networ controlled by ER-alpha and -beta that mediate their ability to transc estrogen signaling in breast cancer cells
Literature	GENERIF_SUMMARY	RC-RNase-induced cell death of ER-positive breast tumors is throu regulation of ER and Bcl-2.
Literature	GENERIF_SUMMARY	Tamoxifen and raloxifene reduce progesterone and estrogen recep alpha expression significantly and to a similar extent in the non-ne breast tissue of women of reproductive age
Literature	GENERIF_SUMMARY	ERalpha, ERbeta and GPER1/GPR30 are involved in preventing beta apoptosis, impeding the loss of critical beta-cell mass.
Literature	GENERIF_SUMMARY	Nylestriol and levonorgestrel can both up-regulate the mRNA expre of ER subtypes in MG-63 cells.
Literature	GENERIF_SUMMARY	royal jelly's components, although structurally entirely different fro mediate estrogen signaling by modulating the recruitment of ERalp ERbeta and co-activators to target genes
Literature	GENERIF_SUMMARY	Low Estrogen receptor alpha expression is associated with ovarian cancer.
Literature	GENERIF_SUMMARY	ER-alpha and ER-beta mRNA were expressed in human tongue squ cancer cell, and the expression of ER-beta was weaker than that of alpha.
Literature	GENERIF_SUMMARY	Estrogen receptor expression in atypical hyperplasia does not seem associated with breast cancer
Literature	GENERIF_SUMMARY	PIK3CA mutation and the Wnt pathway have roles in ERalpha-posit breast cancer
Literature	GENERIF_SUMMARY	The expressions of ERalpha and p-ERalpha were negatively correla with the expression of IL-6 in hepatitis B-related hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	ESR1 as a candidate regulator of proopiomelanocortin in hypothal
Literature	GENERIF_SUMMARY	results suggest that calcium mediates the cross-talk between ERa activating signaling pathways
Literature	GENERIF_SUMMARY	genetic variations within DNA promoter elements may fine-tune th physiological action of ERalpha and other nuclear receptors
Literature	GENERIF_SUMMARY	Our results suggest that the less frequent CC genotype in obese premenopausal women is related to narrower femoral neck and hig cortical density at the tibial shaft compared to the TC and TT geno
Literature	GENERIF_SUMMARY	RAC3 is a novel ERalpha co-activator that promotes cell migration has prognostic value for ERalpha-positive breast cancer metastasi
Literature	GENERIF_SUMMARY	Data show that in conjunction with ER-related genes, expression o protein may provide important predictive outcomes for responses neoadjuvant chemotherapy and may allow for the identification of patients who could benefit from chemotherapy.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data suggest that P53 dysfunction is prognostically most relevant cancers and supports the hypothesis that different predictive or prognostic markers will be needed for different molecular subsets breast cancer.
Literature	GENERIF_SUMMARY	polymorphisms associated with family environment affect on men age
Literature	GENERIF_SUMMARY	these findings offer new insights into the regulation of the breast c stem cell phenotype by ERalpha.
Literature	GENERIF_SUMMARY	lower in endometrium of polycystic ovary syndrome group compar controls
Literature	GENERIF_SUMMARY	The higher PR-A and ER-alpha gene expression in granulosa cells i patients with endometriosis, compared with patients with tubal inf might be a leading cause of ovarian dysfunction due to endometrio
Literature	GENERIF_SUMMARY	Estrogen Receptor positivity in non gastrointestinal soft tissue sar was a significant favourable indicator for disease specific survival women
Literature	GENERIF_SUMMARY	Lower ER alpha expression is associated with tamoxifen-resistanc breast cancer.
Literature	GENERIF_SUMMARY	Findings suggest that coexpression of CARM1 and ERalpha may p a better biomarker of well-differentiated breast cancer.
Literature	GENERIF_SUMMARY	The studies reveal the sequence(s) in the ERalpha hinge region tha involved in tethered-mediated actions as well as nuclear localizatio attribute important functionality to this region of the receptor
Literature	GENERIF_SUMMARY	Data show that that bortezomib enhances the fulvestrant-mediate aggregation of the ER in the cytoplasm without blocking the degra of the ER in the nucleus.
Literature	GENERIF_SUMMARY	The interaction between ERalpha and Akt plays a crucial role in the regulation of proliferative activity in postmenopausal endometrium
Literature	GENERIF_SUMMARY	Data show that the L858R mutation of EGFR is associated with SN estrogen biosynthesis and metabolism genes CYP17, CYP19A1, E and COMT in never-smoking female lung adenocarcinoma patients
Literature	GENERIF_SUMMARY	estrogen receptor alpha mediates gene expression changes and g responses induced by 25 HC in breast and ovarian cancer cells
Literature	GENERIF_SUMMARY	Results demonstrate that ROSC negatively affects the functional s ER-alpha, making it potentially useful in the treatment of estrogen-dependent breast cancer cells.
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha polymorphism is associated with fracture
Literature	GENERIF_SUMMARY	Results showed that changes in both ERalpha and ERbeta express characterize the transition from normal epithelium to cancer in pat with vulvar squamous cell carcinoma
Literature	GENERIF_SUMMARY	ER-alpha recruitment in the tethering pathway is dependent on the induced formation of transcription factor complexes that involves interplay between the proteins from different transcription factor fa
Literature	GENERIF_SUMMARY	Estrogen receptor alpha/beta influence sperm metabolism and ma involved in the pathophysiology of varicocele-associated male infe

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	This observation suggests that a switch from ERalpha66 to a predominant expression of ERalpha36 may be associated with development and/or progression of human HCC.
Literature	GENERIF_SUMMARY	ERalpha expression was reduced in in Sertoli, myoid and Leydig ce infertile men.
Literature	GENERIF_SUMMARY	an inherent role for monoubiquitination in estrogen receptor alpha signaling and physiological function of receptor monoubiquitinatio regulation of estrogen induced cell proliferation
Literature	GENERIF_SUMMARY	gen up-regulates ATBF1 transcription but causes its protein degrad in estrogen receptor-alpha-positive breast cancer cells
Literature	GENERIF_SUMMARY	histone modifications governing TFF1 transcription are mediated b estrogen receptor
Literature	GENERIF_SUMMARY	Differential expression of 4 out of 5 genes related to estrogen met and action (ESR1, ESR2, PGR and BGN) was confirmed in endomet
Literature	GENERIF_SUMMARY	in breast cancer cells, IGF-2 activates ER-alpha and ER-beta, and modulates their translocation to nucleus, membrane organelles an mitochondria; IGF-2 actions are mediated by the IGF-1R and the ins receptor
Literature	GENERIF_SUMMARY	ERalpha and some growth factors may be involved in the tumorige of prolactinoma. ERalpha could potentially be an effective therapy for treating prolactinoma.
Literature	GENERIF_SUMMARY	The results indicate that the PvuII and the XbaI polymorphisms of gene are associated with bone mineral density in premenopausal w with GD and may help to estimate the risk of bone loss particularly lumbar spine.
Literature	GENERIF_SUMMARY	CYP17A knockdown in H295R cells resulted in less production of b 17alpha hydroxyprogesterone and androstenedione.
Literature	GENERIF_SUMMARY	A comprehensive immunohistochemical study was performed to investigate the protein expression of eight CT genes in 454 invasiv ductal carcinomas, including 225 ER/PR/HER2-negative (triple-neg carcinomas.
Literature	GENERIF_SUMMARY	ESR1 expression did change through the menstrual cycle or in uter adenomyosis.
Literature	GENERIF_SUMMARY	JMJD2B is recruited to ERalpha target sites, demethylates H3K9m facilitates transcription of ER responsive genes including MYB, MY CCND1
Literature	GENERIF_SUMMARY	ERalpha gene CG haplotype could be associated with a reduced ris primary knee osteoarthritis in Mexican mestizo population.
Literature	GENERIF_SUMMARY	these observations show that hypoxia-stabilized HIF-1a induces th repression of the ESR1 gene, which may facilitate hormone insens the tumor microenvironment.
Literature	GENERIF_SUMMARY	Overexpression of ERalpha gene and enhanced activation of calcin response to estrogen in peripheral blood lymphocytes may contrib the pathogenesis of female dominant in systemic lupus erythemat
Literature	GENERIF_SUMMARY	These results suggest that ERalpha induces the down-regulation o through PI3K activation in breast cancer cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results suggest that co-targeting c-ABL and ERalpha as a combina approach may have therapeutic potential for breast cancer.
Literature	GENERIF_SUMMARY	Upon IMS stress, overproduction of reactive oxygen species (ROS) phosphorylation of AKT triggers estrogen receptor (ER) activity, wh further upregulates the transcription of the mitochondrial regulato and the IMS protease HTRA2.
Literature	GENERIF_SUMMARY	Studies suggest that the development of ERbeta-based therapies r lead to improved drug efficacy.
Literature	GENERIF_SUMMARY	analysis of estrogen and progesterone receptor expression in HPV positive and HPV-negative cervical carcinomas
Literature	GENERIF_SUMMARY	ERalpha phosphorylation at serine 118 affects recruitment of uniq protein complexes at gene-specific promoters.
Literature	GENERIF_SUMMARY	Significant elevation of ERalpha and MTCYB transcript levels in premenopausal leiomyomas and its association with ERalpha, -397 genotype suggests the mitochondrial-mediated role of estrogen as promoter of leiomyoma tumorigenesis.
Literature	GENERIF_SUMMARY	Combination of Src kinase inhibitor KX-01 with tamoxifen may pres promising new strategy for clinical management of ERalpha positi breast cancer.
Literature	GENERIF_SUMMARY	no association of rs9340799 with adolescent idiopathic scoliosis predisposition or curve severity
Literature	GENERIF_SUMMARY	homocysteine promoted hypermethylation of the promoter A regio estrogen receptor-alpha gene, which may be an important mechan the pathogenesis of postmenopausal osteoporosis
Literature	GENERIF_SUMMARY	TLE1, is a necessary transcriptional component of the ER complex it facilitates ER-chromatin interactions.
Literature	GENERIF_SUMMARY	ESR1 expression is correlated with three open reading frames on chromosome 6 in tumours.
Literature	GENERIF_SUMMARY	An association between ESR1 and global anxiety in older women, however these associations varied depending on the anxiety syndr and the use of hormone treatment.
Literature	GENERIF_SUMMARY	AP-2gamma is a novel collaborative factor in ERalpha-mediated transcription.
Literature	GENERIF_SUMMARY	Differential expression of estrogen receptor alpha and beta transcr tissues and in primary culture cells from pubertal gynecomastia.
Literature	GENERIF_SUMMARY	Expression of TLR9 isoforms A and B can be detected in clinical br cancer specimens. The ERalpha and sex steroid hormones regulat expression and invasive effects in the breast cancer cells.
Literature	GENERIF_SUMMARY	Devised a novel flow cytometric method to quantify the expression ERalpha, HER2, and the activation of ERK1/2 simultaneously in sin cells.
Literature	GENERIF_SUMMARY	ER-alpha, which is regulated by miR-26a, is important for liver tumo growth.
Literature	GENERIF_SUMMARY	determined the status of ERalpha and ERbeta in the myometrium a leiomyomas, atypical leiomyomas and leiomyosarcomas, concom

Main Category	Category	Terms
		with the expression of NTS/NTS receptor 1 in these tumors
Literature	GENERIF_SUMMARY	The ERalpha methylation status in basal-like breast cancer was significantly higher than in sporadic non-basal-like breast cancer
Literature	GENERIF_SUMMARY	ERalpha and ERbeta may act in opposite directions to regulate the cadherin pathway and to affect invasiveness of nonfunctional pitu adenoma
Literature	GENERIF_SUMMARY	Two ER subtypes exhibit different expression patterns in spinal trig nucleus, with higher expression levels of ERalpha than ERbeta, sug that estrogens may directly affect trigeminal neuron excitability in humans.
Literature	GENERIF_SUMMARY	This study established WBP2 as a tyrosine phosphorylation target estrogen signaling via EGFR crosstalk.
Literature	GENERIF_SUMMARY	CYP17 gene and ERalpha gene may not be genetic risk factors for endometriosis among southern women in China
Literature	GENERIF_SUMMARY	The prognostic value of the correlations between the expression o estrogen and progesterone receptors, Ki67 expression and the histological stage, tumor stage, the degree of myometrial and vasc invasion, was investigated.
Literature	GENERIF_SUMMARY	Studying each locus alone, no significant results were drawn for ES ESR2 genes.
Literature	GENERIF_SUMMARY	single nucleotide polymorphisms in the ESR1 or ESR2 genes do no to play a major role in the genetic susceptibility of Alzheimer's dise
Literature	GENERIF_SUMMARY	17beta-estradiol induces the expression of KCNK5 via ERalpha(+) breast cancer cells, and this channel plays a role in regulating proli in these cell lines.
Literature	GENERIF_SUMMARY	The menopausal women having heterozygous and homozygous m of PvuII or XbaI genotypes were strongly correlated with breast ca
Literature	GENERIF_SUMMARY	associations between PvuII-XbaI haplotype 2 polymorphism and b mass in girls were found in recent cohort, non-carriers of this polymorphism are likely to be more vulnerable to fat accumulation today's obesity promoting environment, than carriers
Literature	GENERIF_SUMMARY	After calcium supplementation, bone mineral density increased in lactating women with some estrogen receptor genotypes.
Literature	GENERIF_SUMMARY	Estrogen regulates human DBS, which could reduce the risk for du ulcer in women and contribute to sex differences in the prevalence duodenal ulcer.
Literature	GENERIF_SUMMARY	The molecular signaling set off by ERalpha and CrkL association m have a central role in pregnancy and cancer.
Literature	GENERIF_SUMMARY	ER-alpha XbaI and VDR BsmI genetic polymorphisms had a signifi association with the risk of prostate cancer.
Literature	GENERIF_SUMMARY	Consistent with its ability to regulate the levels of ERalpha, express RUNX3 inversely correlates with the expression of ERalpha in brea cancer cell lines, human breast cancer tissues and Runx3(+/-) mou mammary tumors
Literature	GENERIF_SUMMARY	ER-alpha might have an important role in human papillary thyroid

Main Category	Category	Terms
		carcinoma, and its overexpression can be studied in routine needle aspirate as a possible marker of malignancy.
Literature	GENERIF_SUMMARY	Data suggest that estrogen receptor loss in estrogen/antiestrogen sensitive cells triggers loss of endocrine dependence and onset of that change the cell from an epithelial to a mesenchymal type.
Literature	GENERIF_SUMMARY	The genetic variants in the ESR, FTO, and UCP2 genes may be cons as a screening tool prior to bariatric surgery to help clinicians pred weight loss or glycemic control outcomes for severely obese patie
Literature	GENERIF_SUMMARY	Our data suggest that polymorphisms of the ER alpha and ER beta may influence tumor growth, particularly in the passage from in sit invasive melanoma.
Literature	GENERIF_SUMMARY	ER and PR do not appear to have a role in how dietary folate and vi B6 intakes are inversely associated with breast cancer risk
Literature	GENERIF_SUMMARY	AR is enriched in ER- breast tumors that overexpress HER2.
Literature	GENERIF_SUMMARY	pancreatic Stat3 has a role in ER-mediated suppression of lipid syn
Literature	GENERIF_SUMMARY	Data show that the triple-negative and HER2 subtypes were associ with poorer outcomes compared with the luminal A subtype.
Literature	GENERIF_SUMMARY	the nongenomic PI3K signaling pathway plays a role in estrogen/ hERalpha-stimulated hSlo1 gene expression; whereas c-Src activity to hSlo1 gene tonic repression independently of estrogen, likely thr ERK activation.
Literature	GENERIF_SUMMARY	These data replicate previous associations found between single nucleotide polymorphisms in the 6q25 locus and bone mineral den [BMD(a)] at the hip.
Literature	GENERIF_SUMMARY	Aurora-A phosphorylation of p53 sequesters the protein in the cyto and enhances its interaction with ERalpha in a ternary complex comprising of the three proteins
Literature	GENERIF_SUMMARY	microRNAs actively control some of the most distinguishing characteristics of the breast cancer subtypes, such as ERalpha itse Dicer, and growth factor receptor levels
Literature	GENERIF_SUMMARY	Sata support the relevance of phosphorylation at p-Y537-ERalpha i breast cancers and add further support to the presence of a phosphorylation code for ERalpha in human breast cancer in vivo.
Literature	GENERIF_SUMMARY	The age at disease diagnosis was lower in carriers of the ESR1 rs9340799 A allele compared with GG homozygotes as well as in p with ESR1 rs2234693 TT and CT genotypes compared with CC homozygotes.
Literature	GENERIF_SUMMARY	Data show that triple-negative breast cancer (TNBC) are larger in s compared with other subtypes and almost all grade 3.
Literature	GENERIF_SUMMARY	Different expression patterns of androgen receptor and ER alpha m in malignant tissues of oral mucosa suggest an involvement of the sexual hormones in oral cancer
Literature	GENERIF_SUMMARY	TNF-alpha activates ER signaling in endometrial epithelial cells.
Literature	GENERIF_SUMMARY	Results found that in the Brazilian male population, genetic variatio both estrogen receptors alpha (PvuII and XbaI) and beta (AluI and

Main Category	Category	Terms
		were not relevant to idiopathic infertility.
Literature	GENERIF_SUMMARY	These results suggested a novel regulatory mechanism of let-7 mi on ER-alpha36 mediated nongenomic estrogen signal pathways.
Literature	GENERIF_SUMMARY	These results demonstrate a novel pathway of estrogen regulation gene expression by cooperation between ER and AP-1 at the distal enhancer element and that AP-1 is involved in estrogen induction c myc oncogene.
Literature	GENERIF_SUMMARY	the ER1 polymorphisms might associate with type 2 diabetes and glucose among Iranian male subjects.
Literature	GENERIF_SUMMARY	A polymorphism of the ESR1 gene is a novel c.453-397C allele-spe and c-myb-dependent enhancer-like cis-acting regulatory variation could be part of the genetic variations underlying the susceptibility hepatitis B virus-related liver cirrhosis.
Literature	GENERIF_SUMMARY	We confirmed evidence that discordance of ER/Her2 expression be primary and metastatic breast carcinoma does occur.
Literature	GENERIF_SUMMARY	Annonacin induced growth arrest and apoptosis via ERalpha-relate pathways in MCF-7 cells.
Literature	GENERIF_SUMMARY	analysis of Aurora A and B kinases, Ki-67 and the estrogen and progesterone receptors from an endometriosis tissue microarray r
Literature	GENERIF_SUMMARY	H3K27 demethylation by JMJD3 at a poised enhancer of anti-apop gene BCL2 determines ERalpha ligand dependency
Literature	GENERIF_SUMMARY	The allele of estrogen receptor-alpha identified by XbaI (X) restricti detected less frequently in the study group compared to the contro group.
Literature	GENERIF_SUMMARY	Data from material procured by laser capture microdissection sug that coordinated expression of ESR1 and XBP1 in breast carcinom to their co-expression in carcinoma cells rather than in the surroun stromal cells.
Literature	GENERIF_SUMMARY	We conclude that Reed-Sternberg cells in classical Hodgkin lymph children seem to be mainly ERbeta positive and ERalpha negative.
Literature	GENERIF_SUMMARY	These studies identify PITX1 as a new ERalpha transcriptional targ
Literature	GENERIF_SUMMARY	All the enhanceosome components comprising estrogen receptor (ERalpha), FOXA1, and GATA3 are necessary for the full repertoire cancer-associated effects of the ERalpha.
Literature	GENERIF_SUMMARY	common single-nucleotide polymorphisms inside ER are related to clinical outcomes of androgen-deprivation therapy in men with pro cancer
Literature	GENERIF_SUMMARY	summarizes recent studies on the structure and function of ER-alp and the relationship of ER-alpha36 with cancer, with special empha its function in the resistance of breast cancer to endocrine therapy
Literature	GENERIF_SUMMARY	Data show that hierarchical clustering based on mRNA expression estrogen receptors, progesterone receptor and neu receptor by quantitative RT-PCR identified patient groups that were comparabl subtypes identified by immunohistochemistry.
Literature	GENERIF_SUMMARY	These results suggest a novel negative regulation of estrogen rece

Main Category	Category	Terms
		alpha activity and protein by JAK2 in breast cancer cells and indica potential new cross-talk
Literature	GENERIF_SUMMARY	demonstrated that ER-alpha36 positively regulates HER2 expressio the population of ALDH1 positive breast cancer cells, and suggest non-genomic estrogen signaling mediated by ER-alpha36 is involve maintenance of breast cancer stem cells
Literature	GENERIF_SUMMARY	ERalpha physically interacted and functionally cooperated with SIR breast cancer cells. ERalpha also bound to the promoter for SIRT1 increased its transcription.
Literature	GENERIF_SUMMARY	Epigenetic alteration of ERalpha gene may play an important role i pathogenesis of familial breast cancer
Literature	GENERIF_SUMMARY	This study described the generation of two isogenic breast cancer lines, Hs578T-ERalphaLuc and Hs578T-ERbetaLuc, with stable inte of an estrogen responsive luciferase reporter gene.
Literature	GENERIF_SUMMARY	the A allele of SNP12 of ESRa may increase the risk of hypospadia cryptorchidism.
Literature	GENERIF_SUMMARY	total internal reflection fluorescence microscopy in endothelial cel illustrates that ER46 can topologically conform to a type I transme protein structure
Literature	GENERIF_SUMMARY	We propose that different variants of IVS1-397 estrogen receptor a polymorphism may become additional genetic factor that influenc regulatory conditions during diabetes type 1 in females.
Literature	GENERIF_SUMMARY	14-3-3beta has oncogenic potential in breast cancer via binding to ERalpha and activation of the transcriptional activity of ERalpha.
Literature	GENERIF_SUMMARY	Data suggest that the low-level expression of ESR1 is associated v tamoxifen resistance in ER-positive breast cancer.
Literature	GENERIF_SUMMARY	There was no compelling evidence of an association between vari alleles of genotypes estrogen receptors alpha and breast density.
Literature	GENERIF_SUMMARY	The median time of survival in patients with lung cancer depending expression of ER alpha was 31 months, while for ER beta was 73 n
Literature	GENERIF_SUMMARY	1 of 8 in
Literature	GENERIF_SUMMARY	aberrantly expressed miRNA-206 may be associated with the development of ERalpha-positive endometrioid adenocarcinoma (E
Literature	GENERIF_SUMMARY	There is little involvement of genetic variation of the ESR1 gene in modifying risk of coronary artery disease.
Literature	GENERIF_SUMMARY	XAP2 exerts a negative effect on ERalpha transcriptional activity a thus prevent ERalpha-dependent events.
Literature	GENERIF_SUMMARY	When combined with HER2, ESR1 may be prognostic but should no used for endocrine treatment selection in postmenopausal women endocrine-responsive early breast cancer.
Literature	GENERIF_SUMMARY	ERalpha gene polymorphisms may be associated with the pathoge of schizophrenia in Korean patients.
Literature	GENERIF_SUMMARY	Reduced estrogen receptor is associated with recurrent breast car

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	A combination of CALM1 and ER1 gene polymorphisms might be r to double curve in patients with AIS.
Literature	GENERIF_SUMMARY	analysis did not reveal any association between the ER1 genetic va and postmenopausal breast cancer risk, either ER-positive or ER-ne disease
Literature	GENERIF_SUMMARY	Gastrointestinal stromal tumours do not show immunopositivity fo or HMB45.
Literature	GENERIF_SUMMARY	Gene expression signatures comprising 24, 51 and 14 genes predi ER, PR and ERBB2 status respectively, were investigated.
Literature	GENERIF_SUMMARY	Data suggest that pelvic organ prolapse is associated with higher expression of ERalpha/ERbeta in the pelvic floor of both pre- and p menopausal women.
Literature	GENERIF_SUMMARY	Chromatin immunoprecipitation identified ERalpha genomic bindin activity in estrogen-deprived cells.
Literature	GENERIF_SUMMARY	our study has identified novel aspects of ligand-mediated estrogen receptor alpha and beta interactions underlying the diversity of est mediated regulation of different cellular functions mediated by liga different biocharacter
Literature	GENERIF_SUMMARY	In breast cancer, estrogen receptor alpha is downregulated by TWI (Twist) through interactions with DNA methyltransferase 3B (DNM and histone deacetylase 1 (HDAC1).
Literature	GENERIF_SUMMARY	regulation of ER by Twist could be an underlying mechanism for th of ER activity observed in breast tumors, and may contribute to the generation of hormone-resistant, ER-negative breast cancer
Literature	GENERIF_SUMMARY	Estrogen receptor alpha unstructured transactivation domain (AF1 activity is regulated by S118 phosphorylation-dependent recruitme PIN1.
Literature	GENERIF_SUMMARY	Data indicate that in myometrium in pregnancy, ER1 and GPR30 m are expressed; ER2 mRNA is undetectable; ER1 protein is undetect unless 26S proteasome activity is inhibited; estradiol acts via ER1 activation of ERK/MAPK pathway.
Literature	GENERIF_SUMMARY	Highlight the importance of cross-regulation between AhR and ERa and a novel mechanism by which AhR controls, through modulatin recruitment of RIP140 to ERalpha-binding sites, the kinetics and magnitude of ERalpha-mediated gene stimulation.
Literature	GENERIF_SUMMARY	a small increase in the expression of p23 amplifies ER-binding gen wide and, in combination with ER, elicits an invasive phenotype in b cancer
Literature	GENERIF_SUMMARY	NF-kappaappaB, acting through two response elements, is require recruitment to an adjacent estrogen response element in the BIRC3 promoter.
Literature	GENERIF_SUMMARY	Genetic variants in ESR1 does not contribute to the development o syndrome.
Literature	GENERIF_SUMMARY	Expression of PR and ESR1 defines a subgroup of ovarian carcinor with a favourable prognosis.
Literature	GENERIF_SUMMARY	ER-alpha and PR may be useful markers predicting therapy respon

Main Category	Category	Terms
		endometrial hyperplasia.
Literature	GENERIF_SUMMARY	The prognostic impact of Akt (Akt1) phosphorylated at threonine3 serine473, Akt2, Akt3, PI3K and PTEN, alone and in coexpression v and PgR in non-gastrointestinal stromal tumor soft tissue sarcoma
Literature	GENERIF_SUMMARY	Invasive ductal mammary carcinoma type was analyzed for compa investigation of hormonal receptors (estrogen receptor ER and progesterone receptor PR) and Her2/neu oncoprotein expression.
Literature	GENERIF_SUMMARY	PBX1 is a novel pioneer factor defining aggressive ERalpha-positiv breast tumors, as it guides ERalpha genomic activity to unique ger regions promoting a transcriptional program favorable to breast ca progression
Literature	GENERIF_SUMMARY	Thyroid hormone is shown to induce human lung cancer cell prolif via crosstalk between integrin alphavbeta3 and ERalpha.
Literature	GENERIF_SUMMARY	Colocalization of estrogen receptor alpha 36 isoform with cytoske microfilaments suggests a role of estrogens in cell motility.
Literature	GENERIF_SUMMARY	It was shown that cocaine- and amphetamine-regulated transcript (CART) increases the transcriptional activity of estrogen receptor a and that CART stimulates an autocrine/paracrine loop within tumo to amplify the CART signal.
Literature	GENERIF_SUMMARY	ER gene PvuII and XbaI polymorphisms have no major influence on mineral density in our group of postmenopausal women.
Literature	GENERIF_SUMMARY	Results demonstrate that estradiol induces growth of human astro cell lines through ERalpha and its interaction with SRC-1 and SRC-3 also suggest differential roles of ERalpha on cell growth depending astrocytoma grade.
Literature	GENERIF_SUMMARY	Amplification of ESR1 in breast carcinoma occurs as low-level amplification.
Literature	GENERIF_SUMMARY	NR2E3 regulates ESR1 via direct binding to the ESR1 promoter with concomitant recruitment of PIAS3 to the promoter in breast cance and is essential for physiological cellular activity of ESR1 in estrog receptor (ER)-positive breast cancer cells
Literature	GENERIF_SUMMARY	Our findings suggest that estrogen signaling could be affected not estrogen receptor alpha (ER A)-negative breast cancer, but also in positive breast cancer due to lack of circadian availability of ER A.
Literature	GENERIF_SUMMARY	Expression of the newly identified estrogen receptor alpha-36 is no limited within breast cancer epithelium, but is equally identified in t vasculature, peritumoral fat tissue, lymphocytic infiltrate and strom fibroblasts.
Literature	GENERIF_SUMMARY	SNPs (HSD17B1 (rs2010750, rs598126 and rs676387), COMT (rs4 UGT1A1 (rs8175347) and ESR1 (rs9340799)) seem to be related to mammographic density in the same direction of their associations breast cancer risk
Literature	GENERIF_SUMMARY	p27KIP1 has a role in node-negative breast cancer, but the interact with ER and PR is only significant in T1a/b stage
Literature	GENERIF_SUMMARY	ER-alpha intrinsic DNA binding free energy is identical to that of progesterone receptor B (PR-B) isoform, as expected; this ability se as a framework for regulating receptor-specific promoter occupan

Main Category	Category	Terms
		thus receptor-specific gene regulation.
Literature	GENERIF_SUMMARY	The predictive value is amplified when the ER status is correlated v PR status.
Literature	GENERIF_SUMMARY	We have determined the estrogen receptor alpha cistrome defined novel method to study DNA-protein interactions in vivo.
Literature	GENERIF_SUMMARY	OD extracts enhance the tumor suppressor p53 expression as a re an increased binding of ERalpha/Sp1 complex to the p53 promote region.
Literature	GENERIF_SUMMARY	siRNA-mediated knockdown of Noxa resulted in cell cycle arrest fo E2 treatment, indicating that Noxa expression is required for cell cy progression in ER-positive breast cancer cells.
Literature	GENERIF_SUMMARY	The findings in this pilot study suggest that estrogen receptor 1 (E gene polymorphisms do not contribute to endometriosis susceptib in women.
Literature	GENERIF_SUMMARY	ERalpha repressively modulates the expression of LKB1 at the transcriptional level.
Literature	GENERIF_SUMMARY	identified a locus on 6q25 associated with breast cancer risk in US Latinas; the peak of the 6q25 signal includes the estrogen recepto (ESR1) gene and 5' region, a locus previously implicated in breast c
Literature	GENERIF_SUMMARY	In the Iranian population, genetic polymorphisms of the ER-alpha a beta genes may be involved in the etiology of prostate cancer
Literature	GENERIF_SUMMARY	Estrogen can repress transcription of HBV genes by up-regulating alpha, which interacts with and alters binding of HNF-4alpha to the enhancer I.
Literature	GENERIF_SUMMARY	The results suggest that polymorphisms in ESR1, ESR2 and LHCGI contribute to the risk of developing testicular germ cell cancer, histological subtype, and risk to metastasis.
Literature	GENERIF_SUMMARY	presence of an ERalpha splice variant, ERalpha36, in ERalpha-posit MCF7 and ERalpha-negative HCC38 breast cancer cells, which loca to plasma membranes and rapidly activated PKC in response to E(
Literature	GENERIF_SUMMARY	Data show that phosphorylation of serine 212 enabled estrogen re alpha (ERalpha) to regulate a unique set of genes and cellular func
Literature	GENERIF_SUMMARY	RASSF1A has a key role in suppressing the transformation of hum breast epithelial cells and estrogen receptor alpha positive breast c initiation.
Literature	GENERIF_SUMMARY	None or very small amounts of ERbeta were detected, and ERalpha ERbeta2 mRNA were elevated in the Puumala virus infection patien group.
Literature	GENERIF_SUMMARY	discussion of tissue-specific regulation of gene expression involvi ERalpha and ERbeta, ER agonists, estrogen response elements, an epigenetic modification of ER genes; discussion of molecular targe anticancer therapy with ER agonists [REVIEW]
Literature	GENERIF_SUMMARY	discussion of role of ERalpha/ERbeta in cardiovascular adaptation uterus during pregnancy; structural basis for functional diversity of ERalpha/ERbeta subtype selectivity of estrogens; and genetic/epig regulation in endothelium [REVIEW]

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results indicate that the 2,4- and 2,5-dimethyl derivatives of pheny estradiols had the highest relative binding affinity with estrogen al receptor.
Literature	GENERIF_SUMMARY	CaM binds two molecules of ER-alpha in a 1
Literature	GENERIF_SUMMARY	ERbeta AluI A allele presence was the most important independen predictor of hormone therapy-induced increase in flow-mediated d while ERalpha XbaI A allele was the only independent predictor of decrease in sCD40L.
Literature	GENERIF_SUMMARY	This study suggested that there was a modest but statistically sig association between the PvuII pp genotype of the ESR1 gene and vertebral fracture in Chinese patients.
Literature	GENERIF_SUMMARY	The ERa PvuII polymorphism is associated with the increased risk in men of a Chinese population.
Literature	GENERIF_SUMMARY	S6K1 directly phosphorylates and regulates ligand-independent ac ERalpha, while ERalpha upregulates S6K1 expression.
Literature	GENERIF_SUMMARY	ER/PR and HER2 statuses between primary and liver metastatic le of breast carcinoma can be modified after treatment but are stable most cases during the natural metastatic process.
Literature	GENERIF_SUMMARY	Among the 1% to 9%, 10%, and more than 10% ER IHC-positive pati 24%, 67%, and 92% were also positive by ESR1 mRNA expression. average ESR1 expression was higher in the >/= 10% ER-positive co compared with the 1% to 9% or ER-negative cohort.
Literature	GENERIF_SUMMARY	PRSS23 is upregulated by estrogen receptor alpha in breast cance
Literature	GENERIF_SUMMARY	reduced microRNA-152 can lose an inhibitory effect on DNA methyltransferase, which leads to hypermethylation of the ERalpha and a decrease of ERalpha level
Literature	GENERIF_SUMMARY	a role of non-genomic signaling through the activation of p-ERK1/2 hormonal dependence of inflammatory breast cancer by a combin estrogen receptors
Literature	GENERIF_SUMMARY	genetic association studies in population of Greek women
Literature	GENERIF_SUMMARY	These results provide new evidence of the mechanisms whereby e regulates leptin expression in placenta and support the importanc leptin in placental physiology.
Literature	GENERIF_SUMMARY	The estrogen receptor alpha PvuII/XbaI and vitamin D receptor Fok polymorphisms may contribute to the determination of bone mine density in Turkish postmenopausal women.
Literature	GENERIF_SUMMARY	The presence of the APOE epsilon4 allele, and not the alleles of ER is a risk factor for MCIa. Furthermore, APOE genotype seems to pr the conversion from MCIa to AD.
Literature	GENERIF_SUMMARY	genetic association studies in Brazilian women
Literature	GENERIF_SUMMARY	Data show that 7alpha, 6-hexanyl derivatives of estradiol (compou functions as a mixed agonist/antagonist, inducing only weak estro receptos (ER) transcriptional activity while preventing coactivator recruitment and stabilizing ER expression.
Literature	GENERIF_SUMMARY	Significant increases in osteoporosis risk with minor alleles of rs2

Main Category	Category	Terms
		located in the first exon and rs2234693 located in the first intron
Literature	GENERIF_SUMMARY	Binding of the ERalpha and ARNT1 AF2 domains to exon 21 of the isoform SRC1e is essential for estrogen- and dioxin-transcription.
Literature	GENERIF_SUMMARY	Our results indicated that expressions of ERalpha and ERbeta corr with each other and with less aggressive phenotypical features of invasive breast ductal carcinoma.
Literature	GENERIF_SUMMARY	ERalpha25p is a new CaM-binding ligand.
Literature	GENERIF_SUMMARY	The regulation of estrogen receptor alpha signaling by target prote SSX2 may explain the metastatic potential of breast cancer cells.
Literature	GENERIF_SUMMARY	ERalpha/beta heterodimers are growth inhibitory in breast and pro cells which co-express the two ER isoforms
Literature	GENERIF_SUMMARY	in ER-positive breast cancer cells, LRH-1 promotes cell proliferatio enhancing ERalpha mediated transcription of target genes such as GREB-1
Literature	GENERIF_SUMMARY	epigenetic silencing of the estrogen receptor (ER) by hypermethyla the ESR1 promoter is seen predominantly in triple-negative breast cancers in Indian women
Literature	GENERIF_SUMMARY	In clinical practice, besides routine detection of ER-alpha and PR in invasive luminal subtype breast cancers, immunohistochemical st of ER-Beta1 and ER-Beta2 should be considered in order to achieve useful information.
Literature	GENERIF_SUMMARY	HADHB is a functional molecular target of estrogen receptor alpha mitochondria, and the interaction may play an important role in the estrogen-mediated lipid metabolism in animals and humans.
Literature	GENERIF_SUMMARY	PIMT may act as a co-activator in ERalpha-mediated transcription through its recruitment to the promoter via interacting with ERalph
Literature	GENERIF_SUMMARY	In the female bipolar disorder a significant association result was for rs6023059, a single nucleotide polymorphism (SNP) placed downstream of the gene coding for transglutaminase 2 (TGM2).
Literature	GENERIF_SUMMARY	EGFR overexpression relates to triple negative profile (ER-/PR-/HE poor prognosis in breast cancer patients in Tunisia
Literature	GENERIF_SUMMARY	phosphorylation at Ser-118 is required for ER to bind both p38gam c-Jun, thereby promoting ER relocation from ERE to AP-1 promoter
Literature	GENERIF_SUMMARY	ER-alpha36 expression was significantly higher in ER-negative tum than in ER-positive tumors
Literature	GENERIF_SUMMARY	This meta-analysis provides evidence that PvuII polymorphisms in ESR1 gene, which encodes for estrogen receptor-alpha, could mod susceptibility to endometrial cancer.
Literature	GENERIF_SUMMARY	68.3% of patients were ER-positive, and 71.5% were PR-positive. A the 28.3% of HER2-positive tumors, the majority (19.2%) stained +- Almost 50% of the tumors were ER+ PR+ HER2-, which makes this predominant tumor type.
Literature	GENERIF_SUMMARY	GPER status adds clinically relevant information to ERalpha status endometrial carcinoma and suggest a potential for new inhibitors treatment of metastatic endometrial cancers with ERalpha express

Main Category	Category	Terms
		and GPER loss.
Literature	GENERIF_SUMMARY	results indicate that in ER-negative breast cancer cells Src functior switch in ERalpha36-mediated biphasic estrogen signaling through EGFR/STAT5 pathway
Literature	GENERIF_SUMMARY	Data suggest that ERalpha colocalizes with activated/phosphoryla JNK1 (c-Jun N-terminal kinase 1) at many target gene promoters i of breast cancer cells exposed to estrogens.
Literature	GENERIF_SUMMARY	Data suggest that lack of palmitoylation renders ERalpha more susceptible to estradiol-dependent degradation, blocks phosphory of ERalpha at Ser118, and prevents estradiol-induced ERalpha estr response element occupancy.
Literature	GENERIF_SUMMARY	TWIST expression was inversely correlated with ERalpha expressio both breast cancer cell lines and human breast ductal carcinomas
Literature	GENERIF_SUMMARY	Her2-tumor-initiating cells-enriched signature was specific to HER2
Literature	GENERIF_SUMMARY	Data indicate that estrogen receptor (ER)alpha and ERbeta immunoreactivity was detected in the nuclei of esophageal squam cell (41.1 and 97.8%, respectively).
Literature	GENERIF_SUMMARY	AR and ESR1 have distinct modes of interaction with chromatin an DNase I hypersensitivity dynamics provides a general approach fo predicting cell-type specific cistromes.
Literature	GENERIF_SUMMARY	Based on a limited selection of gene variants, no involvement of SI SRD5A2, ESR1, ESR2 or PGR in female pattern hair loss is evident.
Literature	GENERIF_SUMMARY	these data suggest that TTF-2 may modulate the function of ERalp corepressor and play a role in ER-dependent proliferation of mamn cells.
Literature	GENERIF_SUMMARY	Aberrant hypermethylation of ER gene is associated with invasive cell carcinoma of the breast.
Literature	GENERIF_SUMMARY	ERalpha positivity, ERbeta negativity, and AR expressions are asso with a more aggressive phenotype of differentiated thyroid cancer.
Literature	GENERIF_SUMMARY	divergent effects of coregulators on the stable association of estro receptor alpha with estrogen response elements associated with t TFF1 and progesterone receptor promoters
Literature	GENERIF_SUMMARY	increased expression is associated with improved overall survival i patients with adenocarcinoma that underwent surgical resection
Literature	GENERIF_SUMMARY	The ER-CXCL12-CXCR4 signaling axis may represent a promising t for development of therapeutics for ER+ ovarian cancers.
Literature	GENERIF_SUMMARY	study establishes a role for AIB1 in aromatase inhibitor-resistant b cancer and describes a new mechanism of ER-alpha/AIB1 gene regulation which could contribute to the development of an aggres tumor phenotype
Literature	GENERIF_SUMMARY	Data show that HEK293T cells transfected with estrogen receptor (ERalpha)+/-Galpha(o) revealed that Galpha(o) stimulated phosphorylation of ERK MAP kinases ERK 1/2 and subsequently increased the phosphorylation of ERalpha on serine 118.
Literature	GENERIF_SUMMARY	CAV-1 affected cell growth of lung cancer NCI-H446 cell through th

Main Category	Category	Terms
		interactions with p-ERK1/2, estrogen receptor and progestin recep
Literature	GENERIF_SUMMARY	The mechanism of estrogen-promoted ovarian cancer metastasis due to the up-regulation of survivin conducted through the ERalpha signalling pathway.
Literature	GENERIF_SUMMARY	A consistent enrichment of PR transcripts, relative to estrogen rec transcripts, in the bipotent progenitor subfraction in the normal hu breast, is reported.
Literature	GENERIF_SUMMARY	There were no statistically significant associations of ESR1 and ES gene polymorphisms with response to treatment.
Literature	GENERIF_SUMMARY	The epithelial component demonstrated the expression for ER-alph (45.6%, 36 of 79), ER-beta (37.2%, 29 of 78), PR (91.1%, 72 of 79), a (10.1%, 8 of 79).
Literature	GENERIF_SUMMARY	measurement of ESR1 mRNA levels may be predictive of response endocrine therapy in a manner that is different from the predictive of ER
Literature	GENERIF_SUMMARY	Estrogen receptor positive (ER+)BRCA1-related breast cancers sho several morphologic and immunophenotypic differences from ER+ sporadic breast cancers.
Literature	GENERIF_SUMMARY	17beta-estradiol treatment increased the proliferation of MCF-7 ce a higher ERalpha/ERbeta ratio.
Literature	GENERIF_SUMMARY	A novel mechanism by which overexpression of estrogen receptor beta through NCoR is able to down regulate ER alpha gene express thus blocking ER alpha's driving role on breast cancer cell growth.
Literature	GENERIF_SUMMARY	Major variations in estrogen receptor (ER) measured by ligand bind assay and immunohistochemistry are observed within all ESR1 co number subgroups and other mechanisms than gene copy numbe to contribute to the ER protein content in the tumors.
Literature	GENERIF_SUMMARY	Estrogen receptor (ER)alpha gene polymorphism using XbaI restric enzyme influences the perception of back pain in patients with degenerative spondylolisthesis.
Literature	GENERIF_SUMMARY	that hinge domain phosphorylation patterns uniquely inform on the various ER alpha activation mechanisms thought to underlie the bi and clinical diversity of hormone-dependent breast cancers
Literature	GENERIF_SUMMARY	Estrogen receptor alpha gene polymorphisms has a role in risk of H related acute liver failure in the Chinese population
Literature	GENERIF_SUMMARY	attenuated phosphorylation of ERalpha at S118 and S167 significa affected cellular physiology and behavior in MCF-7 breast cancer c resulting in increased growth, migration/invasion, an altered gene expression patterns.
Literature	GENERIF_SUMMARY	The ESR1-PvuII polymorphism emerges as a potential candidate fo early prediction of infertility due to premature ovarian aging.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha gene polymorphism was related to unkno endometrium, in which P allele may be the risk and X allele may be guard factor. The expression of estrogen receptor alpha was decre thin endometrium.
Literature	GENERIF_SUMMARY	ESR1 454-397T>C mutation is significantly associated with increas

Main Category	Category	Terms
		of ischemic stroke. [Meta-analysis]
Literature	GENERIF_SUMMARY	Presence of membrane-bound ERalpha expression together with E mutation was found to be an independent prognostic factor for su patients with lung adenocarcinoma, suggesting cross-talk betwee membrane-bound ERalpha and EGFR mutation.
Literature	GENERIF_SUMMARY	Differences in uterine immunoexpression of PR, ERalpha and OTR comparing prostaglandin- to progestagen-based protocols for ovin estrus synchronization.
Literature	GENERIF_SUMMARY	an insight into a feedback loop between ERalpha and p53 and a bi role of p53 in the DNA damage response in ER-positive breast can
Literature	GENERIF_SUMMARY	show that c-Fos and ERalpha can cooperate in regulating E2F1 ge expression by binding to regulatory elements in the E2F1 promote
Literature	GENERIF_SUMMARY	The microsatellite ESR1_TA in the ESR1 gene seems to have no association with breast cancer among African American or Nigeri women.
Literature	GENERIF_SUMMARY	The XbaI polymorphism was clearly associated with a reduction in endometrial thickness in women with breast cancer treated with aromatase inhibitors.
Literature	GENERIF_SUMMARY	ERalpha gene polymorphism is correlated with the pathogenesis o missed abortion. Alleles P and X may be susceptibility genes.
Literature	GENERIF_SUMMARY	This study found a significant effect of estrogen receptor alpha (E microsatellite repeat number on left hand 2nd
Literature	GENERIF_SUMMARY	Increased TRPV6 expression is a feature of estrogen receptor (ER) negative breast tumors.
Literature	GENERIF_SUMMARY	The associations of 6 selected polymorphisms in three genes
Literature	GENERIF_SUMMARY	The results indicates that there appears to be an influence of ESR1 594 genotypes on genetic susceptibility to breast cancer.
Literature	GENERIF_SUMMARY	testable hypotheses regarding a role of genetic variants in ESR1 in dysfunctional human maternal parenting.
Literature	GENERIF_SUMMARY	expression of PES1 correlated positively with ERalpha expression negatively with ERbeta expression and predicted good clinical out breast cancer
Literature	GENERIF_SUMMARY	BCA2 is therefore a newly identified transcriptional target of estrog receptor.
Literature	GENERIF_SUMMARY	17beta-estradiol stimulation induces the recruitment of PAD2 to ta promoters by ERalpha, whereby PAD2 then citrullinates H3R26, wh leads to local chromatin decondensation and transcriptional activa
Literature	GENERIF_SUMMARY	ERalpha phosphorylation at Y537 by Src triggers E6-AP-ERalpha bi ERalpha ubiquitylation, promoter occupancy, and target gene expre
Literature	GENERIF_SUMMARY	The XbaI polymorphism is correlated to BMD at diverse skeletal si had a protective role for the lumbar spine but might be a risk facto femoral neck.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha mRNA/protein expression levels were inc in ectopic tissue in peritoneal endometriosis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha gene PvuII (T/C) and XbaI (A/G) polymorp are not associated with endometriosis risk.
Literature	GENERIF_SUMMARY	These results suggest that ERalpha is a critical regulator of chondr proliferation and maturation during skeletal development, mediatir longitudinal bone growth in vivo.
Literature	GENERIF_SUMMARY	Activation of the PKA signaling pathway alters the transcriptome b redirecting ERalpha to new transcriptional start sites, resulting in a transcription and tamoxifen resistance.
Literature	GENERIF_SUMMARY	Both ACTN4 (full length) and ACTN4 (Iso) potentiate basal transcr activity and directly interact with estrogen receptor alpha, although ACTN4 (Iso) binds ERalpha more strongly.
Literature	GENERIF_SUMMARY	interaction of Delta(5)-androstenediol with ERalpha and ERbeta
Literature	GENERIF_SUMMARY	These experiments further define the molecular mechanism of est bone-protective effects by inducing osteoclast apoptosis through upregulation of MMP3 and FasL cleavage.
Literature	GENERIF_SUMMARY	ASPP 049 from C. comosa induced osteoblastic cell proliferation a differentiation through ERalpha-, Akt-, and GSK-3beta-dependent activation of beta-catenin signaling.
Literature	GENERIF_SUMMARY	Characterize an estradiol-regulated interaction of NPPA with ERalp cardiomyocytes.
Literature	GENERIF_SUMMARY	findings unveil a potential tumor-suppressor role of ERbeta in ovar carcinogenesis, which could be of potential clinical relevance for t selection of the most appropriate treatment for patients
Literature	GENERIF_SUMMARY	Expression of ER-a is positively correlated with positive rates of AM antibody, Sp100 and gP210 in tissues of primary biliary cirrhosis p
Literature	GENERIF_SUMMARY	The predictive capacity of 32 genome wide association studies-de common variants for breast cancer risk, alone and in combination classical risk factors, and for tumours with different hormone rece status, was investigated.
Literature	GENERIF_SUMMARY	Overexpression of antizyme (AZ), an ornithine decarboxylase (ODC inhibitory protein, suppressed ERalpha expression, suggesting that plays an important role in regulation of ERalpha expression.
Literature	GENERIF_SUMMARY	Immunohistochemical evaluation of TRAP1 together with ERalpha provides significant prognostic information. TRAP1 alone is signifi associated with chemotherapy response and overall survival.
Literature	GENERIF_SUMMARY	Data suggest that estradiol counters oxidized LDL by restoring dimethylargininase activity and preventing loss of eNOS (nitric oxi synthase 3) in arterial endothelium acting through ERalpha (but no ERbeta or G protein-coupled estrogen receptor).
Literature	GENERIF_SUMMARY	The aim of this study was to assess the expression of estrogen rec alpha and beta in paratesticular tissues in a group of boys with and without cryptorchidism.
Literature	GENERIF_SUMMARY	The genomic and transcriptional cross talk between AR and ERalp signaling in a luminal breast cancer cell line, was investigated.
Literature	GENERIF_SUMMARY	Immunohistochemistry revealed very low expression of ERalpha, w all informative tumors expressed ERbeta1 and ERbeta2 in parathyr

Main Category	Category	Terms
		adenomas.
Literature	GENERIF_SUMMARY	Data indicate that the persistent organic pollutants (POPs)-related receptors estrogen-, androgen- and aryl hydrocarbon-receptortransactivities can be used as effect biomarkers.
Literature	GENERIF_SUMMARY	the 5'-UTRs of the E isoforms post-transcriptionally control ERalph expression mainly through translational repression
Literature	GENERIF_SUMMARY	The presence of ERalpha in Sertoli and Leydig cells in normal hum testis demonstrated in this study suggests that estrogens may aff testicular function.
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha, RBCK1, and protein kinase C beta 1 coop regulate estrogen receptor-alpha gene expression.
Literature	GENERIF_SUMMARY	The relationship between ERRgamma and ERalpha status could be predictive marker for the treatment of uterine endometrial cancer.
Literature	GENERIF_SUMMARY	Data suggest a role for the ERalpha, ERbeta and GPER in the estrog disruption of acinar formation.
Literature	GENERIF_SUMMARY	Data suggest that 4-methyl-2,4-bis(4-hydroxyphenyl)pent-1-ene [M key contacts with amino acids in estrogen receptors ERalpha and
Literature	GENERIF_SUMMARY	Estrogen receptor alpha signaling regulates breast tumor-initiating by down-regulating miR-140 which targets the transcription factor
Literature	GENERIF_SUMMARY	ERalpha binds to the DEK promoter.
Literature	GENERIF_SUMMARY	Analyses of femoral neck BMD also supported association with SN WNT16 and ESR1/C6orf97
Literature	GENERIF_SUMMARY	Polymorphic variants of ESR1 gene with TA-repeats could be relate dysfunction of the ovaries and infertility, which requires additional studies.
Literature	GENERIF_SUMMARY	the XBAI polymorphism in the estrogen receptor alpha gene was observed at a higher mutated homozygous frequency in women w dense breasts
Literature	GENERIF_SUMMARY	Interaction between genetic and environmental factors may help to explain nonreplication in genetic studies of hypospadias.
Literature	GENERIF_SUMMARY	Variants on ESR1 gene is associated with prostate cancer.
Literature	GENERIF_SUMMARY	findings show that ERalpha methylation is correlated with poor pro subtypes of breast tumors in Iranian patients and may play an imp role in pathogenesis of the more aggressive breast tumors
Literature	GENERIF_SUMMARY	Study sequence rules of estrogen receptor alpha-DNA interactions dual polarization interferometry and computational modeling.
Literature	GENERIF_SUMMARY	genetic association study in population in China
Literature	GENERIF_SUMMARY	p allele of the ESR1 Pvu II polymorphism is a potential biomarker o logical memory decline among aging women.
Literature	GENERIF_SUMMARY	Breast cancer mRNA levels of ESR1 and HER2 predict response to neoadjuvant chemotherapy and are significantly associated with lo term outcome.
Literature	GENERIF_SUMMARY	DOX could modulate the expression of GCS through the Sp1 site o

Main Category	Category	Terms
		promoter in ERalpha-positive breast cancer cells
Literature	GENERIF_SUMMARY	the polymorphic genotypes of BGLAP, ER1, Col1A1 and CALCR are found to be associated with osteoporosis in a single form but four associated in combined forms.
Literature	GENERIF_SUMMARY	these results demonstrate a novel role for cohesin in estrogen-med regulation of MYC and the first evidence that cohesin plays a role i ERalpha binding
Literature	GENERIF_SUMMARY	In Brazilian population evaluated results have demonstrated that th genetic variation in ESR1 gene (PvuII polymorphism) is associated risk.
Literature	GENERIF_SUMMARY	these data suggest that ZNF131 inhibits estrogen signaling by acti an ERalpha-co-repressor.
Literature	GENERIF_SUMMARY	The ESR1 polymorphism rs3798573 A/G was associated with risk endometriosis and infertile endometriosis in Han women from cen China.
Literature	GENERIF_SUMMARY	Data suggest that CAC1 (cullin domain-containing protein 1), in association with LSD1 (lysine-specific histone demethylase 1), fun as an ERalpha corepressor and may be involved in drug resistance breast cancer.
Literature	GENERIF_SUMMARY	Data indicate that estrogen receptor alpha, estrogen receptor beta, progesterone receptor and androgen receptor may be partly involv gastric carcinogenesis but their clinicopathological and prognostic significance appears to be limited.
Literature	GENERIF_SUMMARY	ESR1 variants have no effect on cardiorespiratory and metabolic variables, while HRV indices are influenced by aerobic capacity and in healthy women.
Literature	GENERIF_SUMMARY	ESPR1 polymorphisms are associated with adolescent behavior st
Literature	GENERIF_SUMMARY	mutant p53 protein binds to the promoter of ESR1 through direct interaction with HDAC1 and indirect interaction with DNMT1, MeCP proteins in the ER-negative MDA-MB-468 cells.
Literature	GENERIF_SUMMARY	Point mutants in the estrogen receptor alpha Galphal-binding dom provide genetic segregation of nonnuclear from nuclear function.
Literature	GENERIF_SUMMARY	ESR1 PvuII (C>T) polymorphism places pre-menopausal breast can women at risk for breast cancer.
Literature	GENERIF_SUMMARY	genetic association studies in population of women in Brazil
Literature	GENERIF_SUMMARY	Data indicate there was a HIF-1alpha-dependent cancer stem cell (increase in ER-alpha-positive cancers following hypoxic exposure.
Literature	GENERIF_SUMMARY	The findings reveal a tumor suppressive role of hsa-miR-196a2 thr regulation of TP63 by ERalpha and/or ERK2 signaling.
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha36 may be an important biomarker for diag prognostication, and treatment choice in endometrial carcinoma.
Literature	GENERIF_SUMMARY	Identifying novel kinases which modulate Estrogen Receptor alpha activity is relevant to therapeutics. LATS2 modulates Estrogen Rec alpha-regulated gene transcription, through direct and/or indirect interactions with Estrogen Receptor alpha.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The frequency of ESR1 amplification in breast cancer is low when measured by FISH in large sections. ESR1 gain and amplification i cancer may be associated with older age and poorer outcomes.
Literature	GENERIF_SUMMARY	we identified Caspase 3, BAG1 and BAG3 as key targets of ERalph neuronal cells that may play a role in ERalpha-mediated neuroprot
Literature	GENERIF_SUMMARY	Our results provide strong evidence to support that the common polymorphism near the ESR1 gene, rs2046210, is associated with increased risk of breast cancer in Asian and European populations in Africans.
Literature	GENERIF_SUMMARY	ESR1 gene and insulin resistance remission are associated with se uric acid decline for severely obese patients undergoing bariatric s
Literature	GENERIF_SUMMARY	the first immunohistochemical study of the MIER1alpha protein expression pattern in human tissues, is reported.
Literature	GENERIF_SUMMARY	The expression of deltaEF1 is up-regulated by 17beta-estradiol (E2 MCF-7 cells in an ER-alpha-dependent manner.
Literature	GENERIF_SUMMARY	Data indicate that the prognostic value of Akt2 increases with high oestrogen receptor (ER) expression, motivating further mechanisti studies on the role of Akt2 in ER+ breast cancer.
Literature	GENERIF_SUMMARY	WNT10B/beta-catenin signalling induces HMGA2 and proliferation metastatic breast cancer tumours devoid of ERalpha, PR and HER2 expression.
Literature	GENERIF_SUMMARY	Lack of estrogen receptor-alpha is associated with epithelial-mesenchymal transition and PI3K activations resulting in endomet carcinoma.
Literature	GENERIF_SUMMARY	Epigallocatechin gallate downregulates ER-alpha and explains the inhibitory action of EGCG on the proliferation of ER+ PR+ cancer ce tested.
Literature	GENERIF_SUMMARY	The expression of ER protein in superficial and deep ectopic tissue significantly lower than in eutopic tissues of adenomyosis.
Literature	GENERIF_SUMMARY	Short-term folate supplementation in physiological doses decreas plasma homocysteine but has no effect on ESR1 and MLH1 methy in colonic mucosa of individuals with adenoma.
Literature	GENERIF_SUMMARY	Genetic predispositions and the short- and long-term effects of ho therapy on bone mineral density in girls with functional hypothalan amenorrhoea.
Literature	GENERIF_SUMMARY	Taken together, our findings offer evidence that in human breast ca RhoB acts as a positive function to promote expression of ERa and a manner correlated with cell proliferation.
Literature	GENERIF_SUMMARY	The study indicates a positive link between hNIS and ER expressio breast cancer.
Literature	GENERIF_SUMMARY	Moreover, subgrouping of Luminal A-type breast cancer samples according to ERE activity revealed that ERalpha mRNA expression correlated with ER target gene mRNA expression in the high-, but n low-, ERE-activity group.
Literature	GENERIF_SUMMARY	The article reviews the relationship between polymorphism of sex hormone-related genes (CYP17, CYP19, CYP1A1, COMPT, ERalpha

Main Category	Category	Terms
		ERbeta and others).
Literature	GENERIF_SUMMARY	Estrogen receptor alpha mediates proliferation of osteoblastic cell stimulated by estrogen and mechanical strain, but their acute dow regulation of the Wnt antagonist Sost is mediated by estrogen rece beta
Literature	GENERIF_SUMMARY	The prolyl isomerase Pin1 acts synergistically with CDK2 to regula basal activity of estrogen receptor alpha in breast cancer.
Literature	GENERIF_SUMMARY	Response to trastuzumab is correlated to HER2 mRNA levels only ESR1-positive tumors. This study adds further evidence to the diffe biology of both subsets within the HER2-positive group.
Literature	GENERIF_SUMMARY	our results suggest that the detection of ESR1/PGR/ERBB2 mRNA by RT-QPCR is a better approach for subtyping breast cancer and predicting the prognosis.
Literature	GENERIF_SUMMARY	HDAC3 is necessary for ERalpha mRNA stability, and is involved in estrogen-dependent proliferation of ERalpha-positive tumors.
Literature	GENERIF_SUMMARY	ERalpha is a positive regulator of Runx2-I transcription.
Literature	GENERIF_SUMMARY	In this study ribosome display has been used to reassess the estro receptor alpha and beta for new and enlarged peptide recognition leading to the discovery of a potent and highly evolved PXLXXLLX binding consensus.
Literature	GENERIF_SUMMARY	polymorphisms of rs3798759 may be a risk factor for female patie with schizophrenia, and haplotypes C-A-G and C-G-G may be risk fa for schizophrenia.
Literature	GENERIF_SUMMARY	Data show that ER-alpha (nuclear) was detected in only 4 (9 %) of and not statistically significant, and presence of ER-beta (nuclear) associated with favourable survival, whereas the presence of ER-b (cytoplasm) was associated with a poor survival.
Literature	GENERIF_SUMMARY	The suppression of ERalpha by the upregulation of let-7 expressior be a promising strategy.
Literature	GENERIF_SUMMARY	ESR1 is associated with a distinct tumor subtype defined by estrog receptor-negative status in male breast cancer patients in Italy.
Literature	GENERIF_SUMMARY	meta-analysis of 10 studies involving Chinese Han population sug the CC genotype of the ESR1 rs2234693 polymorphism is significa associated with increased risk of coronary heart disease(CHD)in n only; there was no evidence of a significant association between th rs9340799 polymorphism and CHD risk
Literature	GENERIF_SUMMARY	ER alpha and ER alpha-isophorm 36 estrogenic transcriptional effe opposite.
Literature	GENERIF_SUMMARY	In female mice, perinatal exposure to low dose of BPA resulted in increased ERalpha mRNA expression and high fat diet resulted in e ERalpha expression among females that were exposed to the high of BPA early in life. There were no changes in males.
Literature	GENERIF_SUMMARY	There is an association between ESR1 polymorphisms and some A parameters such as the level of E2 on the day of hCG administratio the quality of the embryos.
Literature	GENERIF_SUMMARY	Haplotypes consisting of these two polymorphisms in ESRalpha m

Main Category	Category	Terms
		strongly associated with schizophrenia
Literature	GENERIF_SUMMARY	MCF-7/tamoxifen resistant cells possess overexpressed ER-alpha EGFR, but down-regulated ER-alpha 66.
Literature	GENERIF_SUMMARY	Glutathione peroxidase, Complex V, Complex III, Complex II, Compl AKT, SAPK, and ERalpha were significantly and positively correlate ERalpha/ERbeta ratio
Literature	GENERIF_SUMMARY	ESR1 single nucleotide polymorphism is a predictor of diabetes m type 2 in hypogonadal men.
Literature	GENERIF_SUMMARY	IGFBP-2 not only modulates IGFs and directly regulates PTEN but has a role in maintaining ER-alpha expression.
Literature	GENERIF_SUMMARY	PvuII polymorphism (rs2334693) is associated with diminished ov reserve.
Literature	GENERIF_SUMMARY	ligand dynamics directs unique signaling pathways and reveals a r of the DBD in allosteric control of ERalpha-mediated signaling
Literature	GENERIF_SUMMARY	Data suggest that the ERalpha status of mammary cells dictates w KISS1R may be a novel clinical target for treating breast cancer metastasis.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha regulatory mechanisms in breast cancer g expression.
Literature	GENERIF_SUMMARY	As Pin1 impacts both ERalpha protein levels and transactivation fu these data implicate Pin1 as a potential surrogate marker for predi outcome of ERalpha-positive breast cancer
Literature	GENERIF_SUMMARY	an intronic variant in ESR1 (rs9322336) was associated with increa risk of musculoskeletal toxicity-related exemestane discontinuatio breast cancer treatment
Literature	GENERIF_SUMMARY	Results indicate that elevated ER-alpha36, a variant of ER-alpha, concentration is one of the mechanisms by which ER-positive brea cancer cells escape tamoxifen (TAM) therapy.
Literature	GENERIF_SUMMARY	Data suggest that 17 beta-estradiol enhances the the glucose upta activates PI3K/Akt signaling pathway, leading to translocation of g transporter 4 to the plasma membrane in an ERalpha-dependent m
Literature	GENERIF_SUMMARY	present study reveals that neither PvuII nor XbaI polymorphisms in estrogen receptor alpha gene are associated with onset and progr of puberty
Literature	GENERIF_SUMMARY	study showed association between ESR polymorphisms and cogni decline or specific areas in cognitive profile.
Literature	GENERIF_SUMMARY	ER+ /PR+ /TFF3+ /IMP3- immunoprofile distinguishes endometrio serous and clear cell carcinomas of the endometrium.
Literature	GENERIF_SUMMARY	Wanted to identify estrogen receptor alpha (ERalpha) interacting p in Tamoxifen treated MCF7 cells. Using a GST-pull down assay wit ERalpha ligand-binding domain and MS-based proteomics approa identified Profilin1 as a novel ERalpha interacting protein.
Literature	GENERIF_SUMMARY	findings show ESR1 is an independent positive prognostic factor f survival in early-stage non-small cell lung cancer patients; patients ESR1 high tumors did not benefit from adjuvant chemotherapy

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ERa and pAR210 are differentially expressed in hormone-naive (HN castration-resistant prostate cancers and that high levels of AR, ER pAR210 in HN prostate cancer correlate with a poor clinical outcor
Literature	GENERIF_SUMMARY	Study provides further evidence of functional interplay among NRF and ERalpha signaling pathways through altered p300 recruitment NRF2-regulated target genes.
Literature	GENERIF_SUMMARY	Findings suggest that estradiol and T regulate arginine-vasopressi expression in SH-SY5Y cells through ERs, exerting a stimulatory ac ERalpha and an inhibitory action via ERbeta.
Literature	GENERIF_SUMMARY	The scanning and typing of ERalpha polymorphism in girls with precocious puberty has uncovered several potentially meaningful polymorphisms.
Literature	GENERIF_SUMMARY	Impairment of ERalpha association to ubiquitinated species strong affects E2-induced ERalpha transcriptional activity.
Literature	GENERIF_SUMMARY	PvuII (C>T) and rs3020314 (C>T) polymorphisms in estrogen recep may be risk factors for endometrial cancer
Literature	GENERIF_SUMMARY	these findings demonstrate that ERalpha is a major driver of growt tamoxifen-resistant breast cancer cells
Literature	GENERIF_SUMMARY	Results suggest that the polymorphisms of ESR1, ESR2 and FSHR did not result in differences in the genotype distribution when obes polycystic ovary syndrome population were compared with the nor population
Literature	GENERIF_SUMMARY	The results suggest that the G-quadruplex formation in the coding of the hERalpha mRNA impacts folding and proteolysis of hERalph protein by slowing down or temporarily stalling the translation elor
Literature	GENERIF_SUMMARY	ER, COX-2 and aromatase are expressed in endometrial neoplasms suggesting their involvement in tumor development and progressio
Literature	GENERIF_SUMMARY	ESR1 can bind to genomic locations with closed chromatin, induci chromatin opening in an estradiol-dependent manner.
Literature	GENERIF_SUMMARY	Variants in ESR1 and PARP1 modify effect of passive smoking on l cancer risk.
Literature	GENERIF_SUMMARY	Data indicate that the menstrual cyclic expression of bcl-2 could b controlled by either direct binding of estrogen receptor alpha (ERal estrogen-responsive element (ERE) in the c-Jun promoter or the interaction of ERalpha with c-Jun.
Literature	GENERIF_SUMMARY	ERalpha gene XbaI polymorphism may modify individual susceptib breast cancer in a Portuguese population.
Literature	GENERIF_SUMMARY	Expression of the progesterone and Estrogen Receptors can signif affect therapeutic decisions in selected patients with endometroio endometrial cancer.
Literature	GENERIF_SUMMARY	A total of 10 markers in the ESR1 gene show association with migr a Norfolk Island pedigree.
Literature	GENERIF_SUMMARY	Mutation of the penultimate Threonine of ERalpha (T594) strongly enhances estradiol-dependent dimerization and transactivation.
Literature	GENERIF_SUMMARY	Expression of JMJD2A in infiltrating duct carcinoma is higher than

Main Category	Category	Terms
		fibroadenoma, and is associated with ARHI, p53 and ER
Literature	GENERIF_SUMMARY	Studies indicate that the biosensing strategy is based on the prote DNA duplex from exonuclease III (Exo III)-mediated digestion by sp binding of estrogen receptor (ER) to its DNA response element.
Literature	GENERIF_SUMMARY	Data show that the expressions of ER-alpha, PR, LIF, VEGF, iNOS ar in fallopian tube and chorionic villi of tubal pregnancy were not alte exposure to levonorgestrel emergency contraception.
Literature	GENERIF_SUMMARY	Lower expression of ER-alpha66 is associated with the developme endometrial hyperplasia in polycystic ovarian syndrome patients.
Literature	GENERIF_SUMMARY	The ESR1 gene was identified as a candidate tumor suppressor ge HCC and further validation is warranted.
Literature	GENERIF_SUMMARY	Estradiol and IGF-I induced the rapid association of ER to IGF-IR.
Literature	GENERIF_SUMMARY	Hepatic nuclear ER-alpha expression percentage (staining index) is in patients with nonalcoholic steatohepatitis when compared to si steatosis and normal liver.
Literature	GENERIF_SUMMARY	Negative estrogen receptor status was strongly associated with locoregional recurrence after accelerated partial breast irradiation early-stage invasive breast cancer.
Literature	GENERIF_SUMMARY	show gender-specific transcriptional control of UGT1A genes in jej and colon, which is repressed by ERalpha and the recruitment of H and 2 to the UGT1A promoter sequence in females
Literature	GENERIF_SUMMARY	overexpression of ERalpha mediates apoptosis in ERalpha-negativ Hep3B cells by the binding of ERalpha to SP1 protein.
Literature	GENERIF_SUMMARY	The aim of this study was to evaluate the effect of two polymorphi the estrogen receptor alpha, PvuII and XbaI, on the development of prostate cancer within Slovak population.
Literature	GENERIF_SUMMARY	The risk factors for breast cancer were more prevalent in women w high breast density but without significant associations with ESR1 polymorphisms.
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha mediates multidrug resistance associated protein 3 expression following 17alpha-ethynylestradiol administra
Literature	GENERIF_SUMMARY	The PvuII polymorphism in the ESR1 gene may be a risk factor for leiomyoma. [Meta-analysis]
Literature	GENERIF_SUMMARY	Estrogen receptor alpha expression is upregulated in the surroundi ovarian tissues of patients with endometrioma compared to contro
Literature	GENERIF_SUMMARY	There was no association between the PvuII polymorphism and str severity and short-term outcome in the studied female stroke popu
Literature	GENERIF_SUMMARY	This work not only identifies these late-activated genes as bona fid ERalpha targets but describes a novel mechanism for their periodi activation.
Literature	GENERIF_SUMMARY	Estrogen receptors alpha and beta and their key coregulators, SRC RIP140, generate overlapping as well as unique chromatin-binding transcription-regulating modules.
Literature	GENERIF_SUMMARY	The two ESR1 SNPs, Pvu II and Xba I, are not commonly associate

Main Category	Category	Terms
		premature ovarian failure in Serbian women and may not contribut genetic basis of the condition.
Literature	GENERIF_SUMMARY	A significant difference in the allele frequencies of XbaI.
Literature	GENERIF_SUMMARY	Data suggest that ESRA activity is modulated in breast cancer via mechanisms
Literature	GENERIF_SUMMARY	Data suggest that FOXP1 (forkhead box P1) and FOXA1 (forkhead A1) are primary ESR1 target genes and critical transcription factor regulate ESR1 activity in breast cancer. [REVIEW]
Literature	GENERIF_SUMMARY	Data indicate that in breast cancers with no lymphatic metastasis, expression of HtrA3 was lower in patients with estrogen receptor (and progesterone receptor (PR)-positive tumors.
Literature	GENERIF_SUMMARY	The recruitment of GR to ERalpha-binding regions (EBRs) requires assistance from ERalpha and FOXA1 and is facilitated by AP1 binc within the EBRs.
Literature	GENERIF_SUMMARY	These results clarify the relationship between ER-alpha and PDZK1 propose a direct relationship between PDZK1 and IGF1R, and ident novel oncogenic activity for PDZK1 in breast cancer.
Literature	GENERIF_SUMMARY	Frequencies of p and x alleles in ER-alpha gene PvuII and XbaI polymorphisms were more prevalent among fertile men compared infertile oligoasthenoteratozoospermic men.
Literature	GENERIF_SUMMARY	the expression of ERalpha prevents cells from apoptosis
Literature	GENERIF_SUMMARY	Glucose-regulated protein 94 is a downstream effector of ER-alpha mediated oestrogen signalling, and may be involved in ER-alpha36 function during gastric carcinogenesis.
Literature	GENERIF_SUMMARY	ERalpha overexpression significantly inhibited cell growth and proliferation, blocked cell entry into the G1/G0 phase and promote apoptosis.
Literature	GENERIF_SUMMARY	Pumpkin seed extract downregulates ER-alpha and upregulates progesterone receptor in the MCF-7 breast cancer cell line.
Literature	GENERIF_SUMMARY	Specific binding of ER (alpha and/or beta) to ERETTR was demons by EMSA and supershift assays confirmed the binding to ERalpha ERbeta
Literature	GENERIF_SUMMARY	Mediator of ERBB2-driven cell motility (MEMO) promotes extranuc estrogen receptor signaling involving the growth factor receptors I and ERBB2.
Literature	GENERIF_SUMMARY	study suggest that SNPs in sex steroid related genes, known to aff gene expression (rs2747648 in ESR1) and enzymatic activity (Leu8 SRD5A2), seem to be associated with ALTs in a general population
Literature	GENERIF_SUMMARY	LXRbeta has nonnuclear function in endothelial cell caveolae/lipid that entails crosstalk with estrogen receptor alpha.
Literature	GENERIF_SUMMARY	Retinoic acid receptor alpha may be a novel therapeutic target and predictive factor for estrogen receptor alpha-positive breast cance patients treated with adjuvant tamoxifen.
Literature	GENERIF_SUMMARY	Data indicate that receptors (ERs) ERalpha and ERbeta showed significantly decreased mRNA and protein expression levels in

Main Category	Category	Terms
		endometrial carcinoma as compared to normal endometrium.
Literature	GENERIF_SUMMARY	Transcription factor Ets1 cooperates with estrogen receptor alpha stimulate estradiol-dependent growth in breast cancer cells and tu
Literature	GENERIF_SUMMARY	Results suggest a possible role of ERalpha in male infertility.
Literature	GENERIF_SUMMARY	Genotypes of these single nucleotide polymorphisms of ESR1 and may help us predict the osteoporosis risk in menopausal women.
Literature	GENERIF_SUMMARY	Our results support a role for ESR1 in the etiology of PPD, possibly through the modulation of serotonin signaling.
Literature	GENERIF_SUMMARY	ESR1 gene deletion and amplification do not constitute per se prog markers, instead they can be classified to distinct prognostic grou according to their protein-mediated functional status.
Literature	GENERIF_SUMMARY	the results show that the cervical malignant cells tend to loss the E alpha but maintain the ER-beta actively expressed
Literature	GENERIF_SUMMARY	motivate a cooperation model of ESR1 and SP1 in regulating cell c arrest, and a hypothesis that their overactivation in non-hyperdiplo multiple myeloma disrupts proper regulation of cell cycle arrest
Literature	GENERIF_SUMMARY	ESR1 rs2234693 and rs9340799 were not significantly associated the risk of decline on any of the cognitive tasks.
Literature	GENERIF_SUMMARY	MEK5/ERK5 signaling suppresses estrogen receptor expression ar promotes hormone-independent tumorigenesis.
Literature	GENERIF_SUMMARY	ESR1 gene dosage in the entire group of breast cancer patients co with higher T stage and lymph node involvement.
Literature	GENERIF_SUMMARY	1p11-rs11249433 polymorphism is involved in breast cancer susceptibility and confer its effect primarily in estrogen receptor-p and progesterone receptor-positive tumors.
Literature	GENERIF_SUMMARY	Data indicate that ZNF703 overexpression in MCF-7 breast cancer activated the Akt/mTOR signaling pathway, downregulated ERalph reduced the antitumor effect of tamoxifen.
Literature	GENERIF_SUMMARY	Significantly increased estrogen receptor level was found in patien juvenile angiofibromas.
Literature	GENERIF_SUMMARY	Findings suggest a novel pathway in which ERalpha signaling regu miR-190a expression, causing inhibition of PAR-1 expression, corre with inhibition of cancer metastasis in breast cancer.
Literature	GENERIF_SUMMARY	chronic cadmium exposure stimulates the expression of SDF-1 by the molecular interactions between ERalpha, c-jun and c-fos.
Literature	GENERIF_SUMMARY	Expression of HER4 JM-a/CYT2 isoform relate to survival of bladd cancer patients only if the tumors do not express ESR1.
Literature	GENERIF_SUMMARY	The expression of the estrogen receptor was found in 18% of saliv duct carcinoma patients.
Literature	GENERIF_SUMMARY	ESR1 amplification is independent of known clinicopathological fa related to poor prognosis and PTEN, p53, HER2, MLH1, and ARID1A protein expression, suggesting ESR1 amplification may be an early in endometrial carcinoma development.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Knockdown of TFAP2C or RET inhibited activation of ERK and AKT MCF-7 cells. Knockdown of TFAP2C, which controls ESR1 (estroge receptor) and RET, had a greater effect on cell growth than either R ESR1 alone.
Literature	GENERIF_SUMMARY	The estrogen receptor alpha is the key regulator of the bifunctiona FoxO3a in breast cancer motility and invasiveness.
Literature	GENERIF_SUMMARY	The GFP-GRER exhibited cytoplasmic localization in the absence o ERalpha ligands and translocated to the nucleus in response to stimulation with ERalpha agonists and antagonists
Literature	GENERIF_SUMMARY	data suggest that signal peptide CUB domain (SCUBE2), like in bre cancer, associates with estrogen receptor alpha and might have a potential as prognostic or predictive marker in endometrial cancer
Literature	GENERIF_SUMMARY	Did not find any significant correlation between ER-alpha expressio histopathological and biochemical findings in chronic hepatitis C p
Literature	GENERIF_SUMMARY	Estrogen-receptor-positive PDXs were associated with ESR1 ligand binding-domain mutations, gene amplification, or an ESR1/YAP1 translocation.
Literature	GENERIF_SUMMARY	anti-ERalpha antibody serum immunoreactivity was detected in 42 patients with systemic sclerosis.
Literature	GENERIF_SUMMARY	CTCs frequently lack ER expression in metastatic breast cancer pa with ER-positive primary tumors and show a considerable intra-pat heterogeneity, which may reflect a mechanism to escape endocrin therapy.
Literature	GENERIF_SUMMARY	The loss of TBK1 expression parallels the loss of ERalpha express turn helping drive an aggressive breast cancer phenotype.
Literature	GENERIF_SUMMARY	Mutations of putative estrogen receptor alpha and/or nuclear facto binding sites markedly reduced promoter activity.
Literature	GENERIF_SUMMARY	RhoA and RhoC play different, but clear, roles in ERalpha signaling breast cancer cells.
Literature	GENERIF_SUMMARY	these results demonstrate that knockdown of HOXA7 in MCF7 cell significantly decreased cell proliferation and ERa expression.
Literature	GENERIF_SUMMARY	ERalpha and ERbeta levels were inversely correlated, and the downregulation of ERalpha and upregulation of ERbeta may indica unfavorable prognosis of esophageal squamous cell carcinoma
Literature	GENERIF_SUMMARY	Our results demonstrate the predictive value of the RS for clinical response to neoadjuvant exemestane therapy in postmenopausal with estrogen receptor-positive breast cancer.
Literature	GENERIF_SUMMARY	Data suggest that SMYD2-mediated estrogen receptor alpha (ERal protein methylation and p300/cAMP response element-binding pro binding protein-dependent ERalpha acetylation play an important r the estrogen-induced gene expression profiles.
Literature	GENERIF_SUMMARY	similarly to the situation for the ovary, a diagnostic panel of HNF-1 and ER may be considered for separating clear cell carcinoma fron endometrioid and serous carcinoma of the endometrium
Literature	GENERIF_SUMMARY	Meta-analytically combined evidence from the present genotyping literature showed that rs2295193 polymorphism in ESR1 is not a m

Main Category	Category	Terms
		genetic susceptibility factor in Anorexia Nervosa.
Literature	GENERIF_SUMMARY	Progesterone receptor and estrogen receptor alpha expression are discordant in different distant breast cancer metastases.
Literature	GENERIF_SUMMARY	Estradiol mediated phosphorylation of ESR1 is mechanism to pote transcription from an estrogen response element.
Literature	GENERIF_SUMMARY	Hbo1 modulated transcription by estrogen receptor-alpha. Hbo1 contributes to degradation of ER-alpha by ubiquitination through ly 48.
Literature	GENERIF_SUMMARY	the present study demonstrated that both ERa and ERbeta were su to inhibit PPARgama and provide a valuable therapeutic option for treatment of Hepatocellular carcinoma patients.
Literature	GENERIF_SUMMARY	Our results provide additional insights into the opposing roles of th and FLJ43663 genes in breast cancer onset and progression.
Literature	GENERIF_SUMMARY	ERalpha XbaI polymorphism is associated with an increased risk o Alzheimer's disease in Chinese population
Literature	GENERIF_SUMMARY	Genetic variants in ERalpha were not associated with metabolic syndrome or its components, in a group of 250 Lebanese Caucasia elderly participants, a group with a high prevalence of metabolic syndrome.
Literature	GENERIF_SUMMARY	It was concluded that estrogenic signals exert malignant transforn capacity of cancer cells, exclusively through non-genomic regulatio female germ cell tumors.
Literature	GENERIF_SUMMARY	A significant effect of ESR1 genotypes was observed for the donep treated alzheimer patients, among which those carrying at least or of P and X alleles showed a significantly lower cognitive decline th noncarriers.
Literature	GENERIF_SUMMARY	No association was found between ESR1 XbaI or ESR1 PvuII single nucleoptide polymorphism and idiopathic scoliosis in Caucasian fe
Literature	GENERIF_SUMMARY	Immunohistochemical expression of ERalpha, ERbeta, and TFF1 in and II ovarian tumors.
Literature	GENERIF_SUMMARY	marker of main subtypes of invasive breast cancer in China
Literature	GENERIF_SUMMARY	It mediates direct and indirect regulations in bone metabolism.(rev
Literature	GENERIF_SUMMARY	Data show that estrogen receptor 1 (ESR1) mRNA as assessed by represents a reliable method for detecting ESR1 expression in non cell lung cancer (NSCLS) and that ESR1 expression is an independ prognostic factor in metastatic NSCLC.
Literature	GENERIF_SUMMARY	Activating mutations in ESR1 are a key mechanism in acquired end resistance in breast cancer therapy.
Literature	GENERIF_SUMMARY	Ligand-binding domain mutant forms of ESR1 mediate clinical resi to hormonal therapy.
Literature	GENERIF_SUMMARY	The discovery of sub-isoforms of ER alpha and beta has further complicated our understanding of how the interaction between ER its ligands contribute to the development of disease
Literature	GENERIF_SUMMARY	IL-6 secreted by ovarian cancer cells may contribute to the refracto

Main Category	Category	Terms
		of these cells to tamoxifen via the estrogen receptor alpha isoform SRC-1 (NCOA1).
Literature	GENERIF_SUMMARY	This study provides a deeper understanding of ERbeta's role in bre tumors that coexpress both ERbeta and ERalpha and supports an emerging bi-faceted role of ERbeta.
Literature	GENERIF_SUMMARY	Experiments in breast cancer cells indicated constitutive, ligand-independent transcriptional activity of the D538G estrogen recepto overexpression of it enhanced proliferation and conferred resistan tamoxifen.
Literature	GENERIF_SUMMARY	The expression of both ERalpha and ERbeta in human fetal brown adipose tissue suggests a role for estrogen in its development, pri via ERalpha.
Literature	GENERIF_SUMMARY	NgBR is highly associated with ESR1-positive/HER2-negative brea cancer.
Literature	GENERIF_SUMMARY	Data indicate that the Cyclopia extracts displayed estrogen recepto (ERalpha) antagonism and estrogen receptor beta (ERbeta) agonis when ER subtypes were expressed separately, however, when co-expressed only agonism was uniformly observed.
Literature	GENERIF_SUMMARY	These studies thus identified PTPH1 as a novel ER phosphatase a further demonstrate a therapeutic potential of enhancing breast ca sensitivity to antiestrogens through dephosphorylating ER by PTPH
Literature	GENERIF_SUMMARY	The potential association of spontaneous abortion with the ESR1 g variations (PvuII and XbaI) in fetal tissue, was studied.
Literature	GENERIF_SUMMARY	Variations in the ERalpha gene affected the BMI in women with Hig breast density, who already are at increased risk for Breast Cancer.
Literature	GENERIF_SUMMARY	Studies indicate that changes in oestrogen receptor (ER), progeste receptor (PgR) and epidermal growth factor receptor 2 (HER2) exp may occur during the natural history of breast cancer.
Literature	GENERIF_SUMMARY	Neither the haplotype nor the three ESR1 SNPs were associated w BN, AN, or restricting type AN.
Literature	GENERIF_SUMMARY	Data suggest that ESR1 (estrogen receptor 1) regulates cell cycle b suppressing expression of p53/p21 (tumor protein p53/cyclin-dep kinase inhibitor 1A protein) and up-regulating proliferating cell nuc antigen (PCNA).
Literature	GENERIF_SUMMARY	NCOA3 is a selective co-activator of estrogen receptor alpha-medi transactivation of PLAC1.
Literature	GENERIF_SUMMARY	calretinin immunoexpression and the hormonal status for estroge receptors (ER), progesterone receptors (PR), and ERRB2, in 33 cas cutaneous metastases of ductal breast carcinoma, as well as in se primary cutaneous apocrine tumors, were studied
Literature	GENERIF_SUMMARY	serum ERalpha-mediated bioactivity is associated with increased b cancer risk in post-menopausal women
Literature	GENERIF_SUMMARY	Adiponectin modulates breast cancer cell growth, providing evider the cell-type dependency of adiponectin action in relationship to E status.
Literature	GENERIF_SUMMARY	ER-alpha expression in uterine serous carcinoma is associated wit

Main Category	Category	Terms
		advanced stage tumors and a significantly worse prognosis
Literature	GENERIF_SUMMARY	results demonstrate that, while ER signaling acts synergistically wi Wnt3A in promoting osteogenic differentiation, Wnt3A may crossta ER signaling by up-regulating ERalpha expression and down-regula ERbeta expression in MPCs.
Literature	GENERIF_SUMMARY	PRKCA activity suppresses the level of ER-alpha expression by inhi c-JUN phosphorylation.
Literature	GENERIF_SUMMARY	ER and HER2 are useful markers in core needle biopsies, providing reliable base for preoperative decisions in neoadjuvant chemother and may be omitted on surgical specimens with certain precautior exceptions.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha in cancer-associated fibroblasts suppress prostate cancer invasion via modulation of thrombospondin 2 and metalloproteinase 3.
Literature	GENERIF_SUMMARY	genistein treatment produces cell cycle arrest and an improvemen mitochondrial functionality in T47D cells with a low ERa/ERb ratio, in MCF-7
Literature	GENERIF_SUMMARY	PES1 may play important role in the progression of ovarian cancer inversely regulating the ERalpha and ERbeta expression.
Literature	GENERIF_SUMMARY	PHB lies downstream of ERalpha and mediates estrogen-depende Paclitaxel resistance signaling cascades
Literature	GENERIF_SUMMARY	ESR1 polymorphisms are associated with susceptibility to breast c
Literature	GENERIF_SUMMARY	ESR1-positive epithelial ovarian cancer tissues had lower expressi lnRNAs compared to ESR1-negative tissues.
Literature	GENERIF_SUMMARY	Rs9340799 of ESR1 and one haplotype formed by the two polymor of the ESR1 were significantly associated with bone mineral densit postmenopausal Mexican-mestizo women.
Literature	GENERIF_SUMMARY	ALTTO ineligibility did not change when HER2 testing was perform dual antibody ER assay had fewer false-negative test results than a assay with a single antibody.
Literature	GENERIF_SUMMARY	Authors first report that these SNPs (rs2077647, rs9340799, rs223 rs1801132 and rs2228480) in estrogen receptor alpha can influenc outcomes of hepatitis C in a high-risk Chinese population.
Literature	GENERIF_SUMMARY	we show evidence for the temporal selection of functional ESR1 mutations as potential drivers of endocrine resistance during the progression of ER(+) breast cancer.
Literature	GENERIF_SUMMARY	The values and the distribution of the ER and PR taken from the bi made in the patients with mammary carcinoma are influenced by t and menopausal status.
Literature	GENERIF_SUMMARY	Letrozole was effective for ovarian cancers with abundant express ERalpha. Inhibition of angiogenesis and of ascites production appe contribute to prolongation of the survival period.
Literature	GENERIF_SUMMARY	MTA1-TFAP2C or the MTA1-IFI16 complex may contribute to the epigenetic regulation of ESR1 expression.
Literature	GENERIF_SUMMARY	meta-analysis indicated elected PvuII polymorphism of ER-alpha w

Main Category	Category	Terms
		risk factor for prostate cancer development
Literature	GENERIF_SUMMARY	In aP2-Cre/ERalpha(flox/flox) mice, increased serum estrogen leve cause over-stimulation in the uterus.
Literature	GENERIF_SUMMARY	ESR1 repression by SENP2 is independent of its SUMO protease a
Literature	GENERIF_SUMMARY	An association between ESR1 rs9340799 polymorphism and infert women with endometriosis.
Literature	GENERIF_SUMMARY	SNPs in ESR1 (rs2234693), ESR2 (rs2987983), CYP17A1 (rs74357 CYP19A1 (rs700519) indeed contribute to breast cancer susceptib among North Indian women.
Literature	GENERIF_SUMMARY	Methylation status of DNA sequences at ERalpha binding sites is t coupled with ERalpha activity.
Literature	GENERIF_SUMMARY	RNF31, via stabilizing ESR1 levels, controls the transcription of est dependent genes linked to breast cancer cell proliferation.
Literature	GENERIF_SUMMARY	SUPT6H is a new epigenetic regulator of ERalpha activity and cellu differentiation.
Literature	GENERIF_SUMMARY	Results demonstrate that degradation of proteins such as ESRa expressed in activated blastocysts is involved in successful implan
Literature	GENERIF_SUMMARY	Regulation of TRIM-21 expression occurs through an ERalpha-dep mechanism, a pathway that was observed to be overactive in SLE patients.
Literature	GENERIF_SUMMARY	The data show a clear difference in ER expression between lobular ductal breast cancer.
Literature	GENERIF_SUMMARY	FOXA1 is a tumor suppressor in endometrial cancer and has a pos interaction with estrogen receptor alpha.
Literature	GENERIF_SUMMARY	Overexpression of DNMT1/3a was correlated with promoter hypermethylation and reduced expression of estrogen receptor-alp BRCA1.
Literature	GENERIF_SUMMARY	PKM2 acts as a transcriptional coactivator for ERalpha and small-molecule PKM2 activators inhibit ERalpha transcriptional activity a reduce E2-induced cell proliferation.
Literature	GENERIF_SUMMARY	Immune escape mechanisms in ERalpha breast cancer may be fac through an ERalpha suppressive mechanism on IFN-gamma signa
Literature	GENERIF_SUMMARY	The present study aimed to gain further insight into the DNA meth patterns of alternative promoters of the steroid receptor genes ES ESR2 and PGR in intestinal deep endometriosis.
Literature	GENERIF_SUMMARY	Our findings suggest that ESR1 and C6orf97 gene polymorphism is associated with fracture and vertebral fracture risk in Chinese postmenopausal women.
Literature	GENERIF_SUMMARY	ESR1 PvuII T allele may increase the risk of hip fracture and that th polymorphism is not associated with hip fracture
Literature	GENERIF_SUMMARY	[review] Estrogen resistance due to mutations in the estrogen rece alpha (ESR1) gene have been described. Their clinical presentation similar to that of aromatase-deficient men and women.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ERbeta and CXCL12/CXCR4 expression in lung adenocarcinoma d on sex and hormonal status, which may partly explain the sex and hormonal differences in lung cancer behavior.
Literature	GENERIF_SUMMARY	Impairment of ERalpha signal-mediated OLFM4 expression promo malignant progression of endometrioid adenocarcinoma.
Literature	GENERIF_SUMMARY	Silencing, by methylation, of the promoter region of the ESR1 affec expression of the estrogen receptor protein in tumors of breast ca patients; high methylation of ESR1-DNA is associated with estroge receptor negative status which.
Literature	GENERIF_SUMMARY	miR-222-3p plays a significant role in the regulation of ERalpha expression and could be potential targets for restoring ERalpha expression and responding to antiestrogen therapy in a subset of E
Literature	GENERIF_SUMMARY	Data show that erythropoietin receptor (EPOR) protein is expresse breast cancer cells, where it appears to promote proliferation by a independent mechanism in estrogen receptor alpha (ERalpha) exp breast cancer cells.
Literature	GENERIF_SUMMARY	Estren and ginsenoside Rg1 activate ESR1 via phosphorylation at S 118 in a ligand-independent manner.
Literature	GENERIF_SUMMARY	TT variant of estrogen receptor alpha polymorphism may be assoc with enhanced inflammatory response
Literature	GENERIF_SUMMARY	IQGAP1 binds to ERalpha and modulates its transcriptional functio
Literature	GENERIF_SUMMARY	ER-alpha36-mediated rapid estrogen signaling plays an important regulation and maintenance of ER-positive breast cancer stem/pro cells
Literature	GENERIF_SUMMARY	There are comparable levels of EESR1, total ERK1/2 and active ER the stroma and epithelium of proliferative phase endometrium.
Literature	GENERIF_SUMMARY	Trim21, an estrogen receptor coactivator, induces MYC oncogene removing repressive CoAA through E2-dependent degradation of C
Literature	GENERIF_SUMMARY	Estrogen receptor 1 gene PvuII and XbaI polymorphisms were ass with unexplained recurrent spontaneous abortion in a Chinese Har population.
Literature	GENERIF_SUMMARY	PR frequently prevails over ER in pulmonary lymphangioleiomyoma
Literature	GENERIF_SUMMARY	Our results indicate that a greater number of dinucleotide repeats promoter region of ESR1 is associated with a significantly increase likelihood of lone atrial fibrillation in men.
Literature	GENERIF_SUMMARY	meta-analysis revealed that the G allele in the XbaI polymorphism estrogen receptor alpha was associated with a statistically signific increase in the risk of prostate cancer
Literature	GENERIF_SUMMARY	ERalpha intracellular concentration is an important target through endocrine disruptors hamper the hormonal milieu of E2 target cells driving cells to different outcomes or mimicking E2.
Literature	GENERIF_SUMMARY	Data indicate that irect estrogen receptor alpha (ERalpha) methyla histone methyltransferase SMYD2 regulates estrogen signaling.
Literature	GENERIF_SUMMARY	A meta-analysis suggests that the ESR1 PvuII polymorphism has a important role in coronary artery disease susceptibility.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the results implicate the XbaI and PvuII polymorphisms of the ESR as possible disease-modifying factors in dermatomyositis in relati the clinical manifestations.
Literature	GENERIF_SUMMARY	Data suggest that expression of ESR1 in omental adipose tissue (collected at term Cesarean section) is down-regulated in subjects gestational diabetes mellitus; expression of MIRN222 is up-regulat these tissue samples.
Literature	GENERIF_SUMMARY	In response to the elevated levels of Estrogen, the ERa signaling w accentuated in the ovarian epithelial cells of ERalphad/d mice, trig increased ERa-dependent gene expression, abnormal cell prolifera and tumorigenesis
Literature	GENERIF_SUMMARY	ERalphaA87 mutation reduces the transactivation efficiency of ERa on an ERE reporter gene.
Literature	GENERIF_SUMMARY	ER-alpha-related bioinformatic data was integrated at different leve systematically explore the mechanistic and therapeutic implicatior ER-alpha.
Literature	GENERIF_SUMMARY	The associations between LKB1 expression and the methylated fo ERalpha (metERalpha), Src and PI3K, have been analyzed both in c cells and in 154 primary breast tumor samples.
Literature	GENERIF_SUMMARY	Data indicate that Fe65 is a positive estrogen receptor alpha (ERal transcriptional regulator.
Literature	GENERIF_SUMMARY	unliganded ERalpha participates, together with other factors, in the maintenance of the luminal-specific cistrome in breast cancer cell
Literature	GENERIF_SUMMARY	Polymorphisms in the ESR1 gene may have differential roles in the predisposition to male infertility according to the different ethnic backgrounds.
Literature	GENERIF_SUMMARY	CONCLUSION
Literature	GENERIF_SUMMARY	estrogen receptor expression was positively correlated with lymphovascular space invasion in high-grade serous ovarian carci This study emphasizes the importance of estrogen pathway in pro lymphatic or vascular spread of ovarian neoplasms.
Literature	GENERIF_SUMMARY	Results indicate a combination of decreased expression of ERs, AF SRC-3 but not SRC-1 may be involved in the tumorigenesis of glion ERalpha/SRC-3 axis may play central role in the regulation these tu
Literature	GENERIF_SUMMARY	The ratio of ER-alpha and PR-A expression in the epithelial glands a stroma of endometrioid carcinoma biopsies may serve as an addit parameter in the histological evaluation.
Literature	GENERIF_SUMMARY	ERalpha and ERbeta genetic variants were associated with ovarian response to standard gonadotropin stimulation of women undergo vitro fertilization affecting mainly the follicular growth.
Literature	GENERIF_SUMMARY	ESR1 intramolecular interactions in agonist and antagonist conformations have been elucidated.
Literature	GENERIF_SUMMARY	ESR1 polymorphisms affect age of onset of AD in women, and risk vary by ethnicity.
Literature	GENERIF_SUMMARY	TTP acts as a bona fide ERalpha corepressor and suggest that this protein may be a contributing factor in the development of E2-depe

Main Category	Category	Terms
		tumors in breast cancer.
Literature	GENERIF_SUMMARY	PRMT6 specifically interacts with ERalpha at its ligand-binding dor
Literature	GENERIF_SUMMARY	We conclude that ER expression can be understood as a Darwinian process and linked to variations in estrogen delivery by temporal a spatial heterogeneity in blood flow
Literature	GENERIF_SUMMARY	ESR1 amplification is not a key genetic abnormality in the tumorige of breast cancer in Taiwan.
Literature	GENERIF_SUMMARY	This may have profound implications for the anti-estrogen treatme ER-positive breast cancers that have increased levels of EGFR.
Literature	GENERIF_SUMMARY	Estrogen receptor expression and vessel density in the vagina wall postmenopausal women with prolapse.
Literature	GENERIF_SUMMARY	ESR1 gene is considerably associated with knee Osteoarthritis etio the Chinese Han population.
Literature	GENERIF_SUMMARY	ESR1 genetic polymorphisms may contribute to the development c cerebral infarction, especially in the female population. [Meta-analy
Literature	GENERIF_SUMMARY	Both OPG and progesterone receptor are independently associated good prognosis in ER positive breast cancer.
Literature	GENERIF_SUMMARY	Our data strongly demonstrated that ER1 was able to increase drug resistance of breast cancer cells through activating MDR1 transcri
Literature	GENERIF_SUMMARY	E2 promoted the binding of ERalpha to the EREs (estrogen-respons elements) of CLOCK promoter, thereby up-regulating the transcript CLOCK
Literature	GENERIF_SUMMARY	methylation of promoter sites of ESR1 and PGR is a relatively unco event in ductal and lobular breast cancer, the determination of epig states of ESR1 and PGR could represent an alternative or complem histopathological expression analysis
Literature	GENERIF_SUMMARY	Insulin-induced apoptotic commitment depended on the down-reg of Erk-1, insulin growth factor-1 receptor (IGF-1R), and fibroblast gr factor receptor-1 (FGFR-1)-mediated signaling.
Literature	GENERIF_SUMMARY	The ESR1 rs9340799 genotype may modify the ischemic arterial d risk related to high endogenous estrogens levels in older postmen women.
Literature	GENERIF_SUMMARY	Association of ESR1 gene polymorphism with preterm rupture of fe membranes.
Literature	GENERIF_SUMMARY	Oral squamous cell carcinoma cells expressed functional ERa whc Ser118 phosphorylation and transcriptional activity were enhanced FAK/AKT signaling, leading to promoted cell growth
Literature	GENERIF_SUMMARY	MiR-4728-3p regulates expression of estrogen receptor alpha.
Literature	GENERIF_SUMMARY	The x allele of rs9340799 might have a protective effect on the risk Alzheimer disease (AD) in Asia and in patients with sporadic AD (S addition, the p allele of rs2234693 might decrease the risk of patie with SAD. [Meta-Analysis]
Literature	GENERIF_SUMMARY	strong associations between ERalpha/ERbeta methylation and ER, and HER2 status, tumor size, clinical stage, and triple negative tum

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	there is an independent positive association between burden comc and breast cancer with estrogen receptor expression
Literature	GENERIF_SUMMARY	The loss of both hormone receptors in carcinoma cells and loss of the myometrium was a predictor of a more aggressive type of endometrial cancer and a poor prognosis.
Literature	GENERIF_SUMMARY	the PvuII variant was negatively associated with Alzheimer disease Caucasian population
Literature	GENERIF_SUMMARY	ESR1 rs9340799 polymorphism was associated with prostate can in overall populations
Literature	GENERIF_SUMMARY	Data indicate that estrogen receptors alpha (ERalpha) promoter unmethylation occurred more frequently in systemic lupus erythen (SLE) and rheumatoid arthritis(RA) than in controls.
Literature	GENERIF_SUMMARY	MCP-1 was shown to have a role in activation of the ER1 by phosphorylating Ser167 via the PI3K/Akt/mTORC1 signaling pathv
Literature	GENERIF_SUMMARY	estrogen-activated ERa may mediate stimulatory effects on papilla thyroid cancergrowth and progression whereas ERb1 has some inh actions
Literature	GENERIF_SUMMARY	Aminoflavone mediates growth inhibition effects in ERalpha-positi negative breast cancer cells.
Literature	GENERIF_SUMMARY	In low-grade ovarian carcinoma and in ovarian tumors of low malig potential, methylation of the ESR1 promoter inversely correlated w survival (p = 0.031).
Literature	GENERIF_SUMMARY	The ERS1 P allele is a novel sporadic breast cancer biomarker whe and x alleles enhanced chemotherapy responsiveness.
Literature	GENERIF_SUMMARY	ERalpha36 had a high expression mainly in adenocarcinoma and t expression of ERalpha36 was strongly correlated with more advan regional lymph node metastasis and poor survival in lung adenocarcinoma
Literature	GENERIF_SUMMARY	TRPS1 gene expressionis associated with different epithelial mesenchymal transformation markers in ERalpha-positive and ERa negative breast cancers.
Literature	GENERIF_SUMMARY	Data suggest expression of ESR1 (but not ESR2) in endometrium i women with endometriosis may be biomarker for subsequent ferti ESR1 levels in normal range or down-regulated are indicative of fer following medical or surgical treatment.
Literature	GENERIF_SUMMARY	Nongenomic activities of ESR1 are unable to mediate proliferative changes or the transcriptional profile following estradiol exposure.
Literature	GENERIF_SUMMARY	Med25 is identified as a new coactivator of ERalpha that is require ERalpha-mediated regulation of CYP2C9 expression
Literature	GENERIF_SUMMARY	ERalpha XbaI and ERalpha PvuII polymorphisms are not obviously associated with risk of idiopathic scoliosis.
Literature	GENERIF_SUMMARY	ERalpha expression highly correlates with ZNF217 in lysates from tumors.
Literature	GENERIF_SUMMARY	SMRT enhances cell growth of estrogen receptor-alpha-positive br cancer cells by promotion of cell cycle progression and inhibition c

Main Category	Category	Terms
		apoptosis.
Literature	GENERIF_SUMMARY	Mutations of p53 increase the stability of the protein, which in turn promotes the biogenesis of miR-18a and decrease the expression ESR1.
Literature	GENERIF_SUMMARY	Oncotype DX RT-qPCR assay for ER and PR correlation with IHC
Literature	GENERIF_SUMMARY	Provide additional prognostic information when evaluating clinical features of ER(+), HER2(-) early breast cancer.
Literature	GENERIF_SUMMARY	Estradiol regulates TNF-alpha expression and secretion in ERalpha positive breast cancer cells.
Literature	GENERIF_SUMMARY	Peroxiredoxin-1 protects estrogen receptor alpha from oxidative st induced suppression and is a protein biomarker of favorable progn breast cancer.
Literature	GENERIF_SUMMARY	We conclude that the CC genotype of PvuII in ESR1 is associated w increased risk of CHB, HBV-related LC and HCC in Guangxi Zhuang populations.
Literature	GENERIF_SUMMARY	Data suggest that estrogen receptor (ER) subtypes (ERalpha and E might play different roles in the development of breast cancer.
Literature	GENERIF_SUMMARY	These results suggested that both CYP17A1 and ESR1 polymorphi may modulate the effects of phthalate exposure on the developme leiomyoma.
Literature	GENERIF_SUMMARY	CCNB1 is a biomarker for the prognosis of ER+ breast cancer and monitoring of hormone therapy efficacy
Literature	GENERIF_SUMMARY	estrogen receptor binding and p300 redistribution have a role in a squelching mechanism for estradiol-repressed genes
Literature	GENERIF_SUMMARY	Therefore, we studied expression of ERalpha and ERbeta protein in trigeminal mesencephalic nucleus of ten human brains (five femal male).
Literature	GENERIF_SUMMARY	Based on this meta-analysis, we conclude that the ESR1 PvuII polymorphism might be a risk factor in Alzheimer's disease in Caucasians.
Literature	GENERIF_SUMMARY	examined in isolation, rs9340799 polymorphism does not appear t likely susceptibility variant for adolescent idiopathic scoliosis predisposition. rs9340799 may be associated with adolescent idio scoliosis severity, progression and treatment.
Literature	GENERIF_SUMMARY	ESR1 XbaI and G894T NOS3 polymorphisms may be useful in acce insulin resistance and type 2 diabetes risks in all women, even bef menopause and occurrence of metabolic disease
Literature	GENERIF_SUMMARY	ESR1-CCDC170 rearrangements are found in an aggressive subset estrogen receptor-positive breast cancers
Literature	GENERIF_SUMMARY	findings suggest that TSC may have therapeutic effects on ERalph positive breast cancers and moreover that TSC may suppress brea epithelial cell proliferation by inhibiting the estrogen pathway
Literature	GENERIF_SUMMARY	High expression of ER-alpha is associated with lymphatic metasta vaginal recurrence in cervical adenocarcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	expression and localization of estrogen receptor in human breast c and its clinical significance
Literature	GENERIF_SUMMARY	Data from case-control genetic association studies suggest that S ERA/ESR1 (rs2234693C; rs9340799) and ERB (rs1256049GA; not rs4986938) are associated with male infertility. [META-ANALYSIS]
Literature	GENERIF_SUMMARY	Epigenetic marks in estrogen receptor alpha CpG island correlate v some reproductive risk factors in breast cancer.
Literature	GENERIF_SUMMARY	data suggest that RIP140 plays an important role in ERalpha-media transcriptional regulation in breast cancer and response to tamoxi treatment
Literature	GENERIF_SUMMARY	These results suggest that ERalpha genotype may influence the relationship between habitual exercise and bone density.
Literature	GENERIF_SUMMARY	Nuclear p-Ser(167) -ERalpha was significantly positively correlated MAPK and p-S6K1, and with significantly shorter relapse-free survi endometrioid endometrial cancer (EEC).
Literature	GENERIF_SUMMARY	ERa variant ERa36 mediates antiestrogen resistance in ER-positive cancer stem/progenitor cells
Literature	GENERIF_SUMMARY	The two compounds inhibited cell growth, reduced expression of t estrogen receptor alpha (ER), Ser118-phosphorylated ER, FOXA1 a and resensitized tamoxifenresistant cells to tamoxifen
Literature	GENERIF_SUMMARY	Androgen receptor collaborates with estrogen receptor alpha to pr resistance to aromatase inhibitors in breast cancer cells.
Literature	GENERIF_SUMMARY	There was a higher number of pxpx and pxPX genotypes in the DJI samples compared to the DJD- group, which suggests the presenc polymorphism possibly modulates the ER-alpha activity in bone an contributes to the degenerative process in the TMJ.
Literature	GENERIF_SUMMARY	ESR1 and maternal sensitivity are both associated with age of mer
Literature	GENERIF_SUMMARY	SERPINA6-rs1998056 regulated by FOXA/ERalpha might be assoc with female HCC risk
Literature	GENERIF_SUMMARY	The results indicated that ESR1 rs1801132 (C>G) was associated v cancer risk in Caucasian populations
Literature	GENERIF_SUMMARY	In the coronary intima from surgically repaired congenital heart dis patients there was a reduction in ESRA expression.
Literature	GENERIF_SUMMARY	Elevated expression of the ER-alpha36-EGFR/HER2 loops is one of mechanisms by which ER-positive breast cancer cells escape tam therapy.
Literature	GENERIF_SUMMARY	ESR1 is strongly expressed in rectosigmoid endometriosis tissue.
Literature	GENERIF_SUMMARY	The potential utility of serum DNA methylation of ERalpha gene pr
Literature	GENERIF_SUMMARY	In breast cancer cells, GSTP1 inhibits the expression of RIP140, a negative regulator of estrogen receptor alpha transcription, at both and protein levels.
Literature	GENERIF_SUMMARY	This review focuses on the new advances made in understanding t complex and important interface between ERRalpha and mitochon physiology. [review]

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Single-nucleotide polymorphism in the ESR1 gene is associated wi racial differences in breast cancer.
Literature	GENERIF_SUMMARY	SGC7901 cells with high levels of ER-a36 expression exhibited estr hypersensitivity, high growth rate and high levels of Cyclin D1 expre while SGC7901 cells with knocked-down levels of ER-a36 expressi insensitive to estrogen stimulation
Literature	GENERIF_SUMMARY	ASH2L enhances ERalpha expression as a coactivator of GATA3 in cancers
Literature	GENERIF_SUMMARY	At clinically effective concentrations, phenytoin is a strong ERalpha antagonist but a many-fold weaker agonist, via interactions with he
Literature	GENERIF_SUMMARY	Data show that estrogen, acting via estrogen receptor alpha (ERalp induces rapid anticipatory activation of the unfolded protein respo (UPR).
Literature	GENERIF_SUMMARY	The ERalpha mRNA levels in tumor tissue were upregulated.
Literature	GENERIF_SUMMARY	GPR30 activation inhibits proliferation of estrogen receptor-negati breast cancer cells.
Literature	GENERIF_SUMMARY	ERalpha plays a protective role in bladder cancer initiation and gro least partly via modulating the INPP4B/Akt pathway.
Literature	GENERIF_SUMMARY	Estradiol inhibits oxidative stress-induced accumulation of AIF in nucleolus and PARP1-dependent cell death via estrogen receptor a MCF-7 cells.
Literature	GENERIF_SUMMARY	mRNA expression profile of runx2 mirrored that of ERalpha36 and a degree subcellular co-localisation with ERalpha36. This study su that ERalpha36 is involved in the process of osteogenesis in BMSC
Literature	GENERIF_SUMMARY	In colorectal carcinoma, pERalpha(Ser167) score was a significant prognostic factor for both disease-free interval and overall survival independent of sex, age, glucose levels, tumor stage, bowel obstru perforation, or adjuvant chemotherapy.
Literature	GENERIF_SUMMARY	miRNA mediated alterations in mTOR and ERalpha signaling estab new mechanism for altered estrogen responses independent of gr factor stimulation.
Literature	GENERIF_SUMMARY	The association between body mass index (BMI) and the risk of developing breast cancer according to the menstruation status an and the correlation between BMI and expression of estrogen recep was examined.
Literature	GENERIF_SUMMARY	Data show that odulation of RNA helicase DDX18 directly affects g of tamoxifen-resistant cells, suggesting that it may be a critical downstream effector of the estrogen receptors (ERs) and high mo group box 2 (HMGB2) complex.
Literature	GENERIF_SUMMARY	Data indicate that serine/threonine-protein kinase D1 (PKD1) stimu estrogen receptor alpha (ERalpha) expression in breast cancer MC cells.
Literature	GENERIF_SUMMARY	Neuroglobin player role in estrogen receptor alpha-positive cancer viability.
Literature	GENERIF_SUMMARY	Study reports a signature of the functionally active estrogen-regula enhancers, particularly the 1,333 most active ERalpha enhancers li

Main Category	Category	Terms
		target coding gene activation; signature of these MegaTrans enhar the selective recruitment in trans of an apparent complex of other transcription factors, including RARalpha/gamma, GATA3, AP2gan STAT1, AP1, and FoxA1.
Literature	GENERIF_SUMMARY	Data show that recruitment of MLL1 (mixed lineage leukaemia hist methylase 1) to the homeobox-containing gene HOXA10 EREs (est response elements) is mediated via ERalpha (estrogen receptor al
Literature	GENERIF_SUMMARY	Data show that estrogen receptor 1 (ESR1) rs2881766 polymorphi increased breast cancer risk and rs3798377 decreased the risk in women.
Literature	GENERIF_SUMMARY	The ESR1 rs9340799 polymorphism might be associated with the osteoarthritis. [Meta-analysis]
Literature	GENERIF_SUMMARY	This study contributes to elucidating the intrinsic relations among cancer subgroups defined by ER, PR and HER2 expression via integ mRNA and miRNA expression
Literature	GENERIF_SUMMARY	role in TLR7 signaling in plasmacytoid dendritic cells
Literature	GENERIF_SUMMARY	Studies indicate that activated CDC42 kinase 1 (ACK1) interacts w estrogen receptor (ER) and (AR) regulates the activity of androgen receptor (AR) to promote the growth of breast cancer and prostate cancer.
Literature	GENERIF_SUMMARY	Immunohistochemical profile of breast cancer with respect to estr receptor and HER2 status.
Literature	GENERIF_SUMMARY	Data suggest that rs2046210 SNPs located in the region of C6ORF (CCDC170)-estrogen receptor alpha (ESR1) might have substantia not only in carcinogenesis but also in progression toward a more aggressive phenotype in breast cancer patients.
Literature	GENERIF_SUMMARY	upon exposure to E2, ERalpha rapidly localizes to mitochondria, in interacts with HSD17B10 to modulate the expression of mitochon RNA transcripts.
Literature	GENERIF_SUMMARY	Circulating tumor cells expression of ER, BCL-2, HER2, and Ki67 ca reproducibly measured with high analytical validity using the CellS System.
Literature	GENERIF_SUMMARY	The ratios of IGF-I/IGFBP-3 were increased in postmenopausal Tai women with breast cancer, irrespective of their ages, ER and PR st and HER2 expression.
Literature	GENERIF_SUMMARY	ERalpha is overexpressed in a significant number of prostate canc cohorts.
Literature	GENERIF_SUMMARY	a polymorphic variant of A-351G gene ESR1 estrogen can be consi as a marker of premenstrual syndrome. Pathological variant GG ge was significantly associated with the presence of edematous and forms of the disease
Literature	GENERIF_SUMMARY	Results show FOXO3a and FOXM1, two transcription factors with antagonistic roles in cancer and DNA damage response, mediating signaling and repair as well as for senescence, cell cycle and cell c control. [review]
Literature	GENERIF_SUMMARY	Our investigation identified miR-219-5p as a negative regulator of p thyroid carcinoma development through targeting of ERalpha.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ESR1 gene PvuII polymorphism is significantly associated with an increased risk of premature ovarian failure.ESR1 gene XbaI polymo is significantly association with risk of premature ovarian failure in population.
Literature	GENERIF_SUMMARY	Data shows that lobular breast neoplasm expresses high levels of ESR1 and ESR2. No tumor infiltrating leucocytes express ESR1. [re
Literature	GENERIF_SUMMARY	No significant differences in ESR1-397C>T and -351A>G polymorp were was found between the patients with premature ovarian failu controls
Literature	GENERIF_SUMMARY	ESR1 single nucleotide polymorphisms and haplotypes influence t of hypospadias in white and Hispanic subjects
Literature	GENERIF_SUMMARY	Study show that miR-221 is upregulated in HBV-related hepatocellu carcinoma (HCC) and modulates cell proliferation by suppressing ERalpha, functioning as a tumor promoter.
Literature	GENERIF_SUMMARY	REGgamma expression is positively correlated with ERalpha status poor clinical prognosis in ERalpha positive breast cancer patients
Literature	GENERIF_SUMMARY	Data indicate that the mRNA level for forkhead box P1 (FOXP1) an estrogen receptor beta (ERbeta) in ovarian carcinoma tissues decr while the expression level of estrogen receptor alpha (ERalpha) mR increased.
Literature	GENERIF_SUMMARY	after administration of abexinostat for three consecutive days, we observe any induction of ESR1 or ESR1-related genes and ERalpha expression
Literature	GENERIF_SUMMARY	ERRalpha is needed for the efficient production of cytomegalovirus progeny, and we propose that the nuclear receptor contributes importantly to the induction of a metabolic environment that suppo optimal cytomegalovirus replication.
Literature	GENERIF_SUMMARY	Data indicate that carnosol could significantly increase estrogen receptors ERalpha and ERbeta expressions of T47D cells, and rem increase ERalpha/ERbeta ratio.
Literature	GENERIF_SUMMARY	low ERb1 expression and high ERb2 expression are markers for identification of AET-treated ERa-positive breast carcinoma patient risk of early and late relapse, respectively.
Literature	GENERIF_SUMMARY	ESR1/2 gene polymorphisms are associated with the results of co ovarian hyperstimulation.
Literature	GENERIF_SUMMARY	Mir-873 inhibits ESR1 activity and cell growth via targeting CDK3.
Literature	GENERIF_SUMMARY	ERalpha regulates chromosome alignment and spindle dynamics b stabilizing microtubules during mitosis.
Literature	GENERIF_SUMMARY	Our data indicates that cohesin is not a universal facilitator of estr receptor alpha-induced transcription and can even restrict enhance promoter communication.
Literature	GENERIF_SUMMARY	Data suggest that, in vaginal agenesis following modified Abbe-Mc neovaginoplasty using regenerated oxidised cellulose, expression is low in neovaginal tissue resulting in local hypo-estrogenism and vaginal trophism (vaginal atrophy).
Literature	GENERIF_SUMMARY	High ESR1 expression is associated with lymph node metastasis a

Main Category	Category	Terms
		recurrence in Thyroid cancer.
Literature	GENERIF_SUMMARY	roles of ESR1-mediated transactivation and transrepression in the regulation of invasiveness of ESR1+/HER2+ breast cancer cells by
Literature	GENERIF_SUMMARY	ERalpha contributes to the growth of papillary thyroid carcinoma (enhancing an important prosurvival catabolic process, autophagy, cells.
Literature	GENERIF_SUMMARY	This study investigetes the functional properties and subcellular localization of alpha human and rainbow trout estrogen receptors.
Literature	GENERIF_SUMMARY	Data identified phosphorylation sites in ESR1 that direct its tempor recruitment to specific genes. Disruption of some of them in breas cancer cells elicit unique gene expression related to cell proliferati morphology and migration. [review]
Literature	GENERIF_SUMMARY	ESR1 mRNA expression was significantly lower on granulosa cells PCOS patients compared to controls.
Literature	GENERIF_SUMMARY	The identification of cytoplasmic proteins interacting with estroge receptor alpha in breast cancer cells
Literature	GENERIF_SUMMARY	No association between the ESR1 gene polymorphisms and the pr of PCOS was observed.
Literature	GENERIF_SUMMARY	None of the four polymorphisms studied in ESR1 and ESR2 were associated with increased risk for cataract.
Literature	GENERIF_SUMMARY	Combined expression of estrogen receptor 1 and VEGF-A is a pote prognostic marker for gallbladder carcinoma patients.
Literature	GENERIF_SUMMARY	our results suggest that ERalpha and ERRalpha cooperate to prom endocrine resistance, and they provide a rationale for the explorati ERRalpha as a candidate drug target to treat endocrine-resistant b cancer
Literature	GENERIF_SUMMARY	Report substituted benzopyran targeting refractory prostate cance behaving as ESR1 antagonist/ESR2 agonist.
Literature	GENERIF_SUMMARY	these data show that DC-SCRIPT acts as a novel regulator of CDK induces cell cycle arrest in ESR1-positive breast cancer cells.
Literature	GENERIF_SUMMARY	Has no impact on the prognosis of patients with ER-negative tumo
Literature	GENERIF_SUMMARY	c-MYC expression is up-regulated through the cross-talk between E HER2 in aromatase inhibitor resistant breast cancer cells.
Literature	GENERIF_SUMMARY	It is shown that there is a connection between the PvuII-gene polymorphism EsRalpha with the degree of proliferation (chi2=43,1
Literature	GENERIF_SUMMARY	estrogen receptor activation under chronic hyperinsulinemic condi increases breast cancer growth through the modulation of cell cyc apoptotic factors and nutrient metabolism.
Literature	GENERIF_SUMMARY	Alterations of ER-alpha in the pubocervical fascia and around the u in postmenopausal women may play an important role in the pathophysiology of pelvic organ prolapse (POP); the risk for develo urodynamic stress urinary incontinence among POP patients seem strongly associated with the reduction of both estrogen receptors alpha and ER-beta) expression

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	At the molecular level, estrogen upregulated hepatic SHP expressi through binding to its proximal promoter. In addition, the roles of e were largely blunted in mice with SHP deficiency.
Literature	GENERIF_SUMMARY	ER-alpha, SRC-3, and p300 form an active protein complex on estro response element-DNA.
Literature	GENERIF_SUMMARY	Data show that loss of ERalpha in breast cancer cells results in a p epithelial to mesenchymal transition characterized by changes in t expression profile of matrix effectors suggesting their potential rol breast cancer endocrine resistance.
Literature	GENERIF_SUMMARY	ESR1 gene PvuII polymorphism may not be associated with fractu in postmenopausal women [meta-analysis]
Literature	GENERIF_SUMMARY	ER positivity was increased in BRCA2 breast cancers compared to BRCA familial and BRCA1 breast cancers.
Literature	GENERIF_SUMMARY	FOXK2 might act as a negative regulator of ERalpha, and its assoc with both ERalpha and BRCA1/BARD1 could lead to the down-regu of ERalpha transcriptional activity, effectively regulating the functio ERalpha.
Literature	GENERIF_SUMMARY	TNFalpha signaling causes redistribution of NF-kappaB and FoxA1 binding sites across the genome, creates latent ERalpha binding si underlie altered patterns of gene expression and clinically relevant responses.
Literature	GENERIF_SUMMARY	Our study revealed novel molecular mechanisms underlying the re of ERa protein and provided insights in understanding the relations between SPOP mutations and the development of endometrial car
Literature	GENERIF_SUMMARY	These results demonstrate that estradiol upregulates REA express recruits REA via ERalpha to the EREs on the RORgammaT promote region, thus inhibiting RORgammaT expression and Th17 different
Literature	GENERIF_SUMMARY	Review/Meta-analysis
Literature	GENERIF_SUMMARY	H4K12ac is regulated by estrogen receptor-alpha and is associate BRD4 function and inducible transcription
Literature	GENERIF_SUMMARY	S6K1, overexpressed in many breast cancers, controls HDAC1-dep transcriptional regulation of ESR1 levels upon mitogenic stimuli.
Literature	GENERIF_SUMMARY	HER2 status was highly concordant between the paired primary an tumors, whereas changes of ER and PR status occurred in a subst proportion of the patients
Literature	GENERIF_SUMMARY	MEL-18 loss in luminal breast cancer cells resulted in the downreg of expression and activity of ER-alpha and the progesterone recept MEL-18 overexpression restored ER-alpha expression in triple-nega breast cancer cells.
Literature	GENERIF_SUMMARY	cells. These results suggest that adipocytes can modify breast car cell ER gene expression through hypoxia and also can promote EM processes in breast cancer cells, supporting an important role of c in aggressive breast cancer development.
Literature	GENERIF_SUMMARY	Data suggest that targeting therapy-resistant estrogen receptor alp (ERalpha)-positive cancer cells by converting the unfolded protein response (UPR) from cytoprotective to cytotoxic may hold significa therapeutic promise.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	rs2077647 polymorphism may be a risk factor for prostate cancer, rs3798577 polymorphism could be a promoting factor in combina with other polymorphisms in estrogen receptor alpha contributing development of high-grade carcinomas.
Literature	GENERIF_SUMMARY	ERalpha XbaI AA polymorphism may intensify the beneficial effect estradiol on the distribution of fatty tissue after menopause; XbaI PuvII TC genotypes may intensify the beneficial effect of estradiol level.
Literature	GENERIF_SUMMARY	negative-ER ,negative-PR ,positive-p53 and higher Ki-67 expression related to a higher mtDNA copy ratio. In addition, advanced T-statu (p=0.019) and negative-HER-2/neu expression (p=0.061) were ass with mtDNA D310 mutations
Literature	GENERIF_SUMMARY	These results suggest that H19 is an estrogen-inducible gene and key role in cell survival and in estrogen-induced cell proliferation in cells, indicating that H19 lncRNA may serve as a biomarker for bre cancer diagnosis and progression
Literature	GENERIF_SUMMARY	Data suggest that ESR1 agonists (here endocrine disruptors 2,4-dihydroxybenzophenone or nonylphenol) alter gene expression, sti cell proliferation, and induce cell migration/metastasis in adenocarcinoma breast cancer cells (MCF-7 cells).
Literature	GENERIF_SUMMARY	this study demonstrated that a celecoxib and luteolin combination treatment yielded synergistic effects in four different breast cance lines and suppressed triple-negative (ER-negative and HER-2 negat breast tumor growth in vivo
Literature	GENERIF_SUMMARY	Suggest that OATP2B1 is involved in cell proliferation by increasing amount of estrogen in ER1-positive breast cancer cells.
Literature	GENERIF_SUMMARY	we describe a TR-FRET assay that monitors the interaction of the estrogen receptor (ER) alpha ligand binding domain (labeled with a terbium chelate via a streptavidin-biotin interaction) with a sequen coactivator protein SRC3
Literature	GENERIF_SUMMARY	Many membrane trafficking proteins contribute to the regulation o degradation, thus unraveling the contribution of endocytic proteins estradiol
Literature	GENERIF_SUMMARY	The anticancer activity of triptolide against MCF-7 cells is partially mediated by downregulation of the ERalpha-mediated signaling pa
Literature	GENERIF_SUMMARY	Pin1 has a role in allosteric regulation of ERalpha function
Literature	GENERIF_SUMMARY	Results showed that HNRNPG and HTRA2-BETA1 were specific antagonistic regulators of ERa exon7 splicing and increased HNRN levels were associated with improved clinical outcome of endome cancer through up-regulation of ERaD7 expression.
Literature	GENERIF_SUMMARY	Among women with breast cancer, those who were highly obese, b mildly obese, around breast cancer diagnosis were more likely to h breast tumors with greater expression of proliferation genes and le expression of ESR1
Literature	GENERIF_SUMMARY	Review/Meta-analysis
Literature	GENERIF_SUMMARY	This study suggested that interactions among impulsiveness, testosterone, sex hormone binding globulin and androgen recepto CAG repeat length

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The present meta-analysis suggests that the rs2234693 polymorpl associated with an increased knee osteoarthritis risk
Literature	GENERIF_SUMMARY	Our findings shed new light on the mechanisms underlying the progression of ER(+) breast cancer and indicate the potential of no therapies targeting CCL2 and CCL5 pathways.
Literature	GENERIF_SUMMARY	Prediction of breast cancer prognosis and treatment response was improved by combining AR and ER status
Literature	GENERIF_SUMMARY	demonstrates that miR-135b regulates ERalpha, AR and HIF1AN pr levels through interaction with their 3'UTR regions, and proliferatio ERalpha-positive BCa and AR-positive PCa cells
Literature	GENERIF_SUMMARY	Regardless of methodology, basal-like cases seldom expressed ER or PR/PGR and were associated with the lowest expression level o HER2/ERBB2 relative to other subtypes.
Literature	GENERIF_SUMMARY	Data shows that ERA36 is a novel ERA that is expressed in breast neoplasm and confers tamoxifen resistance. It is localized in the cytoplasm and plasma membrane and seems playing a role in brea cancer stem cells. [review]
Literature	GENERIF_SUMMARY	Estrogen deprivation induced a cluster of noncoding RNAs that de large chromatin domain containing the ESR1 locus.
Literature	GENERIF_SUMMARY	Intragenic methylation levels correlate with ESR1 expression in homogenous cell populations but not in heterogeneous tumor biop highlighting the significant differences between the in vivo tumor microenvironment and individual homogenous cell types.
Literature	GENERIF_SUMMARY	ESR1 mutations in advanced, endocrine therapy-resistant breast cancers(Review)
Literature	GENERIF_SUMMARY	SNAI2/SLUG is prognostic of patients' outcome. The strong invers correlation with ESR1 indicates a significant impact of estrogen re pathway regarding these malignant features.
Literature	GENERIF_SUMMARY	IL6 secreted by ovarian cancer cells may contribute to the refracto of these cells to tamoxifen via the crosstalk between ESR1 and IL6 mediated intracellular signal transduction cascades.
Literature	GENERIF_SUMMARY	The present results indicate that the estrogen-ERalpha-H19 signali plays a role in regulating the proliferation and differentiation poten the normal luminal progenitor which may also be important in the development of ER(+) breast cancer tumors
Literature	GENERIF_SUMMARY	Gestational diabetes mellitus affects expression of placental ERal cell type-dependent way, on epigenetic level. Gestational diabetes downregulated ERalpha in decidual vessels only in pregnancies wi but not female fetuses.
Literature	GENERIF_SUMMARY	miR-18a expression suppression led to a decrease in trophoblast c invasion and an increase in trophoblast cell apoptosis, by inducing estrogen receptor - alpha expression.
Literature	GENERIF_SUMMARY	In localized colorectal cancer carriers, mRNA-based CD3Z/CD8 pro of tumor immune response may have stage, site and tissue-specifi prognostic significance, along with ESR1 expression.
Literature	GENERIF_SUMMARY	The loss of heterozygosity in the regulatory regions of ESRa, ESRb, AR genes is an early genetic alteration in breast carcinoma and ha

Main Category	Category	Terms
		possibly synergistic effects on tumor proliferation, histological aggressiveness and down-regulation of these genes in tumor sam
Literature	GENERIF_SUMMARY	Our study suggests that ESR1 gene amplification is an early event breast pathology and might be a helpful predictive marker to identi patients at high risk of developing breast cancer.
Literature	GENERIF_SUMMARY	SERD- or SSH-palbociclib combination has therapeutic potential in tumors resistant to endocrine therapies or those expressing ESR1 mutations
Literature	GENERIF_SUMMARY	Digital PCR results showed mutations in ESR1 gene significantly associated with aromatase inhibitor resistant metastatic breast neoplasm.
Literature	GENERIF_SUMMARY	genetic association study in population of postmenopausal wome southern China
Literature	GENERIF_SUMMARY	Conclude that not only ERalpha, but also basal activities of EGFR a kinase are essential for Cd-induced signal transduction and activat MAPK/ERK pathway for breast cancer cell proliferation.
Literature	GENERIF_SUMMARY	This study illustrates for the first time that, in contrast to hepatic a adipose tissue, estrogen induces SCD-1 expression and activity in carcinoma cells. These results support SCD-1 as a therapeutic targ estrogen-sensitive breast cancer.
Literature	GENERIF_SUMMARY	Data suggest that differential transactivation of ESR1 and ESR2 is exhibited by the various isoflavone phytoestrogens present in soy red clover commercial dietary supplements available on the Germa market as capsules.
Literature	GENERIF_SUMMARY	ERalpha inhibits epithelial-mesenchymal transition and stemness the downregulation of Bmi1 in breast cancer
Literature	GENERIF_SUMMARY	Trp383 in hERalpha and Trp335 in hERbeta interact with the NR-bo of the H12 helix.
Literature	GENERIF_SUMMARY	Data show that ESR1 promotes obesity and metabolic dysfunction animal model; its tissue-specific actions control feeding, physical and insulin sensitivity. [review]
Literature	GENERIF_SUMMARY	The expression of ERalpha and ERbeta was inversely correlated wi menopausal age. No significant association was observed betwee or PR expression and benign or malignant endometrial pathology.
Literature	GENERIF_SUMMARY	ESR1 XbaI and PvuII polymorphisms are associated with precocio puberty susceptibility (meta-analysis).
Literature	GENERIF_SUMMARY	Coordinate pattern of Rb1+ and ER+ in HGSC and LGSC is 19 and 5 respectively. Rb1 and P16 show inverse expression pattern accord tumor grade with more frequent Rb1 in low grade vs. more frequen in grade 3 tumors
Literature	GENERIF_SUMMARY	Ulipristal acetate antagonizes the inhibitory effect of progesterone ciliary beat frequency and upregulates the expression levels of est receptor alpha and the progesterone receptor in the fallopian tube
Literature	GENERIF_SUMMARY	results show that PHB2 binds to the ligand binding domain of ERal with a conformational change in the helix 12 of ERalpha
Literature	GENERIF_SUMMARY	results indicate that estrogen signaling in cervical cancer has dram

Main Category	Category	Terms
		differences from ERalpha+ breast cancers, and imply that estroger signaling increasingly proceeds indirectly through ERalpha in tumo associated stromal fibroblasts
Literature	GENERIF_SUMMARY	Results show that ESR1 and ERBB2 were found to modulate the m numbers of gene pairs in breast neoplasm and they share complex interplay between each other suggesting a role in breast cancer pathogenesis through gene regulation.
Literature	GENERIF_SUMMARY	By investigating the role of ERa in the ETS-1 activity regulation, we demonstrated that ERalpha may be a novel ETS-1 co-activator and potential therapeutic target in human neuroblastoma treatment
Literature	GENERIF_SUMMARY	the contribution of ESR1 mutations to the development of acquired resistance to endocrine therapy in breast cancer patients (Review)
Literature	GENERIF_SUMMARY	ERalpha36 expression is valuable for differential diagnosis of mali renal tumors from benign ones.
Literature	GENERIF_SUMMARY	ERa-dependent expression of LRIG1 dampens ErbB3 signaling in lu breast cancer cells, and by blocking ERa activity with fulvestrant, L decreased thus permitting ErbB3 accumulation, enhanced ErbB3 signaling to cell survival pathways
Literature	GENERIF_SUMMARY	PR functions as a molecular rheostat to control ERalpha chromatin binding and transcriptional activity, which has important implicatio prognosis and therapeutic interventions in breast cancer
Literature	GENERIF_SUMMARY	This newly described ERa-mediated and estrogen response eleme (ERE)-independent non-canonical autophagy pathway, which involv function of BAG3 and provides stress resistance in our model syst
Literature	GENERIF_SUMMARY	10.7% of this Pakistani breast cancer cohort were estrogen recept positive only, though 51% were ER and PR positive.
Literature	GENERIF_SUMMARY	The ER-/PgR+ phenotype is associated with higher grade with HER overexpression/amplification and occurs more commonly in young women.
Literature	GENERIF_SUMMARY	ESR1 polymorphisms, PvuII and XbaI and VDR polymorphisms (Ta ApaI) did not significantly influence fractional calcium absorption
Literature	GENERIF_SUMMARY	HAND2-mediated proteolysis negatively regulates the function of estrogen receptor alpha.
Literature	GENERIF_SUMMARY	condensin complexes exhibit an unexpected, dramatic estrogen-in recruitment to estrogen receptor alpha (ER-alpha)-bound eRNA(+) enhancers in interphase breast cancer cells, exhibiting non-canoni interaction with ER-alpha via its DNA-binding domain
Literature	GENERIF_SUMMARY	Our findings support the hypothesis that estrogen receptor polymorphisms influence risk for major depression.
Literature	GENERIF_SUMMARY	Tissue factor isoform and estrogen signaling share downstream ta in BrCa; the concomitant presence of asTF and estrogen signaling required to promote BrCa cell proliferation.
Literature	GENERIF_SUMMARY	HOXB7 Is an ERalpha Cofactor in the Activation of HER2 and Multi Target Genes Leading to Endocrine Resistance
Literature	GENERIF_SUMMARY	PARP regulates estradiol-mediated cell growth by controlling the E IGF-1R/PDZK1 axis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Functional roles of ESR1 mutations in breast cancer (review).
Literature	GENERIF_SUMMARY	ESR1 directly interacts with TRAF3 and promotes K48-linked prote degradation of TRAF3.
Literature	GENERIF_SUMMARY	Results suggest that expression levels of protease activated recep and 2 (PAR-1 and PAR-2) associate with breast cancer outcome in estrogen receptor (ER)-dependent manner.
Literature	GENERIF_SUMMARY	ESR2 rs4986938 AG and AG + AA genotypes were associated with significantly decreased risk of prostate cancer (PCa). However, the no significant association between ESR1 rs2234693 and PCa risk.
Literature	GENERIF_SUMMARY	the morphological features and ER-alpha/PRA/p53 immunohistoch expression of a sample of 45 endometrial endometrioid carcinoma metastases compared to matched primary tumors, were evaluated
Literature	GENERIF_SUMMARY	ESR1 mutations can occur at a high frequency in human breast ca
Literature	GENERIF_SUMMARY	The structural and functional importance of estrogen receptor and aromatase in endocrine-responsive and -resistant breast cancers v discussed in more detail. [review]
Literature	GENERIF_SUMMARY	TLX has a role in regulating growth and invasion in ERalpha-negati breast cancer cells
Literature	GENERIF_SUMMARY	ER-alpha status of endometrioid adenocarcinomas could restrict e proapoptotic or anti-apoptotic impact of aberrantly expressed Cx4 Cx26 in these tumors.
Literature	GENERIF_SUMMARY	the conformational dynamics of ERalpha36 in the absence and pre of Hsp90 and gammaSyn, is reported.
Literature	GENERIF_SUMMARY	estrogen-responsive transcription in ERalpha-positive breast cance
Literature	GENERIF_SUMMARY	ESR1 sequence variability may regulate in statin therapy response sex-specific manner.
Literature	GENERIF_SUMMARY	Nuclear ERalpha expression is an independent predictor of recurre pT1a lung adenocarcinomas and correlates with poor prognostic i microenvironments.
Literature	GENERIF_SUMMARY	Data indicate that high proto-oncogene protein HER-2 (HER2) and estrogen receptor alpha (ESR1) mRNA levels are required for achie trastuzumab response.
Literature	GENERIF_SUMMARY	ESR1 gene polymorphisms may have major contributions to the pathogenesis of migraine in Caucasian populations.
Literature	GENERIF_SUMMARY	Unmethylated ESR1 gene is associated with chemoradiation resist Invasive Cervical Carcinoma.
Literature	GENERIF_SUMMARY	the ESR1 PvuII polymorphism is associated with AD susceptibility males, and that ESR1 XbaI polymorphism is associated with AD ris Japanese population. the PP/XX haplotype is associated with AD susceptibility in the Chinese population.
Literature	GENERIF_SUMMARY	The ESR1 ubiquitin binding surface contributes to the modulation estradiol-induced ESR1-mediated cell proliferation.
Literature	GENERIF_SUMMARY	our results show the involvement of the variants of ESR1, LPL and genes in metabolic events related to MetS or some of its features.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	AMPK-alpha-2 binds to ER-alpha, as well as ER-beta.
Literature	GENERIF_SUMMARY	This study demonstrates that ER-alpha36-mediated rapid estrogen signaling is involved in the neuroprotection activity of estrogen aga oxidative toxicity.
Literature	GENERIF_SUMMARY	Collectively, the results demonstrate a novel ligand-dependent acti ERalpha that stimulates the activity of an E3 ligase.
Literature	GENERIF_SUMMARY	Data suggest 17beta-estradiol stimulates odontoblastic differentia dental pulp via coordinated regulation of estrogen receptor alpha, and MAP kinase signaling pathways, which may play key role in calcification/regeneration of dentin.
Literature	GENERIF_SUMMARY	In estradiol-treated monocytes, there is a direct physical interactio ligand-activated estrogen receptor-alpha 36-kDa splice variant with p65 component in the nucleus. This is not changed during differen into macrophages or dendritic cells.
Literature	GENERIF_SUMMARY	CORRELATION BETWEEN CLINICAL PATHOLOGY OF LUMINAL B E CANCER AND DETERMINATION OF ESTROGEN RECEPTOR, PROGESTERONE RECEPTOR AND HER2 EXPRESSION COMBINED NUCLEAR MORPHOLOGY.
Literature	GENERIF_SUMMARY	Results show that genetic variants in the oestrogen receptor alpha the oxytocin receptor may be associated with an increased risk of oesophageal adenocarcinoma and Barrett's oesophagus.
Literature	GENERIF_SUMMARY	APOBEC3B is required for regulation of gene expression by ERalph acts by causing C-to-U deamination at ERalpha binding regions.
Literature	GENERIF_SUMMARY	Data show that estrogen receptor alpha (ESR1) is a predicted targe microRNA miR-19.
Literature	GENERIF_SUMMARY	Uni- and multivariate analyses included age, tumor size, grade, ER, Her2; in multivariate analysis, MammaPrint reached near-significan 3.01; p 0.08).
Literature	GENERIF_SUMMARY	Expression of seven ERalpha splice variants, total progesterone re (PR) and of five PR transcript isoforms was found to be significant decreased in endometrial cancer
Literature	GENERIF_SUMMARY	ddPCR technique could be a promising tool for the next-generation sequencing-free precise detection of ESR1 mutations in endocrine therapy resistant cases and may assist in determining the treatme strategy
Literature	GENERIF_SUMMARY	ESR1 PvuII, XbaI, and P325P polymorphisms are associated with b cancer susceptibility [meta-analysis]
Literature	GENERIF_SUMMARY	Results of the meta-analysis revealed a potential benefit of positiv mRNA expression in survival in patients with NSCLC, especially in of advanced stage. No statistically significant association was fou between ESR2 mRNA expression and NSCLC patients' prognosis.
Literature	GENERIF_SUMMARY	Cryptotanshinone and wogonin up-regulate eNOS in vascular endo cells via ERalpha and down-regulate iNOS in LPS stimulated vascu smooth muscle cells via ERbeta.(
Literature	GENERIF_SUMMARY	Our results thus demonstrated that ER-alpha36 is involved in neuroprotective activities mediated by both estrogen and tamoxife

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	A Lower PBMC Estrogen Receptor alpha Gene Expression in Chron Hepatitis B Is Associated with a Sustained Virological Response to Pegylated Interferon
Literature	GENERIF_SUMMARY	Knockdown of METTL17 reduces transcriptional activities of ESR1 ESR2 in breast cancer cells.
Literature	GENERIF_SUMMARY	ESR1 mediated signaling in cancer-associated fibroblasts regulate differential gene expression affecting angiogenesis and cancer progression.
Literature	GENERIF_SUMMARY	ESR1 mutations were detected at very low allele frequencies in so primary breast cancers, and at high allele frequency in metastases suggesting that in some tumors rare ESR1-mutant clones are enric endocrine therapy
Literature	GENERIF_SUMMARY	Inhibition of MTA1 by ERalpha contributes to protection hepatocell carcinoma from tumor proliferation and metastasis
Literature	GENERIF_SUMMARY	Studies indicate that prostate cancer stem cells lack the androgen receptor, and express classical alpha and/or beta and GPR30 estro receptors.
Literature	GENERIF_SUMMARY	Data show that mTOR protein forms a complex with Raptor and es receptor-alpha (ERalpha).
Literature	GENERIF_SUMMARY	Cells traditionally considered as being deprived of estrogen recept able to respond to hormonal stimulation via the ERalpha36 recepto [Review]
Literature	GENERIF_SUMMARY	ERa mRNA and protein levels of in multiple leiomyomas were signi lower than those of solitary leiomyomas, whereas ERb mRNA and levels in multiple leiomyomas were significantly higher than those solitary leiomyomas.
Literature	GENERIF_SUMMARY	a G2014A SNP genotype in exon 8 of the estrogen receptor was associated with postmenopausal women who had osteoporotic hi fracture.
Literature	GENERIF_SUMMARY	For key genes ERBB2 and ESR1, we demonstrate that overexpressi linked to the production of highly modified and truncated splice va in tumours, but not controls, correlated with tumour subtype
Literature	GENERIF_SUMMARY	Our meta-analysis results strongly suggest that the ESR1 gene polymorphisms PvuII (rs2234693T>C) and XbaI (rs9340799A>G) a associated with CVD risk.
Literature	GENERIF_SUMMARY	the consistency test results indicated the inter-observer agreemen more robust in MS images for HER2 , CK5/6 and ER than that in RG images for HER2 , CK5/6 and ER .
Literature	GENERIF_SUMMARY	Estrogen receptor, progesterone receptor or human epidermal grov factor receptor 2 samples with high-methylation risk score had the 5-year survival rate and overall survival time.
Literature	GENERIF_SUMMARY	ESR1 mutations can be robustly identified with circulating DNA ana and predict for resistance to aromatase inhibitors therapy.
Literature	GENERIF_SUMMARY	ESR1 is essential for prostate carcinogenesis and implicated in Pr Cancer growth control. Polymorphism of ESR1 is associated with Prostate Cancer prognosis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Inhibition of ERalpha signaling in prostate cancer cells may be a n therapeutic strategy to inhibit the osteoblastic lesion development as lung metastasis in patients with advanced prostate cancer.
Literature	GENERIF_SUMMARY	Activation of ERalpha but not ERbeta increases ADAM9 expressio cultured human neuronal cells.
Literature	GENERIF_SUMMARY	findings define, for the first time to our knowledge, a direct regulat pathway between ERalpha and HIF-1alpha, which might modulate hormone response in treatment
Literature	GENERIF_SUMMARY	The C allele at ESR1 rs2234693 was associated with abnormal alk phosphatase (OR = 5.2469, 95% CI = 1.3704-20.0895) and gamma- glutamyl transferase (OR = 3.4286, 95% CI = 1.0083-13.6578) level PBC patients.
Literature	GENERIF_SUMMARY	Our study offers evidence that the catechol-O-methyltransferase a estrogen receptor alpha genes may also play a role in the suscepti depressive mood in postmenopausal women.
Literature	GENERIF_SUMMARY	ESR1 may be associated with low mineral osseous densitometry, CYP17 and COMT gene polymorphisms were not associated with osseous densitometry.
Literature	GENERIF_SUMMARY	the genomic amplification of ESRRA causes its upregulation in a s of oral squamous cell carcinoma tumor samples.
Literature	GENERIF_SUMMARY	Data indicate higher ASXL2 expression in ERalpha-positive patient suggesting that ASXL2 could be a prognostic marker in breast can
Literature	GENERIF_SUMMARY	it is concluded that the ESR1 haplotypes may influence the Th2 cy profile and susceptibility to SLE among the South Indians
Literature	GENERIF_SUMMARY	ER-alpha regulates the osteoblast differentiation through modulati Dlx3 expression and/or interaction with Dlx3.
Literature	GENERIF_SUMMARY	ESR1/TUBA1B gene interaction might play pivotal roles in the occu and development of Postmenopausal Osteoporosis.
Literature	GENERIF_SUMMARY	ESR1 has a tissue-dependent and differentially methylated region.
Literature	GENERIF_SUMMARY	homozygote (XX) genotype of ER-alpha gene XbaI polymorphism, PvuII mutation, may increase the risk of chronic periodontitis, spec in the Chinese Han female population.
Literature	GENERIF_SUMMARY	expressed in prolactinomas, especially among men and post-men women
Literature	GENERIF_SUMMARY	we found no trend of increasing risk for primary dysmenorrhea in association with the number of putative high-risk genotypes of IL6 -597 G>A and -572 G>C and ESR1 gene XbaI polymorphisms.
Literature	GENERIF_SUMMARY	We found that elevated levels of estradiol and SHBG in HT [Hormo therapy] users were associated with elevated levels of CRP and lo levels of TFPI, both of which may be related to a prothrombotic mi HT [Hormone therapy]users.
Literature	GENERIF_SUMMARY	Protein kinase CK2 expression predicts relapse survival in ERalpha dependent breast cancer, and modulates ERalpha expression in vi
Literature	GENERIF_SUMMARY	miR-26b-3p played an important role in the regulation ofHuman u cord-derived mesenchymal stem cells proliferation in vitro by targe

Main Category	Category	Terms
		the ESR-CCND1 pathway.
Literature	GENERIF_SUMMARY	the augmentation of endogenous MTA1 expression during neurona ischemic injury acts additionally to an endocrinous cascade orche intimate interactions between ERalpha and BCL2 pathways.
Literature	GENERIF_SUMMARY	Data show that Cyclin D1 has a higher expression in breast cancer, positively correlated with estrogen receptors (ER) and better progn
Literature	GENERIF_SUMMARY	the results indicate that there is a crosstalk between ERalpha and to regulate the expression of each other, and suggest the involvem other receptors, such as ER-alpha36, in the rapid ERK1/2 activation
Literature	GENERIF_SUMMARY	These findings expand our understanding of hormone responsiven the tumor microenvironment; however, additional in-depth studies required to elucidate the detailed mechanisms of TSA-induced ERa regulation under hypoxia.
Literature	GENERIF_SUMMARY	ERalpha decreased CCL5 and IL-6 expression in conditioned media was collected from Cancer associated fibroblast cell.
Literature	GENERIF_SUMMARY	TargetPrint ER had a positive agreement of 98 %, and a negative agreement of 95 % with central pathology. Corresponding figures f were 85 and 94 % and for HER-2 72 and 99 %. Agreement of mRNA central protein was not different when the same or a different port the tumor tissue was analyzed or when DCIS and/or normal tissue included in the sample subjected to mRNA assays
Literature	GENERIF_SUMMARY	A review of ESR1 mediated mechanisms in the context of the fema reproductive system. [review]
Literature	GENERIF_SUMMARY	the MapQuanttrade mark assay, based on mRNA expression assay provides an objective and quantitative assessment of Estrogen rec Progesterone receptor and HER2 status in invasive breast cancer.
Literature	GENERIF_SUMMARY	Study shows the structural and functional features of novel C-term truncated ERa variants and revealed the mechanisms underlying constitutive transactivation by C-terminally truncated nuclear recep variants.
Literature	GENERIF_SUMMARY	Data support the idea that sequestration of ERalpha by tau protein underlies the loss of estrogen neuroprotection during the course o Alzheimer's disease.
Literature	GENERIF_SUMMARY	In the setting of highly sensitive and robust IHC methodology, cuto ER status determination and subsequent systemic therapy should revisited.
Literature	GENERIF_SUMMARY	modulation of ESR1-MED1 interactions by cAMP signaling plays a role in human decidualization.
Literature	GENERIF_SUMMARY	ESR1 rs1062577 polymorphism increases breast cancer risk in Ha Chinese women.
Literature	GENERIF_SUMMARY	ERalpha gene expression biomarker can accurately identify ERalph modulators in large collections of microarray data derived from MC cells.
Literature	GENERIF_SUMMARY	OLFM4 is downregulated by miR-486-5p, which contributes to ovar cancer tumorigenesis. Conversely, estrogen receptor signaling downregulates miR-486-5p and upregulates OLFM4 expression, sl the development and progression of ovarian cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	an efficient ERa-RNA aptamer is identified using a non-SELEX proc of aptamer selection. The high-affinity and specificity can be utilize detection of ERalpha in breast cancer and related diseases.
Literature	GENERIF_SUMMARY	This study found a significant interaction between rs4680 in COMT rs4986936 in ESR1 (P = .007) on opioid consumption.
Literature	GENERIF_SUMMARY	Biological insights into ERalpha genomics and behavior in breast c and cancer cell lines. Review.
Literature	GENERIF_SUMMARY	we have herein demonstrated that ERalpha expression associates MDM4 and MDM2 gene expression in primary breast invasive carc samples
Literature	GENERIF_SUMMARY	Genetic variation at the WLS and CCDC170/ESR1 loci were found t significantly associated with bone mineral density
Literature	GENERIF_SUMMARY	In this trial, maintenance therapy with E + BEV in ER-positive, HER2 negative MBC patients with no evidence of progression after first-l BEV did not achieve longer PFS compared with continuation of T +
Literature	GENERIF_SUMMARY	Expression of ESR1, RMND1 and CCDC170 associated with varian separate enhancer elements predisposing breast cancer. [meta-an
Literature	GENERIF_SUMMARY	In a recent study we reported associations between ALTs and sing nucleotide polymorphisms (SNPs) in the genes encoding estrogen receptor 1 (ESR1), steroid-5-alpha-reductase, type 2 (SRD5A2) and hormone-binding globulin (SHBG). No associations between the previously associated SNPs in the genes ESR1 and SRD5A2 and au like traits could be seen in the large replication sample.
Literature	GENERIF_SUMMARY	There was no difference in distribution of polymorphic genes ESR1 VKORC1 in peptic ulcer hemorrhage patients of both sexes, with th exception of A/A VKORC1 genotype found in women.
Literature	GENERIF_SUMMARY	these data strongly support a critical interplay between prolactin a estrogen via PAK1 and suggest that ligand-independent activation ERalpha through prolactin/PAK1 may impart resistance to anti-est therapies.
Literature	GENERIF_SUMMARY	Associations in women between recognition of identity through vo sounds and variation in ESR1.
Literature	GENERIF_SUMMARY	Lymph node involvement is more common with lack of ERalpha expression, and may be able to help triage which patients should u lymphadenectomy. Mutations in ESR1 might explain why some lov women with low BMI develop endometrial cancer.
Literature	GENERIF_SUMMARY	Along with the PHF20/MOF complex, G9a links the crosstalk betw ERalpha methylation and histone acetylation that governs the epig regulation of hormonal gene expression.
Literature	GENERIF_SUMMARY	the aberrant global DNA hypermethylation was dominantly induce ERalpha-activated-DNMT1, since DNMT1 over-expression significa increased global DNA methylation and DNMT1 knockdown reverse ERalpha-induced global DNA methylation
Literature	GENERIF_SUMMARY	The AIC value for this model was 3333.06, while for a model using higher as the cutoff for ER positivity, it was 3333.38, indicating tha model fits were nearly identical. The improvement in fit of the mod ERBB2 status was added was highly significant, with 5-year diseas specific survival estimates ranging from 48% to 99% (P

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ER-alpha expression increased in duodenal familial adenomatous polyposis.
Literature	GENERIF_SUMMARY	This study demonstrates that over-expression of WHSC1L1 is linke over-expression of ER-alpha in SUM-44 breast cancer cells and in p human breast cancers.
Literature	GENERIF_SUMMARY	Tumor-associated macrophage-secreted CXCL8 downregulated th ERalpha expression of endometrial cancer cells via HOXB13, which be associated with cancer invasion, metastasis and poor prognosi
Literature	GENERIF_SUMMARY	results indicated that there was a significant association between migraine and gene-gene interaction among the CYP19A1, FSHR, ES NRIP1.
Literature	GENERIF_SUMMARY	ESR1 mutation is associated with Breast Tumors.
Literature	GENERIF_SUMMARY	We investigate the impact of central histology (n = 772), immunohistochemistry for intrinsic subtyping and IHC4, and dicho (GG) or continuous (GGI) genomic grade (n = 472) on patient outco and benefit from taxane chemotherapy, focusing on HR+/HER2(HF
Literature	GENERIF_SUMMARY	Human influenza hemagglutinin-tagged PPARGC1B coprecipitated intensely with ERalpha in the +102525A than the +102525G constr after 17beta estradiol treatment
Literature	GENERIF_SUMMARY	it was observed that ER stimulated gene expression by interacting MEIS1 and FOXP3, and ER inhibited gene expression by interacting THRB and GRHL1.
Literature	GENERIF_SUMMARY	rapid signaling through ERalpha is essential for many of the transcriptional and physiological responses of ECs to E2, and that ERalpha rapid signaling in ECs, in vivo, may be critical for the vasculoprotective and anti-inflammatory effects of estrogen
Literature	GENERIF_SUMMARY	Here, we demonstrate a novel FOXC1-driven mechanism that supp ERa expression in breast cancer. We find that FOXC1 competes wi GATA-binding protein 3 (GATA3) for the same binding regions in th regulatory elements upstream of the ERa gene and thereby downregulates ERa expression and consequently its transcriptiona activity
Literature	GENERIF_SUMMARY	Estrogen receptor alpha is cell cycle-regulated and regulates the c in a ligand-dependent fashion.
Literature	GENERIF_SUMMARY	AGR2, through interaction with ER-alpha, enhances fulvestrant resi and IGF-1-induced carcinogenesis in breast cancer cells.
Literature	GENERIF_SUMMARY	Overexpressed miR-1280 levels in thyroid carcinoma cells may pro cell proliferation and invasion by inhibiting ER-alpha.
Literature	GENERIF_SUMMARY	Three SNPs of the ESR1 gene were not associated with increased
Literature	GENERIF_SUMMARY	increased expression of stromal colXalpha1 and low TILs correlate poor pathologic response in ER+/HER2+ breast tumors. Further st are needed to confirm their predictive value and impact on long-ter outcomes, and to determine whether this collagen exerts a protect effect on the cancer cells or simply reflects other factors within th microenvironment
Literature	GENERIF_SUMMARY	Data show that the combination of Bacillus Calmette-Guerin (BCG) the anti-estrogen/estrogen receptor ICI 182,780 led to better suppr

Main Category	Category	Terms
		of bladder cancer (BCa) than BCG alone.
Literature	GENERIF_SUMMARY	To investigate any changes in each cfDNA ESR1 mutation before a after treatment, authors analyzed the difference with cfDNA ESR1 mutations ratio in the first blood sample using droplet digital polyn chain reaction (ddPCR).
Literature	GENERIF_SUMMARY	Enhanced KLF17 expression sensitizes ERalpha-positive breast ca cells to endocrine therapy.KLF17-ERalpha interaction plays a poter role in inhibition of ERalpha-dependent breast cancer progression.
Literature	GENERIF_SUMMARY	Treatment with estrogens further increased PI-9 level while decrea that of ERalpha66 isoform thus excluding the involvement of this r isoform in the event. Moreover, our studies also provided evidence tertiary tumorspheres express elevated levels of CXCR4 and phosp p38, suggesting that the high PI-9 content might be ascribed to the activation of the proliferative CXCR4/phospho-p38 axis.
Literature	GENERIF_SUMMARY	Further mechanism study proved ERRF to be an interacting partner ERalpha. In total, these data revealed that ERRF is essential for the activity of E2-ERalpha pathway.
Literature	GENERIF_SUMMARY	Neoadjuvant tamoxifen treatment synchronized ERalpha/chromati interactions and downstream gene expression related to treatmen outcome and cell proliferation.
Literature	GENERIF_SUMMARY	LOX is a prognostic factor for poor progression free survival in pat with ER- breast cancer. LOX overexpression was positively correlat resistance to radiation and drug therapy.
Literature	GENERIF_SUMMARY	Myocardin inhibited the transcription and expression of ER alpha in cancer cells.
Literature	GENERIF_SUMMARY	A novel mechanism whereby GPR30 limits ERalpha expression via inhibition of the cAMP/protein kinase A signaling pathway. This su that stem-like cells within the stroma are a possible source of pros Cancer-associated fibroblasts and that the negative regulation of E expression by GPR30 is centrally involved in prostate stromal cell activation.
Literature	GENERIF_SUMMARY	GPR30, ERalpha, and ERbeta mRNA expression in testis of Klinefel syndrome men, were evaluated.
Literature	GENERIF_SUMMARY	Liposaccharides affects ERalpha but not ERbeta activation status.
Literature	GENERIF_SUMMARY	ESR1 mutations are not associated with clinical resistance to fulve in breast cancer patients.
Literature	GENERIF_SUMMARY	Among the proteins reported here, some well-known breast cancer markers (e.g., annexin A1, annexin A2 and vimentin) were identifie MDA-MB-231 cell line and thus we were able to distinguish both ce sufficiently.
Literature	GENERIF_SUMMARY	Report shows that sensitivity to the anti-proliferative effects of tam in some ER-positive breast cancer cells expressing the hormone-bi hotspot ESR1 mutations is dependent on cross-talk with the IGF1R signaling pathway.
Literature	GENERIF_SUMMARY	These findings identify a molecular mechanism by which ERalpha signaling reduces the aggressiveness of cancer cells, and demons that C/EBPdelta can have different functions in different types of c

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Considering the importance of the UBD
Literature	GENERIF_SUMMARY	Review of the role of estrogen receptor alpha in obesity and diabet
Literature	GENERIF_SUMMARY	High ESR1 expression is associated with breast Cancers.
Literature	GENERIF_SUMMARY	Estradiol can weakly modulate the motility and this effect is strictl associated with GPER and not with ESR1 and ESR2. The subcellula localization of GPER in the neck on stallion sperm is coherent with effect.
Literature	GENERIF_SUMMARY	In the presence of oestrogen, the oestrogen receptor (ESR1) regulo of ESR1 target genes) is in an active state. However, signalling by F is able to reverse the activity of the ESR1 regulon.
Literature	GENERIF_SUMMARY	Infiltrating lobular carcinoma (ILC)of the breast represents a biolog distinct subset of breast cancer, a biology defined by specific gene aberrations in E-cadherin, the high prevalence of ER-positive diseas relatively low frequency of HER2-positive disease, and specific mu events revealed by deep sequencing of ILC genomes
Literature	GENERIF_SUMMARY	ERR-alpha is involved in estrogen-mediated obstructive sleep apne hypopnea syndrome by regulating Myhc-slow expression.
Literature	GENERIF_SUMMARY	All neuromuscular choristoma (NMC) and NMC-fibromatosis show aberrant nuclear localization of beta-catenin, nuclear ERbeta expre and no ERalpha expression. The presence of CTNNB1 mutations b NMC and NMC-fibromatosis may be a shared molecular genetic abnormality underlying their pathogenesis.
Literature	GENERIF_SUMMARY	ESR1 inhibits senescence-like phenotype and facilitates transform induced by oncogenic ras in human mammary epithelial cells
Literature	GENERIF_SUMMARY	estrogen-stimulated SAOS-2 cells showed a significant increase of ERalpha expression
Literature	GENERIF_SUMMARY	Med25, a variable member of the Mediator complex, is a coactivat ligand-activated ERalpha that interacts with ERalpha through its C- terminal LXXLL motif after BPA exposure, and is functionally involv BPA-induced transcriptional regulation of CYP2C9 expression and enzyme activity.
Literature	GENERIF_SUMMARY	miR-513a-5p, miR-22-3p and miR-625-5p may have an impact on th regulation of the immune response and inflammatory cytokine pat through the regulation of their target gene(s), CBL, PPARGC1B and which may then lead to a dust mite-induced asthma attack
Literature	GENERIF_SUMMARY	miR-192-5p, induced by LY6K, downregulates ERalpha directly and induced tamoxifen resistance in ERalpha-positive breast cancer ce expression of ERalpha in ERalpha-negative breast cancer cells incr miR-500a-3p expression and directly inhibits LY6K expression
Literature	GENERIF_SUMMARY	these results highlight the involvement of ERa activation and of E2 dependent NGB upregulation in the insensitivity of MCF-7 to paclit These novel findings could have important implications in the development of targeted therapeutics for overcoming paclitaxel insensitivity in ERa-positive human breast cancer
Literature	GENERIF_SUMMARY	we genotyped five hormone-related genes
Literature	GENERIF_SUMMARY	ESR1 polymorphisms at rs2234693 and rs9340799 may be associ with OP, and could be used as markers to screen those with high ri

Main Category	Category	Terms
		postmenopausal OP in Chinese women.
Literature	GENERIF_SUMMARY	Tert expression was substantially increased, whereas ER alpha expression significantly decreased in the endometrium with RIF. N change was observed in PR expression. Tert expression was inver associated with ER alpha expression.
Literature	GENERIF_SUMMARY	The von Hippel-Lindau protein (pVHL)-dependent degradation of H participates in the regulation of ERalpha expression. Additionally, H forms a protein complex with ERa, and amino acids 396-823 of HI physically interact with the ligand-binding domain of ERa
Literature	GENERIF_SUMMARY	ERalpha engages not only LSD1, but its partners of the CoREST corepressor complex and the molecular chaperone Hsp90.
Literature	GENERIF_SUMMARY	A wide range of therapy regimens are available for treatment of ER metastatic breast cancer, but no guidelines are established for sequencing these therapies
Literature	GENERIF_SUMMARY	Results indicate that automated systems enable objective, precise quantification of estrogen receptor (ER).
Literature	GENERIF_SUMMARY	ESR1 Gene Polymorphism is associated with Lymphovascular Inva and Extranodal Extension in Breast Cancer.
Literature	GENERIF_SUMMARY	Results show that ESR1 expression is modulated by GGNBP2 in br cancer cells.
Literature	GENERIF_SUMMARY	Polymorphisms of the ESR1 gene are associated with the pathoge of endometriosis. [review]
Literature	GENERIF_SUMMARY	verexpression of COPS5, through its isopeptidase activity, leads to ubiquitination and proteasome-mediated degradation of NCoR, a k corepressor for ERalpha and tamoxifen-mediated suppression of E target genes.
Literature	GENERIF_SUMMARY	The estrogenic effects of environmental endocrine disruptors are primarily mediated by two estrogen receptor (ESR) forms
Literature	GENERIF_SUMMARY	For overall breast cancer risk, three single-nucleotide polymorphisr (SNPs) in miRNA biogenesis genes DROSHA rs78393591 (OR = 0. % CI
Literature	GENERIF_SUMMARY	the significance of posttranscriptional gene regulation by miR-19 i preventing the irregular migration of adult-born neurons that may contribute to the etiology of schizophrenia
Literature	GENERIF_SUMMARY	These results suggest that the ERalpha Pvu II polymorphism influe the prevalence of degenerative lumbar scoliosis
Literature	GENERIF_SUMMARY	the systematic absence of biological responses with an ESTZ deri unable to induce ERa dimerization stresses the importance of this the action of the compound, as reported for conventional estrogen
Literature	GENERIF_SUMMARY	Tax effect on ERalpha induced activation of genes by the classical pathway by testing its influence on E2-induced expression of Eestr response element promoter-driven luciferase reporter , is reported
Literature	GENERIF_SUMMARY	ESR1 gene polymorphism is associated with the risk of breast can Chinese population.
Literature	GENERIF_SUMMARY	ESR1 and VKORC1 single nucleotide polymorphisms were used to

Main Category	Category	Terms
		determine the vitamin K dosage in patients with ulcer-related hemorrhage.
Literature	GENERIF_SUMMARY	Na+/H+ exchanger regulatory factor (NHERF1) expression level po associates with G protein-coupled estrogen receptor (GPER) activa ER-positive breast cancer.
Literature	GENERIF_SUMMARY	Beta protein 1 (BP1) binds to and regulates estrogen receptor.
Literature	GENERIF_SUMMARY	ERalpha promotes metabolic alterations in breast cancer cells med by its target CHPT1, which this study implicates as a candidate therapeutic target.
Literature	GENERIF_SUMMARY	The number of dark cells progressively increased in DES-treated pa indicating their androgen-independence. The localization of AR and some light and dark basal cells indicated that the supply of androg estrogen was not inhibited during androgen deprivation (ablation) therapy .
Literature	GENERIF_SUMMARY	Calculation of the PGR/ESR1 gene-expression ratio and its immunohistochemical surrogate could be a useful and simple add routine breast cancer diagnostics. A high PGR/ESR1 ratio could be indicative of a favorable clinical outcome.
Literature	GENERIF_SUMMARY	Memo facilitates ER-alpha and c-Src interaction, ER-alpha Y537 phosphorylation, and has the ability to control ER-alpha extra-nucle localization in breast cancer cells.
Literature	GENERIF_SUMMARY	Choroidal metastases from BC are associated with ER and PR exp in the primary tumor and the luminal B molecular subtype.
Literature	GENERIF_SUMMARY	Systematic proteomic identification of the heat shock proteins tha interact with estrogen receptor alpha and biochemical characteriza the ERalpha-hsp70 interaction has been reported.
Literature	GENERIF_SUMMARY	The Hippo signaling pathway was significantly associated with ER- negative breast cancer (pathway level P = 0.02). Gene-based analy revealed that CDH1 was responsible for the pathway association (
Literature	GENERIF_SUMMARY	Transcriptional down-regulation of ESR1 mRNA could contribute to selective activity of neo-tanshinlactone on ER+ breast cancer cells
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha methylation or expression is a reversible s breast carcinogenesis which affected biological behavior of cells.
Literature	GENERIF_SUMMARY	Innon-muscle-invasive bladder cancer expression of ESR1, ERBB2 MKI67 are significantly different between stage and grade
Literature	GENERIF_SUMMARY	This study investigated the GPER expression in different types of n and abnormal proliferative endometrium, and the correlation with t presence of ERalpha.
Literature	GENERIF_SUMMARY	We have identified the AA genotype of SNP rs1643821 (ESR1 gene risk factor for dysfunctional worsening after orthognathic surgery.
Literature	GENERIF_SUMMARY	BMP4 inhibited ERalpha signaling by increasing the degradation of ERalpha, and that the expression of BMP4 itself was suppressed b estrogens.
Literature	GENERIF_SUMMARY	SNPs (rs2046210 and rs3734805) were strongly associated with susceptibility to breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Meta-analysis shows that the ESR1 rs9340799 polymorphism is no significant, independent contributing factor to age at menarche.
Literature	GENERIF_SUMMARY	Both Y537S and D538G mutations are associated with more aggre breast cancer disease biology.
Literature	GENERIF_SUMMARY	Combined targeting of XPO1 and ERalpha in several tamoxifen-res cell lines and tumor xenografts with the XPO1 inhibitor, Selinexor, a tamoxifen restored tamoxifen sensitivity and prevented recurrence
Literature	GENERIF_SUMMARY	Inherited ESR1 and PGR genotypes may affect tumor ESR1/ERalph PGR/PgR expression.
Literature	GENERIF_SUMMARY	ESR1 mutations may be detected by genomic sequencing of tissue biopsies of the metastatic tumor or by sequencing the circulating t cells or tumor DNA (ctDNA).Sequencing results may lead to a thera ""match"" with an existing FDA-approved drug or match with an experimental agent that fits the clinical setting"
Literature	GENERIF_SUMMARY	The aim of this study was to analyze the role of estrogen receptor (ESR1) genotypes in female infertility.The study revealed the prese two single nucleotide polymorphisms in the ESR1 gene, PvuII and X Individual analyses of these two polymorphisms showed that the X heterozygote was significantly increased in controls compared to c (odds ratio-0.39, confidence interval- 0.21 to 0.74, p-0.005).
Literature	GENERIF_SUMMARY	the most extensive molecular dynamics simulations to date of ER- with over 1mus of combined simulations for the monomer and dim forms, are reported.
Literature	GENERIF_SUMMARY	our findings identified a novel regulatory mechanism of MSI2 as an upstream regulator of ESR1 and revealed the clinical relevance of t RNA-binding protein MSI2 in breast cancer.
Literature	GENERIF_SUMMARY	significant prognostic roles of ERalpha, ERbeta and aromatase wer discovered in the in prostate cancer specimens of our large multic cohort.
Literature	GENERIF_SUMMARY	Data indicate the necessity of EPRS for proliferation of tamoxifen-resistant estrogen receptor (ER+) breast cancer, but not ER- breast cells.
Literature	GENERIF_SUMMARY	As BRCA1-mutated ER(+) tumors show a BRCAness copy number and LOH, it is likely that the loss of a functional BRCA1 protein play in tumorigenesis in BRCA1-mutated ER(+) tumors. Therefore, we hypothesize that these tumors are sensitive to drugs targeting the gene defect, providing new targeted treatment modalities for adva BRCA-deficient, ER(+) breast cancer
Literature	GENERIF_SUMMARY	Interplay of nuclear receptors (ER, PR, and GR) and their steroid ho in MCF-7 cells.
Literature	GENERIF_SUMMARY	Cy-3-glu directly binds to the ligand-binding domain (LBD) of ERalp inhibits EGFR/AKT signaling, and promotes EGFR degradation
Literature	GENERIF_SUMMARY	a novel mechanism of estrogen receptor in controlling osteogenes human cells
Literature	GENERIF_SUMMARY	Despite the large concordance between P and matched M for the evaluated molecular alterations, potential actionable targets such ESR1 mutations were found only in M. This supports the importan characterizing the M disease. Other targets we identified, such as

Main Category	Category	Terms
		and IDO1, warrant further investigation in this patient population.
Literature	GENERIF_SUMMARY	The Estrogen receptor alpha gene polymorphism SNP rs2234693 v associated with Systemic lupus erythematosus per se but the min rs2234693-C was correlated with the presence of nephritis and dis skin rash. On the other hand, the rs2234693-CC genotype was corr with the absence of arthritis as well as anti-ANA and anti-RNP autoantibodies.
Literature	GENERIF_SUMMARY	Report a significant survival benefit in lung adenocarcinoma patier positive expression of one of the investigated hormonal receptors
Literature	GENERIF_SUMMARY	Interaction of estrogen receptor alpha gene and fluoride exposure influence women's serum reproductive hormone concentrations
Literature	GENERIF_SUMMARY	We demonstrate that ER(+) breast cancers contain two CAF subtyp defined by CD146 expression. CD146(neg) CAFs suppress ER expr in ER(+) breast cancer cells, decrease tumor cell sensitivity to estr and increase tumor cell resistance to tamoxifen therapy
Literature	GENERIF_SUMMARY	Estradiol-induced DDP chemoresistance depends on the balance b ERa and ERb expression and the p53 pathway.
Literature	GENERIF_SUMMARY	The results of this meta-analysis confirmed that there was no corr between polymorphisms in ESRalpha and coronary heart disease susceptibility in the Chinese population
Literature	GENERIF_SUMMARY	Our results support an association between eight selected SNPs in ESR1 gene and percent mammographic density.
Literature	GENERIF_SUMMARY	Loss-of-Function ESR1 Mutation is associated with Estrogen Inser
Literature	GENERIF_SUMMARY	ESR1 variants associated with anomalous uterine bleeding in adol girls born with low birth weight.
Literature	GENERIF_SUMMARY	Findings show that ESR1 selectively down-regulates HYAL1 expres breast cancer cells.
Literature	GENERIF_SUMMARY	Taken together, these results indicate that E2 exerts inhibitory effe VSMC apoptosis and calcification through ERalpha-mediated Gas6 transactivation. These findings indicate a potential therapeutic stra for the prevention of vascular calcification, especially in postmeno women.
Literature	GENERIF_SUMMARY	Association study investigated the ESR1, ESR2, and CYP19A1 gene polymorphisms in Chinese migraine patients, data suggested that rs2234693 plays a potential role in migraine susceptibility in a Chi population, especially for menstrually related migraine
Literature	GENERIF_SUMMARY	Expression of E-cadherin was significantly associated with the diff type familial EOGCs and early stage, whereas expression of ERalph might have little role in EOGC tumorigenesis.
Literature	GENERIF_SUMMARY	Estrogen Enhances the Expression of the Polyunsaturated Fatty Ac Elongase Elovl2 via ERalpha in Breast Cancer Cells
Literature	GENERIF_SUMMARY	FOXA1 overexpression mediates endocrine resistance by altering t transcriptome and IL-8 expression in ER-positive breast cancer.
Literature	GENERIF_SUMMARY	ESR1 circulating mutations are independent risk factors for poor o after AI failure, and are frequently detectable before clinical progre Interventional studies based on ESR1 circulating status are warran

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results show that KMT2B interacts with ERalpha to bind the ERalp binding sites of IL-20 and other ERalpha target genes with H3K4 modifications.
Literature	GENERIF_SUMMARY	variations in airway smooth muscle ER expression profiles occur w inflammation.
Literature	GENERIF_SUMMARY	analysis of interplay between hypoxia, HIF stabilization and the ER- beta-ratio in several ER-positive breast cancer cell lines
Literature	GENERIF_SUMMARY	review of literature indicates ERa polymorphisms are of great impo with respect to the effect of estrogens on the functioning of the bo woman after menopause, and may imply the development of many pathological states, including the prevention or development of me disorders
Literature	GENERIF_SUMMARY	Our results suggest that calcitriol anti-cancer therapy is more likely induce higher levels of CAMP in ERalpha- breast cancer cells, wher compared to ERa+ breast cancer cells.
Literature	GENERIF_SUMMARY	MCAM promotes tamoxifen resistance by transcriptionally suppres ERalpha expression and activating the AKT pathway, followed by induction of epithelial-mesenchymal transition.
Literature	GENERIF_SUMMARY	analysis of epigenetic mechanisms of ERalpha implicated in the fa fulvestrant-resistant breast cancer
Literature	GENERIF_SUMMARY	This study showed that phosphorylation of oestrogen receptor at s 118 and 167 were associated with several microenvironmental fac including increased microvessel density in estrogen receptor-posit operable ductal breast cancers
Literature	GENERIF_SUMMARY	MYST3 binds to the proximal promoter region of ERalpha gene, an inactivating mutations in its HAT domain abolished its ability to reg ERalpha, suggesting MYST3 functioning as a histone acetyltransfe that activates ERalpha promoter.
Literature	GENERIF_SUMMARY	Alternative splicing of estrogen receptor alpha is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	The 1-, 3-, and 5-year survival rates were higher for patients with es receptors (ER)-positive tumors than for patients with ER-negative t
Literature	GENERIF_SUMMARY	besides the full-length ERalpha66, the overlooked ERalpha46 isofo also expressed in a majority of breast tumors. This finding highligh importance of the choice of antibodies used for the diagnosis of b cancer, which are able or not to detect the ERalpha46 isoform.
Literature	GENERIF_SUMMARY	the direct binding of ERalpha-36 to ERK2 prevents its dephosphory by MKP3 and enhances the downstream signaling. These findings improve our understanding of the regulation of non-genomic estro signaling and open new avenues for personalized therapeutic appr targeting Src or MEK in ERalpha-36-positive patients.
Literature	GENERIF_SUMMARY	The altered expression of ER and PR may be associated with the expression variation of integrin and pinopode formation in endome of luteal phase deficit women.
Literature	GENERIF_SUMMARY	Analysis of ER ChIP-seq data from MCF7 cells identified an introni in the IGF1R gene, rs62022087, predicted to increase ER binding. Functional studies confirmed that ER binds preferentially to rs6202 versus the wild-type allele.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Novel ESR1 mutations that constitutively activate the receptor and promote breast cancer phenotypes.
Literature	GENERIF_SUMMARY	A Novel Functional Domain of Tab2 Involved in the Interaction with Estrogen Receptor Alpha in Breast Cancer Cells
Literature	GENERIF_SUMMARY	RNAscope may be a complementary method in the detection of ER status in breast cancer patients.
Literature	GENERIF_SUMMARY	estrogen may promote the progression of ER-negative breast canc stimulating cancer-associated fibroblasts to secrete SDF-1alpha, w can recruit MDSCs to the tumor microenvironment to exert tumor-promoting effects
Literature	GENERIF_SUMMARY	IKKalpha is an important determinant of poor outcome in patients ER-positive invasive ductal breast cancer and thus may represent a potential therapeutic target
Literature	GENERIF_SUMMARY	Next-generation sequencing analysis of the MCF-7 breast cancer c overexpressing miR-335-5p and miR-335-3p demonstrated that the miRNA duplex repressed genes involved in the ERalpha signaling pathway, and enhanced resistance of MCF-7 cells to the growth inh effects of tamoxifen.
Literature	GENERIF_SUMMARY	FES PET/CT was a useful biomarker of ER occupancy and/or downregulation in a phase I dose escalation trial of GDC-0810 and select the dosage of the ER antagonist/degrader for phase II trials
Literature	GENERIF_SUMMARY	Results show that ERalpha expression level is regulated by FOXC1 breast cancer cells affecting their sensitivity to tamoxifen.
Literature	GENERIF_SUMMARY	ESR1 can be a potential target for Ppituitary adenoma.
Literature	GENERIF_SUMMARY	RUNX1 and AXIN1 proteins are strongly correlated in ER(-) tumors
Literature	GENERIF_SUMMARY	Data suggest that, under agonism of estrogens, ESR1 exhibits a ge anxiogenic effect, ESR2 exhibits a generally anxiolytic effect, and G exhibits both anxiogenic and anxiolytic effects. (ESR1 = estrogen r 1/alpha; ESR2 = estrogen receptor 2/beta; GPR30 = G protein-coup estrogen receptor 1) [REVIEW]
Literature	GENERIF_SUMMARY	the Hippo kinases LATS1 and 2 control human breast cell fate via crosstalk with ERalpha
Literature	GENERIF_SUMMARY	Laboratory models corroborate intertumor heterogeneity of endoc response in HGSOC but identify features associated with functiona ERalpha and endocrine responsiveness. Assessing ERalpha functi IGFBP3 expression) in conjunction with H-score may help select pa who would benefit from endocrine therapy. Preclinical data sugges SERDs might be more effective than tamoxifen
Literature	GENERIF_SUMMARY	Hormone receptor expression was concentrated in the fibroblasts, RXFP1 was also evident in blood vessels and nerves
Literature	GENERIF_SUMMARY	Here the authors report a nodal role of IGF-IR in the regulation of E positive breast cancer cell aggressiveness and the regulation of expression levels of several extracellular matrix molecules.
Literature	GENERIF_SUMMARY	Estradiol induced the expression of ABCG2 through ERalpha and th expressed ABCG2 made MCF7 more tolerant to methotrexate.
Literature	GENERIF_SUMMARY	The rs2228480 (ESR1) and rs4986938 (ESR2) variants did not alte

Main Category	Category	Terms
		sporadic Breast cancer risk, but they did modulate the Breast canc severity.
Literature	GENERIF_SUMMARY	The molecular-level exploration into the signaling mechanism was investigated for the respective role of the residues from the two im proteins from Homo sapiens ERalpha and XBP-1 (bZIP domain)) fo breast tumors metastasis via proliferation. It further dealt with the analysis and examination of the changes that are caused due to th mutation K214R in the ERalpha protein.
Literature	GENERIF_SUMMARY	Genotypes and Haplotypes of the Estrogen Receptor alpha Gene (Are Associated With Female-to-Male Gender Dysphoria
Literature	GENERIF_SUMMARY	Suggest a possible link between ESR1/2 SNPs and the occurrence Takotsubo syndrome.
Literature	GENERIF_SUMMARY	We found an approximately approximately 3 fold increased rate of RS among BRCA1 and 2 carriers with ER positive breast cancer (BC)compared with the general BC population
Literature	GENERIF_SUMMARY	High ESR1 expression is associated with Papillary Thyroid Carcino
Literature	GENERIF_SUMMARY	stable hyaluronate synthase 2 (HAS2) overexpression in estrogen receptor alpha (ERalpha) -positive MCF7 cells oppositely altered es dependence of cell growth and its sensitivity towards antiestrogen
Literature	GENERIF_SUMMARY	Promoter methylation of MCAM, ERalpha and ERbeta have a poten be utilized as biomarker for the early detection of prostate cancer their sensitivity and specificity seem to be better than serum PSA.
Literature	GENERIF_SUMMARY	Genetic Polymorphisms in the ESR1 Genes Do Not Correlate With Osteoporosis in Patients With Familial Dysautonomia.
Literature	GENERIF_SUMMARY	Found a striking resemblance in the ERalpha patterns of the two ti types.
Literature	GENERIF_SUMMARY	SGK3 a kinase transcriptionally regulated by estrogen receptor alp (ERalpha) in breast cancer, sustains ERalpha signaling and drives t acquired aromatase inhibitors resistance by protecting against endoplasmic reticulum (EnR) stress-induced ERalpha downregulat cell death through preserving SERCA2b function.
Literature	GENERIF_SUMMARY	Calmodulin-induced dimerization of ER-alpha is required for estrog stimulated transcriptional activation by the receptor.
Literature	GENERIF_SUMMARY	Our data uncovered an important role of Erbin in regulating hepato carcinoma (HCC) tumorigenesis through inactivating ERalpha-med tumor-suppressive signaling, suggesting a new strategy for tamox therapy in HCC by targeting Erbin/ERalpha signaling axis.
Literature	GENERIF_SUMMARY	the increasing of TG and TL serum concentrations associated to S XbaI c454-351A>G on ESR1 gene is enhanced in both obese (highe and elder women. Taken together, these results suggested that Xb c454-351A>G SNP is associated to changes in lipid profile, particu serum TG and TL, in this cohort of post-menopausal woman
Literature	GENERIF_SUMMARY	Estrogen receptor 1 (ESR1) single nucleotide polymorphism (SNP) rs1062577 A allele is significantly associated with the elevated risk greater stages of breast cancer.
Literature	GENERIF_SUMMARY	study describes RNF8 as a co-activator of ERalpha increases ERalp stability via post-transcriptional pathway, and provides a new insig

Main Category	Category	Terms
		mechanisms for RNF8 to promote cell growth of ERalpha-positive cancer.
Literature	GENERIF_SUMMARY	High ERalpha expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	The single nucleotide variant rs12722489 determines differential E binding and enhancer properties of an IL2RA intronic region.
Literature	GENERIF_SUMMARY	PAX8-positive Biphasic Synovial Sarcoma Expressing Hormonal Receptors.
Literature	GENERIF_SUMMARY	ERalpha upregulates vinculin expression in breast cancer cells; Los vinculin promotes amoeboid features of cancer cells
Literature	GENERIF_SUMMARY	PHB promotes AR activation in ER-positive breast cancer.
Literature	GENERIF_SUMMARY	Analysis of ESR1 mutations in metastatic breast cancer.
Literature	GENERIF_SUMMARY	results also suggest that, in the presence of adult endogenous ste levels, ERalpha36 overexpression in vivo contributes to alter mamm gland architecture which may support pre-neoplastic lesion and au breast cancer risk
Literature	GENERIF_SUMMARY	ERalpha functions as a transcriptional effector of cytokine-induced IKKbeta signaling, suggesting a mechanism through which the tum microenvironment controls tumor progression and endocrine resis
Literature	GENERIF_SUMMARY	disease re-occurrence in patients with ER-alpha positive tumors m from tumor cells chronically resident in hypoxic environments.
Literature	GENERIF_SUMMARY	Activation of HIF1A by prolonged in vivo hyperinsulinemia potentia cancerous progression in estrogen receptor-positive breast cancer
Literature	GENERIF_SUMMARY	polymorphisms affect susceptibility to prostate cancer and benign prostatic hyperplasia
Literature	GENERIF_SUMMARY	ESR-1 could be associated with one-year cognitive decline in healt oldest-old individuals, since the estrogen pathway may be involved neuroprotection, even in healthy brain aging.
Literature	GENERIF_SUMMARY	17beta-estradiol induced up-regulation of apoM in HepG2 cells is t an ER-alpha-dependent pathway involving ER-alpha binding elemen promoter of the apoM gene.
Literature	GENERIF_SUMMARY	Patients with Estrogen- and progesterone receptors-positive invasi lobular carcinoma (ILC) and invasive ductal carcinoma (IDC)have s quantitative ER and PR expression profiles, implicating that ER/PR expression is unlikely to be a confounding factor in studies concer chemo-sensitivity of ILC and IDC
Literature	GENERIF_SUMMARY	we are lacking prospective studies to confirm these results and to determine the optimal treatment combinations for patients with th mutations. In addition, the clinical development of novel agents to overcome resistance engendered by these mutations is also neede
Literature	GENERIF_SUMMARY	breast tumors that highly express ZEB1 exhibit ER-alpha promoter hypermethylation.
Literature	GENERIF_SUMMARY	polymorphisms not associated with late miscarriage
Literature	GENERIF_SUMMARY	Data suggest marked changes in DNA methylation occur at a singl site (CpG4) in promoter region of ESR1 as breast cancer cells beco

Main Category	Category	Terms
		resistant to hormonal/aromatase inhibitor antineoplastic agents; t region is completely conserved among species, suggesting that it a methylation-sensitive ETS2 transcription factor binding site/resp element. (ESR1 = estrogen receptor 1; ETS2 = ETS proto-oncogene
Literature	GENERIF_SUMMARY	Cytokeratin 5/6 protein expression as a frequent feature of high-gr serous ovarian carcinoma with various staining patterns. Cytokera in combination with ER-alpha proved to be a negative prognostic m
Literature	GENERIF_SUMMARY	ClC-3 is a potential target of 17beta-estradiol and is modulated by ERalpha in breast cancer cells.
Literature	GENERIF_SUMMARY	ER-alpha36 is localized on the plasma membrane and cytoplasm i cervical cancer tissues and cell lines. ER-alpha36 mediates estrog stimulated MAPK/ERK activation and regulates migration, invasion proliferation in cervical cancer cells.
Literature	GENERIF_SUMMARY	No association was found between the ERalpha gene polymorphis Premature thelarche in girls. However, XbaI polymorphisms may contribute to early breast budding.
Literature	GENERIF_SUMMARY	Folate deficiency impaired spermatogenesis and reduced sperm concentration may partly due to inhibiting the expression of the th molecules (Esr1, Cav1 and Elavl1) essential for sperm production.
Literature	GENERIF_SUMMARY	The GG genotype of ERalpha XbaI polymorphism could be a geneti factor for PE predisposition.
Literature	GENERIF_SUMMARY	Estrogen receptor-Src signaling plays an important role in ER (+) b cancer, which shows a high potential for bone metastasis.
Literature	GENERIF_SUMMARY	In MCF-7 cells SUMOylated glucocorticoid receptor represses ERa dependent gene and enhancer activity via disassembly of the Meg complex, which requires effective recruitment of the N-CoR/SMRT- complex.
Literature	GENERIF_SUMMARY	Ten international pathology institutions participated in this study a determined messenger RNA expression levels of ERBB2, ESR1, PG MKI67 in both centrally and locally extracted RNA from formalin-fix paraffin-embedded breast cancer specimens with the MammaTyp test. Samples were measured repeatedly on different days within t laboratories, and reproducibility was assessed by means of varian comp...
Literature	GENERIF_SUMMARY	data have identified BASP1 as an ERalpha cofactor that has a cent in the transcriptional and antitumourigenic effects of tamoxifen.
Literature	GENERIF_SUMMARY	Combining ER and IGF1R targeting with HER2 targeted therapies m an alternative to HER2 targeted therapy and chemotherapy for pati with HER2/ER/IGF1R positive breast cancer.
Literature	GENERIF_SUMMARY	Our results revealed a lack of significant association between ESR polymorphisms and calcaneal quantitative ultrasound in a cohort o young adults suggesting that ESR1 gene do not play a major role i acquisition of bone mass during early adulthood.
Literature	GENERIF_SUMMARY	Taken together, ESR1 mutations in genome-edited breast cancer c confer ligand-independent growth and endocrine resistance. These biologically relevant models can be used for further mechanistic a translational studies, including context-specific and mutation site- analysis of the ESR1 mutations.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	AR, ESR1 and ESR2 methylation were correlated with anthropomet metabolic and hormonal parameters in patients with gender dystro under cross-sex hormone treatment.
Literature	GENERIF_SUMMARY	TNBC cells are more resistant to radiation than luminal-type breast cells; ERalpha expression may have major roles in modulating brea cancer cell radiosensitivity
Literature	GENERIF_SUMMARY	Primary and secondary phthalate metabolites appear to exert diffe effects on ESR1 receptors.
Literature	GENERIF_SUMMARY	ERalpha is not regulated by methylation of the promotor region in endometriosis.
Literature	GENERIF_SUMMARY	The negative correlation between high-grade expression HCMV-IE hormone receptor expression suggests a role for HCMV in hormon receptor-negative BC tumors, possibly by forcing BC cells into a m aggressive phenotype.
Literature	GENERIF_SUMMARY	Results identify here a common ESR1 SNP, rs2144025-C>T, as a pc regulatory variant affecting ESR1 expression, and behavioral traits mental diseases.
Literature	GENERIF_SUMMARY	polymorphisms of the ESR1 gene do not contribute significantly to genetic risk for obesity phenotypes in a population of young Cauca adults
Literature	GENERIF_SUMMARY	Estrogen receptor alpha gene polymorphism and vascular complic in girls with type 1 diabetes mellitus.
Literature	GENERIF_SUMMARY	IL-1B-stimulated breast cancer cells down regulate ER-alpha expre ER-alpha promoter methylation is proposed as a regulatory mecha Promoter methylation confers resistance to tamoxifen.
Literature	GENERIF_SUMMARY	sLZIP is a novel co-repressor of ERalpha, and plays a negative role ERalpha-mediated cell proliferation in breast cancer
Literature	GENERIF_SUMMARY	this study demonstrated the feasibility of our protocol to detect an monitor ESR1 gene mutations at the single CTC level in MBCs.
Literature	GENERIF_SUMMARY	Findings suggest ESR1 and PITX2 promoter methylation may be correlated with a worse survival of patients with breast cancer. The clinical utility of aberrantly methylated ESR1 and PITX2 could be a promising factor for the prognosis of breast cancer.
Literature	GENERIF_SUMMARY	Study identified consensus binding motifs for the bile acid-respons FXR in the BRCA1 and estrogen receptors 1 and 2 (ESR1 and ESR2 genes. Bile acids treatment resulted in induction of BRCA1 transcr levels and decrease in ESR1 expression in OVCAR3 ovarian cancer
Literature	GENERIF_SUMMARY	this study found that C allele of rs2234693 and G allele of rs93407 within ESR1 gene, their interaction between rs2234693 and curren smoking were all associated with increased systemic lupus erythematosus risk
Literature	GENERIF_SUMMARY	The methylation level of the ESR1 promoter observed in women wi body mass index (BMI) >/=30 kg/m(2) was higher than in the subg of women of BMI
Literature	GENERIF_SUMMARY	findings revealed a new mechanism through which Bisphenol A pr cell proliferation by reinforcing glycolysis-based energy production dependent on ER signaling.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Based on these findings, we infer that high PPFIA1 expression mig an independent prognostic indicator of increased metastatic relap in patients with estrogen receptor+/N- breast cancer, but not in est receptor+/N+ or estrogen receptor- patients.
Literature	GENERIF_SUMMARY	These data suggest the ERalpha pathway remains active in estrog deprived ER(+)/FGFR1-amplified breast cancers. Therefore, these t are endocrine resistant and should be candidates for treatment wi combinations of ER and FGFR antagonists. Clin
Literature	GENERIF_SUMMARY	the performance of the assay, in terms of sensitivity and reproduci was studied using undiluted calf serum, and excellent recoveries i range of 94.7-108% were achieved for the detection of ERalpha in cell lysate.
Literature	GENERIF_SUMMARY	Hbo1 is another ERalpha coactivator that ubiquitinates ERalpha
Literature	GENERIF_SUMMARY	Patients with ER(+)/HER2(-) breast cancer were consecutively incl for retrospective analysis. The contralateral parenchyma was segn automatically on dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI), where upon the average of the top-10% late enhancement was calculated
Literature	GENERIF_SUMMARY	ESR1 mutation is associated with estrogen receptor expression an proliferative activity and affects about 14% of estrogen receptor-po bone metastases from breast cancer.
Literature	GENERIF_SUMMARY	These studies identify functional EREs in the IKKB promoter and id IKKB as an ERalpha and NSC35446.HCl-regulated gene, and they s that NF-kappaB and IKKB, which were previously linked to antiestro resistance, are targets for NSC35446.HCl in reversing antiestrogen resistance.
Literature	GENERIF_SUMMARY	The identified variants on DNA level might impair receptor functio through different molecular mechanisms. Mutations of ESR1 and are associated with MRKHS. Thus, we consider these genes poten candidates associated with the manifestation of MRKHS.
Literature	GENERIF_SUMMARY	We identified calmodulin-like protein 3 (CALML3) as a key sensor SNP and a coregulator of ERalpha, which contributes to differentia transcription regulation in an estrogen and SERM-dependent fashi Furthermore, using CRISPR/Cas9-engineered ZR75-1 breast cance with different ZNF423 SNP genotypes, striking differences in cellu responses to SERMs and PARP inhibitors, alone or in combination
Literature	GENERIF_SUMMARY	Serum ERbeta/ERalpha ratio predicts endoscopic activity in Crohn' disease patients.
Literature	GENERIF_SUMMARY	Estrogen receptor percent positive staining is not independently re breast cancer survival after adjustment for other survival-related fa ER intensity, in contrast, demonstrates promise for prognostic utili
Literature	GENERIF_SUMMARY	Results of our present study suggest that both IL-6 and HGF derive Cancer-associated fibroblasts (CAFs)could contribute to the intrat androgen metabolism in ER-negative breast carcinoma patients.
Literature	GENERIF_SUMMARY	Estrogen receptor recruits steroid receptor coactivator-3 primary coactivator and secondary coactivators, p300/CBP and CARM1 to regulate genetic transcription.
Literature	GENERIF_SUMMARY	Hypoxia phosphorylates the ER S118 residue in the absence of est via MAPK and regulates breast cancer cell proliferation and migrat

Main Category	Category	Terms
		This suggests a potential role for ERalpha S118 phosphorylation in cancer progression and may provide a prognostic marker and targ endocrine therapies.
Literature	GENERIF_SUMMARY	Functional enrichment analysis of DEGs indicated their potential involvement in cancer, and cell activity-related processes. Fifteen c were identified between DEGs and target genes of ESR1, of which found to be regulated by miRNA. Several known estrogen response and novel genes were obtained in this study and they might provid potential therapeutic targets for osteosarcoma
Literature	GENERIF_SUMMARY	Our findings suggest the possibility for changing the treatment par of patients with ER+HER2-MBC, so a randomized prospective stud warranted to determine the optimum sequence of systemic therap
Literature	GENERIF_SUMMARY	African American patients with breast cancer had worse survival in certain subtype and stage, especially in ER+ breast cancer
Literature	GENERIF_SUMMARY	We found higher nuclear (n)ERalpha and nERbeta5 and lower cytop (c)ERalpha expression in advanced ovarian cancers. Loss of nERal and cERbeta2 expression were observed in clear cell histological subtypes. Positive cERalpha and higher cERbeta1 expression were associated with better disease-free and overall survival.
Literature	GENERIF_SUMMARY	Sp1 induces leptin expression in cooperation with estradiol action through estrogen receptor alpha
Literature	GENERIF_SUMMARY	The role of estrogens in the regulation of ZO-1 and estrogen recept and 2 was evaluated in human primary gut tissues by immunohistochemistry, immunofluorescence and qPCR.
Literature	GENERIF_SUMMARY	higher CXCL8 mRNA levels in breast cancer tissues together with estrogen receptor negativity was associated with significantly sho overall survival.
Literature	GENERIF_SUMMARY	the estrogen- (ER), progesterone- (PgR) and HER2/neu receptor sta the primary tumor with brain metastases in a series of 24 consecu breast cancer patients, is reported.
Literature	GENERIF_SUMMARY	nhibition of miR-18a increases the expression of ER1 and promote apoptosis in HTR8 human trophoblasts.
Literature	GENERIF_SUMMARY	the P2X7R rs3751143 and ER-alpha PvuII two-locus interaction co significantly high susceptibility to osteoporosis in Chinese postmenopausal women.
Literature	GENERIF_SUMMARY	these results reveal an ordered and cooperative assembly of ERalp enhancers requiring functional interplay among p300, Mediator, an which has implications for hormone-dependent gene regulation in cancers.
Literature	GENERIF_SUMMARY	results suggest different interactions may occur in a gender and ti dependent manner among rs9340799 and other potential factors c depression or predictors of its severity, and influence the developm and natural rehabilitation of depression.
Literature	GENERIF_SUMMARY	Taken together, these findings demonstrated that Twist was upreg in high invasion and metastasis cell lines as well as invasive ducta carcinoma (IDC) tissues companioned with downregulated expres E-cadherin and ER, which provides important clues for the deeper s breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	all the patients without these mutations by molecular barcode nex generation sequencing (MB-NGS) were found to have no mutation ddPCR. In conclusion, MB-NGS could successfully detect ESR1 mutations in cfDNA with a higher sensitivity of 0.1% than conventi NGS and was considered as clinically useful as ddPCR
Literature	GENERIF_SUMMARY	General obesity, indicated by BMI, is more strongly associated with PR+ subtype, especially among premenopausal women, whereas c obesity, indicated by WHR, is more specific for ER-/PR- subtype, independent of menopausal status. These results suggest that dif chemoprevention strategies may be appropriate in selected indivic
Literature	GENERIF_SUMMARY	These results suggest that miR-125a-3p can function as a novel tu suppressor in ER(+) breast cancer by targeting CDK3, which may b potential therapeutic approach for TamR breast cancer therapy
Literature	GENERIF_SUMMARY	Concomitant high expression of ERalpha36, EGFR and HER2 was s associated with aggressive behaviors including extrathyroidal exte lymph node metastasis and high TNM stage in papillary thyroid carcinomas.
Literature	GENERIF_SUMMARY	We conducted a replication study using a dataset outside of the Challenge, and found the association between TP53INP2 and ER-n breast cancer was significant (p = 5.07x10-3). Expression of HP (1 showed a suggestive association with ER-negative breast cancer i discovery phase (Z = 4.30, p = 1.70x10-5) although the association not significant after Bonferroni adjustment.
Literature	GENERIF_SUMMARY	By Estrogen Receptor alpha-Bound Enhancers.
Literature	GENERIF_SUMMARY	These results indicate an interaction between Brf1 and ER alpha, w synergistically regulates the transcription of Pol III genes.
Literature	GENERIF_SUMMARY	Oral administration of RAD140 substantially inhibited the growth o ER(+) breast cancer patient-derived xenografts (PDX). Activation o and suppression of ER pathway, including the ESR1 gene, were see RAD140 treatment.
Literature	GENERIF_SUMMARY	A major finding of our study is that one out of five (20%) patients w breast cancer BM had a receptor discrepancy between the primary and the subsequent BM, with loss of hormone receptors (ER and/c expression, and gain of HER2 overexpression as the most commo observed changes
Literature	GENERIF_SUMMARY	the cellular localization of ERalpha36 in estrogen-responsive huma uterine smooth muscle (ht-UtSMC) and leiomyoma, is reported.
Literature	GENERIF_SUMMARY	Data suggest that Y537S/D538G ESR1 mutant breast cancer cell li exhibit enhanced proliferation in response to IGF1/IGF1R signaling mechanism involves IRS1; here, knockdown of IRS1 attenuates enh IGF1/IGF1R signaling response in ESR1 mutant cells. (ESR1 = estro receptor 1; IGF1 = insulin like growth factor 1; IGF1R = IGF1 recepto = insulin receptor substrate 1)
Literature	GENERIF_SUMMARY	the EMS indicated a more malignant subgroup of the ER- cells by discriminating the chemoresistant ER- cells from the chemosensit ones. In addition, the classified chemoresistant ER- patients demonstrated worse prognosis. In conclusion, we developed a nev method to discriminate subgroups of ER- breast cancer cells.
Literature	GENERIF_SUMMARY	Further study revealed AP000439.3 can regulate expression of CC through enhancing estrogen receptor induction of CCND1. This fin

Main Category	Category	Terms
		revealed lncRNAs may serve as important effectors of ER in regula gene expression and cell phenotype in breast cancer
Literature	GENERIF_SUMMARY	ESR1 mutations detected in cell free DNA, rarely present at the sta first-line endocrine therapy, were enriched at progression, strongly suggesting a role in conferring endocrine resistance in metastatic cancer.
Literature	GENERIF_SUMMARY	Estrogen receptor (ER) and progesterone receptor (PR) expression endometrial carcinoma (EC) were significantly higher than those in paracarcinoma tissue and control.
Literature	GENERIF_SUMMARY	ESR1 promoter methylation was an independent risk factor and ha high value to predict 28-day mortality from acute-on-chronic hepat liver failure
Literature	GENERIF_SUMMARY	Gender differences were observed in ERalpha expression in norma Sub-cellular expression of both ER subtypes is altered in hepatitis (HCV) cirrhosis and HCV-related hepatocellular carcinoma livers a compared to normals and correlates with the expression of inflam and oncogenic markers.
Literature	GENERIF_SUMMARY	We investigated whether ESR1 gene which contribute to sagittal ar vertical malocclusions also contribute to facial asymmetries and temporomandibular disorders before and after orthodontic and orthognathic surgery treatment
Literature	GENERIF_SUMMARY	The results demonstrate the association between estrogen and an premenstrual dysphoric disorder (PMDD), supporting the claim tha women with PMDD differ in their responses to normal estrogen lev Furthermore, this association and dysfunctional emotional regulat PMDD existed only among the G carriers of ESR alpha-Xbal polymorphism.
Literature	GENERIF_SUMMARY	We demonstrated the presence of a distinct population of ESR1 LE mutations (E380Q, Y537S, Y537N, Y537C, and D538G) in metastat tissue and plasma using ddPCR assay.
Literature	GENERIF_SUMMARY	The TT (timin/timin) genotype for the ESR1 (rs2273206) gene was strongly associated with the risk of developing muscle TMDs and temporomandibular joint pain (P = .04).
Literature	GENERIF_SUMMARY	High ESR1 expression is associated with metastasis in breast Can
Literature	GENERIF_SUMMARY	Studies indicate that in the presence of estrogen receptor alpha (E may antagonize the ERalpha-induced effects, whereas in the abser estrogens, AR may act as an ERalpha-mimic, promoting tumor.
Literature	GENERIF_SUMMARY	Study reports the first discovery of naturally occurring ESR1 (Y537 ESR1 (Y537S) mutations in MCF7 and SUM44 ESR1-positive cell li after acquisition of resistance to long-term-estrogen-deprivation (L and subsequent resistance to fulvestrant (ICIR). Mutations were er with time, impacted on ESR1 binding to the genome and altered th interactome.
Literature	GENERIF_SUMMARY	There is a strong positive correlation between NGB and Estrogen r alpha(ERa) expression and activity in breast cancer cells
Literature	GENERIF_SUMMARY	Polymorphism in the ERalpha gene is associated with an increase for advanced Pelvic Organ Prolapse. However, polymorphism in th LAMC1 gene does not seem to be associated with such risk.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ESR1 gene fusion and expression is associated with poor outcome hormone receptor-positive breast cancer.
Literature	GENERIF_SUMMARY	role of cystatin A in breast cancer and its functional link with ERalp
Literature	GENERIF_SUMMARY	Estrogen-induced miR-191 was identified as a direct upstream regu of DAB2 in ER-positive breast cancer cells.
Literature	GENERIF_SUMMARY	The present study explores the longitudinal changes of prevalence severity of PTSD in adolescents with different genotypes of estrog receptor alpha gene rs9340799 after the 2008 Wenchuan earthqua
Literature	GENERIF_SUMMARY	Taken together, our results suggest that DHCA may be developed a efficient therapeutic for osteoporosis by regulating osteoblastoger through its estrogenic effects.
Literature	GENERIF_SUMMARY	Data suggest that estrogen receptor 1 (ESR1) methylation should b further evaluated as a potential liquid biopsy-based biomarker.
Literature	GENERIF_SUMMARY	These observations provide a tenable hypothesis for how D538G E expressing clones can contribute to shorter progression-free survi
Literature	GENERIF_SUMMARY	Data show that silencing of fatty acid synthase (FASN) and the downstream estrogen receptor alpha (ERalpha) resulted in suppres cell growth via a caveolin-1 dependent mechanism.
Literature	GENERIF_SUMMARY	findings establish RSK2 as an obligate participant of ERalpha-med transcriptional programs, tumorigenesis, and divergent patient resp to antiestrogen therapies
Literature	GENERIF_SUMMARY	Alcohol consumption may have differential effects on concordant discordant receptor subtypes of breast cancer.
Literature	GENERIF_SUMMARY	ESR alpha PvuII and XbaI polymorphisms have no association with systemic lupus erythematosus. The combination of the TC/AA and GG genotypes were associated with SLE susceptibility.
Literature	GENERIF_SUMMARY	Concomitant high expression of ERalpha36, GRP78 and GRP94 is associated with aggressive papillary thyroid cancer behavior and n used as a predictor for extrathyroid extension, lymph node metasta and distant metastasis.
Literature	GENERIF_SUMMARY	Higher RDI of chemotherapy is associated with better survival for E PR+, HER2- patients and triple-negative breast cancer(TNBC) patie optimize survival benefits, RDI should be maintained >/= 85% in ER HER2- patients, and >/= 75% in TNBC patients.
Literature	GENERIF_SUMMARY	by analyzing different estrogen receptor-alpha(ER-a)-positive and E negative breast cancer cell lines, we defined the role of CCN5 in th mediated regulation of growth and invasive capacity.
Literature	GENERIF_SUMMARY	High ERalpha expression is associated with neuroblastoma.
Literature	GENERIF_SUMMARY	HER2 amplification is a distinct driver event seen in all breast canc subtypes, rather than a subtype marker, with major characteristics restricted to amplification and overexpression of HER2 and neighb genes. The HER2E subtype has a distinctive transcriptional landsc independent of HER2A that reflects androgen receptor signaling as replacement for estrogen receptor (ER)-driven tumorigenesis.
Literature	GENERIF_SUMMARY	heterogeneous distribution in deep infiltrating endometriosis

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ERalpha and ERbeta mRNA expression was significantly higher (p
Literature	GENERIF_SUMMARY	Relative to ER, PR and HER2 from medical records, central IHC stai and the addition of Ki67 or combined tumor grade improved accur classifying PAM50-based luminal subtypes.
Literature	GENERIF_SUMMARY	the association between four ESR1 gene variants, and six ESR2 sir nucleotide polymorphisms and breast cancer risk in Tunisian wom
Literature	GENERIF_SUMMARY	High concentrations of DINCH urinary metabolites activate human receptors.
Literature	GENERIF_SUMMARY	Our study reveals a novel positive feedback between SMURF1 and alpha signaling in supporting breast cancer growth.
Literature	GENERIF_SUMMARY	Analysis of the genome-wide ER binding sites identified mutant ER recruitment mediating the allele-specific transcriptional program
Literature	GENERIF_SUMMARY	chlordecone induces endothelial cells angiogenesis by a cross-talk involving NADPH oxidase and mitochondrial O2(-)via a NO sensitiv pathways through activation of ERalpha.
Literature	GENERIF_SUMMARY	The data show a wide extranuclear ERalpha-variant expression in r lung and NSCLC that is not reported by routine pathology ER evalu criteria, limited in the nuclear wild type receptor.
Literature	GENERIF_SUMMARY	Each estrogen receptor alpha and estrogen receptor beta gene polymorphism might have different impact on postmenopausal osteoporosis risk and bone mineral density in various ethnicities.
Literature	GENERIF_SUMMARY	this study reports the development of a novel class of ERa AF2 inh which have the potential to effectively inhibit ERa activity by a uniq mechanism and to circumvent the issue of mutation-driven resista breast cancer
Literature	GENERIF_SUMMARY	ER-alpha36/EGFR signaling loop promotes growth of hepatocellula carcinoma cells
Literature	GENERIF_SUMMARY	The analysis revealed that severities in the impairment of social interaction and emotional regulation were linked to SNPs in ESR1 rs11155819.The effect of genotype was not observed for the other aspects of Autism spectrum disorders symptoms in Japanese individuals.
Literature	GENERIF_SUMMARY	The frequency of alleles and genotypes of polymorphisms FSHR(-2 and ESRI (XbaI A/G) in women with normal to poor response did n significant correlation.
Literature	GENERIF_SUMMARY	miR-221 may impair the protective effect of estrogen in degenerat cartilaginous endplate cells through targeting estrogen receptor al
Literature	GENERIF_SUMMARY	were able to detect ESR1 mutations in circulating free DNA from th plasma of 56 patients with metastatic ER-positive breast cancer. F of these patients were found to have ESR1 mutations at codons 53
Literature	GENERIF_SUMMARY	Down-regulation of ESR1 gene expression was enhanced by the development of the breast cancer.
Literature	GENERIF_SUMMARY	Results provide evidence that ESR1 is transcriptionally activated by TIMELESS protein.
Literature	GENERIF_SUMMARY	The aim of the present study was to assess whether fibrosis mark

Main Category	Category	Terms
		estrogen receptor (ER)alpha and the stromal derived factor (SDF)1 chemokine receptor type 4 (CXCR4) axis are abnormally expressed Intrauterine adhesions endometrium.
Literature	GENERIF_SUMMARY	Study found that tamoxifen treatment induced a decrease in PRMT an increase in ER-alpha36 as well as ER-alpha36-mediated non-ge effect in MDA-MB-231 breast cancer cell line.
Literature	GENERIF_SUMMARY	Consistent with its role in transcriptional activation, JMJD6 is requ estrogen/ERalpha-induced breast cancer cell growth and tumorige via modulation of recruitment of CARM1/MED12 co-activator com
Literature	GENERIF_SUMMARY	Human Endometriosis Tissue Microarray Reveals Site-specific Exp of Estrogen Receptors, Progesterone Receptor, and Ki67.
Literature	GENERIF_SUMMARY	Reduced expression of ERbeta1 in female ERalpha-negative papilla thyroid carcinoma patients is associated with greater progression disease.
Literature	GENERIF_SUMMARY	The results suggest that minor allele A of ESR1 gene is associated the development of arterial hypertension in men.
Literature	GENERIF_SUMMARY	Observed significant increase in HSD3B2 and ESR1 transcripts in f eutropic endometrium from infertile women with endometriosis m related to abnormal biological effect of E2 in endometrium, further affecting the development of human embryos.
Literature	GENERIF_SUMMARY	ESR1 mutations represent an exciting field with a rapidly increasin number of recent publications that will likely advance the knowled treatment resistance mechanisms and pave the way into more individualized patient endocrine treatment.
Literature	GENERIF_SUMMARY	higher frequency of ESR1 and PIK3CA mutations in the plasma tha serum in 33 MBC patients; therefore, serum samples should not be considered the preferred source of cfDNA.
Literature	GENERIF_SUMMARY	Study demonstrated for the first time, that the expression of the m frequently-occurring hormone binding domain (HBD)-ESR1 mutatio breast cancer (BC) cells promotes the acquisition of a stem-cell lik disease progression driving phenotype and demonstrated that the expression of HBD-ESR1 mutations may drive BC cells to acquire s cell traits through ER/NOTCH4 interplay.
Literature	GENERIF_SUMMARY	Report yeast estrogen screen for androgen- and estrogen-receptor mediated activities of 4-hydroxytestosterone, 4-hydroxyandrostene and their human metabolites.
Literature	GENERIF_SUMMARY	ERalpha gene Pvu II polymorphism exerts an important function in progression of breast cancer and the polymorphism was in signific association with the decrease of breast cancer risk in Asians rathe in Caucasians
Literature	GENERIF_SUMMARY	These studies reveal an unexpected role of the axon guidance rece UNC5A in fine-tuning ERalpha and EGFR signaling and the luminal progenitor status of hormone-sensitive breast cancers. Furthermo UNC5A knockdown cells provide an ideal model system to investig metastasis of ERalpha(+) breast cancers
Literature	GENERIF_SUMMARY	Via a feedback loop involving miR-375, RASD1, and ERalpha.
Literature	GENERIF_SUMMARY	Bisphenol A (BPA) binding on full-length architectures of estrogen receptor.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The expression levels of ER, CD34 and p53 in breast cancer tissue evaluated by color Doppler ultrasonographic features, which is con to assessing the prognosis of these patients.
Literature	GENERIF_SUMMARY	Inhibition of these coactivators decreased the ability of ESR1 muta activate transcription.
Literature	GENERIF_SUMMARY	these results identified a negative cross-regulatory loop between E and CHES1 that was required for growth of breast cancer cells, it n uncover novel insight into molecular mechanism of CHES1 involve breast cancer and provide new avenues for molecular-targeted the hormone-regulated breast cancer.
Literature	GENERIF_SUMMARY	KMT2C is a key regulator of ERalpha activity whose loss uncouples cancer proliferation from hormone abundance.
Literature	GENERIF_SUMMARY	Computational analysis of publicly available microarray datasets o small cell lung cancer (NSCC) treated with EGFR targeted therapy identified ESR1 and ELAV1 as the most highly ranked hub genes in in the NSCLC acquired TKI-resistance.
Literature	GENERIF_SUMMARY	This study identified ESR1 as a direct target of miR-301a-3p.
Literature	GENERIF_SUMMARY	Circulating ESR1 mutation detection at the end of AI-based adjuva treatment is not clinically useful. Circulating ESR1 mutation could assessed as soon as first relapse to guide interventional studies
Literature	GENERIF_SUMMARY	The ERalpha gene seems to play a key role in stress urinary incont in the premenopausal period.
Literature	GENERIF_SUMMARY	These results indicated that ERa-mediated signaling may be involv the regulation of Col1A1, and its regulatory effect may be depende the ERa expression level. The accurate evaluation of ERa expressio in the subsynovial connective tissue (SSCT) of individual patients w idiopathic carpal tunnel syndrome (CTS) might guide the effective new estrogen replacement therapy.
Literature	GENERIF_SUMMARY	Both buterol and K-1047 (at 10(-6)M) effectively suppressed ERa m gene expression in HeLa cell culture by 83.4 - 9 8.6%.
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha was expressed only in women and showe positive correlation with the amount of fungi in oral paracoccidioidomycosis, while progesterone receptor was observe both genders and exhibited no correlation with estrogen receptor-a fungi counting.
Literature	GENERIF_SUMMARY	Results provide evidence that the PvuII and XbaI polymorphisms in gene are not related to development of central precocious puberty
Literature	GENERIF_SUMMARY	Authors report for the first time the presence of ESR1 methylation plasma ctDNA of patients with HGSC. The agreement between ESI methylation in primary tumors and paired ctDNA is statistically significant.
Literature	GENERIF_SUMMARY	Our data support the combination of therapies targeting ERBB2/3 mTORC1 signalling, together with fulvestrant, in patients who relap endocrine therapy and retain a functional ER.
Literature	GENERIF_SUMMARY	The associations between PvuII (T>C) and XbaI (A>G) polymorphis estrogen receptor alpha (ESR1) gene with type 2 diabetes mellitus (T2DM) or metabolic syndrome (MetS), are reported.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The G/G XbaI genotype of ESR1 gene is associated with breast ca risk.
Literature	GENERIF_SUMMARY	Results showed that NAT1 and ESR1 expression were increased in primary breast tumor samples compared with normal breast tissu samples, and in ER+ primary breast tumors compared with ER- tun Also, NAT1 and ESR1 expression seems to have overlapping regula
Literature	GENERIF_SUMMARY	polymorphisms do not predict in vitro fertilization outcome
Literature	GENERIF_SUMMARY	WBP2 directly promoted P-glycoprotein expression through bindin ERalpha to increase the doxorubicin resistance of ERalpha-positive cells.
Literature	GENERIF_SUMMARY	The results for the first time demonstrated that ER-alpha36 mediat cisplatin resistance in breast cancer cells through nongenomic est signaling, suggesting that ER-alpha36 may serve as a novel target cisplatin resistance and a potential indicator of cisplatin sensitivity breast cancer treatment.
Literature	GENERIF_SUMMARY	these results indicate that deletion of the MIR135A1 locus and dec miR-135a expression promote ERa(+) breast cancer progression a tamoxifen resistance.Significance
Literature	GENERIF_SUMMARY	No association between polymorphisms in genes encoding estrog receptors (ESR1 and ESR2) and excreted BPA levels was found in orthodontic patients after bracket bonding.
Literature	GENERIF_SUMMARY	Up-regulates expression of GREB1 in ovarian cancer
Literature	GENERIF_SUMMARY	RNF168 is required for ESR1-positive breast cancer cell proliferatio facilitates ESR1 signaling activity possibly through promoting its transcription.
Literature	GENERIF_SUMMARY	Our study confirms the oestrogenic activity of abiraterone and high its differential impact on cells dependent on oestrogen for their proliferation vs. those that are ligand-independent and harbour wt mutant ESR1. These properties could impact the clinical efficacy o abiraterone in breast cancer.
Literature	GENERIF_SUMMARY	Estrogen activated EMT independent of ER in Brca1-deficient, but n Brca1-proficient, tumor cells. Estrogen activated the AKT pathway BRCA1-deficient tumor cells independent of ER, and pharmaceutic inhibition of AKT activity suppressed EMT and cell proliferation pre BRCA1 deficient tumor progression.
Literature	GENERIF_SUMMARY	In summary, an association between the presence of the particular genotypes at the three ESR1 polymorphisms (rs2234693, rs69027 rs7774230) and one ESR2 polymorphism (rs3020449), and the pre of metabolic syndrome in postmenopausal women was found.
Literature	GENERIF_SUMMARY	These findings identify a feed-forward mechanism that sustains activation of the CXCR7/CXCL11 axis under ERalpha control to ind epithelial-mesenchymal transition pathway and metastatic behavio ovarian cancer cells.
Literature	GENERIF_SUMMARY	ESR1 rs9340799 polymorphism was related to increased Recurren Spontaneous Abortion risk in the non-Asian group while associate decreased risk in Asian group (Meta-Analysis).
Literature	GENERIF_SUMMARY	Estrogen receptor-1 is a key regulator of HIV-1 latency that imparts gender-specific restrictions on the latent reservoir.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	that Oestrogen receptor alpha can enhance the odonto/osteogenic differentiation of stem cells from apical papilla via ERK and JNK M pathways
Literature	GENERIF_SUMMARY	there was a relationship between rs2046210 and rs3803662, and t of developing this disease in Vietnamese women. The A allele is th allele for both rs2046210 (OR [95% CI] = 1.43 [1.14 - 1.78], P = 0.00 rs3803662 (OR [95% CI] = 1.45 [1.16 - 1.83], P = 0.001). We conclu two polymorphisms, rs2046210 in ESR1 and rs3803662 in TNRC9, associated with breast cancer risk in the Vietnamese population.
Literature	GENERIF_SUMMARY	The present study thus indicates the possible association of PVUII polymorphism of ER-alpha gene in the etiology of ovarian cancer.
Literature	GENERIF_SUMMARY	Our results suggest that the ESR1 rs2234693 C allele, in contrast t allele, provides protection against muscle injury by lowering muscl stiffness.
Literature	GENERIF_SUMMARY	estimates for gene-expression (GE) were added for ER (gene ESR1 PGR (gene PGR) and for HER2 (gene ERBB2). Combined results we obtained in each patient via a scoring system based on all three re
Literature	GENERIF_SUMMARY	The unexpected signature for estradiol-induced program resides in indirect recruitment of ERa to a large cohort of basally active FOXA bound enhancers that lack cognate ERa DNA-binding elements.
Literature	GENERIF_SUMMARY	ERalpha-pSer118 is a prognostic factor in endometrial but not brea tumours. Genes associated with ERalpha-pSer118 are predominan unique between endometrial and breast tumours.
Literature	GENERIF_SUMMARY	we found that the control groups of rs746432 deviated from HWE
Literature	GENERIF_SUMMARY	n summary, these data suggest that hypermethylation of the RUNX promoter may play an important role in ER-positive breast tumor progression.
Literature	GENERIF_SUMMARY	Data inidicate CUE domain containing 1 (CUEDC1) expression corr positively with estrogen receptor alpha (ERalpha) in breast cancer, suggest CUEDC1 is a functional target gene of ERalpha and is requ breast cancer cell proliferation.
Literature	GENERIF_SUMMARY	ESR1 (TA)(n) was on the borderline of statistical differences betwe groups. Microsatellite polymorphisms of gonadal steroid receptor might contribute to the genetic basis of POI in Serbian women.
Literature	GENERIF_SUMMARY	In the group of patients with positive ESR1 promoter methylation in primary breast carcinoma survival was lower compared to the grou patients without methylation (38.1 months vs. 54.3 months, n.s.). S out of 19 (37%) of those patients with positive ESR1 promoter methylation developed loss of ER expression in metastatic disease
Literature	GENERIF_SUMMARY	our data demonstrated for the first time that IBC could decrease C expression level via the ERa pathway and make ER+ breast cancer sensitive to paclitaxel treatment.
Literature	GENERIF_SUMMARY	Hypogonadal men with lower serum estradiol have higher subcuta adipocyte ESR1 expression and lose more fat mass in response to months of testosterone therapy.
Literature	GENERIF_SUMMARY	the observed complex formation between SMYD2 and HSP90/P23 contribute to ERalpha regulation.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Role of SUMOylation in differential ERalpha transcriptional repress tamoxifen and fulvestrant in breast cancer cells.
Literature	GENERIF_SUMMARY	SETD1A may serve as a molecular target and prognostic indicator ERalpha-positive breast cancer.
Literature	GENERIF_SUMMARY	An essential role of GALNT6-mediated O-glycosylation for the nucl localization of ER-alpha in breast cancer cells.
Literature	GENERIF_SUMMARY	In post-menopausal Chinese women, ESR1 polymorphism is assoc with metabolic syndrome.
Literature	GENERIF_SUMMARY	identified a susceptible variant of ESR1 which might contribute to developing mild cognitive impairment at earlier stage and Alzheim Disease later
Literature	GENERIF_SUMMARY	ESR1 gene expression affects bone mineral density and osteoporo treatment efficiency in Slovak postmenopausal women.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha (ERalpha) could function as a negative reg of programmed death-ligand 1 (PD-L1) gene transcription in breast cells.
Literature	GENERIF_SUMMARY	This study aimed to determine the presence and localization of oe receptors (ERs), progesterone receptors (PRs), and androgen recep (ARs) in both healthy and varicose vein wall cells and their relation with gender.
Literature	GENERIF_SUMMARY	chromatin immunoprecipitation showed that higher methylation w associated with reduced SP7, RUNX2, and ERalpha binding to the S promoter in patients with osteoporosis; studies provide new insigh the role of SOST methylation in osteoporosis
Literature	GENERIF_SUMMARY	ZBTB7A transcriptionally regulates ER-alpha expression in ER-alph positive breast cancer cell lines by binding to the ESR1 promoter le to increased transcription of ER-alpha.
Literature	GENERIF_SUMMARY	ESR1 activation in adipocytes increased the nuclear content of SP1 protein, the SP1/ESR1 interaction and SP1 binding into the Slc2a4 promoter, culminating with increased Slc2a4/GLUT4 expression
Literature	GENERIF_SUMMARY	A higher expression of ERalpha [Estrogen Receptor alpha] in EC ca [Endometrial carcinomas] was associated with a lower histopath grade.
Literature	GENERIF_SUMMARY	Findings suggest that the methylation profile of ER-alpha may mod adolescent response to estrogen and breast composition, which m influence breast cancer risk in adulthood.
Literature	GENERIF_SUMMARY	Evidence for the saliency of genetic variation in sex hormone recep stress reactivity in humans and highlights their important role as mediators of hormonal activity.
Literature	GENERIF_SUMMARY	the whole-genome insights carried in this work can help fully understanding biological roles of ER1 in breast cancer.
Literature	GENERIF_SUMMARY	Gene expression data suggest that ERalpha signaling becomes inactivated throughout the progression of endometriosis to endometriosis-associated ovarian cancer
Literature	GENERIF_SUMMARY	ERBB2 mutations hyperactivate the HER3/PI3K/AKT/mTOR axis, le to antiestrogen resistance in ER+ breast cancer. Dual blockade of t

Main Category	Category	Terms
		HER2 and ER pathways is required for the treatment of ER+/HER2 breast cancers.
Literature	GENERIF_SUMMARY	TCGA analysis reveals high expression levels of BTF3 in luminal/E breast cancer and cell line models harboring ERalpha overexpressi Concordantly, BTF3 expression is highly and strongly associated w ESR1 expression in multiple breast cancer cohorts.
Literature	GENERIF_SUMMARY	Various studies find that ubiquitin can bind covalently or non-coval ERalpha resulting in either a proteolytic or non-proteolytic fate for t receptor. Thereby, Ub-dependent molecular pathways that modulat ERalpha signaling may play a central role in breast cancer progress [review]
Literature	GENERIF_SUMMARY	The present study reveals a possible endocrine-resistant mechanis which STAT1 modulates ERa signalling and confers tamoxifen resi Targeting of STAT1 is a potential treatment strategy for endocrine- resistant breast cancers
Literature	GENERIF_SUMMARY	Copy-number alterations of ESR1 and key CDK pathway genes are frequent in metastatic breast cancers, and their clinical relevance s be tested further.
Literature	GENERIF_SUMMARY	This pilot study identify the presence of aromatase and estrogen r in coronary arteries. This is the first evidence for the local formatio estrogen in the coronary arteries.
Literature	GENERIF_SUMMARY	In resected PDAC, expression of ER beta seems to correlate with p prognosis. These data may help to identify patients who may bene additional systemic therapy including selective estrogen receptor modulators.
Literature	GENERIF_SUMMARY	we performed whole-exome sequencing of 16 breast cancer tissue classified according to ESR1 expression and 12 samples of whole and detected 310 somatic mutations in cancer tissues with high le ESR1 expression
Literature	GENERIF_SUMMARY	Analysis of docked conformations in terms of binding energy disc estrogen receptor-beta (ERb) as more promising than estrogen rec alpha (ERa). Evaluation of MD trajectories of ZER bound to both ER ERb showed appreciable stability with minimum Calpha-atom root square deviation shifts
Literature	GENERIF_SUMMARY	Recurrent mutations at estrogen receptor binding sites mediates r response to estradiol-induced cell proliferation but increased resis to tamoxifen-induced growth inhibition.
Literature	GENERIF_SUMMARY	Data show that estrogen receptor variant (ER-alpha36) is highly expressed in glioblastoma specimens, and that ER-alpha36 knocko increased sensitivity of glioblastoma U87 cells to tamoxifen (TAM) decreased autophagy in the cells.
Literature	GENERIF_SUMMARY	Fbxo22-mediated KDM4B degradation determines selective estrog receptor modulator activity in breast cancer.
Literature	GENERIF_SUMMARY	AR expression was associated with improved prognosis in ER+ tur and worse prognosis in ER- tumors in the first 5-10 years postdiag These findings support the continued evaluation of AR-targeted th for AR+/ER- breast cancers.
Literature	GENERIF_SUMMARY	phenome-wide association study (PheWAS) demonstrates associa estrogen receptor alpha (ESR1) polymorphism rs1999805 and asth

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results indicate that phosphorylation of ER alpha at serine 118 pro direct DNA binding at active enhancers and is a distinguishing mar associated transcription factor complexes on chromatin.
Literature	GENERIF_SUMMARY	Genome-wide ER-alpha activation is sustained, not cyclical.
Literature	GENERIF_SUMMARY	The data suggest that rs9340799 polymorphisms in ESR1 may cor to digit length and 2nd and 4th digits (2D
Literature	GENERIF_SUMMARY	These results suggest the interplays of ESR1 rs2234693 with post- traumatic stress disorder influence serum glucose and lipids profil a gender dependent manner.
Literature	GENERIF_SUMMARY	In the endocrine therapy resistance (ETR) cells, H19 expression ac an ER modulator and that its levels and subsequently ERalpha leve be substantially decreased by blocking Notch and c-MET receptor signaling. Consequently, treating ETR cells with these pharmacolo inhibitors helps overcome resistance to Fulvestrant and Tamoxifer
Literature	GENERIF_SUMMARY	Study demonstrates that alternatively spliced myometrial ERDelta7 ESR1 acts as a dominant negative repressor of myometrial ERalph uterotonic function through the suppression of the gap junction pr GJA1.
Literature	GENERIF_SUMMARY	MiR-22 downregulated the expression of NK1R-Truncated and ER-a delay and weaken phosphorylation of ERK1/2 to inhibit proliferatio metastasis of breast cancer cells.
Literature	GENERIF_SUMMARY	Identification of SEPTIN12 as a novel target of the androgen and e receptors in human testicular cells.
Literature	GENERIF_SUMMARY	The binding affinity with estrogen receptor alpha (ERalpha) presen order of human >Rattus norvegicus > rainbow trout.
Literature	GENERIF_SUMMARY	the estrogen receptor (ER) and progesterone receptor (PR), as well human epidermal growth factor receptor 2 (Her2neu) expressions breast tumor cells, were evaluated.
Literature	GENERIF_SUMMARY	The XbaI single nucleotide polymorphism in the estrogen receptor contributes to tendinopathy in postmenopausal women.
Literature	GENERIF_SUMMARY	This study provides evidence that ESR1 is associated with dental fluorosis in Brazilian children
Literature	GENERIF_SUMMARY	Metaplastic breast cancer (MpBC) group had significantly more ca with no hormone receptors (ER, PR) and HER2 overexpression/ger amplification compared with the other invasive breast cancer type (ER-, 69% versus 23%, p
Literature	GENERIF_SUMMARY	These data suggest that AZD9496 is a potent anti-oestrogen that antagonises and degrades estrogen receptor (ER) with anti-tumou activity in both endocrine-sensitive and endocrine-resistant model
Literature	GENERIF_SUMMARY	In Han Chinese women, susceptibility to osteoporosis may be affe ERalpha Pvu-II polymorphisms and ERbeta Alu-I polymorphisms; th carrying genotypes containing A and P alleles may have a higher ri osteoporosis.
Literature	GENERIF_SUMMARY	Study suggests that the ER S118P variant does not affect risk for b cancer or response to hormone therapy.
Literature	GENERIF_SUMMARY	Major genotypic and phenotypic heterogeneity exists between AI-r

Main Category	Category	Terms
		disease. ESR1 mutations appear to drive oestrogen-regulated proc in resistant tumours.
Literature	GENERIF_SUMMARY	Study determined that the expression of miR1075p in endometrial carcinoma (EC) was significantly higher than that in normal endom and higher miR1075p expression was related to advanced stages, node metastasis and myometrial invasion. The results of bioinforn algorithms and luciferase reporter assays in human endometrial ca cell lines revealed that ERalpha was a direct target of miR1075p.
Literature	GENERIF_SUMMARY	This meta-analysis indicated that the ERalpha PvuII polymorphism significantly associated with systemic lupus erythematosus susce in the overall and Asian populations, while the ERalpha XbaI GA ge only played a key role in systemic lupus erythematosus susceptibil Asian populations.
Literature	GENERIF_SUMMARY	results underscore the importance of estrogen-ERalpha-mediated of eIF3f expression for the proliferation and survival of ER-positive cancer cells. These findings may provide rationale for the developr new therapies to treat ER-positive breast cancer.
Literature	GENERIF_SUMMARY	Results unraveled that the transactivation domain of estrogen rece structurally disordered, yet unexpectedly compact, and that a meta Ile33- Ser118 contact is observed to inhibit coactivator binding. Dis by mutagenesis alters ensemble-structures and coactivator bindin
Literature	GENERIF_SUMMARY	Low ESR1 expression is associated with Osteosarcoma Growth an Metastasis.
Literature	GENERIF_SUMMARY	data demonstrate that Ajuba functions as a novel co-activator of E and that Ajuba/DBC1/CBP/p300 ternary complex may be a new ta developing therapeutics to treat breast cancer
Literature	GENERIF_SUMMARY	High ERalpha expression was associated with improved survival in ovarian cancer patients.
Literature	GENERIF_SUMMARY	MOF promotes endometrial cancer cell growth and proliferation in and in vivo. Clinical evidence indicates that expression MOF is dec and positively correlated with that of ER-alpha in endometrial canc tissues.
Literature	GENERIF_SUMMARY	The study shows that an association of theRMND1/CCDC170-ESR nucleotide polymorphisms can exist with osteopenia, osteoporosis fragility fracture.
Literature	GENERIF_SUMMARY	The association between ESR1 and LIF polymorphisms can help in prediction of recurrent implantation failure.
Literature	GENERIF_SUMMARY	Results show that low doses of BPA significantly induced gene and protein expression of ERalpha36 and triggered entry of cells into S and promoted cell proliferation suggesting a nongenomic pathway triggering that led to activation of mitogenic signaling molecules a initiation of uterine leiomyoma cell line proliferation .
Literature	GENERIF_SUMMARY	We confirmed the co-expression of RET and ER, but we did not find expression to be an independent prognostic factor in human breas cancer
Literature	GENERIF_SUMMARY	GG genotype of the ESR1 XbaI polymorphism may be a genetic risk for severe pre-eclampsia susceptibility. However, there is no assoc between the ESR1 gene PvuII and XbaI promoter polymorphism an risk of mild pre-eclampsia.[meta-analysis]

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ERalpha and ERbeta are located in the extranuclear region of the androgen-independent prostate cancer cells DU-145, differing from usual nuclear localization. This localization is important for activat pathways, as ERK1/2 phosphorylation, but not AKT. A possible cro between extranuclear and nuclear ERs and their partners as an imp mechanism to control ER function in prostate cancer cells.
Literature	GENERIF_SUMMARY	Arsenic exposure in estrogen receptor-alpha (ERalpha)-positive bre cancer MCF7 cells reduces breast cancer 1 (BRCA1) and ERalpha expression. Xenografts of arsenic-exposed MCF7 cells into the ma fat pads of nude mice produce a larger tumor volume and are mor refractory to tamoxifen in association with the reduced expression BRCA1 and ERalpha, CpG hypermethylation of estrogen receptor 1 and BRCA1.
Literature	GENERIF_SUMMARY	findings support the role of ESR1 genetics in alcohol dependence a show for the first time that estradiol activities may sex-specifically alcohol-related sequelae and outcome following in-patient withdra treatment.
Literature	GENERIF_SUMMARY	The study found a significant association between the severity of v matter hyperintensity (WMH) and CpG island methylation of ERsm Cyrillic-A gene. Multiple regression analysis showed that serum Ho methylation of ERsmall a, Cyrillic-A gene and WMH severity were significant determining factors for cognitive impairment.
Literature	GENERIF_SUMMARY	PES1 promotes the occurrence and development of papillary thyro cancer by upregulating the ERalpha/ERbeta protein ratio.
Literature	GENERIF_SUMMARY	Androgen and Estrogen receptor expression increases with increas prostate volume in benign prostatic hyperplasia cases.
Literature	GENERIF_SUMMARY	In two independent cohorts, ERalpha is a potential biomarker for improved survival that also may represent a therapeutic target in C
Literature	GENERIF_SUMMARY	Gene expression levels of estrogen receptor 1 (ESR1) is significant lower in ovarian endometriomas (OE) than in eutopic endometrium DNA methylation at the tissue-dependent and differentially methyl regions (T-DMRs) of ESR1 are significantly higher in OE than in EE.
Literature	GENERIF_SUMMARY	537S-estrogen receptor alpha were not addicted to glucos
Literature	GENERIF_SUMMARY	Differential expression of estrogen receptor-alpha on follicular den cells from patients with grade 1-2 and grade 3 follicular lymphoma
Literature	GENERIF_SUMMARY	The ERalpha-PvuII polymorphism could be considered as a possib protective factor against male infertility. [meta-analysis]
Literature	GENERIF_SUMMARY	this study identified that ER alpha expression on the macrophages human endometrial cancer was positively correlated with cancer progression
Literature	GENERIF_SUMMARY	Sirtuin 6 in preosteoclasts suppresses age- and estrogen deficienc related bone loss by stabilizing estrogen receptor alpha.
Literature	GENERIF_SUMMARY	BCAS2 is an AF-1 coactivator of ERalpha whose overexpression pr carcinogenic processes, suggesting an important role in the develo of estrogen-receptor positive breast cancer.
Literature	GENERIF_SUMMARY	Our data uncover the ERalpha-miR-1271-SNAI2 feedback loop and a mechanism to explain the TGF-beta network in breast cancer progression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ESR1 mutations are associated with recurrent breast cancer.
Literature	GENERIF_SUMMARY	The study demonstrates that ERa upregulates LINC00472 which suppresses the phosphorylation of NF-kappaB, and suggests that endocrine treatment may lower LINC00472 and increase NF-kappa activities, leading to tumor progression and disease recurrence.
Literature	GENERIF_SUMMARY	High ESR1 expression is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	In a model where the human ERalpha is exclusively expressed in th pituitary on the background of a global ERalpha-null (PitERtgKO) m the ovaries of PitERtgKO mice had an overt cystic, hemorrhagic phenotype but serum E2 and LH levels were normalized. Anomalou sporadic LH secretion caused the severe ovarian phenotype of PitE mice. Pituitary ERalpha controls estrogen-negative feedback. Hypothalamic ERalpha con...
Literature	GENERIF_SUMMARY	dynamics of breast-cancer relapse reveal late-recurring estrogen-receptor-positive genomic subgroups
Literature	GENERIF_SUMMARY	Estradiol stimulates cell proliferation in human renal tubular epithe cells through both ER-alpha and GPER-1 estrogen receptors and in beta-catenin activation.
Literature	GENERIF_SUMMARY	Estrogen Receptor-alpha Quantification in Breast Cancer
Literature	GENERIF_SUMMARY	ERbeta and ERalpha pathways have opposite effects on LCN2 exp and they interact with glucocorticoid action.
Literature	GENERIF_SUMMARY	CUX1 activated by Snail-Cat L signaling may contribute to TNBC vi alpha repression.
Literature	GENERIF_SUMMARY	The enhanced expression of estrogen-related receptor alpha in hu bladder cancer tissues and the effects of estrogen-related recepto knockdown on bladder cancer cells.
Literature	GENERIF_SUMMARY	This study shows that increased ESR1 gene dosage is found more frequently in ERalpha66-negative breast cancer patients.
Literature	GENERIF_SUMMARY	our results indicated several resveratrol analogues to be active in ERnegative phenotypes, acting through an ER receptorindependen manner, supporting further investigation into their mechanism of a and use as potential chemotherapeutics in higherrisk breast cance
Literature	GENERIF_SUMMARY	Metformin could down-regulate the mRNA expression level of aro in the uterine leiomyoma cells. These results indicate that metforn inhibit the local estrogen synthesis and therefore suppress the development of uterine leiomyoma
Literature	GENERIF_SUMMARY	OCT4 is dependent on ERalpha to suppress the proliferation of bre cancer cells through DNMT1/ISL1/ERK axis.
Literature	GENERIF_SUMMARY	Despite the presumed significance of the rs2234693 and rs934079 polymorphisms on mechanisms inherent to hypertension and vasc diseases, the study did not show an association between these polymorphisms and preeclampsia (PE) occurrence.
Literature	GENERIF_SUMMARY	Study presents a systematic analysis of how ERalpha is using diffe classes of TF binding sites in MCF-7 breast cancer cell line. While majority of these sites are bound by ERalpha prior to ligand activat only a small fraction of these genomic loci have a classical estrog response element (ERE) and in most cases, other transcription fac

Main Category	Category	Terms
		help to stabilize ERalpha binding.
Literature	GENERIF_SUMMARY	NEAT1 played an important role in the activation of ERalpha to reg AQP7-mediated hepatic steatosis
Literature	GENERIF_SUMMARY	findings demonstrated that ERalpha, but not ERbeta, is involved in induced ovarian cancer in an E2-independent manner, providing ne evidence for cancer progression in obesity-associated ovarian can
Literature	GENERIF_SUMMARY	Tid1-S attenuates LPS-induced cardiac hypertrophy and apoptosis through ER-a mediated modulation of p-PI3K/p-Akt signaling casca
Literature	GENERIF_SUMMARY	the presence of ER mutations was associated with bone metastas castration resistance
Literature	GENERIF_SUMMARY	Investigated the relationships of pathological response after neoa chemo-endocrine therapy with alterations in the Ki67 labeling inde expression of cyclin D1 (CCND1) and progesterone receptor (PgR) estrogen receptor (ER) activity in breast cancer.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha regulates the Wnt/beta-catenin signaling pathway in colon cancer by targeting the NOD-like receptors.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha activates MAPK signaling pathway to pro the development of endometrial cancer.
Literature	GENERIF_SUMMARY	ERalpha36 expression in breast cancer cells is regulated by mir-13 and methylation status of ERalpha36 promoter.
Literature	GENERIF_SUMMARY	Results observed that polymorphisms in ESR1 gene were not asso with bone mineral density (BMD) in both pre and postmenopausal except X allele of XbaI polymorphism which was found to be prote for BMD at in both pre and postmenopausal women.
Literature	GENERIF_SUMMARY	Modeling the Effect of the Metastatic Microenvironment on Pheno Conferred by Estrogen Receptor Mutations Using a Human Liver Microphysiological System.
Literature	GENERIF_SUMMARY	results demonstrate a relationship between BRCA1 and steroid ho signal transduction pathways in breast cancer cells and point out t importance of complex BRCA1 and ERa expression regulation mechanisms studies including epigenetic gene expression regulat
Literature	GENERIF_SUMMARY	Molecular docking provided quantitative information on strength o binding to the ERalpha receptor.
Literature	GENERIF_SUMMARY	findings suggest that A3A is associated with a good prognosis in ERalpha-positive oropharyngeal carcinoma (OPC), and indicate the prognostic significance of ERalpha in HPV-positive OPC
Literature	GENERIF_SUMMARY	Frequent amplifications of ESR1 is associated with primary invasiv lobular breast carcinoma.
Literature	GENERIF_SUMMARY	COL1A1, CCDC170, and ESR1 single nucleotide polymorphisms associated with distal radius fracture in postmenopausal Mexican women.
Literature	GENERIF_SUMMARY	This preliminary study showed the presence of ER I gene polymorp osteoporosis periodontitis patients without dental fluorosis. Furthe studies with large sample size are needed to confirm the associati shown in this preliminary study

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Identification of pelvic organ prolapse risk susceptibility gene SNP in Xinjiang women.
Literature	GENERIF_SUMMARY	Studies indicate that estrogen receptor 1 (ESR1) mutations expres unique transcriptional profile that favors tumor progression, sugge that selected ESR1 mutations may influence metastasis [Review].
Literature	GENERIF_SUMMARY	intensity of ERalpha in the neuronal nuclei was significantly correla with both nuclear and perikaryal sizes in Vascular Dementia and Alzheimer's Disease
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in the ESR1 gene is associated v primary open angle glaucoma.
Literature	GENERIF_SUMMARY	Study discovered estrogen-independent mutant ESR1 genomic bin that is significantly altered from wild-type ESR1. The D538G mutat impacted expression, including a large set of nonestrogen-regulate genes, and chromatin accessibility. Mutant ESR1 is distinct from constitutive ESR1 activity because mutant-specific changes are no recapitulated with prolonged estrogen exposure.
Literature	GENERIF_SUMMARY	FGFR1 Amplification Mediates Endocrine Resistance but Retains T Sensitivity in Metastatic Hormone Receptor-Positive (HR(+)) Breas Cancer.
Literature	GENERIF_SUMMARY	our data show that distinct CTCF-mediated chromatin structures a required for ERalpha- driven breast cancer cell proliferation
Literature	GENERIF_SUMMARY	Results find a hierarchical regulatory network for CYP3A4 expressi directed by ESR1 through self-regulation, cross regulation, and transcription factors (TF)-TF interactions. ESR1 regulates also the expression of other P450 enzymes, suggesting broad influence of on xenobiotics metabolism in human liver.
Literature	GENERIF_SUMMARY	Trans-acting non-synonymous variant of FOXA1 predisposes to hepatocellular carcinoma through modulating FOXA1-ERalpha transcriptional program and may have undergone natural selection
Literature	GENERIF_SUMMARY	The studies demonstrate for the first time that the function of MM MMP9 in breast cancer cell migration, which is mediated by intera between ERalpha-36 and STAT3.
Literature	GENERIF_SUMMARY	Chronic cadmium exposure decreases the dependency of MCF7 b cancer cells on ERalpha.
Literature	GENERIF_SUMMARY	ERalpha and ERbeta membrane-initiated signaling in normal and cancerous human prostate stem/progenitor cells elicited nongeno actions by activating downstream signaling pathways
Literature	GENERIF_SUMMARY	Single nucleotide polymorphism (SNP) of the estrogen receptor al (ERalpha) gene and abnormal expression of ERalpha in placenta ti were associated with gestational diabetes mellitus(GDM). In addit SNP of the PvuII site in pregnant women with GDM was related to degree of insulin resistance and to the upregulation of ERalpha expression in placental tissues, which may play an important role i patho...
Literature	GENERIF_SUMMARY	Several mechanisms of endocrine resistance in ER+/HER2- advanc breast cancers, through ER-alpha itself, receptor tyrosine signaling cycle pathway, have been identified to be pivotal in endocrine thera [review]

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Findings indicated that increased endogenous estrogen, catalysed elevated aromatase levels, enhanced MMP12 expression via ERalp participated in castration-resistant prostate cancer progression an promoted tumor metastasis.
Literature	GENERIF_SUMMARY	TMPO-AS1 is closely associated with the estrogen signaling pathw TMPO-AS1 could positively regulate estrogen receptor 1 (ESR1) m expression by stabilizing ESR1 mRNA through interaction with ESR mRNA.
Literature	GENERIF_SUMMARY	These studies demonstrate that bone marrow stromal cells (BMSC transcriptionally corepress ESR1 with neighboring genes and inhib receptor activity, but the functional consequences of the BMSC bo tumor microenvironment can be limited by metastasis-associated mutations.
Literature	GENERIF_SUMMARY	Estrogen receptors promote NSCLC progression by modulating the membrane receptor signaling network
Literature	GENERIF_SUMMARY	Data found that the expression of ESR1 increased in human breast (BC) tissues and positively correlated with that of USP7 protein. M USP7 expression was an independent poor prognostic factor in su analysis. USP7 physically interacted with the ERalpha, thereby med the deubiquitination and stabilization of ERalpha.
Literature	GENERIF_SUMMARY	ESR1 rs2234693 polymorphism is significantly correlated with an increased risk of stroke, especially ischemic stroke. There was no evidence of a significant association between ESR1 rs9340799 polymorphism and stroke risk (Meta-Analysis)
Literature	GENERIF_SUMMARY	Effect of Estrogen Receptor- Alpha Gene Polymorphism (IVS1-397 on Microvascular Complications of Type 1 Diabetes Mellitus.
Literature	GENERIF_SUMMARY	E2 signaling may therefore act upstream of RSK4 to promote canc progression. The results obtained in the current study suggested t RSK4 inhibited breast cancer cell invasiveness and that RSK4 pron hypomethylation may serve as a novel prognostic marker for patie breast cancer.
Literature	GENERIF_SUMMARY	Early Enrichment of ESR1 Mutations and the Impact on Gene Expre in Presurgical Primary Breast Cancer Treated with Aromatase Inhib
Literature	GENERIF_SUMMARY	BRD3 is recruited to and controls the activity of a subset ERa transcriptional enhancers, providing a therapeutic opportunity to ta BRD3 with BET inhibitors in ERa-positive breast cancers.
Literature	GENERIF_SUMMARY	These results indicate that the heterogeneous expression of ERalp an accurate predictor of tamoxifen response and survival in lumina breast cancer patients. ESR1 rs2228480 may act as a marker with prognostic potential in luminal-B-like tumors.
Literature	GENERIF_SUMMARY	A substantial discordance in ER, PR, and HER2 status were observ between primary breast cancer tissues and recurrent lesions. Rebi could bring new therapeutic opportunities in the management of p with recurrent breast cancer.
Literature	GENERIF_SUMMARY	Cooperativity of co-factor NR2F2 with Pioneer Factors GATA3, FOX promoting ERalpha function.
Literature	GENERIF_SUMMARY	ER-alpha expression was a biomarker of poor prognosis and it mig inform treatment decision for high risk colorectal cancer patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Association between oestrogen receptors and female temporoma disorders.
Literature	GENERIF_SUMMARY	Present meta-analysis showed that the overall incidence rate of ES mutations in advanced ER+ breast cancers was 24%, with D538G a most frequently found ESR1 mutation. A complete resistance coul developed in aromatase inhibitors-containing regimes, whereas pa resistance will occur in case of tamoxifen or fulvestrant.
Literature	GENERIF_SUMMARY	It seems that the PvuII and XbaI polymorphisms of ESR1 could be associated with bone mineral content and bone mineral density va in Iranian children and adolescents.
Literature	GENERIF_SUMMARY	Docking simulation studies of compound 5d with the ERalpha BD r that the large hydrophobic moiety of compound 5d fit well into the hydrophobic pocket of the ERalpha LBD and that the sulfur atom o compound 5d formed a sulfur-pi interaction with the amino acid re His524 of the ERalpha LBD
Literature	GENERIF_SUMMARY	Both ESR1 and PGR were highly expressed in BC cell lines.
Literature	GENERIF_SUMMARY	Rs2234693 and rs9340799 polymorphisms in the ESR1 gene were associated with susceptibility for BC.
Literature	GENERIF_SUMMARY	Anti-proliferative transcriptional effects of medroxyprogesterone a in estrogen receptor positive breast cancer cells are predominantly mediated by the progesterone receptor.
Literature	GENERIF_SUMMARY	Identification of a novel C-terminally truncated estrogen receptor a variant (ERalphai34) with constitutive transactivation and estrogen receptor antagonist resistance.
Literature	GENERIF_SUMMARY	these findings show a significant association between breast canc the allelic and genotypic frequencies of the ESR1 polymorphism rs3798577
Literature	GENERIF_SUMMARY	MiR-203 is critical in the onset and progression of osteoarthritis (O least in part, caused by estrogen deficiency and ERalpha instability patients.
Literature	GENERIF_SUMMARY	Suppression of miR-203-3p inhibits lipopolysaccharide induced hu intervertebral disc inflammation and degeneration through upregul estrogen receptor alpha.
Literature	GENERIF_SUMMARY	Racial Differences in the Association Between Luminal Master Reg Gene Expression Levels and Breast Cancer Survival.
Literature	GENERIF_SUMMARY	Interplay of ERalpha binding and DNA methylation in the intron-2 determines the expression and estrogen regulation of cystatin A in cancer cells.
Literature	GENERIF_SUMMARY	Neurotoxic effects of aluminum are associated with its interferenc estrogen receptors signaling.
Literature	GENERIF_SUMMARY	Real-time measurement of E2
Literature	GENERIF_SUMMARY	Targeting the estrogen receptor alpha (ERalpha)-mediated circ-SM miR-141-3p/Gelsolin signaling to better suppress the HCC cell inva
Literature	GENERIF_SUMMARY	Genetic variation in ER genes (ESR)1 and ESR2 is associated with susceptibility to unexplained recurrent pregnancy loss.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ESR1 mutations are frequent in newly diagnosed metastatic and lo regional recurrence of endocrine-treated breast cancer and carry w prognosis.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha gene variant, PvuII (rs2234693), as a pote pharmacogenetic biomarker for aneurysmal subarachnoid hemorr postmenopausal women.
Literature	GENERIF_SUMMARY	Expression patterns of ERalpha66 and its novel variant isoform ERalpha36 in lactotroph pituitary adenomas and associations with clinicopathological characteristics.
Literature	GENERIF_SUMMARY	Estradiol-Induced MMP-9 Expression via PELP1-Mediated Membra Initiated Signaling in ERalpha-Positive Breast Cancer Cells.
Literature	GENERIF_SUMMARY	A single droplet digital PCR for ESR1 activating mutations detectio plasma.
Literature	GENERIF_SUMMARY	ETV4 Is Necessary for Estrogen Signaling and Growth in Endometr Cancer Cells.
Literature	GENERIF_SUMMARY	Folate receptor-targeted RNAi nanoparticles for silencing STAT3 in associated macrophages and tumor cells.
Literature	GENERIF_SUMMARY	Survey of low ER-positive expression and its correlation with other and pathological factors in breast cancer.
Literature	GENERIF_SUMMARY	Profiling the expression and function of oestrogen receptor isoform in human endometrial tissues and uterine natural killer cells.
Literature	GENERIF_SUMMARY	New insights into the expression of androgen and estrogen recept the appendix testis in congenital cryptorchidism.
Literature	GENERIF_SUMMARY	Alternatively Constructed Estrogen Receptor Alpha-Driven Super-Enhancers Result in Similar Gene Expression in Breast and Endom Cell Lines.
Literature	GENERIF_SUMMARY	A Systematic Study of the Impact of Estrogens and Selective Estro Receptor Modulators on Prostate Cancer Cell Proliferation.
Literature	GENERIF_SUMMARY	NF1 is a repressor for estrogen receptor alpha signaling in breast c
Literature	GENERIF_SUMMARY	Genetic estrogen receptor alpha gene PvuII polymorphism in susceptibility to knee osteoarthritis in a Chinese Han population
Literature	GENERIF_SUMMARY	Loss of Estrogen Receptors is Associated with Increased Tumor Aggression in Laryngeal Squamous Cell Carcinoma.
Literature	GENERIF_SUMMARY	genetic polymorphisms in ESR1 is not associated with persistent a periodontitis
Literature	GENERIF_SUMMARY	Logistic regression revealed association of rs2234693, rs3798577 rs3020314 (ESR1) and rs1256049 (ESR2), the association of rs223 (C/T) being the strongest with P
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha in female skeletal muscle is not required f regulation of muscle insulin sensitivity and mitochondrial regulatio
Literature	GENERIF_SUMMARY	YB-1 interplays with ERalpha to regulate the stemness and differen of ER-positive breast cancer stem cells.
Literature	GENERIF_SUMMARY	The Role of CHK1 Varies with the Status of Oestrogen-receptor and

Main Category	Category	Terms
		Progesterone-receptor in the Targeted Therapy for Breast Cancer.
Literature	GENERIF_SUMMARY	PSF Promotes ER-Positive Breast Cancer Progression via Posttranscriptional Regulation of ESR1 and SCFD2.
Literature	GENERIF_SUMMARY	Estrogen Receptors (ESRs) Mutations in Adolescent Idiopathic Sco
Literature	GENERIF_SUMMARY	Cancer-specific mutation of GATA3 disrupts the transcriptional reg network governed by Estrogen Receptor alpha, FOXA1 and GATA3.
Literature	GENERIF_SUMMARY	A Randomized, Open-label, Presurgical, Window-of-Opportunity Stu Comparing the Pharmacodynamic Effects of the Novel Oral SERD AZD9496 with Fulvestrant in Patients with Newly Diagnosed ER(+) HER2(-) Primary Breast Cancer.
Literature	GENERIF_SUMMARY	High expression of estrogen receptor alpha and aromatase in glial cells is associated with gender-independent survival benefits in glioblastoma patients.
Literature	GENERIF_SUMMARY	ESR1 Mutations Associated With Estrogen Insensitivity Syndrome Conformation of Ligand-Receptor Complex and Altered Transcript Profile.
Literature	GENERIF_SUMMARY	Acetylation of histone H3K27 signals the transcriptional elongation estrogen receptor alpha.
Literature	GENERIF_SUMMARY	Monitoring the Crosstalk Between the Estrogen Receptor and Hum Epidermal Growth Factor Receptor 2 with PET.
Literature	GENERIF_SUMMARY	Estrogen Receptors and Endometriosis.
Literature	GENERIF_SUMMARY	Quantitative mast cell analysis and hormone receptor study (ER, P HER2/neu) in invasive carcinoma of breast.
Literature	GENERIF_SUMMARY	AR/ER Ratio Correlates with Expression of Proliferation Markers an Distinct Subset of Breast Tumors.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha promotes lung cancer cell invasion via inc of and cross-talk with infiltrated macrophages through the CCL2/C MMP9 and CXCL12/CXCR4 signaling pathways.
Literature	GENERIF_SUMMARY	The AGT Haplotype of the ESR2 Gene Containing the Polymorphism rs2077647A, rs4986938G, and rs1256049T Increases the Suscepti Unexplained Recurrent Spontaneous Abortion in Women in the Chi Hui Population.
Literature	GENERIF_SUMMARY	Bisphenolic compounds alter gene expression in MCF-7 cells throu interaction with estrogen receptor alpha.
Literature	GENERIF_SUMMARY	A Mechanistic High-Content Analysis Assay Using a Chimeric Andr Receptor That Rapidly Characterizes Androgenic Chemicals.
Literature	GENERIF_SUMMARY	ERalpha-related chromothripsis enhances concordant gene transc on chromosome 17q11.1-q24.1 in luminal breast cancer.
Literature	GENERIF_SUMMARY	Estrogen receptor-alpha immunoreactivity predicts symptom seve pain recurrence in deep endometriosis.
Literature	GENERIF_SUMMARY	Identification and Analysis of Estrogen Receptor alpha Promoting Tamoxifen Resistance-Related lncRNAs.
Literature	GENERIF_SUMMARY	Bioinformatics Analysis Identifies the Estrogen Receptor 1 (ESR1)

Main Category	Category	Terms
		and hsa-miR-26a-5p as Potential Prognostic Biomarkers in Patient Intrahepatic Cholangiocarcinoma.
Literature	GENERIF_SUMMARY	Oestrogen receptors and hypoxia inducible factor 1 alpha expressi abdominal wall endometriosis.
Literature	GENERIF_SUMMARY	HDAC3-ERalpha Selectively Regulates TNF-alpha-Induced Apoptot Death in MCF-7 Human Breast Cancer Cells via the p53 Signaling Pathway.
Literature	GENERIF_SUMMARY	Differential ESR1 Promoter Methylation in the Peripheral Blood-Fin from the Women 40+ Healthy Aging Study.
Literature	GENERIF_SUMMARY	Risk of early progression according to circulating ESR1 mutation, C and cfDNA increases under first-line anti-aromatase treatment in metastatic breast cancer.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha plays an important role in Cushing's syndr during pregnancy.
Literature	GENERIF_SUMMARY	Regulatory sharing between estrogen receptor alpha bound enhan
Literature	GENERIF_SUMMARY	Association of ESR1 polymorphism rs2234693 and rs9340799 wit postmenopausal osteoporosis in a Chinese population.
Literature	GENERIF_SUMMARY	Src couples estrogen receptor to the anticipatory unfolded protein response and regulates cancer cell fate under stress.
Literature	GENERIF_SUMMARY	Estrogen-ERalpha signaling and DNA hypomethylation co-regulate expression of stem cell protein PIWIL1 in ERalpha-positive endome cancer cells.
Literature	GENERIF_SUMMARY	R269C variant of ESR1
Literature	GENERIF_SUMMARY	DEC1 directly interacts with estrogen receptor (ER) alpha to suppre proliferation of ER-positive breast cancer cells.
Literature	GENERIF_SUMMARY	The Role of ERalpha36 in Development and Tumor Malignancy.
Literature	GENERIF_SUMMARY	Rapid Effects of Oestrogen on Intracellular Ca(2+) in the Uterine Junctional Myometrium of Patients With and Without Adenomyosi Different Phases of the Menstrual Cycle.
Literature	GENERIF_SUMMARY	Estrogen receptor (ESR1 and ESR2)-mediated activation of eNOS-N cGMP pathway facilitates high altitude acclimatization.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha as a predictive biomarker for survival in hu papillomavirus-positive oropharyngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	ESR1 Mutations and Overall Survival on Fulvestrant versus Exemes Advanced Hormone Receptor-Positive Breast Cancer
Literature	GENERIF_SUMMARY	Chaperone-mediated Autophagy Governs Progression of Papillary Carcinoma via PPARgamma-SDF1/CXCR4 Signaling.
Literature	GENERIF_SUMMARY	The crosstalk between IGF-1R and ER-alpha in the proliferation and inflammation of nucleus pulposus cells.
Literature	GENERIF_SUMMARY	The miR-455-5p/ERalpha36 axis regulates mammalian neuronal vi and axonal regeneration.
Literature	GENERIF_SUMMARY	Enhancer RNAs Mediate Estrogen-Induced Decommissioning of S Enhancers by Recruiting ERalpha and Its Cofactor.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Intrinsic and Extrinsic Factors Governing the Transcriptional Regul ESR1.
Literature	GENERIF_SUMMARY	The influence of microsatellite polymorphisms in sex steroid recep genes ESR1, ESR2 and AR on sex differences in brain structure.
Literature	GENERIF_SUMMARY	Differential Regulation and Targeting of Estrogen Receptor alpha T in Invasive Lobular Breast Carcinoma.
Literature	GENERIF_SUMMARY	Genetic and clinical predictors of arthralgia during letrozole or anastrozole therapy in breast cancer patients.
Literature	GENERIF_SUMMARY	The rs2175898 Polymorphism in the ESR1 Gene has a Significant S Specific Effect on Obesity.
Literature	GENERIF_SUMMARY	Immunohistochemical expression of estrogen receptor alpha, Bcl-2 NF-kappaB P65 in the polyps of patients with and without endome
Literature	GENERIF_SUMMARY	Structural Insights of Transcriptionally Active, Full-Length Androge Receptor Coactivator Complexes.
Literature	GENERIF_SUMMARY	IL6/STAT3 Signaling Hijacks Estrogen Receptor alpha Enhancers t Breast Cancer Metastasis.
Literature	GENERIF_SUMMARY	Association of ESR1 (rs2234693 and rs9340799), CETP (rs708272 MTHFR (rs1801133 and rs2274976) and MS (rs185087) polymorp with Coronary Artery Disease (CAD).
Literature	GENERIF_SUMMARY	Monitoring ligand-mediated helix 12 transitions within the human estrogen receptor alpha using bipartite tetracysteine display.
Literature	GENERIF_SUMMARY	Assessment of estrogen receptor gene polymorphism (t-397c vari patients with premenstrual syndrome.
Literature	GENERIF_SUMMARY	Therapeutic role of recurrent ESR1-CCDC170 gene fusions in breas cancer endocrine resistance.
Literature	GENERIF_SUMMARY	Functionally significant polymorphisms of ESR1and PGR and risk o intrauterine growth restriction in population of Central Russia.
Literature	GENERIF_SUMMARY	ESR1 polymorphisms and risk of preeclampsia.
Literature	GENERIF_SUMMARY	Ether lipid and sphingolipid expression patterns are estrogen recep dependently altered in breast cancer cells.
Literature	GENERIF_SUMMARY	Roles of estrogen receptor alpha and beta in the regulation of proli in endometrial carcinoma.
Literature	GENERIF_SUMMARY	Influence of ESR1 Variants on Clinical Characteristics and Fibromy Syndrome in Turkish Women.
Literature	GENERIF_SUMMARY	Accurate assessment of estrogen receptor profiles in non-function pituitary adenomas using RT-digital PCR and immunohistochemist
Literature	GENERIF_SUMMARY	The ERalpha-miR-575-p27 feedback loop regulates tamoxifen sens in ER-positive Breast Cancer.
Literature	GENERIF_SUMMARY	Elastosis in ERalpha-positive male breast cancer.
Literature	GENERIF_SUMMARY	A mutant form of ERalpha associated with estrogen insensitivity a the coupling between ligand binding and coactivator recruitment.
Literature	GENERIF_SUMMARY	The ubiquitin ligase RNF181 stabilizes ERalpha and modulates bre

Main Category	Category	Terms
		cancer progression.
Literature	GENERIF_SUMMARY	Human permanent tooth sizes are associated with genes encodin oestrogen receptors.
Literature	GENERIF_SUMMARY	Estrogens and development of the mouse and human external ger
Literature	GENERIF_SUMMARY	Association of Estrogen Receptor 1 and Tumor Necrosis Factor alp Polymorphisms with Temporomandibular Joint Anterior Disc Displacement without Reduction.
Literature	GENERIF_SUMMARY	miR-9-5p facilitates hepatocellular carcinoma cell proliferation, mig and invasion by targeting ESR1.
Literature	GENERIF_SUMMARY	Triple-Positive Breast Carcinoma
Literature	GENERIF_SUMMARY	Transportin-2 plays a critical role in nucleocytoplasmic shuttling of oestrogen receptor-alpha.
Literature	GENERIF_SUMMARY	Exploration in the Mechanism of Kaempferol for the Treatment of Cancer Based on Network Pharmacology.
Literature	GENERIF_SUMMARY	Calcium and Phosphorus Levels in Saliva are Influenced by Geneti Polymorphisms in Estrogen Receptor Alpha and Microrna17.
Literature	GENERIF_SUMMARY	ESR1 genetic alterations and their association with clinicopatholog characteristics in advanced breast cancer
Literature	GENERIF_SUMMARY	Tamoxifen induces hypercoagulation and alterations in ERalpha ar ERbeta dependent on breast cancer sub-phenotype ex vivo.
Literature	GENERIF_SUMMARY	Ontogeny of estrogen receptors in human male and female fetal reproductive tracts.
Literature	GENERIF_SUMMARY	Relationships of physical and breast cancer phenotypes with three nucleotide polymorphisms (rs2046210, rs3757318, and rs3803662 associated with breast cancer risk in Japanese women.
Literature	GENERIF_SUMMARY	New frontiers of developmental endocrinology opened by research connecting irreversible effects of sex hormones on developing org
Literature	GENERIF_SUMMARY	Oestrogen actions contribute to female gender-specific risks in the development of lung carcinoma.
Literature	GENERIF_SUMMARY	The role of ERalpha36 in cell type-specific functions of estrogen ar cancer development.
Literature	GENERIF_SUMMARY	Estrogen Receptor 1 Gene (ESR1) rs2234693 Polymorphism and B Cancer Risk in Saudi Women.
Literature	GENERIF_SUMMARY	Deregulated estrogen receptor signaling and DNA damage respons breast tumorigenesis.
Literature	GENERIF_SUMMARY	Discovery of a multi-target compound for estrogen receptor-positiv (ER(+)) breast cancer
Literature	GENERIF_SUMMARY	Impact of ESR1 Polymorphisms on Risk of Breast Cancer in the Ch Han Population.
Literature	GENERIF_SUMMARY	Sex-dependent expression of prostatic markers and hormone rece cystic tumor of the atrioventricular node
Literature	GENERIF_SUMMARY	Hormonal modulation of ESR1 mutant metastasis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Donor variation and sex hormone receptors in periodontal ligamen
Literature	GENERIF_SUMMARY	Estrogen receptor-associated receptor alpha and peroxisome proli activated receptor gamma in metabolism and disease (Review).
Literature	GENERIF_SUMMARY	Expression of Estrogen Receptor-alpha in Nasal Polyps and the Eff Dexamethasone on Estrogen Receptor-alpha Expression in RPMI 2 Cells.
Literature	GENERIF_SUMMARY	Nonlinear relationship between chromatin accessibility and estrad regulated gene expression.
Literature	GENERIF_SUMMARY	Estrogen Receptors
Literature	GENERIF_SUMMARY	Nuclear receptor SHP dampens transcription function and abrogat mitotic chromatin association of PXR and ERalpha via intermolecu interactions.
Literature	GENERIF_SUMMARY	The androgen receptor is a tumor suppressor in estrogen receptor- positive breast cancer.
Literature	GENERIF_SUMMARY	ESR1 ChIP-Seq Identifies Distinct Ligand-Free ESR1 Genomic Bindi Sites in Human Hepatocytes and Liver Tissue.
Literature	GENERIF_SUMMARY	Novel Biphenyl Amines Inhibit Oestrogen Receptor (ER)-alpha in ER Positive Mammary Carcinoma Cells.
Literature	GENERIF_SUMMARY	The variant T allele of PvuII in ESR1 gene is a prognostic marker in breast cancer survival.
Literature	GENERIF_SUMMARY	ESTRADIOL BLOOD LEVEL AND ESR1 GENE POLYMORPHISM IN W WITH PREMENSTRUAL SYNDROME.
Literature	GENERIF_SUMMARY	Twist activates miR-22 to suppress estrogen receptor alpha in brea cancer.
Literature	GENERIF_SUMMARY	Expression of Estrogen and Progesterone Receptors in Papillary Th Carcinoma in Korea.
Literature	GENERIF_SUMMARY	Modulating HSF1 levels impacts expression of the estrogen recept alpha and antiestrogen response.
Literature	GENERIF_SUMMARY	Association of estrogen receptor alpha 1 and TMJ dysfunction
Literature	GENERIF_SUMMARY	TET2 is a component of the estrogen receptor complex and contro to 5hmC conversion at estrogen receptor cis-regulatory regions.
Literature	GENERIF_SUMMARY	Sensitive quantitation of ESR1 mutations in cell-free DNA from bre cancer patients using base-specific invasive reaction assisted qPC
Literature	GENERIF_SUMMARY	Role of estrogen receptor alpha in MEHP-induced proliferation and invasion of SH-SY5Y cells.
Literature	GENERIF_SUMMARY	CUL4B renders breast cancer cells tamoxifen-resistant via miR-32- alpha36 axis.
Literature	GENERIF_SUMMARY	ERalpha down-regulates carbohydrate responsive element binding and decreases aerobic glycolysis in liver cancer cells.
Literature	GENERIF_SUMMARY	Hippo signalling maintains ER expression and ER(+) breast cancer growth.
Literature	GENERIF_SUMMARY	Protective effect of estrogen receptors (ERalpha/beta) against the

Main Category	Category	Terms
		intervertebral disc degeneration involves activating CCN5 via the promoter.
Literature	GENERIF_SUMMARY	Molecular Biomarkers for Contemporary Therapies in Hormone Re Positive Breast Cancer.
Literature	GENERIF_SUMMARY	Methylation of ERbeta 5'-untranslated region attenuates its inhibito effect on ERalpha gene transcription and promotes the initiation ar progression of papillary thyroid cancer.
Literature	GENERIF_SUMMARY	High gene expression of estrogen and progesterone receptors is associated with decreased t cell infiltration in patients with NSCLC
Literature	GENERIF_SUMMARY	Functional estrogen receptor signaling pathway activity in high-gra serous ovarian carcinoma as compared to estrogen receptor prote expression by immunohistochemistry.
Literature	GENERIF_SUMMARY	ESRalpha Promoter Methylation May Modify the Association Betw Lipid Metabolism and Type 2 Diabetes in Chinese Farmers.
Literature	GENERIF_SUMMARY	The deubiquitinating enzyme USP15 stabilizes ERalpha and promo breast cancer progression.
Literature	GENERIF_SUMMARY	Expression Patterns of ERalpha, ERbeta, AR, SIRT1, SIRT2, and SIR Prostate Cancer Tissue and Normal Prostate Tissue.
Literature	GENERIF_SUMMARY	PARP7 and Mono-ADP-Ribosylation Negatively Regulate Estrogen Receptor alpha Signaling in Human Breast Cancer Cells.
Literature	GENERIF_SUMMARY	The Decisive Case-Control Study Elaborates the Null Association b ESR1 XbaI and Osteoarthritis in Asians
Literature	GENERIF_SUMMARY	Co-expression of drug metabolizing cytochrome P450 enzymes ar estrogen receptor alpha (ESR1) in human liver
Literature	GENERIF_SUMMARY	CSNK1G2 differently sensitizes tamoxifen-induced decrease in PI3 mTOR/S6K and ERK signaling according to the estrogen receptor existence in breast cancer cells.
Literature	GENERIF_SUMMARY	High estrogen receptor alpha activation confers resistance to estro deprivation and is required for therapeutic response to estrogen in cancer.
Literature	GENERIF_SUMMARY	Expression of SCUBE2 and BCL2 Predicts Favorable Response in E Positive Breast Cancer.
Literature	GENERIF_SUMMARY	Oestrogen receptor alpha in papillary thyroid carcinoma
Literature	GENERIF_SUMMARY	Nuclear translocation of MRTFA in MCF7 breast cancer cells shifts ERalpha nuclear/genomic to extra-nuclear/non genomic actions.
Literature	GENERIF_SUMMARY	Crosstalk between progesterone receptor membrane component 1 estrogen receptor alpha promotes breast cancer cell proliferation.
Literature	GENERIF_SUMMARY	Preexisting Somatic Mutations of Estrogen Receptor Alpha (ESR1) Early-Stage Primary Breast Cancer.
Literature	GENERIF_SUMMARY	Study of XbaI and PvuII polymorphisms of estrogen receptor alpha (ERalpha) gene in girls with precocious/early puberty.
Literature	GENERIF_SUMMARY	Cell lineage tracing links ERalpha loss in Erbb2-positive breast can the arising of a highly aggressive breast cancer subtype.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	OTUD7B stabilizes estrogen receptor alpha and promotes breast c cell proliferation.
Literature	GENERIF_SUMMARY	CCL23 suppresses liver cancer progression through the CCR1/AKT feedback loop.
Literature	GENERIF_SUMMARY	Association of ERalpha-36 expression with the de novo resistance tamoxifen in ER-positive breast cancer.
Literature	GENERIF_SUMMARY	Estrogen Receptor 1 Gene rs22346939 and rs9340799 Variants ar Associated with Major Depressive Disorder and its Clinical Feature
Literature	GENERIF_SUMMARY	The correlation of estrogen receptor 1 and progesterone receptor g polymorphisms with recurrent pregnancy loss in a cohort of Egypti women.
Literature	GENERIF_SUMMARY	Estrogen suppresses HOXB2 expression via ERalpha in breast can cells.
Literature	GENERIF_SUMMARY	Methylation of Estrogen Receptor 1 Gene in the Paraspinal Muscle Girls with Idiopathic Scoliosis and Its Association with Disease Sev
Literature	GENERIF_SUMMARY	Estradiol Regulates mRNA Levels of Estrogen Receptor Beta 4 and Isoforms and Modulates Human Granulosa Cell Apoptosis.
Literature	GENERIF_SUMMARY	USP35, regulated by estrogen and AKT, promotes breast tumoriger stabilizing and enhancing transcriptional activity of estrogen recep alpha.
Literature	GENERIF_SUMMARY	EVI1 overexpression promotes ovarian cancer progression by regu estrogen signaling.
Literature	GENERIF_SUMMARY	Long Noncoding RNA LINC00899/miR-944/ESR1 Axis Regulates C Cancer Cell Proliferation, Migration, and Invasion.
Literature	GENERIF_SUMMARY	Methylation level of the regulatory regions of the estrogen recepto gene in paravertebral muscles of girls with idiopathic scoliosis.
Literature	GENERIF_SUMMARY	Assessment of 25-Year Survival of Women With Estrogen Recepto Positive/ERBB2-Negative Breast Cancer Treated With and Without Tamoxifen Therapy
Literature	GENERIF_SUMMARY	Previous estradiol treatment during midlife maintains transcription regulation of memory-related proteins by ERalpha in the hippocam a rat model of menopause.
Literature	GENERIF_SUMMARY	The estrogen receptor/GATA3/FOXA1 transcriptional network
Literature	GENERIF_SUMMARY	RUNX1T1, a potential prognostic marker in breast cancer, is co-orc expressed with ERalpha, and regulated by estrogen receptor signal breast cancer cells.
Literature	GENERIF_SUMMARY	ERalpha36-GPER1 Collaboration Inhibits TLR4/NFkappaB-Induced Inflammatory Activity in Breast Cancer Cells.
Literature	GENERIF_SUMMARY	Galectin-3 and Estrogen Receptor Alpha as Prognostic Markers in Prolactinoma
Literature	GENERIF_SUMMARY	ER+ Breast Cancer Strongly Depends on MCL-1 and BCL-xL Anti-Ap Proteins.
Literature	GENERIF_SUMMARY	ESR1 mutation as an emerging clinical biomarker in metastatic ho

Main Category	Category	Terms
		receptor-positive breast cancer.
Literature	GENERIF_SUMMARY	Associations of serum estradiol level, serum estrogen receptor-alp level, and estrogen receptor-alpha polymorphism with male infertil
Literature	GENERIF_SUMMARY	17beta-Estradiol activates Cl(-) channels via the estrogen receptor pathway in human thyroid cells.
Literature	GENERIF_SUMMARY	ESR1 mediated circ_0004018 suppresses angiogenesis in hepatoc carcinoma via recruiting FUS and stabilizing TIMP2 expression.
Literature	GENERIF_SUMMARY	Genetic variations of DNA bindings of FOXA1 and co-factors in bre cancer susceptibility.
Literature	GENERIF_SUMMARY	The rs9340799 polymorphism of the estrogen receptor alpha (ESR and its association with breast cancer susceptibility.
Literature	GENERIF_SUMMARY	Exercise effects on DNA methylation in EVL, CDKN2A (p14, ARF), a ESR1 in colon tissue from healthy men and women.
Literature	GENERIF_SUMMARY	ERalpha is an RNA-binding protein sustaining tumor cell survival a resistance.
Literature	GENERIF_SUMMARY	Reduced menin expression leads to decreased ERalpha expression correlated with the occurrence of human luminal B-like and ER-neg breast cancer subtypes.
Literature	GENERIF_SUMMARY	ESR1 gene variants affect FSHR-depended risk of fibrocystic mast in infertile women.
Literature	GENERIF_SUMMARY	ESR1 mutations and therapeutic resistance in metastatic breast ca
Literature	GENERIF_SUMMARY	MINDY1 promotes breast cancer cell proliferation by stabilizing es receptor alpha.
Literature	GENERIF_SUMMARY	Estrogenic hormones receptors in Alzheimer's disease.
Literature	GENERIF_SUMMARY	Association of Estrogen Receptor Genes Polymorphisms With Poly Ovary Syndrome
Literature	GENERIF_SUMMARY	Association Between Estrogen Receptors and GATA3 in Bladder Ca
Literature	GENERIF_SUMMARY	Disrupted PGR-B and ESR1 signaling underlies defective decidualiz linked to severe preeclampsia.
Literature	GENERIF_SUMMARY	Transcriptional Reprogramming Differentiates Active from Inactive Fusions in Endocrine Therapy-Refractory Metastatic Breast Cancer
Literature	GENERIF_SUMMARY	Calcineurin regulates the stability and activity of estrogen receptor
Literature	GENERIF_SUMMARY	The E3 ligase COP1 promotes ERalpha signaling and suppresses E breast cancer.
Literature	GENERIF_SUMMARY	Expression of GALNT8 and O-glycosylation of BMP receptor 1A su breast cancer cell proliferation by upregulating ERalpha levels.
Literature	GENERIF_SUMMARY	The deep infiltrating endometriosis tissue has lower T-cadherin, E-cadherin, progesterone receptor and oestrogen receptor than endometrioma tissue.
Literature	GENERIF_SUMMARY	Manipulating Estrogenic/Anti-Estrogenic Activity of Triphenylethyle towards Development of Novel Anti-Neoplastic SERMs.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	RASSF1A-Mediated Suppression of Estrogen Receptor Alpha (ERa Driven Breast Cancer Cell Growth Depends on the Hippo-Kinases L and 2.
Literature	GENERIF_SUMMARY	Compressive stress-mediated p38 activation required for ERalpha phenotype in breast cancer.
Literature	GENERIF_SUMMARY	Circulating Tumor DNA Profiling From Breast Cancer Screening Th to Metastatic Disease.
Literature	GENERIF_SUMMARY	Estrogen Receptor Expression in Pancreatic Adenocarcinoma
Literature	GENERIF_SUMMARY	Estrogen receptor alpha and NFATc1 bind to a bone mineral densit associated SNP to repress WNT5B in osteoblasts.
Literature	GENERIF_SUMMARY	ESR1 PvuII polymorphism
Literature	GENERIF_SUMMARY	Expression of Estrogen Receptor Alpha and Evaluation of Histolog Degeneration Scores in Fibroblasts of Hypertrophied Ligamentum
Literature	GENERIF_SUMMARY	BRCA2 represses the transcriptional activity of pS2 by E2-ERalpha.
Literature	GENERIF_SUMMARY	ESR1 hotspot mutations in endometrial stromal sarcoma with high transformation and endocrine treatment.
Literature	GENERIF_SUMMARY	Lack of association between delayed tooth emergence and single nucleotide polymorphisms in estrogen receptors.
Literature	GENERIF_SUMMARY	ERalpha inhibits mesenchymal and amoeboidal movement of liver cell via Galpha12.
Literature	GENERIF_SUMMARY	Downregulation of estrogen receptor-alpha36 expression attenuat metastasis of hepatocellular carcinoma cells.
Literature	GENERIF_SUMMARY	Hotspot ESR1 Mutations Are Multimodal and Contextual Modulato Breast Cancer Metastasis.
Literature	GENERIF_SUMMARY	DDX3X functionally and physically interacts with Estrogen Recepto
Literature	GENERIF_SUMMARY	Estrogen Receptor alpha Inactivation in 2 Sisters
Literature	GENERIF_SUMMARY	Widespread association of ERalpha with RMRP and tRNA genes in cells and breast cancers.
Literature	GENERIF_SUMMARY	The prognostic and predictive value of ESR1 fusion gene transcript primary breast cancer.
Literature	GENERIF_SUMMARY	Decreased Expression of Estrogen Receptors Is Associated with Tumorigenesis in Papillary Thyroid Carcinoma.
Literature	GENERIF_SUMMARY	Genetic polymorphisms in CYP19A1 and ESR1 are associated with CK activity after prolonged running in men.
Literature	GENERIF_SUMMARY	Estrogen Receptor Gene Polymorphism as a Possible Genetic Risk for Treatment Response in ER-Positive Breast Cancer Patients.
Literature	GENERIF_SUMMARY	Osteoporosis and its Association with Vitamin D Receptor, Oestrog alpha Receptor, Parathyroid Receptor and Collagen Type I alpha Re Gene Polymorphisms with Bone Mineral Density
Literature	GENERIF_SUMMARY	Association of exposure to prenatal per fl uoroalkyl substances an estrogen receptor 1 polymorphisms with the second to fourth digit school-aged children

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Transcriptional repression of estrogen receptor alpha by YAP revea Hippo pathway as therapeutic target for ER(+) breast cancer.
Literature	GENERIF_SUMMARY	GAS5/METTL14/ESR1 genetic variants are related to the susceptib coronary heart disease.
Literature	GENERIF_SUMMARY	pH and the Breast Cancer Recurrent Mutation D538G Affect the Pr of Activation of Estrogen Receptor alpha.
Literature	GENERIF_SUMMARY	Association of Polymorphisms in Estrogen Receptor Genes (ESR1 ESR2) with Osteoporosis and Fracture-Involvement of Comorbiditi Epistasis.
Literature	GENERIF_SUMMARY	Association Between ESR1 XBAI and Breast Cancer Susceptibility
Literature	GENERIF_SUMMARY	FKBP52 and FKBP51 differentially regulate the stability of estroge receptor in breast cancer.
Literature	GENERIF_SUMMARY	Overexpression of lncRNA HCP5 in human umbilical cord mesenc stem cell-derived exosomes promoted the proliferation and inhibit apoptosis of ovarian granulosa cells via the musashi RNA-binding 2/oestrogen receptor alpha 1 axis.
Literature	GENERIF_SUMMARY	Genetic polymorphisms related to muscular strength and flexibility associated with artistic gymnastic performance in the Japanese population.
Literature	GENERIF_SUMMARY	Oestrogen receptor expression distinguishes non-ossifying fibrom other giant cell containing bone tumours.
Literature	GENERIF_SUMMARY	ESR1 inhibits ionizing radiation-induced ferroptosis in breast canc via the NEDD4L/CD71 pathway.
Literature	GENERIF_SUMMARY	Androgen and oestrogen receptor co-expression determines the ef of hormone receptor-mediated radiosensitisation in breast cancer.
Literature	GENERIF_SUMMARY	The haplotypes GCA and ACA in ESR1 gene are associated with th susceptibility of recurrent spontaneous abortion (RSA) in Chinese
Literature	GENERIF_SUMMARY	Estrogen Receptor Type 1 and Type 2 Presence in Paravertebral Sk Muscles
Literature	GENERIF_SUMMARY	ESR1 Methylation Measured in Cell-Free DNA to Evaluate Endocrin Resistance in Metastatic Breast Cancer Patients.
Literature	GENERIF_SUMMARY	Modification of ERalpha by UFM1 Increases Its Stability and Trans for Breast Cancer Development.
Literature	GENERIF_SUMMARY	PSPC1 is a potential prognostic marker for hormone-dependent br cancer patients and modulates RNA processing of ESR1 and SCFD
Literature	GENERIF_SUMMARY	Relationship between Coffee Consumption and Osteoporosis Risk Determined by the ESR1 Polymorphism rs2982573.
Literature	GENERIF_SUMMARY	Molecular crosstalk between CUEDC2 and ERalpha influences the outcome by regulating mitosis in breast cancer.
Literature	GENERIF_SUMMARY	Effect of HIF-1alpha on estrogen/ERalpha signaling pathway in pit adenoma cells in hypoxic environment.
Literature	GENERIF_SUMMARY	Activation of estrogen receptor ESR1 and ESR2 induces proliferati the human testicular embryonal carcinoma NT2/D1 cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Association of the estrogen receptor gene with oral health-related of life in patients with dentofacial deformities.
Literature	GENERIF_SUMMARY	The E3 Ligase TRIM4 Facilitates SET Ubiquitin-Mediated Degradati Enhance ER-alpha Action in Breast Cancer.
Literature	GENERIF_SUMMARY	ERalpha and ERbeta Homodimers in the Same Cellular Context Re Distinct Transcriptomes and Functions.
Literature	GENERIF_SUMMARY	Targeting ESR1 mutation-induced transcriptional addiction in brea cancer with BET inhibition.
Literature	GENERIF_SUMMARY	The Estrogen Receptor alpha Cistrome in Human Endometrium an Epithelial Organoids.
Literature	GENERIF_SUMMARY	Estrogen receptor variant ER-alpha36 facilitates estrogen signalin EGFR in glioblastoma.
Literature	GENERIF_SUMMARY	Estrogen Receptor 1 (ESR1) and the Wnt/beta-Catenin Pathway M the Effect of the Coumarin Derivative Umbelliferon on Bone Mineralization.
Literature	GENERIF_SUMMARY	ERalpha-Dependent Regulation of Adropin Predicts Sex Difference Liver Homeostasis during High-Fat Diet.
Literature	GENERIF_SUMMARY	Estrogen receptor status has no prognostic relevance in metaplas breast carcinoma.
Literature	GENERIF_SUMMARY	Modulation of SOCS3 Levels via STAT3 and Estrogen-ERalphap66 Signaling during Hepatitis E Virus Replication in Hepatocellular Carcinoma Cells.
Literature	GENERIF_SUMMARY	Estrogen Receptor Alpha and ESR1 Mutations in Breast Cancer.
Literature	GENERIF_SUMMARY	Molecular characterization of ESR1 variants in breast cancer.
Literature	GENERIF_SUMMARY	Association of ESR1 and ESR2 Polymorphisms with Osteoporosis
Literature	GENERIF_SUMMARY	Estrogen receptor alpha revised
Literature	GENERIF_SUMMARY	Genome engineering for estrogen receptor mutations reveals diffe responses to anti-estrogens and new prognostic gene signatures f breast cancer.
Literature	GENERIF_SUMMARY	Contemporary evaluation of estrogen receptor and progesterone re expression in breast cancer-associated stroma.
Literature	GENERIF_SUMMARY	14-3-3tau drives estrogen receptor loss via ERalpha36 induction a GATA3 inhibition in breast cancer.
Literature	GENERIF_SUMMARY	Correlation between ERalpha gene polymorphism and multiple scl and neuromyelitis optica.
Literature	GENERIF_SUMMARY	Unique ESR1 and ESR2 estrogen receptor gene variants associate altered risk of triple-negative breast cancer
Literature	GENERIF_SUMMARY	TLR-8, TNF-alpha, and ESR-1alpha Gene Polymorphism Susceptibi Onset of Arthritis.
Literature	GENERIF_SUMMARY	Haplotype of ESR1 and PPARD Genes Is Associated with Higher Anthropometric Changes in Han Chinese Obesity by Adjusting Diet Factors-An 18-Month Follow-Up.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Evaluation of the relationship between vitamin D level and adropin, IL-1beta, IL-6, and oxidative status in women.
Literature	GENERIF_SUMMARY	ESR1 activating mutations
Literature	GENERIF_SUMMARY	The ERalpha-NRF2 signalling axis promotes bicalutamide resistan prostate cancer.
Literature	GENERIF_SUMMARY	The influence of ESR1 polymorphisms on selected hormonal, meta and mineral balance markers in women with hyperandrogenism.
Literature	GENERIF_SUMMARY	Ubiquitin-specific peptidase 14 maintains estrogen receptor alpha stability via its deubiquitination activity in endometrial cancer.
Literature	GENERIF_SUMMARY	Estrogen-Inducible LncRNA BNAT1 Functions as a Modulator for E Receptor Signaling in Endocrine-Resistant Breast Cancer Cells.
Literature	GENERIF_SUMMARY	The Clinical Utility of ESR1 Mutations in Hormone Receptor-Positiv HER2-Negative Advanced Breast Cancer.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha K303R mutation reorganizes its binding to forkhead box protein A1 regions and induces chromatin opening.
Literature	GENERIF_SUMMARY	An Association Study of ESR1-XbaI and PvuII Gene Polymorphism Migraine Susceptibility in the Jammu Region.
Literature	GENERIF_SUMMARY	The AF-2 cofactor binding region is key for the selective SUMOylat estrogen receptor alpha by antiestrogens.
Literature	GENERIF_SUMMARY	A Significant Association Between rs2295190 Polymorphism of th Gene and Fibromyalgia Syndrome.
Literature	GENERIF_SUMMARY	Clinical implication of low estrogen receptor (ER-low) expression i cancer.
Literature	GENERIF_SUMMARY	Radiosensitivity is associated with antitumor immunity in estrogen receptor-negative breast cancer.
Literature	GENERIF_SUMMARY	Variation in IL6ST cytokine family function and the potential of IL6 signalling in ERalpha positive breast cancer cells.
Literature	GENERIF_SUMMARY	Expression of estrogen receptors, PELP1, and SRC in human spern and their associations with semen quality.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha (ESR1) gene polymorphism (rs2234693 a rs2046210) with breast cancer risk in pashtun population of Khybe Pakhtunkhwa.
Literature	GENERIF_SUMMARY	ESR1 gene amplification and MAP3K mutations are selected durin adjuvant endocrine therapies in relapsing Hormone Receptor-posit HER2-negative breast cancer (HR+ HER2- BC).
Literature	GENERIF_SUMMARY	Selective modulation of estrogen receptor in obese men with andr deficiency
Literature	GENERIF_SUMMARY	Expression patterns and clinical significance of estrogen receptor small cell lung cancer.
Literature	GENERIF_SUMMARY	Association of ESR1 gene polymorphisms with the susceptibility to Hepatitis B virus infection and the clinical outcomes.
Literature	GENERIF_SUMMARY	Identification of a human estrogen receptor alpha tetrapeptidic fra with dual antiproliferative and anti-nociceptive action.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The Co-Expression of Estrogen Receptors ERalpha, ERbeta, and GP Endometrial Cancer.
Literature	GENERIF_SUMMARY	Variation in second breast cancer risk after primary invasive cance time since primary cancer diagnosis and estrogen receptor status.
Literature	GENERIF_SUMMARY	Association of Social Determinants and Tumor Biology With Racia Disparity in Survival From Early-Stage, Hormone-Dependent Breast Cancer.
Literature	GENERIF_SUMMARY	Codon adaptation by synonymous mutations impacts the function properties of the estrogen receptor-alpha protein in breast cancer
Literature	GENERIF_SUMMARY	Association of the KISS1, LIN28B, VDR and ERalpha gene polymor with early and fast puberty in Chinese girls.
Literature	GENERIF_SUMMARY	Association of the Estrogen Receptor 1 Polymorphisms rs204621 rs9383590 with the Risk, Age at Onset and Prognosis of Breast Ca
Literature	GENERIF_SUMMARY	ERbeta1 Sensitizes and ERbeta2 Desensitizes ERalpha-Positive Br Cancer Cells to the Inhibitory Effects of Tamoxifen, Fulvestrant and Combination with All-Trans Retinoic Acid.
Literature	GENERIF_SUMMARY	Integrating Network Pharmacology and Bioinformatics to Explore t Effects of Dangshen (Codonopsis pilosula) Against Hepatocellular Carcinoma
Literature	GENERIF_SUMMARY	Impact of MIR31HG polymorphisms on risk of breast cancer in Chi women.
Literature	GENERIF_SUMMARY	Association between ESR1 rs2234693 single nucleotide polymorp and uterine fibroids in Taiwanese premenopausal and postmenopa women.
Literature	GENERIF_SUMMARY	The location of estrogen receptor variant ER-alpha36 is associated the invasion of glioblastoma.
Literature	GENERIF_SUMMARY	FOXA1 Reprogramming Dictates Retinoid X Receptor Response in Mutant Breast Cancer.
Literature	GENERIF_SUMMARY	CircPVT1 promotes ER-positive breast tumorigenesis and drug res by targeting ESR1 and MAVS.
Literature	GENERIF_SUMMARY	Microdeletion at ESR1 Intron 6 (DEL_6_75504) Is a Susceptibility F for Cryptorchidism and Hypospadias.
Literature	GENERIF_SUMMARY	Loss of the receptors ER, PR and HER2 promotes USP15-depender stabilization of PARP1 in triple-negative breast cancer.
Literature	GENERIF_SUMMARY	Regulation of ERalpha-dependent breast cancer metastasis by a m signaling.
Literature	GENERIF_SUMMARY	The correlation of ESR1 genetic aberrations with estrogen recepto progesterone receptor status in metastatic and primary estrogen receptor-positive breast carcinomas.
Literature	GENERIF_SUMMARY	Interferon-gamma is quintessential for NOS2 and COX2 expressior ER[-] breast tumors that lead to poor outcome.
Literature	GENERIF_SUMMARY	The Transcriptomic Profiles of ESR1 and MMP3 Stratify the Risk o Biochemical Recurrence in Primary Prostate Cancer beyond Clinica Features.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The G protein-coupled oestrogen receptor GPER in health and dise
Literature	GENERIF_SUMMARY	ERalpha-associated translocations underlie oncogene amplificatio breast cancer.
Literature	GENERIF_SUMMARY	A prospective cohort study of clinical characteristics and outcome Chinese patients with estrogen receptor-negative/progesterone re positive early breast cancer.
Literature	GENERIF_SUMMARY	Molecular basis and dual ligand regulation of tetrameric estrogen receptor alpha/14-3-3zeta protein complex.
Literature	GENERIF_SUMMARY	Plasma ESR1 mutations and outcome to first-line paclitaxel and bevacizumab in patients with advanced ER-positive/HER2-negative cancer.
Literature	GENERIF_SUMMARY	Differential Regulation of POC5 by ERalpha in Human Normal and Scoliotic Cells.
Literature	GENERIF_SUMMARY	[The relationship between the expression of amplified in breast ca 1androgen receptor and tamoxifen resistance in breast cancer].
Literature	GENERIF_SUMMARY	Casein kinase 1 controls the shuttling of epidermal growth factor r and estrogen receptor in endometrial carcinoma induced by breast hormonal therapy
Literature	GENERIF_SUMMARY	ERalpha prevents tumorigenesis of both liver and breast cancer ce through CCN5.
Literature	GENERIF_SUMMARY	Correlation of Estrogen Receptor Alpha Serum Level with Gene Polymorphism and Its Effect on Women with Unexplained Infertilit Iraq.
Literature	GENERIF_SUMMARY	ESR1 Rearrangement as a Diagnostic and Predictive Biomarker in Tumor Resembling Ovarian Sex Cord Tumor
Literature	GENERIF_SUMMARY	The PAPSS1 gene is a modulator of response to cisplatin by regula estrogen receptor alpha signaling activity in ovarian cancer cells.
Literature	GENERIF_SUMMARY	Estrogen receptor alpha-mediated mitochondrial damage in intrah bile duct epithelial cells leading to the pathogenesis of primary bili cholangitis.
Literature	GENERIF_SUMMARY	Interaction between estrogen receptor-alpha and PNPLA3 p.I148M drives fatty liver disease susceptibility in women.
Literature	GENERIF_SUMMARY	Targeting estrogen receptor signaling for treating heart failure.
Literature	GENERIF_SUMMARY	Interplay between ESR1/PIK3CA codon variants, oncogenic pathwa alterations and clinical phenotype in patients with metastatic brea cancer (MBC)
Literature	GENERIF_SUMMARY	Estrogen receptor and programmed death ligand-1 expression in ty endometrial cancer and its associated clinicopathological charact
Literature	GENERIF_SUMMARY	Association between the ESR1 and ESR2 polymorphisms and osteoporosis risk
Literature	GENERIF_SUMMARY	Proteomics on malignant pleural effusions reveals ERalpha loss in metastatic breast cancer associates with SGK1-NDRG1 deregulati
Literature	GENERIF_SUMMARY	Somatic estrogen receptor alpha mutations that induce dimerizati

Main Category	Category	Terms
		promote receptor activity and breast cancer proliferation.
Literature	GENERIF_SUMMARY	Endometrial cancer (EC) derived G3BP1 overexpression and mutar promote EC tumorigenesis and metastasis via SPOP/ERalpha axis
Literature	GENERIF_SUMMARY	Clinical relevance of circulating ESR1 mutations during endocrine t for advanced hormone-dependent endometrial carcinoma.
Literature	GENERIF_SUMMARY	Increasing expression of STING by ERalpha antagonizes LCN2 downregulation during chronic endometritis.
Literature	GENERIF_SUMMARY	Activation of estrogen receptor induces differential proteomic resp mainly involving migration, invasion, and tumor development pathv human testicular embryonal carcinoma NT2/D1 cells.
Literature	GENERIF_SUMMARY	Correlation between ESR1 and APOE gene polymorphisms and risk osteonecrosis of the femoral head
Literature	GENERIF_SUMMARY	Circulating ESR1, long non-coding RNA HOTAIR and microRNA-130 expression as biomarkers for breast cancer stage and metastasis.
Literature	GENERIF_SUMMARY	Exploring the Spatial Landscape of the Estrogen Receptor Proxima Proteome With Antibody-Based Proximity Labeling.
Literature	GENERIF_SUMMARY	Differential DNA methylation and CTCF binding between the ESR1 promoter a of MCF-7 and MDA-MB-231 breast cancer cells.
Literature	GENERIF_SUMMARY	ER-1 deficiency induces inflammation and lipid deposition in meibo gland and lacrimal gland.
Literature	GENERIF_SUMMARY	Exploring the association of ESR1 and ESR2 gene SNPs with polyc ovary syndrome in human females
Literature	GENERIF_SUMMARY	Allelic variants of the estrogen receptor genes and frailty phenotyp postmenopausal women.
Literature	GENERIF_SUMMARY	TRPS1 modulates chromatin accessibility to regulate estrogen rec alpha (ER) binding and ER target gene expression in luminal breast cells.
Literature	GENERIF_SUMMARY	MCM2 is involved in subtyping and tamoxifen resistance of ERalph positive breast cancer by acting as the downstream factor of ERalp
Literature	GENERIF_SUMMARY	Association between estrogen receptor alpha and aryl hydrocarbon receptor gene polymorphisms in the prognosis of breast cancer in
Literature	GENERIF_SUMMARY	PFKFB3 regulates breast cancer tumorigenesis and Fulvestrant se by affecting ERalpha stability.
Literature	GENERIF_SUMMARY	An estrogen receptor alpha-derived peptide improves glucose homeostasis during obesity.
Literature	GENERIF_SUMMARY	[Estrogen, estrogen receptor and miR-21 in adenomyosis
Literature	GENERIF_SUMMARY	Estrogen receptor 1 chromatin profiling in human breast tumors re high inter-patient heterogeneity with enrichment of risk SNPs and enhancer activity at most-conserved regions.
Literature	GENERIF_SUMMARY	Transcription factors ERalpha and Sox2 have differing multiphasic and RNA-binding mechanisms.
Literature	GENERIF_SUMMARY	Asymmetric allostery in estrogen receptor-alpha homodimers drive responses to the ensemble of estrogens in the hormonal milieu.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Association of ESR1 Germline Variants with TP53 Somatic Variant Breast Tumors in a Genome-wide Study.
Literature	GENERIF_SUMMARY	Genomic analysis uncovers that cold-inducible RNA binding protei associated with estrogen receptor in breast cancer.
Literature	GENERIF_SUMMARY	Analysis of Single Nucleotide Polymorphisms (SNPs) rs2234693 a rs9340799 of the ESR1 Gene and the Risk of Breast Cancer.
Literature	GENERIF_SUMMARY	USP36 promotes tumorigenesis and tamoxifen resistance in breas cancer by deubiquitinating and stabilizing ERalpha.
Literature	GENERIF_SUMMARY	Association of Estrogen Receptor-alpha and Aryl Hydrocarbon Rec Gene Polymorphisms with Ischemic Stroke in an Egyptian Populati
Literature	GENERIF_SUMMARY	The N-terminal activation function AF-1 domain of ERalpha interac directly with the C-terminal AF-2-holding ligand-binding domain to the coactivator proteins.
Pathways	KEGG_PATHWAY	Endocrine resistance
Pathways	KEGG_PATHWAY	Hormone signaling
Pathways	KEGG_PATHWAY	Estrogen signaling pathway
Pathways	KEGG_PATHWAY	Prolactin signaling pathway
Pathways	KEGG_PATHWAY	Thyroid hormone signaling pathway
Pathways	KEGG_PATHWAY	Endocrine and other factor-regulated calcium reabsorption
Pathways	KEGG_PATHWAY	Pathways in cancer
Pathways	KEGG_PATHWAY	Proteoglycans in cancer
Pathways	KEGG_PATHWAY	Chemical carcinogenesis - receptor activation
Pathways	KEGG_PATHWAY	Breast cancer
Pathways	REACTOME_PATHWAY	Signaling by ERBB4
Pathways	REACTOME_PATHWAY	Nuclear signaling by ERBB4
Pathways	REACTOME_PATHWAY	PIP3 activates AKT signaling
Pathways	REACTOME_PATHWAY	Developmental Biology
Pathways	REACTOME_PATHWAY	Signal Transduction
Pathways	REACTOME_PATHWAY	Disease
Pathways	REACTOME_PATHWAY	Negative regulation of the PI3K/AKT network
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway
Pathways	REACTOME_PATHWAY	PI3K/AKT Signaling in Cancer
Pathways	REACTOME_PATHWAY	Constitutive Signaling by Aberrant PI3K in Cancer
Pathways	REACTOME_PATHWAY	Cellular responses to stress
Pathways	REACTOME_PATHWAY	SUMOylation

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	SUMO E3 ligases SUMOylate target proteins
Pathways	REACTOME_PATHWAY	Nuclear Receptor transcription pathway
Pathways	REACTOME_PATHWAY	Metabolism of proteins
Pathways	REACTOME_PATHWAY	SUMOylation of intracellular receptors
Pathways	REACTOME_PATHWAY	Diseases of signal transduction by growth factor receptors and se messengers
Pathways	REACTOME_PATHWAY	Deubiquitination
Pathways	REACTOME_PATHWAY	Ovarian tumor domain proteases
Pathways	REACTOME_PATHWAY	Post-translational protein modification
Pathways	REACTOME_PATHWAY	PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)
Pathways	REACTOME_PATHWAY	Transcriptional regulation by the AP-2 (TFAP2) family of transcripti factors
Pathways	REACTOME_PATHWAY	TFAP2 (AP-2) family regulates transcription of growth factors and receptors
Pathways	REACTOME_PATHWAY	Transcriptional regulation by RUNX2
Pathways	REACTOME_PATHWAY	Transcriptional regulation by RUNX1
Pathways	REACTOME_PATHWAY	RUNX1 regulates estrogen receptor mediated transcription
Pathways	REACTOME_PATHWAY	ESR-mediated signaling
Pathways	REACTOME_PATHWAY	RUNX1 regulates transcription of genes involved in WNT signaling
Pathways	REACTOME_PATHWAY	Regulation of RUNX2 expression and activity
Pathways	REACTOME_PATHWAY	Cellular responses to stimuli
Pathways	REACTOME_PATHWAY	Intracellular signaling by second messengers
Pathways	REACTOME_PATHWAY	Signaling by Nuclear Receptors
Pathways	REACTOME_PATHWAY	Signaling by Receptor Tyrosine Kinases
Pathways	REACTOME_PATHWAY	Extra-nuclear estrogen signaling
Pathways	REACTOME_PATHWAY	Estrogen-dependent gene expression
Pathways	REACTOME_PATHWAY	Developmental Cell Lineages
Pathways	REACTOME_PATHWAY	Developmental Cell Lineages of the Integumentary System
Pathways	REACTOME_PATHWAY	Mitochondrial unfolded protein response (UPRmt)
Pathways	REACTOME_PATHWAY	Developmental Lineages of the Mammary Gland
Pathways	REACTOME_PATHWAY	Developmental Lineage of Mammary Gland Luminal Epithelial Cell
Pathways	REACTOME_PATHWAY	Developmental Lineage of Mammary Gland Alveolar Cells

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Nuclear receptors
Pathways	WIKIPATHWAYS	Integrated breast cancer pathway
Pathways	WIKIPATHWAYS	miR targeted genes in muscle cell
Pathways	WIKIPATHWAYS	Leptin signaling
Pathways	WIKIPATHWAYS	Aryl hydrocarbon receptor pathway
Pathways	WIKIPATHWAYS	Mammary gland development puberty stage 2 of 4
Pathways	WIKIPATHWAYS	Mammary gland development pregnancy and lactation stage 3 of
Pathways	WIKIPATHWAYS	Estrogen receptor pathway
Pathways	WIKIPATHWAYS	Nuclear receptors meta pathway
Pathways	WIKIPATHWAYS	Breast cancer pathway
Pathways	WIKIPATHWAYS	Relationship between inflammation COX 2 and EGFR
Pathways	WIKIPATHWAYS	Oxysterols derived from cholesterol
Pathways	WIKIPATHWAYS	Male infertility
Pathways	WIKIPATHWAYS	16p11 2 proximal deletion syndrome
Pathways	WIKIPATHWAYS	Estradiol regulation in porto sinusoidal vascular disease
Pathways	WIKIPATHWAYS	Genetic causes of porto sinusoidal vascular disease
Pathways	WIKIPATHWAYS	Female steroid hormones in cardiomyocyte energy metabolism
Pathways	WIKIPATHWAYS	1p36 copy number variation syndrome
Pathways	WIKIPATHWAYS	Sulfatase and aromatase pathway
Pathways	WIKIPATHWAYS	Cancer pathways
Pathways	WIKIPATHWAYS	Primary ovarian insufficiency
Pathways	WIKIPATHWAYS	Cyclin dependent kinase 4 6 inhibitors in breast cancer
Pathways	WIKIPATHWAYS	Estrogen signaling
Protein Domains	INTERPRO	Nucl_hrmn_rcpt_lig-bd
Protein Domains	INTERPRO	Estr_rcpt
Protein Domains	INTERPRO	Znf_hrmn_rcpt
Protein Domains	INTERPRO	Nuclear_hrmn_rcpt
Protein Domains	INTERPRO	Znf_NHR/GATA
Protein Domains	INTERPRO	Est_rcpt/est-rel_rcp

Main Category	Category	Terms
Protein Domains	INTERPRO	Oestrogen-typ_rcpt_final_C_dom
Protein Domains	INTERPRO	NHR-like_dom_sf
Protein Domains	INTERPRO	Estr_rcpt_N
Protein Domains	INTERPRO	Nuclear_hormone_rcpt_NR3
Protein Domains	PIR_SUPERFAMILY	ER-like_NR
Protein Domains	PIR_SUPERFAMILY	ER-a
Protein Domains	SMART	ZnF_C4
Protein Domains	SMART	HOLI
Protein Domains	UP_KW_DOMAIN	Transmembrane
Protein Domains	UP_KW_DOMAIN	Zinc-finger